



PINNACLE

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edition**

Complete Arithmetic and Advance maths

Maths Formula Book

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- Smart methods
- solved examples
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- Calculation tricks

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PINNACLE Publications

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Geometry

Topic of Geometry

Lines & Angles Triangle Polygon Quadrilateral Circle

Lines

POINT :- Point is a circle having zero radius. It is usually represented by a dot (.) .

Type of Points

Collinear Points	Non - Collinear Point
Points are said to be collinear when a set of three or more points exist on the same straight line.	The points do not exist on the same straight line.
	

Note :- From a point, an infinite number of lines can be drawn.

LINE :- A line is a straight one - dimensional figure that does not have a thickness, and it extends endlessly in both directions.

Types of Line

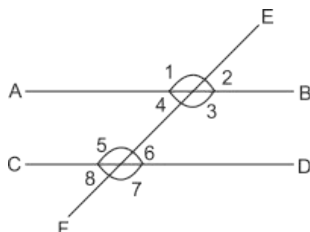
1. Parallel Line :- A pair of two lines that are on the same plane and the distance between them is equal and remains constant.

Note: These lines do not intersect with each other.

i.e.



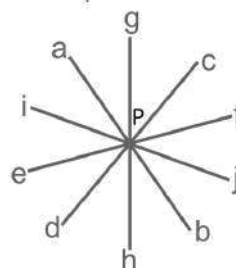
2. Transversal Lines :- A line which cuts parallel lines at a distinct point, as shown in the figure below.



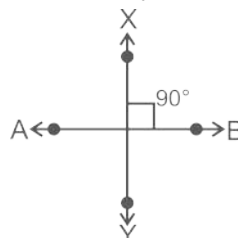
- EF is the transversal line. AB and CD are parallel lines. Symbol of parallel lines is '||'. To denote a line symbols like $\overline{AB}$ or simply AB are used.

3. Intersecting Line :- When two or more pairs of lines are on the same plane and intersect each other at one given point, they are known as intersecting lines.

P is the point of intersection



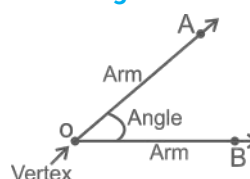
4. Perpendicular Line :- When two lines on the same plane intersect each other and form a 90° angle at the point of intersection, they are known to be perpendicular lines.



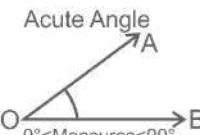
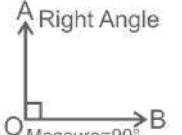
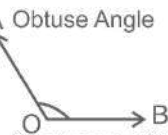
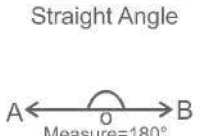
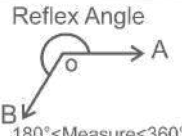
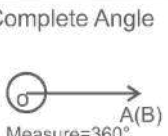
Angles

The measure of the 'opening' between two lines/rays is called an 'angle'. It is represented by the symbol " $\angle$ ".

Parts of Angle :-

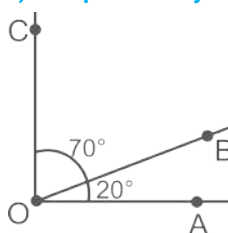


Types of Angle

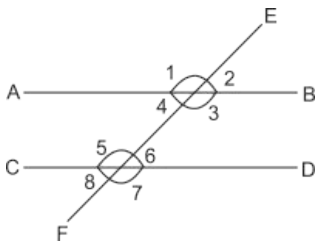
Acute Angle  $0^\circ < \text{Measure} < 90^\circ$	Right Angle  $\text{Measure} = 90^\circ$	Obtuse Angle  $90^\circ < \text{Measure} < 180^\circ$
Straight Angle  $\text{Measure} = 180^\circ$	Reflex Angle  $180^\circ < \text{Measure} < 360^\circ$	Complete Angle  $\text{Measure} = 360^\circ$

Pair of Angles

1) Complementary Angles :- Sum of two angles is equal to 90° .

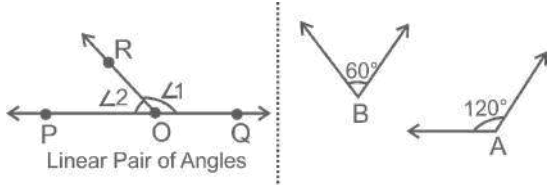


2) Supplementary Angles :- Sum of the two angles is equal to 180° . E.g. $\angle 1 + \angle 2 = 180^\circ$, $\angle 4 + \angle 3 = 180^\circ$ etc.

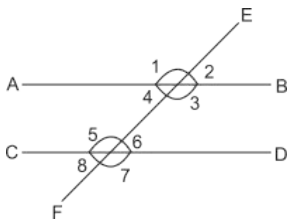


3) Linear pair of Angles :- A linear pair of angles is a pair of adjacent angles formed when two lines intersect each other at a point.

Linear pair of angles V/S Supplementary Angles :-



When a transversal line intersects two parallel lines :-



1) Corresponding Angles :-

$$\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 4 = \angle 8, \angle 3 = \angle 7$$

2) Alternate Angles :-

$$(\angle 1 = \angle 7), (\angle 2 = \angle 8), (\angle 4 = \angle 6), (\angle 3 = \angle 5)$$

3) Vertically opposite angles :-

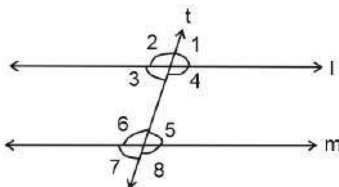
$$(\angle 1 = \angle 3), (\angle 2 = \angle 4), (\angle 6 = \angle 8), (\angle 5 = \angle 7)$$

4) Adjacent angles:- $\angle 1 + \angle 2 = 180^\circ$ in this case as these are linear pairs. It is not necessary that their sum should be 180° .

5) Co-Interior angles :- Sum of Interior angles on same side = 180° . E.g. $\angle 3 + \angle 6 = 180^\circ$, $\angle 4 + \angle 5 = 180^\circ$. The angle made by bisectors of interior angle will be 90° .

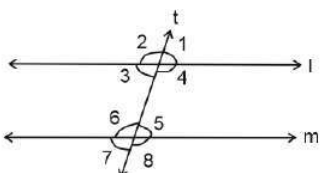
Example :- In the given figure, $l \parallel m$ and t is a transversal. If $\angle 1$ and $\angle 2$ are in the ratio 4 : 11, the measures of the angles $\angle 7$ and $\angle 8$, respectively, are :

RRC Group D 09/09/2022 (Morning)



- (a) 110° and 70° (b) 87° and 93°
(c) 132° and 48° (d) 65° and 115°

Solution :-



$$\text{Ratio of } \angle 1 \text{ and } \angle 2 = 4 : 11$$

$$4x + 11x = 180$$

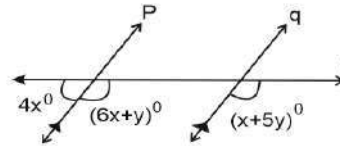
$$\Rightarrow 15x = 180 \Rightarrow x = 12$$

$$\angle 1 = 48^\circ, \text{ and } \angle 2 = 132^\circ$$

$$\angle 7 = \angle 3 = \angle 1 = 48^\circ$$

$$\angle 8 = \angle 4 = \angle 2 = 132^\circ$$

Example :- In the given figure, $p \parallel q$ and l is a transversal cutting p and q with angles as specified. The value of $(5x - y)$ is :



RRC Group - D 06/09/2022 (Evening)

- (a) 60 (b) 40 (c) 96 (d) 116

Solution :- When two lines are parallel then corresponding angles are equal

$$\text{So, } 6x + y = x + 5y$$

$$5x = 4y \Rightarrow x = \frac{4y}{5}$$

On a straight line sum of angles = 180°

$$\text{So, } 4x + 6x + y = 180^\circ$$

$$10x + y = 180^\circ \dots (1)$$

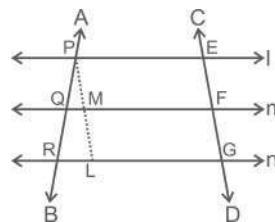
By putting value of x in equation (1)

$$8y + y = 180^\circ \text{ so, } y = 20^\circ$$

$$\text{And } x = \frac{4 \times 20^\circ}{5} = 16^\circ$$

$$\text{Then the value of } 5x - y = 16^\circ \times 5 - 20^\circ = 60^\circ$$

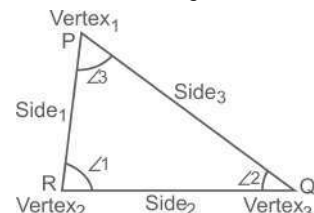
If three or more parallel lines are intersected by two transversals -



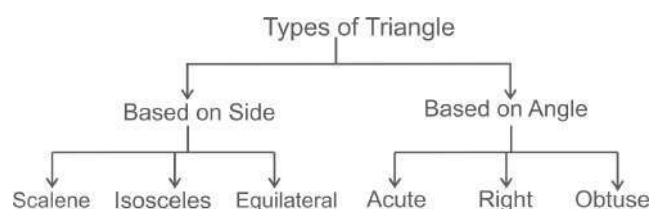
$$\text{Results} \rightarrow \frac{PQ}{QR} = \frac{EF}{FG}$$

Triangles

A triangle has three sides, three angles, and three vertices.



Types of Triangles



By Side	By Angle
 Scalene Triangle has no equal sides	 Acute Triangle has three angles $< 90^\circ$
 Isosceles Triangle has two equal sides	 Right Triangle has one angle $= 90^\circ$
 Equilateral Triangle has three equal sides	 Obtuse Triangle has one angle $> 90^\circ$

Example:- In a triangle ABC, the three angles are x , y and $y + 10$. Also, $2x - 4y = 20^\circ$. Which type of triangle is ABC?
SSC CGL 26/07/2023 (3rd shift)

(a) Equilateral (b) Obtuse (c) Acute (d) Right-angled

Solution :- According to question,

$$x + y + (y + 10) = 180^\circ$$

$$x + 2y = 170^\circ \dots\dots(1)$$

$$2x - 4y = 20^\circ \Rightarrow x - 2y = 10^\circ \dots\dots(2)$$

By solving both equations, we get

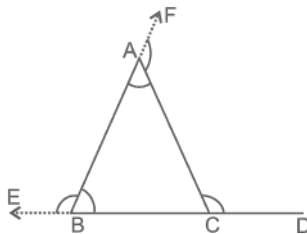
$$x = \frac{170^\circ + 10^\circ}{2} = 90^\circ$$

So, triangle ABC is a right angled triangle.

Special cases of a Right-angled triangle :-

Isosceles Right - angled triangle	Scalene Right - angled triangle
• $45^\circ - 45^\circ - 90^\circ$ triangle	• $30^\circ - 60^\circ - 90^\circ$ triangle
• The angles of this triangle are in the ratio – $1 : 1 : 2$ • The sides of this triangle will be in the ratio – $1 : 1 : \sqrt{2}$ respectively.	• The angles of this triangle are in the ratio – $1 : 2 : 3$ • The sides opposite to these angles will be in the ratio – $1 : \sqrt{3} : 2$ respectively

Properties of Triangles



Angle Sum Property :- The sum of angles of a triangle is always 180° .

$$\angle ABC + \angle BAC + \angle ACB = 180^\circ \text{ (Angle sum property)}$$

Triangle inequality :-

(a) The sum of the length of any two sides of a triangle is greater than the length of the third side.

- $AB + BC > AC$
- $AB + AC > BC$
- $BC + AC > AB$

(b) The difference between the lengths of any two sides of a triangle is always less than the length of the third side

- $AB - BC < AC$
- $AB - AC < BC$
- $BC - AC < AB$

(c) Let ABC be a triangle with sides a , b , c .

If $c^2 = a^2 + b^2$, then the angle at C is a right angle.

If $c^2 < a^2 + b^2$, then the angle at C is acute angle.

If $c^2 > a^2 + b^2$, then the angle at C is obtuse angle.

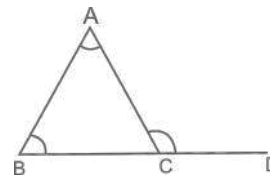
Angle Side Inequality:-

(a) The side opposite to the largest angle of a triangle is the largest side.

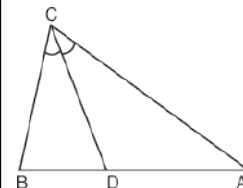
(b) The side opposite to the smallest angle is the shortest side.

Exterior Angle property:- Any exterior angle of the triangle is equal to the sum of its interior opposite angles. This is called the exterior angle property of a triangle.

$$\angle ACD = \angle ABC + \angle CAB \text{ (Exterior Angle Property)}$$



Example :- In the given triangle, CD is the bisector of $\angle BCA$. $CD = DA$. If $\angle BDC = 76^\circ$, what is the degree measure of $\angle CBD$?

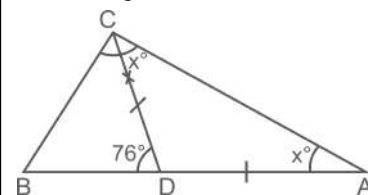


SSC CGL 01/12/2022 (4th Shift)

(a) 32° (b) 76° (c) 80° (d) 66°

Solution :-

Exterior angle of a triangle is equal to Sum of two opposite interior angles.



In $\triangle CDA$

$$\angle DCA = \angle DAC \dots\dots (DC = DA)$$

$$\angle DCA + \angle DAC = 76^\circ \dots\dots (\text{exterior angle})$$

$$x^\circ + x^\circ = 76^\circ \Rightarrow x^\circ = 38^\circ$$

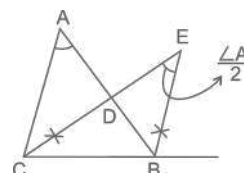
Now, $\angle DCA = \angle DCB = 38^\circ \dots (CD \text{ is angle bisector of } \angle C)$

Now, In $\triangle CBD$

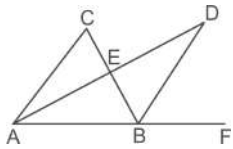
$$\angle DCB + \angle CBD + \angle BDC = 180^\circ$$

$$\Rightarrow 38^\circ + \angle CBD + 76^\circ = 180^\circ \Rightarrow \angle CBD = 66^\circ$$

- In a triangle ABC, angle bisector of interior angle C and exterior angle B meet at E, then $\angle BEC = \frac{\angle A}{2}$



Example :- In the given figure, AD is bisector of angle $\angle CAB$ and BD is bisector of angle $\angle CBF$. If the angle at C is 34° , the angle $\angle ADB$ is:



SSC CHSL 13/10/2020 (Morning)

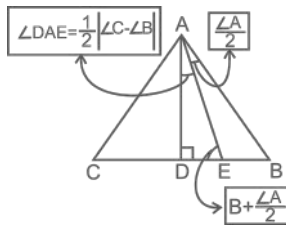
(a) 34° (b) 32° (c) 17° (d) 16°

Solution :- $\angle ADB = \frac{1}{2} \angle ACB$

$$\angle ADB = \frac{1}{2} \times 34^\circ = 17^\circ$$

- In $\triangle ABC$, AE is angle Bisector of $\angle A$ and $AD \perp BC$

$$\text{then } \angle DAE = \frac{1}{2} |\angle C - \angle B|$$

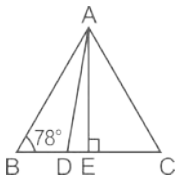


Example :- In $\triangle ABC$, $\angle B = 78^\circ$, AD is a bisector of $\angle A$ meeting BC at D, and $AE \perp BC$ at E. If $\angle DAE = 24^\circ$, then the measure of $\angle ACB$ is:

SSC CGL Tier II (29/01/2022)

(a) 32° (b) 38° (c) 30° (d) 42°

Solution :-



As we know,

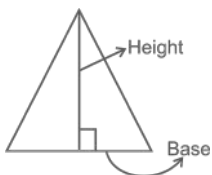
$$\frac{1}{2} (\angle ABC - \angle ACB) = \angle DAE$$

$$\Rightarrow \frac{1}{2} (78^\circ - \angle ACB) = 24^\circ$$

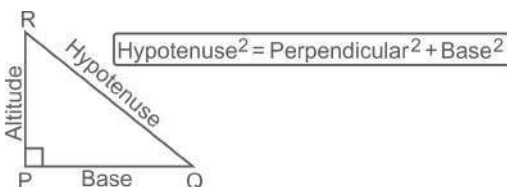
$$78^\circ - \angle ACB = 48^\circ$$

$$\angle ACB = 78^\circ - 48^\circ = 30^\circ$$

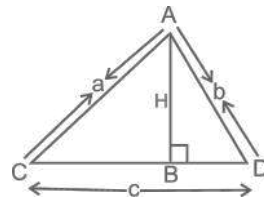
Height and Base of Triangle :- The height of a triangle is equal to the length of the perpendicular dropped from a vertex to its opposite side, and this side is considered the base.



Pythagoras Theorem :- In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.



Area of Triangle :- The area of a triangle is the region occupied by the triangle in 2d space.



$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

In $\triangle ABD$,

$$\sin D = \frac{H}{AD} \text{ or } AD = \frac{H}{\sin D}$$

General Formula:

$$\text{Area} = \frac{1}{2} \times b \times c \times \sin D$$

$$= \frac{1}{2} \times a \times c \times \sin C$$

$$= \frac{1}{2} \times a \times b \times \sin A$$

Example :- Find the area of a triangle whose length of two sides are 4 cm and 5 cm and the angle between them is 45° .
SSC CPO 04/10/2023 (2nd Shift)

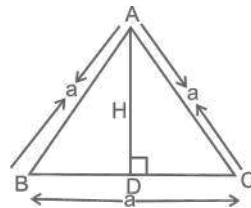
(a) $7\sqrt{2} \text{ cm}^2$ (b) $4\sqrt{2} \text{ cm}^2$ (c) $6\sqrt{2} \text{ cm}^2$ (d) $5\sqrt{2} \text{ cm}^2$

Solution :- According to the question,

Sine rule = $\frac{1}{2} \times a \times b \times \sin \theta$ (By using the formula)

$$\text{Area of triangle} = \frac{1}{2} \times 4 \times 5 \times \frac{1}{\sqrt{2}} = 5\sqrt{2}$$

Formulas of triangle :-



In case of equilateral triangle,

$$\bullet \text{ Area}(A) = \frac{\sqrt{3}}{4} (\text{side})^2$$

$$\bullet \text{ Inradius}(r) = \frac{\text{side}}{2\sqrt{3}}$$

$$\bullet \text{ circumradius}(R) = \frac{\text{side}}{\sqrt{3}}$$

$$\bullet \text{ height}(H) = \frac{\sqrt{3}}{2} \times \text{side}$$

Example:- The perimeter of an equilateral triangle is 36 m and the length of its altitude is $6\sqrt{3}$ m. The area of triangle is:
RRB NTPC CBT - I 03/02/2021 (Morning)

(a) $36\sqrt{3} \text{ m}^2$ (b) 18 m^2 (c) 24 m^2 (d) 12 m^2

Solution :-

Perimeter of equilateral triangle = 36 m

$$\text{Each Side of triangle} = \frac{36}{3} = 12 \text{ m,}$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} (\text{side})^2 = \frac{\sqrt{3}}{4} \times 12 \times 12 = 36\sqrt{3} \text{ m}^2$$

Example :- An equilateral triangle has sides of 18 cm each. The ratio of the inradius to circumradius of the triangle is:
SSC CHSL 14/08/2023 (4th Shift)

(a) 2 : 1 (b) 3 : 2 (c) 3 : 4 (d) 1 : 2

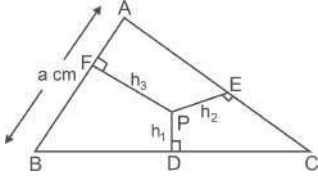
Solution :-

$$\text{Inradius of equilateral } \Delta = \frac{\text{side}}{2\sqrt{3}}$$

$$\text{circumradius of equilateral } \Delta = \frac{\text{side}}{\sqrt{3}}$$

Required ratio = 1 : 2

- If P is any point inside an equilateral triangle, then the sum of perpendiculars drawn from point P to sides AB, BC and AC is equal to the height of the triangle.



$$h_1 + h_2 + h_3 = \frac{\sqrt{3}}{2} a$$

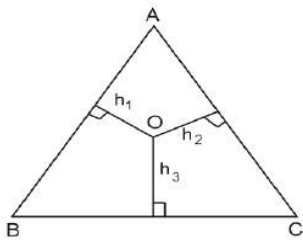
Where a = side of equilateral triangle

Example:- 'O' is a point in the interior of an equilateral triangle. The perpendicular distance from 'O' to the sides are $\sqrt{3}$ cm, $2\sqrt{3}$ cm, $5\sqrt{3}$ cm. The perimeter of the triangle is:

SSC CGL 13/12/2022 (4th Shift)

- (a) 48 cm (b) 32 cm (c) 24 cm (d) 64 cm

Solution :-

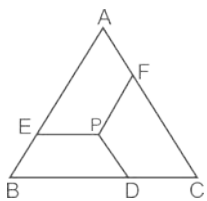


$$\text{Side (a)} = \frac{2}{\sqrt{3}} (h_1 + h_2 + h_3)$$

$$= \frac{2}{\sqrt{3}} (\sqrt{3} + 2\sqrt{3} + 5\sqrt{3}) = 16 \text{ cm}$$

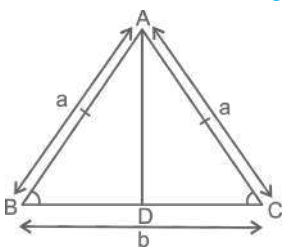
$$\text{Perimeter of equilateral triangle} = 3 \times a = 48 \text{ cm}$$

- If P is any point inside the equilateral triangle and PF || AB, PD || AC, PE || CB.



$$PD + PE + PF = AB = BC = CA.$$

In case of isosceles triangle :-



$$\text{Area(A)} = \frac{BC}{4} \sqrt{4AB^2 - BC^2}$$

$$\text{Inradius(r)} = \frac{\text{area of triangle}}{\text{Semi perimeter}}$$

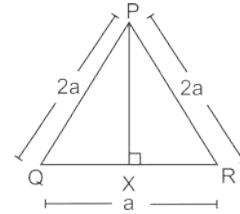
$$\text{circumradius(R)} = \frac{abc}{4 \times \text{Area of triangle}}$$

$$\text{height(h)} = \sqrt{a^2 - \frac{b^2}{4}}$$

Example :- ΔPQR is an isosceles triangle and $PQ = PR = 2a$ unit, $QR = a$ unit. Draw $PX \perp QR$, and find the length of PX .
SSC CGL 06/12/2022 (4th Shift)

- (a) $\sqrt{5}a$ (b) $\frac{\sqrt{5}a}{2}$ (c) $\frac{\sqrt{10}a}{2}$ (d) $\frac{\sqrt{15}a}{2}$

Solution :-



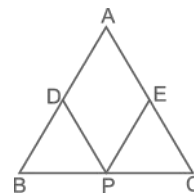
In an isosceles triangle, perpendicular drawn on the base from the opposite vertex bisect the base.

$$QX = XR = \frac{a}{2}$$

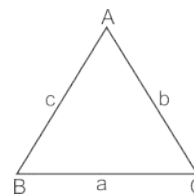
In right angle triangle PQX,

$$PX = \sqrt{(2a)^2 - \left(\frac{a}{2}\right)^2} = \sqrt{\frac{15a^2}{4}} = \frac{\sqrt{15}a}{2}$$

- If P is a point on the side BC and $DP \parallel AC$ & $EP \parallel AB$ then $DP + EP = AB = AC$



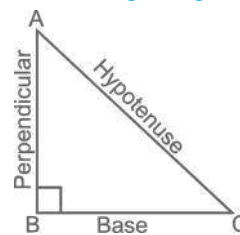
In case of scalene triangle :-



$$\text{Semi-perimeter (s)} = \frac{a + b + c}{2}$$

$$\text{Area(A)} = \sqrt{s(s-a)(s-b)(s-c)}$$

In case of right angle triangle :-

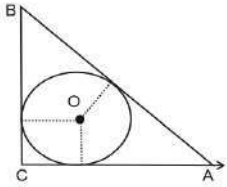


$$\text{Area(A)} = \frac{1}{2} \times \text{Base} \times \text{height}$$

$$\text{Inradius(r)} = \frac{\text{Perpendicular} + \text{base} - \text{hypotenuse}}{2}$$

$$\text{circumradius(R)} = \frac{\text{hypotenuse}}{2}$$

Example:- A circle of radius 4 cm is drawn inscribed in a right angle triangle ABC, right angled at C. If AC = 12 cm, then the value of CB is:



SSC CGL 25/07/2023 (4th shift)

(a) 8 cm (b) 12 cm (c) 20 cm (d) 16 cm

Solution :-

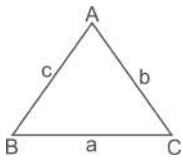
Pythagoras triplets :- (12, 16, 20)

Using hit and trial method,

AC = 12 cm, BC = 16 cm and AB = 20 cm

Inradius = $\frac{16+12-20}{2} = 4$ cm(satisfy)

Sine rule :-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Where R = circumradius of triangle

Example :- In a triangle, if the angles are in the ratio 1 : 2 : 3, then the ratio of the corresponding sides is:

Selection Post 04/08/2022 (4th Shift)

(a) 1 : 1 : 2 (b) 1 : 2 : $\sqrt{3}$ (c) 1 : $\sqrt{3}$: 2 (d) 1 : 2 : 3

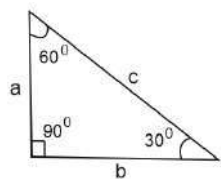
Solution :- Let the angles of a triangle be x, 2x and 3x, $\angle A + \angle B + \angle C = 180^\circ$

$$6x = 180^\circ \Rightarrow x = \frac{180}{6} = 30^\circ$$

Now,

$$2x = 2 \times 30^\circ = 60^\circ \Rightarrow 3x = 3 \times 30^\circ = 90^\circ$$

So, we have following triangle;



Using sine rule, we have;

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 60^\circ} = \frac{c}{\sin 90^\circ} = K$$

$$a : b : c = \frac{1}{2}K : \frac{\sqrt{3}}{2}K : K = 1 : \sqrt{3} : 2$$

So, the required ratio = $1 : \sqrt{3} : 2$

Cosine rule :-

$$\cos A = \frac{c^2 + b^2 - a^2}{2bc}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Example:- Side AB of a triangle ABC is 80 cm long, whose

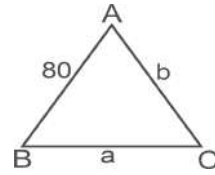
perimeter is 170 cm. If angle ABC = 60° , the shortest side of triangle ABC measures _____ cm.

SSC CPO 16/03/2019 (Evening)

(a) 17 (b) 15 (c) 25 (d) 21

Solution :-

$$a + b + 80 = 170$$



$$a + b = 90 \Rightarrow a = 90 - b$$

From cosine rule

$$\cos 60^\circ = \frac{a^2 + 80^2 - b^2}{2 \times 80 \times a}$$

$$80(90 - b) = (90 - b)^2 + 80^2 - b^2$$

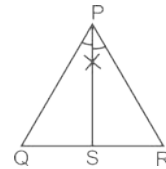
$$7200 - 80b = 8100 - 180b + 6400$$

$$100b = 7300 \Rightarrow b = 73$$

$$a = 90 - 73 = 17$$

Some Important Result of Angle Bisector

When bisect internally :



$$\frac{PQ}{PR} = \frac{QS}{SR} \text{ or } \frac{PQ}{QS} = \frac{PR}{SR}$$

Here PS is the internal angle bisector and a common side of the triangles $\triangle PQS$ and $\triangle PSR$.

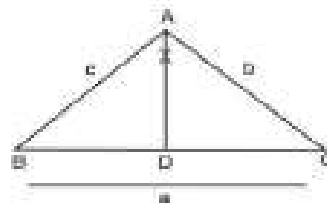
Example:- In $\triangle ABC$, AD, the bisector of $\angle A$, meets BC at D. If BC = a, AC = b and AB = c, then BD - DC =

SSC CHSL 03/07/2019 (Afternoon)

(a) $\frac{ac}{b+c}$ (b) $\frac{a(c+b)}{c-b}$ (c) $\frac{a(c-b)}{c+b}$ (d) $\frac{ab}{b+c}$

Solution:- BC = a cm

Let BD = x, DC = a - x



We know that, if AD is bisector of $\angle BAC$ then,

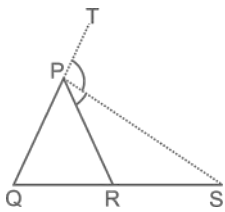
$$\frac{AB}{AC} = \frac{BD}{DC} \Rightarrow \frac{c}{b} = \frac{x}{a-x}$$

$$\Rightarrow ac - cx = bx \Rightarrow x = \frac{ac}{b+c}$$

$$\Rightarrow DC = a - x \Rightarrow DC = a - \frac{ac}{b+c} = \frac{ab}{b+c}$$

$$BD - DC = \frac{ac}{b+c} - \frac{ab}{b+c} = \frac{a(c-b)}{c+b}$$

When bisect externally :-



$$\frac{PQ}{PR} = \frac{QS}{RS}$$

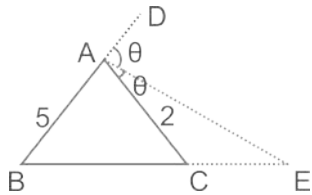
PS is the external angle bisector.

Example :- In a triangle ABC AB : AC = 5 : 2. BC = 9 cm. BA is produced to D, and the bisector of the Angle CAD meets BC produced at E. What is the length (in cm) of CE?

SSC CGL 13/08/2021 (Afternoon)

(a) 9 (b) 10 (c) 6 (d) 3

Solution :-



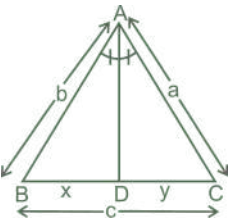
As AE is the external angle bisector so it divide the opposite sides in the ratio of the corresponding sides

Therefore,

$$\frac{AB}{AC} = \frac{BE}{CE} \Rightarrow \frac{5}{2} = \frac{CE + 9}{CE} \Rightarrow CE = 6 \text{ cm}$$

How to find length of angle bisector :-

(i)



AD = angle bisector of $\angle A$

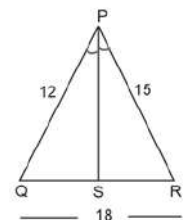
$$AD^2 = AB \times AC - BD \times DC$$

Example :- The bisector of $\angle QPR$ of $\triangle PQR$ meets the side QR at S. If PQ = 12 cm, PR = 15 cm and QR = 18 cm, then the length of SR and PS is:-

NTPC CBT II (14/06/2022) 2nd Shift

(a) 10, 10 (b) 12, 5 (c) 8, 7 (d) 13, 15

Solution :-



By angle bisector theorem,

$$\begin{aligned} \frac{PQ}{QS} &= \frac{PR}{SR} \\ \frac{12}{x} &= \frac{15}{18-x} \\ 216 - 12x &= 15x \end{aligned}$$

$$27x = 216 \Rightarrow x = 8$$

So, length of SR = 18 - 8 = 10 cm

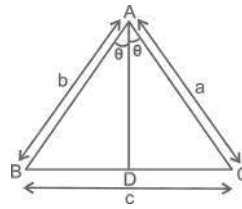
length of angle bisector $\rightarrow$

$$PS^2 = PQ \times PR - QS \times SR$$

$$= 12 \times 15 - 8 \times 10 \Rightarrow PS^2 = 180 - 80$$

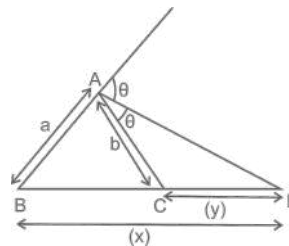
$$\Rightarrow PS = \sqrt{100} \Rightarrow PS = 10 \text{ cm}$$

(ii)



$$\text{Length of angle bisector}(AD) = \frac{2ab \cos \theta}{a + b}$$

(iii)



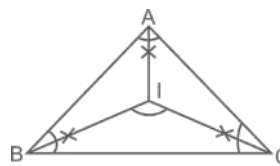
AD = angle bisector of $\angle A$

$$AD^2 = BD \times DC - AB \times AC$$

Centers of The Triangle

Incentre(r): The point of intersection of angle bisectors of a triangle.

1.



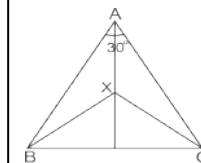
$$\angle BIC = 90^\circ + \frac{\angle A}{2}, \angle AIC = 90^\circ + \frac{\angle B}{2}, \angle AIB = 90^\circ + \frac{\angle C}{2}$$

Example :- In $\triangle ABC$, the internal bisectors of $\angle ABC$ and $\angle ACB$ meet at X and $\angle BAC = 30^\circ$. The measure of $\angle BXC$ is:

SSC CHSL 03/08/2023 (4th Shift)

(a) 120° (b) 115° (c) 105° (d) 150°

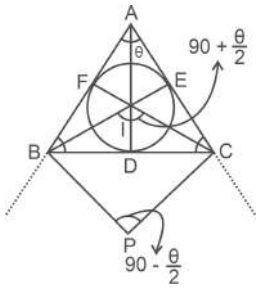
Solution :-



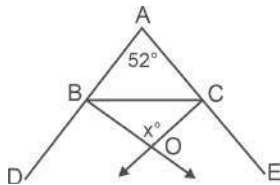
We know that :-

$$\angle BXC = 90^\circ + \frac{\angle BAC}{2} = 90^\circ + 15^\circ = 105^\circ$$

2.

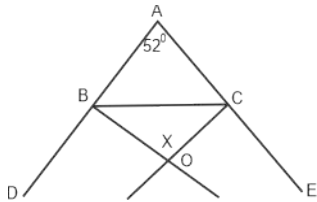


Example:- In the above figure, BO and CO are the angular bisectors of $\angle DBC$ and $\angle BCE$ respectively. The measure of x is _____.



RRB NTPC 01/04/2021 (Evening)
(a) 52° (b) 58° (c) 64° (d) 54°

Solution :-



Given, $\angle BAC = 52^\circ$, BO and CO are the angle bisectors of $\angle DBC$ and $\angle BCE$

$$\text{Now, } \angle X = \left(90^\circ - \frac{52^\circ}{2}\right) = (90^\circ - 26^\circ) = 64^\circ$$

3. Inradius (r):

- For any triangle :

$$r = \frac{A}{S}, \text{ (where A = area of triangle, S = semi perimeter of triangle)}$$

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

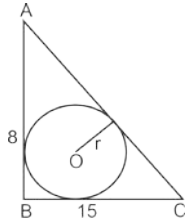
Example :- In a triangle ABC, right angled at B, $BC=15\text{cm}$, and $AB=8\text{ cm}$. A circle is inscribed in triangle ABC. Then the radius of the circle is:

RRB ALP 13/08/2018 (Evening)

(a) 2 cm (b) 4 cm (c) 3 cm (d) 1 cm

Solution :-

According to the question,



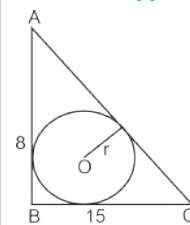
By using triplet (8, 15, 17) = AC = 17 cm

$$\Rightarrow S = \frac{AB+BC+AC}{2} = \frac{8+15+17}{2} = \frac{40}{2} = 20 \text{ cm}$$

Now,

$$\begin{aligned} \text{Inradius of circle (r)} &= \frac{A}{S} \\ &= \frac{\frac{1}{2} \times 8 \times 15}{20} = \frac{4 \times 15}{20} = 3 \text{ cm} \end{aligned}$$

Exam hall approach:-

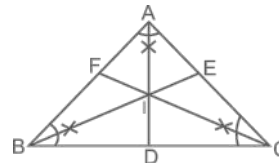


Using pythagoras theorem, we have :

$$AC = \sqrt{15^2 + 8^2} = \sqrt{289} = 17 \text{ cm}$$

$$\begin{aligned} \text{Inradius} &= \frac{AB+BC-AC}{2} = \frac{15+8-17}{2} \\ &= \frac{6}{2} = 3 \text{ cm} \end{aligned}$$

Concept :- Incentre divides each angle bisector in the ratio of length of sum of two adjacent sides and opposite side.

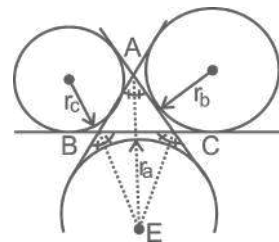


$$AI : ID = AB + AC : BC$$

$$BI : IE = AB + BC : AC$$

$$CI : IF = AC + BC : AB$$

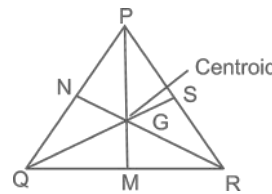
Excentre :- The intersection point of the internal angle bisector of one angle and bisectors of the other two opposite exterior angles.



$$(i) \angle BEC = 90^\circ - \frac{\angle A}{2}$$

$$(ii) \quad r_a = \frac{\Delta}{s-a}, \quad r_b = \frac{\Delta}{s-b}, \quad r_c = \frac{\Delta}{s-c}; \quad \text{where S is semiperimeter of triangle}$$

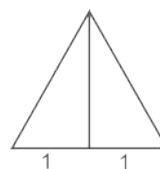
Centroid (G) :- It is the intersection point of all medians of a triangle. It is also called the center of mass.



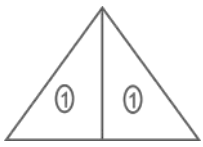
PM, QS, RN are median

Properties: of medians :-

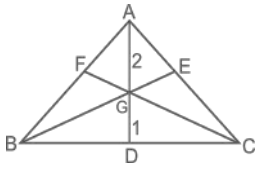
a) Median divides any side into two equal parts.



b) Median divides the triangle into two equal areas.

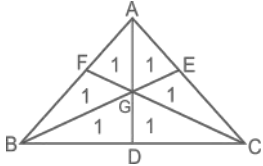


c) Centroid divides each median in the ratio of 2 : 1.

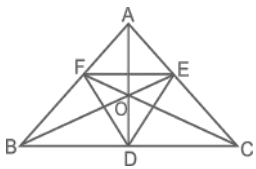


AG : GD = BG : GE = CG : GF = 2 : 1

d) Medians divide the triangle into 6 equal Parts.



e). area of triangle formed by joining midpoints of two sides and centroid is $\frac{1}{12}$ of area of triangle.



ar $\triangle OFE$ = ar $\triangle OFD$ = ar $\triangle OED$ = $\frac{1}{12}$ area of $\triangle ABC$

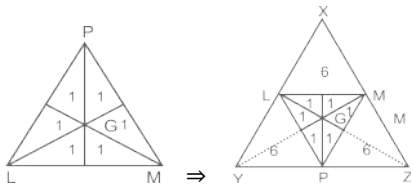
Example :- XYZ is a triangle. If the medians ZL and YM intersect each other at G, then (Area of $\triangle GLM$: Area of $\triangle XYZ$)

SSC CHSL 12/10/2020 (Afternoon)

(a) 1 : 14 (b) 1 : 12 (c) 1 : 11 (d) 1 : 10

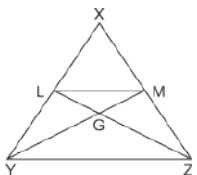
Solution :-

Median divides triangle in two equal parts



Area of $\triangle GLM$: Area of $\triangle XYZ = 2 : 24 = 1 : 12$

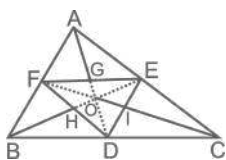
Exam hall approach :-



We know,

Area of $\triangle GLM$: Area of $\triangle XYZ = 1 : 12$

f). The line segment joining the midpoints of two sides divides the line joining of vertex in between line to the centroid in the ratio 3 : 1.



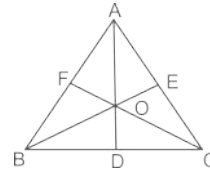
AG : GO = BH : HO = CI : IO = 3 : 1

g) Area of triangles formed by medians

$$= \frac{4}{3} \sqrt{s(s-a)(s-b)(s-c)}$$

$$S = \frac{\text{sum of the medians}}{2}$$

h) Sum of any two sides of a triangle is greater than twice the median with respect to the third side.



AB + AC > 2AD

BC + AC > 2CF

AB + BC > 2BE

So we can say that Sum of the perimeter of a triangle is greater than the sum of the medians.

AB + BC + AC > AD + BE + CF

also

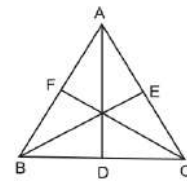
$$3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$$

Example:- If $\triangle ABC$ is an equilateral triangle in which D, E and F are the points on sides BC, AC, and AB respectively such that $AD \perp BC$, $BE \perp AC$ and $CF \perp AB$, which of the following is true ?
SSC CPO 11/12/2019 (Morning)

(a) $7AB^2 = 9AD^2$ (b) $2AB^2 = 3AD^2$

(c) $4AC^2 = 5BE^2$ (d) $3AC^2 = 4BE^2$

Solution :-



Since, ABC is an equilateral triangle

Altitude = Angle bisector = Median

We know that the following relation satisfies in case of an equilateral triangle.

$$3[AB^2 + BC^2 + AC^2] =$$

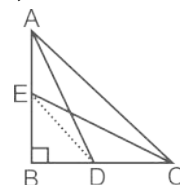
$$4[AD^2 + BE^2 + CF^2] \dots \text{eq. (1)}$$

$\Rightarrow AB = BC = AC$ ($\triangle ABC$ is an equilateral triangle)

Putting the values in eq. (1)

$$3[3AC^2] = 4[3BE^2] \Rightarrow 3AC^2 = 4BE^2$$

i)

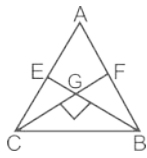


$$4(AD^2 + CE^2) = 5(AC^2)$$

$$(AD^2 + CE^2) = (AC^2 + ED^2)$$

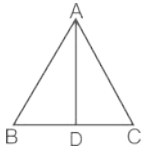
$$(AD^2 + CE^2) = 5(ED^2)$$

j) If two medians bisect at 90°



$$(AC^2 + AB^2) = 5(BC^2)$$

k) Apollonius theorem (used to evaluate length of median)

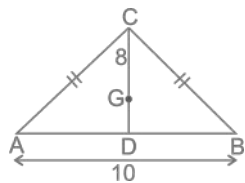


- $(AB^2 + AC^2) = 2(AD^2 + DC^2)$
- $(AB^2 + AC^2) = 2(AD^2 + \frac{BC^2}{4})$

Example :- In $\triangle ABC$, $AC = BC$, and the length of the base AB is 10 cm. If $CG = 8$ cm, where G is the centroid, then what is the length of AC (in cm) ?
SSC CHSL 12/08/2021 (Evening)

- (a) 13 (b) 15 (c) $\sqrt{91}$ (d) 12

Solution :-



$AC = BC$

G is the centroid so $GD = 4$ cm

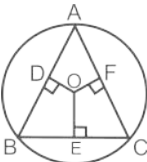
Therefore $CD = 8 + 4 = 12$ cm

By Apollonius theorem,

$$(AC^2 + BC^2) = 2(CD^2 + AD^2)$$

$$2AC^2 = 2(5^2 + 12^2) \Rightarrow AC^2 = 13^2 \Rightarrow AC = 13$$

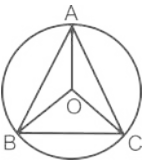
Circumcentre (O) : point of intersection of perpendicular bisector of sides.



$OD \perp AB$, $OF \perp AC$, $OE \perp BC$

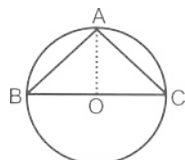
$AD = BD$, $AF = CF$, $BE = EC$

ii)



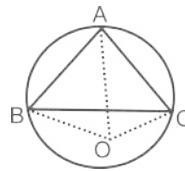
$$\angle BOC = 2\angle A, \angle AOB = 2\angle C, \angle AOC = 2\angle B$$

iii)



$$\angle BOC = 2\angle A, \angle AOB = 2\angle C, \angle AOC = 2\angle B$$

iv) In an obtuse angled triangle,



$$\angle BOC = 2(180^\circ - \angle A), \angle COA = 2\angle B \text{ and } \angle BOA = 2\angle C$$

Position of circumcentre in different triangles :-

(i) In an acute angled triangle it lies inside the triangle.

(ii) In a right angled triangle it is the midpoint of the hypotenuse.

(iii) In an obtuse triangle, it lies outside and in front of the obtuse angle of that triangle.

Circumradius (R) :-

- For any triangle

$$R = \frac{abc}{4 \times \text{area of triangle}} \text{ where } a, b, c \text{ is sides of the triangle}$$

- For right angled triangle

$$R = \frac{H}{2} \text{ where } H \text{ is hypotenuse}$$

- For Equilateral triangle

$$R = \frac{a}{\sqrt{3}} \text{ where } a = \text{side}$$

- Distance between circumcentre and incentre of any triangle

$$= \sqrt{R^2 - 2Rr}$$

Example :- The circumradius of a triangle is 9 cm while the inradius of it is 4 cm. What is the distance between the circumcentre and the incentre of the triangle?

RRB NTPC 27/03/2021 (Evening)

- (a) 4 cm (b) 2 cm (c) 3 cm (d) 5 cm

Solution :- Given, Radius of triangle (R) = 9 cm

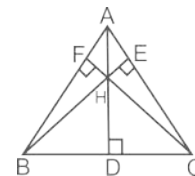
And the radius of the triangle (r) = 4 cm

Distance between circumcentre and center of triangle =

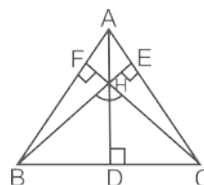
$$\sqrt{R^2 - 2Rr}$$

$$= \sqrt{9^2 - 2 \times 9 \times 4} = 3 \text{ cm}$$

Orthocentre (H):- point of intersection of altitudes of a triangle.



Useful results :-



(i) $\angle BHC = 180^\circ - \angle A$, $\angle BHA = 180^\circ - \angle C$, $\angle CHA = 180^\circ - \angle B$

(ii) $BH \times EH = FH \times HC$,

$BF \times EH = FH \times CE$

$BF \times HC = HB \times CE$ (by similarity of $\triangle BFH$ & $\triangle CEH$)

(iii) $BD \times DC = AD \times AH$

$BE \times EH = AE \times CE$

$CF \times FH = AF \times BF$

(iv) The perimeter of any triangle is greater than the sum of altitudes of the triangle.

$AB + BC + AC > AD + BE + CF$

Position of Orthocentre in different triangles :-

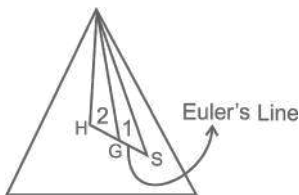
(i) In an acute angled triangle it lies inside the triangle.

(ii) In a right angled triangle it is at the vertex of the right angled triangle.

(iii) In an obtuse angle triangle, it lies outside and in the back side of the obtuse angle of that triangle.

Euler's Line :- According to the Euler's theory, in a triangle, there exists a straight line called the Euler's line, which passes through the orthocenter, the circumcentre, and the centroid of the triangle. Hence, these given points of concurrencies of the triangle are the collinear points in a triangle.

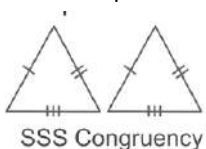
Relation between orthocentre(s), centroid (G) and circumcentre(H)



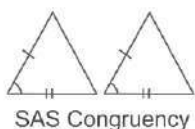
Congruency of Triangles

Two triangles will be congruent if:

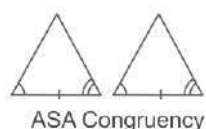
1. SSS: (Side – Side – Side rule) :- When all three corresponding sides are equal.



2. SAS: (Side – Angle – Side rule) :- When two corresponding sides and one corresponding angle are equal.

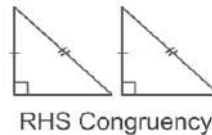


3. ASA: (Angle – Side – Angle rule): When two corresponding angles and one corresponding side are equal.



4. RHS: (Right angle – Hypotenuse – Side rule) :- When one corresponding side and corresponding hypotenuse of the right

angled triangle are equal.



5. AAS: (angle – angle – Side rule) :- When any two pairs of corresponding angles and corresponding sides are equal.

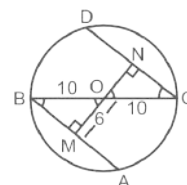
Example :- In a circle of diameter 20 cm, chords AB and CD are parallel to each other. BC is diameter. If AB is 6 cm from the centre of the circle, what is the length

(in cm) of the chord CD?

SSC CGL 13/04/2022 (Morning)

(a) 8 (b) 12 (c) 20 (d) 16

Solution :-



radius = $\frac{20}{2} = 10$ cm.

As $AB \parallel CD$, $\angle MBO = \angle NCO$ and

$\angle BOM = \angle CON$ (vertically opposite angles)

$BO = CO = 10$ (radius)

So, by ASA congruency $\triangle BOM \cong \triangle CON$

Now, by c.p.c.t., $ON = OM = 6$ cm

And $\angle OMB = \angle ONC = 90^\circ$

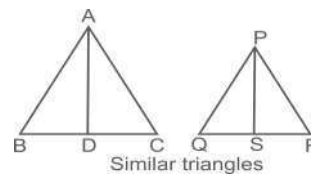
$NC = \sqrt{10^2 - 6^2} = 8$ cm

$CD = 2 \times NC = 16$ cm

Similarity of Triangles

1). Two triangles are similar if they have the same shape but vary in size.

2). In congruency the triangles are mirror images of each other. So, We can say that all congruent triangles are similar but the vice versa is not true.



3). Important properties of similar triangles :

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR} = \frac{AB^2}{PQ^2} = \frac{AC^2}{PR^2} = \frac{BC^2}{QR^2} = \frac{AD^2}{PS^2}$$

$$\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR} = \frac{AD}{PS}$$

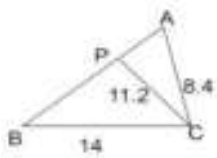
Example :- In $\triangle ABC$, $AC = 8.4$ cm, $BC = 14$ cm. P is a point on AB such that $CP = 11.2$ cm and $\angle ACP = \angle B$. What is the length (in cm) of BP ?

SSC CGL 04/03/2020 (Evening)

(a) 4.12 (b) 2.8 (c) 3.78 (d) 3.6

Solution :-

In $\triangle ABC$, $\angle ACP = \angle B$



In $\triangle ABC$ and $\triangle ACP$:

$\angle ACP = \angle B$,

$\angle A$ and side AC is common

$$\text{Hence, } \frac{AB}{AC} = \frac{BC}{PC} = \frac{AC}{AP}$$

$$\Rightarrow \frac{BC}{PC} = \frac{AC}{AP} \Rightarrow \frac{14}{11.2} = \frac{8.4}{AP} \Rightarrow AP = 6.72$$

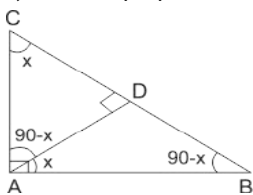
Similarly, in $\triangle ABC$ and $\triangle CBP$:

$$\Rightarrow AB \times PC = AC \times BC$$

$$\Rightarrow AB \times 11.2 = 8.4 \times 14$$

$$\Rightarrow AB = 10.5, BP = 10.5 - 6.72 = 3.78 \text{ cm}$$

4). When a perpendicular is drawn in right angled triangle



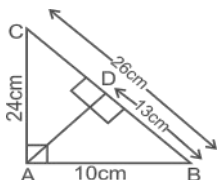
- $AD^2 = BD \times DC$
- $AB^2 = BD \times BC$
- $AC^2 = CD \times BC$
- $AD = \frac{AC \times AB}{BC}$

Example :- In $\triangle CAB$, $\angle CAB = 90^\circ$ and $AD \perp BC$. If $AC = 24$ cm, $AB = 10$ cm, then find the value of AD (in cm).

SSC CHSL 17/08/2023 (1st Shift)

(a) 9.23 (b) 8.23 (c) 7.14 (d) 10.23

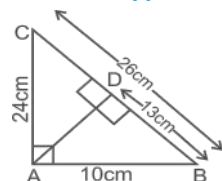
Solution :- According to the question,



By using triplet $(10, 24, 26) = CB = 26$ cm

$$\text{So, } AD = \frac{AC \times AB}{BC} = \frac{24 \times 10}{26} = 9.23 \text{ cm}$$

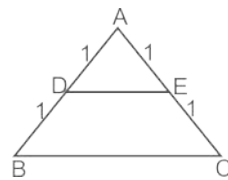
Exam Hall Approach :-



$\triangle CAB$ and $\triangle ADB$ is similar

$$\frac{26}{10} = \frac{24}{AD} \Rightarrow AD = \frac{240}{26} = 9.23$$

[1.] Basic proportionality theorem :- A line drawn parallel to one side divides the other two sides in the same ratio.



$$\frac{AD}{DB} = \frac{AE}{EC} \text{ and } \frac{AD}{AB} = \frac{DE}{BC}$$

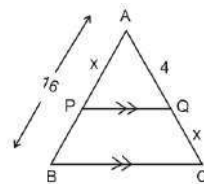
Example :- In $\triangle ABC$, the straight line parallel to the side BC meets AB and AC at the points P and Q, respectively. If $AP = QC$, the length of AB is 16 cm and the length of AQ is 4 cm, then the length (in cm) CQ is:

SSC CHSL 02/08/2023 (2nd Shift)

(a) $(2\sqrt{21} + 2)$ (b) $(2\sqrt{18} - 2)$

(c) $(2\sqrt{17} - 2)$ (d) $(2\sqrt{19} + 2)$

Solution :-



By using Thales theorem

$$\frac{AP}{AB} = \frac{AQ}{AC} \Rightarrow \frac{x}{16} = \frac{4}{x+4}$$

$$= x^2 + 4x - 64 = 0$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4 \times 1 \times (-64)}}{2 \times 1}$$

$$x = \frac{-4 \pm \sqrt{272}}{2 \times 1} = (2\sqrt{17} - 2)$$

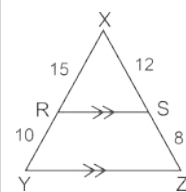
Example :- R and S are the points of the sides XY and XZ, respectively, of

$\triangle XYZ$. Also, $XR = 15$ cm, $XY = 25$ cm, $XS = 12$ cm and $XZ = 20$ cm. RS is equal to :

Selection Post 27/06/2023 (4th Shift)

(a) $\frac{2}{5} YZ$ (b) $\frac{5}{3} YZ$ (c) $\frac{3}{5} YZ$ (d) $\frac{3}{4} YZ$

Solution :-



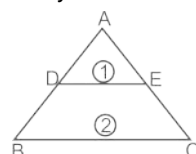
$$\frac{XR}{XY} = \frac{XS}{XZ} \Rightarrow \frac{15}{25} = \frac{12}{20} \Rightarrow \frac{3}{5}$$

So, $RS \parallel YZ$ and $\triangle XRS \sim \triangle XYZ$

$$\text{Using thales theorem, } \frac{XR}{XY} = \frac{RS}{YZ} = \frac{15}{25}$$

$$\frac{RS}{YZ} = \frac{3}{5} \Rightarrow RS = \frac{3}{5} \times YZ$$

[2.] Mid Point theorem: The line segment joining the midpoints of any two sides is parallel to and half of the third side.



$$DE = \frac{BC}{2}$$

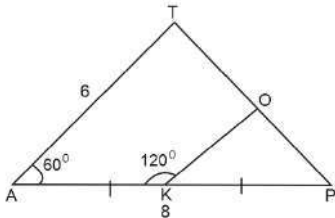
DE || BC.

Example :- $\triangle TAP$, $\angle TAP = 60^\circ$, $TA = 6\text{ cm}$, $AP = 8\text{ cm}$. K is the midpoint of AP. A line from K is produced to meet TP at O such that $\angle AKO = 120^\circ$. Find the length of OK.

Graduate Level 01/08/2022 (Shift - 4)

(a) 3 cm (b) 4 cm (c) 5 cm (d) 6 cm

Solution :-



Since,
 $\angle TAP + \angle OKA = 60^\circ + 120^\circ = 180^\circ$
 (pair of interior angles on the same side of transversal is supplementary)

So,

OK || AT and also K is the midpoint of AP. $\Rightarrow OK = \frac{AT}{2}$ (Mid point theorem)

$$\text{So, } OK = \frac{6}{2} = 3\text{ cm}$$

Mass point Theorem :-



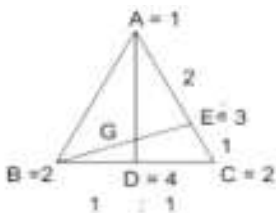
$$M_1X = M_2Y \Rightarrow \frac{M_1}{M_2} = \frac{Y}{X}$$

Example :- D is the midpoint of BC of $\triangle ABC$. Point E lies on AC such that $CE = \frac{1}{3}AC$. BE and AD intersect at G. What is $\frac{AG}{GD}$?

SSC CGL 05/03/2020 (Morning)

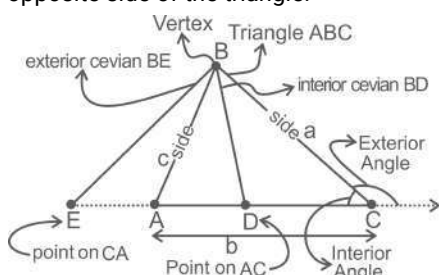
(a) 8 : 3 (b) 4 : 1 (c) 5 : 2 (d) 3 : 1

Solution :- Apply mass point theorem in the following diagram:

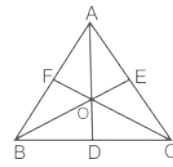


$$AG : GD = 4 : 1$$

Cevians : A line segment which joins a vertex of a triangle to a point on the opposite side of the triangle.



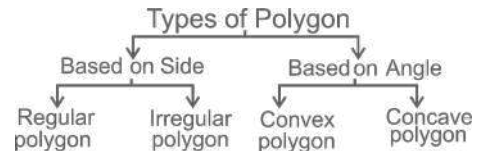
Some important result :-



$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = 1$$

Polygons

A two-dimensional closed shape bounded with minimum three sides. Triangle is the simplest form of polygon.

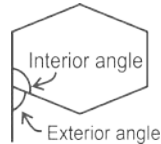


By side	By Angle
Regular polygon - all the sides and interior angles of the polygon are equal.	Concave angle - one interior angle must be greater than 180°
Irregular Polygon - polygons that do not have equal sides and not equal angles.	Convex angle - whose all interior angles are less than 180° .

Types of Polygon

Triangle - Has 3 sides and 3 vertices - Has no diagonals - Sum of the interior angles is 180°	Quadrilateral - Has 4 sides and 4 vertices - Has two diagonals - Sum of the interior angles is 360°
Pentagon - Has 5 sides and 5 vertices - Has 5 diagonals - Sum of the interior angles is 540°	Hexagon - Has 6 sides and 6 vertices - Has 9 diagonals - Sum of the interior angles is 720°
Heptagon - Has 7 sides and 7 vertices - Has 14 diagonals - Sum of the interior angles is 900°	Octagon - Has 8 sides and 8 vertices - Has 20 diagonals - Sum of the interior angles is 1080° - Area = $2a^2(1 + \sqrt{2})$
Nonagon - Has 9 sides and 9 vertices - Has 27 diagonals - Sum of the interior angles is 1260°	Decagon - Has 10 sides and 10 vertices - Has 35 diagonals - Sum of the interior angles is 1440°

Some Important Results of Polygon



n = number of sides.

* Sum of all exterior angles = 360° .

* Each exterior angle = $\frac{360^\circ}{n}$.

* Exterior angle + Interior angle = 180°

* In case of convex polygon, sum of all interior angles = $(2n - 4) \times 90^\circ$

* Number of diagonals = $\frac{n(n-3)}{2}$

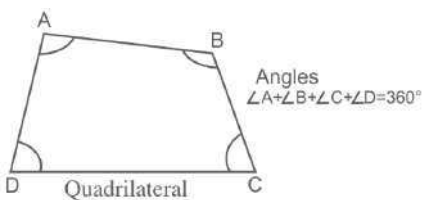
* area of regular polygon = $\frac{na^2}{4} \cot \frac{\pi}{n}$

* inradius = $\frac{a}{2} \cot \frac{\pi}{n}$

* circumradius = $\frac{a}{2} \operatorname{cosec} \frac{\pi}{n}$

Quadrilateral

A two-dimensional shape with four sides, four vertices, and four angles.



Example :- If the angles of a quadrilateral are in the ratio of 4 : 9 : 11 : 12, then the largest of these angles is:

RRB NTPC CBT - I 29/01/2021 (Evening)

(a) 168° (b) 72° (c) 166° (d) 120°

Solution :- Sum of angles of quadrilateral = 360°

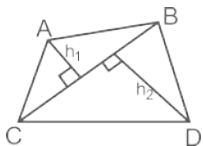
$$4x + 9x + 11x + 12x = 360^\circ$$

$$36x = 360^\circ \Rightarrow x = 10^\circ$$

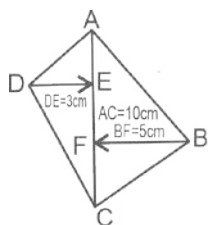
$$\text{Largest Angle}(12x) = 120^\circ$$

Area of quadrilateral

(i) Area of quadrilateral = $\frac{1}{2} \times \text{diagonal} \times (h_1 + h_2)$



Example :- Find the area (in cm^2) of the given quadrilateral ABCD.



SSC CHSL 31/05/2022 (Afternoon)

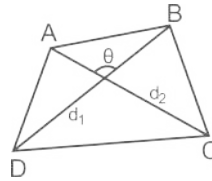
(a) 50 (b) 35 (c) 45 (d) 40

Solution :- Area of quadrilateral ABCD =

$$\Rightarrow \frac{1}{2} \times AC (DE + BF)$$

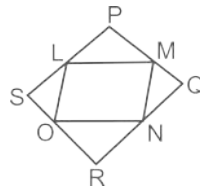
$$= \frac{1}{2} \times 10 \times (3 + 5) = 5 \times 8 = 40 \text{ cm}^2$$

(ii) Area of quadrilateral = $\frac{1}{2} \times d_1 \times d_2 \times \sin(\theta)$

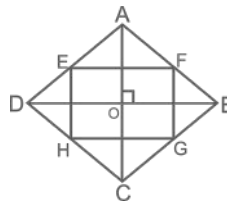


(iii) Line joining the midpoints of the adjacent sides of the quadrilateral formed a parallelogram.

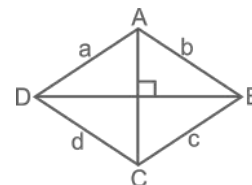
Where L, M, N, O are the mid points of PS, PQ, QR and RS.



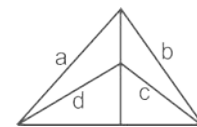
(iv) when the midpoints of adjacent sides of quadrilateral is joined and its diagonal intersects at 90° then a rectangle is formed.



Special results of quadrilateral :-

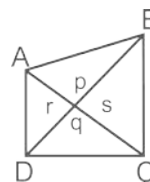


$$a^2 + c^2 = b^2 + d^2$$

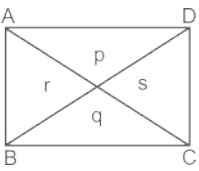


$$a^2 + c^2 = b^2 + d^2$$

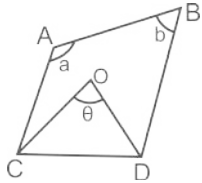
- If p, q, r and s be the area of respective triangles then $P \times q = r \times s$



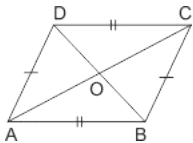
- If ABCD is a rectangle/square, then
- $\text{ar.}(P) + \text{ar.}(q) = \text{ar.}(r) + \text{ar.}(s)$



- If DO and CO are angle bisectors of $\angle ADC$ and $\angle BCD$ then,
- $\angle DOC = \frac{a+b}{2}$



1). Parallelogram : Opposite sides and angles of a parallelogram are equal.



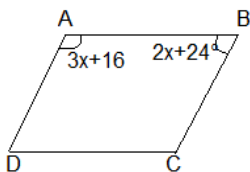
- $AB = DC$ and $AD = BC$
- $\angle A + \angle B = \angle B + \angle C = \angle C + \angle D = \angle D + \angle A = 180^\circ$

Example :- In a parallelogram ABCD, if $m\angle A = (3x + 16)^\circ$ and $m\angle B = (2x + 24)^\circ$, then find $m\angle C$.

RRC Group D 30/09/2022 (Evening)

(a) 56° (b) 28° (c) 100° (d) 110°

Solution :-



We know,

In parallelogram, sum of two adjacent angles = 180°

$$3x + 16^\circ + 2x + 24^\circ = 180^\circ$$

$$5x + 40^\circ = 180^\circ$$

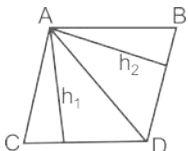
$$5x = 140^\circ \Rightarrow x = 28^\circ$$

$$\angle A = 3x + 16^\circ = 3 \times 28 + 16 = 100^\circ$$

$$\angle A = \angle C = 100^\circ \text{ (in parallelogram, opposite angles are equal)}$$

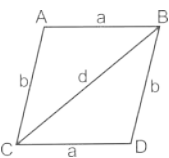
$$\angle C = 100^\circ$$

- $\angle A = \angle C$ and $\angle B = \angle D$
- The diagonals bisect each other, i.e. $AO = OC$ and $OB = OD$
- Area of parallelogram = Base $\times$ Altitude

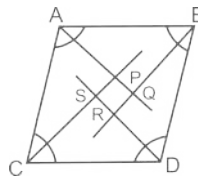


$$\text{Area of parallelogram} = 2 \sqrt{s(s-a)(s-b)(s-c)}$$

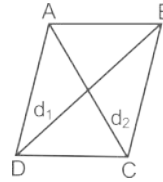
$$S = \frac{a+b+c}{2}$$



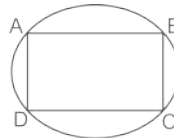
- Bisectors of a parallelogram(ABCD) form a rectangle(PQRS)



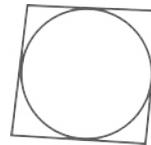
- Sum of squares of all four sides = Sum of squares of diagonals.
- $AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2$



- Parallelogram inscribed in a circle is a rectangle.

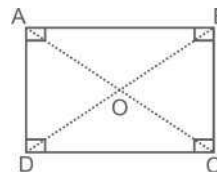


- Parallelogram that circumscribes a circle is a rhombus.



- Square, Rectangle and Rhombus are parallelogram.
- All rectangles are parallelogram but all parallelogram are not rectangles.

Rectangle



Opposite Side
 $AB = CD$ & $AD = BC$
Diagonals
 $AO = OC$ & $DO = OB$
Angles
 $\angle A = \angle B = \angle C = \angle D = 90^\circ$

- $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- Diagonals bisect each other i.e. $OA = OC$, $OB = OD$
- Opposite sides are equal i.e. $AB = DC$, $AD = BC$.
- Area = length $\times$ breadth
- Perimeter = 2 (length + breadth)

Example :- The two unequal sides of a rectangle are in the ratio 3:4. If the perimeter is 42 cm, then the length of the diagonal will be:

RRB NTPC CBT - I 02/02/2021 (Morning)

(a) 30 cm (b) 25 cm (c) 15 cm (d) 35 cm

Solution :-

Perimeter of rectangle

$$= 2(3 + 4) = 14 \text{ unit}$$

$$14 \text{ unit} = 42 \text{ cm} \Rightarrow 1 \text{ unit} = 3 \text{ cm}$$

Sides are 9 cm and 12 cm

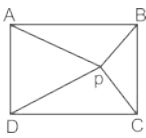
Diagonal of rectangle

$$= \sqrt{9^2 + 12^2} = \sqrt{81 + 144}$$

$$= \sqrt{225} = 15 \text{ cm}$$

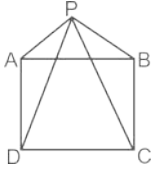
- If P be any point inside the rectangle.

$$PA^2 + PC^2 = PB^2 + PD^2$$

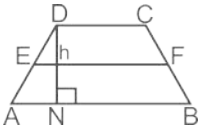


- If P be any point outside the rectangle.

$$PA^2 + PC^2 = PB^2 + PD^2$$



2). Trapezium: one pair of opposite sides are parallel.



Area of trapezium ABCD

$$= \frac{1}{2} \times \text{Sum of parallel side} \times \text{height}$$

$$= \frac{1}{2} \times (AB + DC) \times h$$

Example :- The lengths of a pair of parallel sides of a trapezium are 20 cm and 25 cm, respectively, and the perpendicular distance between these two sides is 14 cm.

What is the area (in cm^2) of the trapezium?

SSC CHSL 02/06/2022 (Afternoon)

- (a) 512 (b) 250 (c) 300 (d) 315

Solution :- Area of trapezium =

$$\frac{1}{2} \times (\text{sum of parallel side}) \times \text{perpendicular distance between them}$$

$$\Rightarrow \frac{1}{2} \times (20 + 25) \times 14 = \frac{1}{2} \times 45 \times 14$$

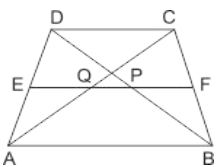
$$= 315 \text{ cm}^2$$

- If E is the midpoint of $\overline{AD}$ and F is the midpoint of $\overline{BC}$, then

$$EF = \frac{AB + DC}{2}.$$

- If P is the midpoint of diagonal $\overline{BD}$ and Q is the midpoint of diagonal $\overline{AC}$, then

$$PQ = \frac{AB - DC}{2}$$

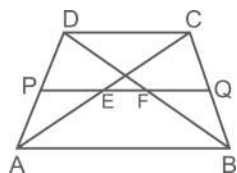


Example:- ABCD is a quadrilateral in which $AB \parallel DC$. E and F are the midpoints of the diagonals AC and BD, respectively. If $AB = 18 \text{ cm}$ and $CD = 6 \text{ cm}$, then $EF = ?$

SSC CGL TIER 2 (29 / 01/22)

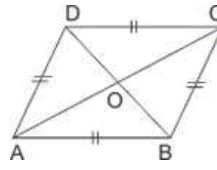
- (a) 8 (b) 9 (c) 6 (d) 12

Solution :-



$$EF = \frac{AB - DC}{2} = \frac{18 - 6}{2} = \frac{12}{2} = 6 \text{ cm}$$

3). Rhombus: All sides are equal and opposite sides are parallel to each other.



$$\angle A + \angle B = \angle B + \angle C = \angle C + \angle D = \angle D + \angle A = 180^\circ$$

- $\angle A = \angle C$ and $\angle B = \angle D$

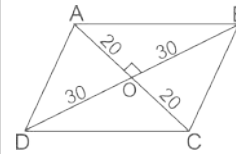
$$4a^2 = d_1^2 + d_2^2$$

i.e. Sum of squares of sides = Sum of squares of diagonals.

Example :- The length of the diagonals of a rhombus is 40 cm and 60 cm. What is the length of the side of the rhombus ?
SSC CHSL 13/03/2023 (4th Shift)

- (a) $50\sqrt{3} \text{ cm}$ (b) $20\sqrt{3} \text{ cm}$ (c) $10\sqrt{13} \text{ cm}$ (d) $40\sqrt{13} \text{ cm}$

Solution :-



Length of the diagonal is 40cm and 60cm

Then, in $\triangle AOB$

$$AB = \sqrt{(20)^2 + (30)^2} = \sqrt{1300} = 10\sqrt{13} \text{ cm}$$

So, the side of rhombus = $10\sqrt{13} \text{ cm}$

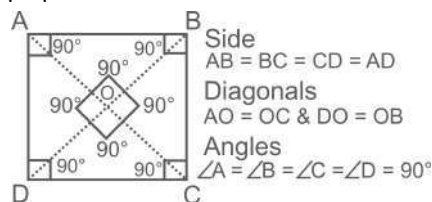
- Diagonals bisect each other at right angles and form four right angled triangles with equal areas.

$$\Rightarrow \text{Area of } \triangle AOB = \text{Area of } \triangle BOC = \text{Area of } \triangle COD$$

$$= \text{Area of } \triangle DOA = \frac{1}{4} \times \text{area of rhombus } ABCD$$

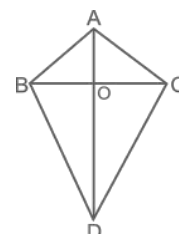
- The diagonals of Rhombus are not of equal length.

4). Square: All sides are equal in length and adjacent sides are perpendicular to each other.



- $AB = BC = CD = AD$ and $\angle A = \angle B = \angle C = \angle D = 90^\circ$
- Diagonals bisect each other at right angles and form four right angled isosceles triangles.
- Diagonals are of equal length i.e. $AC = BD$.

5) KITE :

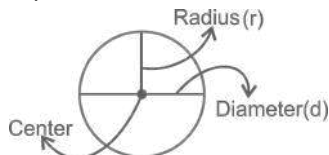


- adjacent sides are equal in length. i.e. $AB = AC$, $BD = DC$

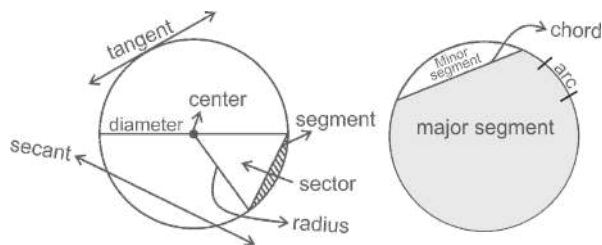
- Diagonal AD bisect BC but diagonal BC does not bisect diagonal AD i.e, BO = OC
- It has one pair of opposite angles (which are obtuse) that are equal. Here, $\angle B = \angle C$
- Diagonals cuts at 90°
- Perimeter = $2(a + b)$
- Area = $\frac{1}{2} \times d_1 \times d_2$

Circle

A **circle** is a two-dimensional figure formed by a set of points that are at a fixed distance (radius) from a fixed point (center) on the plane.



Parts of Circle:



Important Formulae:

- 1) Circumference of circle = $2\pi r$,
- 2) Area of circle = πr^2
- 3) Length of Arc (l) = $2\pi r \times \frac{\theta}{360}$ (θ is in degree)

Or Length of Arc (l) = $r \times \theta$ (θ is in radian)

Change radian to degree = θ in radian $\times \frac{180^\circ}{\pi}$

Example :- In a circle of radius 14 cm, an arc subtends an angle of 90° at the center. The length of arc (in cm) is equal to:

Take ($\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (3rd Shift)

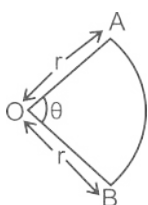
(a) 22 (b) 18 (c) 20 (d) 24

Solution :- Length of the arc = $\frac{\theta}{360^\circ} \times 2\pi r$

$$= \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 14$$

$$= \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22 \text{ cm}$$

Sector :-



$$1) \text{ Perimeter of sector} = [2r + \frac{\theta}{360} \times (2\pi r)]$$

$$2) \text{ Area of Sector} = \pi r^2 \times \frac{\theta}{360} \text{}(\theta \text{ is in degree})$$

$$\text{Or Area of Sector} = \frac{1}{2} \times l \times r \text{}(l \text{ is length of arc and } r \text{ is radius})$$

Example :- The area of a sector of a circle with central angle 60° is A. The circumference of the circle is C. Then A is equal to :

SSC CHSL 10/07/2019 (Evening)

$$(a) \frac{c^2}{6\pi} \quad (b) \frac{c^2}{18\pi} \quad (c) \frac{c^2}{24\pi} \quad (d) \frac{c^2}{4\pi}$$

Solution :- We know that

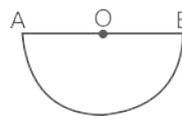
$$\text{Circumference of a circle (C)} = 2\pi r$$

$$\Rightarrow r = \frac{c}{2\pi}$$

$$\text{Area of the sector (A)} = \frac{\theta}{360} \times \pi r^2$$

$$\Rightarrow A = \frac{60}{360} \times \pi \times \frac{c}{2\pi} \times \frac{c}{2\pi} = \frac{c^2}{24\pi}$$

Semi - Circle :-

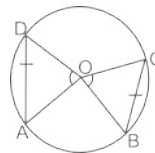


$$\text{Diameter} = 2 \times \text{radius}, \text{ Perimeter} = (\pi + 2)r, \text{ Area} = \frac{\pi r^2}{2}$$

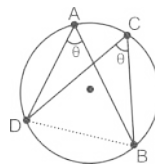
Properties of circles

1). Equal chords of a circle subtend equal angle at the centre.

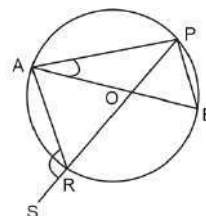
If AD = BC then, $\angle AOD = \angle COB$



2). Angles in the same segment of a circle are equal.



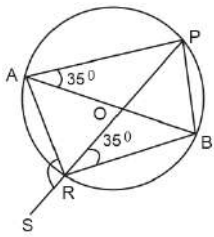
Example:- In the given figure O is the centre of the circle. If $\angle PAB = 35^\circ$, then find $\angle ARS$.



RRB NTPC 11/01/2021 (Evening)

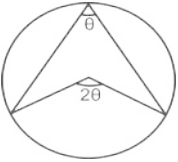
(a) 125° (b) 55° (c) 65° (d) 115°

Solution :-



$\angle PRB = 35^\circ$ (angle made by same chord)
 $\angle ARB = 90^\circ$ (Angle made by diameter)
 $\angle ARP = 90 - 35 = 55^\circ$
 $\angle ARS = 180 - 55 = 125^\circ$

- 3). The angle subtended by an arc at the centre is double the angle subtended by it at any point on the circumference of circle.

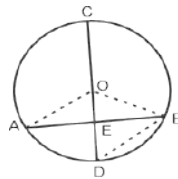


Example :- In a circle with centre O, Chord AB and diameter CD intersect each other at point E, inside the circle. If $\angle AOD = 42^\circ$ and $\angle BOC = 104^\circ$ then what is the measure (in degrees) of $\angle AED$?

SSC CHSL 04/08/2021 (Evening)

(a) 73° (b) 62° (c) 58° (d) 84°

Solution :-



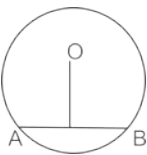
$\angle AOD = 42^\circ$, $\angle BOC = 104^\circ$

$\angle ABD = \frac{1}{2} \angle AOD$ (Angle subtended by an arc at the circumference is half of angle subtended at the center)

$\angle ABD = 21^\circ$, $\angle BDC = \frac{1}{2} \angle BOC = 52^\circ$

$\angle AED = \angle ABD + \angle BDC = 73^\circ$ (External angle theorem)

- 4) The perpendicular from the centre of a circle to a chord bisects the chord.

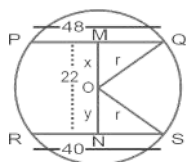


Example :- PQ and RS are two parallel chords of a circle such that PQ is 48 cm and RS is 40 cm. If the chords are on the opposite sides of the centre and the distance between them is 22 cm, what is the radius (in cm) of the circle?

SSC CPO 09/11/2022 (Afternoon)

(a) 25 (b) 24 (c) 35 (d) 22

Solution :-



From the given figure, $MQ = \frac{48}{2} = 24$ cm

and $NS = \frac{40}{2} = 20$ cm

$x + y = 22$ ----- (1)

In $\triangle MOQ$, we have :

$r^2 = x^2 + 24^2$ ----- (2)

In $\triangle NOS$, we have :

$r^2 = y^2 + 20^2$ ----- (3)

From equation (2) and (3) we have :

$x^2 + 24^2 = y^2 + 20^2$

$\Rightarrow x^2 + 576 = y^2 + 400$

$\Rightarrow x^2 - y^2 = -176 \Rightarrow (x - y)(x + y) = -176$

$(x - y)22 = -176$ (from eqn(1))

$x - y = -8$ ----- (4)

Solving eqn(1) and (4) we have :

$x = 7$ and $y = 15$

Now putting $x = 7$ in eqn(2), we get :

$r = \sqrt{49 + 576} = 25$ cm

Exam hall approach :-

Pythagoras triplets $\rightarrow$

(15, 20, 25) and (7, 24, 25)

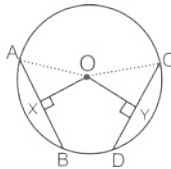
20 and 24 are the given chords of the circle .

Distance between the chord = $(15 + 7)$
 $= 22$ cm

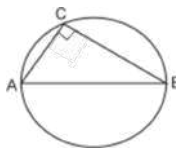
Given conditions satisfied .

Clearly , radius of the circle = 25 cm

- 5) Chords equidistant from the centre of a circle are equal in length. $AB = CD$



- 6). Angle dropped on the circumference of a circle with the diameter as base is always a right angle.

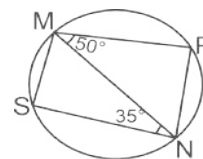


Example:- MN is the diameter of a circle with centre O. P and S are two points on the circumference of the circle on either side of MN, such that $\angle PMN = 50^\circ$ and $\angle MNS = 35^\circ$. What is the degree measure of the difference of $\angle PMS$ and $\angle PNS$?

SSC CHSL 04/08/2023 (3rd Shift)

(a) 25° (b) 55° (c) 65° (d) 30°

Solution:-



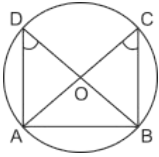
$\angle SMN = 180^\circ - (90^\circ + 35^\circ) = 55^\circ$

$\angle MNP = 180^\circ - (90^\circ + 50^\circ) = 40^\circ$

Now , $\angle PMS - \angle PNS$

$= (55^\circ + 50^\circ) - (40^\circ + 35^\circ) = 30^\circ$

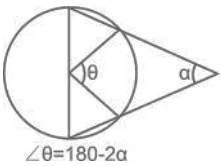
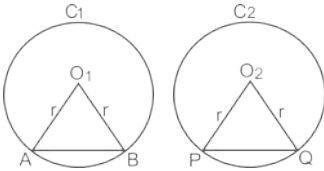
7). The angles in the same area of a circle with the same base are always equal.



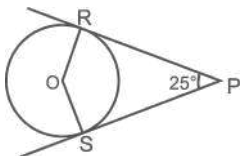
8). Equal chords of two equal circles subtend equal angles at their centres and vice versa.

If $AB = PQ$ then, $\angle AOB = \angle POQ$

If $\angle AOB = \angle POQ$ then, $AB = PQ$



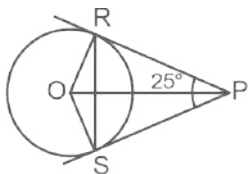
Example :- In the given figure, if $\angle RPS = 25^\circ$, the value of $\angle ROS$ is _____.



SSC CHSL 27/05/2022 (Afternoon)

(a) 155° (b) 145° (c) 135° (d) 165°

Solution :-

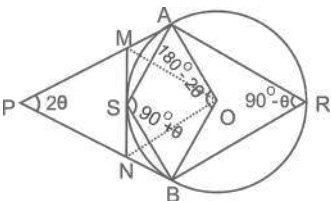


$$\angle ORS = \angle OSR = \frac{25^\circ}{2}$$

In $\triangle ORS$,

$$\begin{aligned}\angle ROS &= 180^\circ - \left(\frac{25^\circ}{2} + \frac{25^\circ}{2}\right) \\ &= 180^\circ - 25^\circ = 155^\circ\end{aligned}$$

9). PA and PB are two tangents, O is the center of the circle and R and S are the points on the circle, then

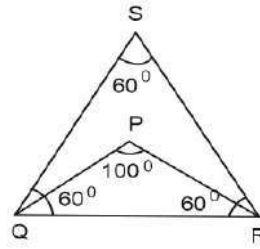


Example :- $\triangle PQR$ and $\triangle SQR$ are both isosceles triangles on a common base QR such that P and S lie on the same side of QR. If $\angle QSR = 60^\circ$ and $\angle QPR = 100^\circ$, then find $\angle SRP$.

SSC CHSL 03/08/2023 (1st Shift)

(a) 80° (b) 60° (c) 100° (d) 20°

Solution :-



$QS = SR$

So, $\angle PQR = \angle PRQ = 40^\circ$

$PQ = PR$

So, $\angle SQR = \angle SRQ = 60^\circ$

Now, $\angle SRP = 60^\circ - 40^\circ = 20^\circ$

10) A circle externally touch side QR of a $\triangle PQR$ at A, PQ produced at S and PR produced at T. Then perimeter of $\triangle PQR$?



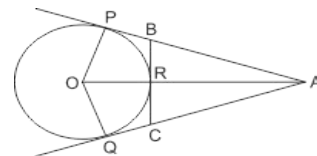
$$\text{Perimeter of } \triangle PQR = 2 \times PS$$

Example :- A is a point at a distance 26 cm from the centre O of a circle of radius 10 cm. AP and AQ are the tangents to the circle at the point of contacts P and Q. If a tangent BC is drawn at a point R lying on the minor arc PQ to intersect AP at B and AQ at C, then the perimeter of $\triangle ABC$ is:

SSC CHSL 20/10/2020 (Morning)

(a) 40 cm (b) 48 cm (c) 46 cm (d) 42 cm

Solution :-



$$AO = 26 \text{ cm}, PO = OQ = 10 \text{ cm}$$

$$AP = \sqrt{26^2 - 10^2} = 24$$

$$AP = AQ = 24 \text{ cm}$$

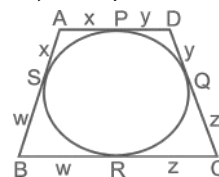
we know,

$$AP + AQ = AB + BC + AC$$

$$24 + 24 = AB + BC + AC$$

$$\text{the perimeter of } \triangle ABC = 24 + 24 = 48 \text{ cm}$$

11) . If a quadrilateral ABCD circumscribe a circle then



$$AB + CD = AD + BC$$

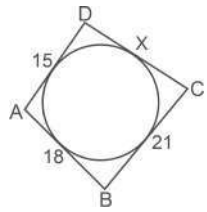
Example :- A circle touches all four sides of a quadrilateral

ABCD. If AB = 18 cm, BC = 21 cm and AD = 15 cm, then length CD is:

SSC CGL Tier II (02/03/2023)

(a) 16 cm (b) 14 cm (c) 12 cm (d) 18 cm

Solution :-

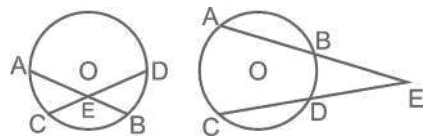


$$AB + CD = BC + AD \quad \text{.....(as per theorem)}$$

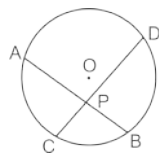
$$18 + x = 21 + 15 \Rightarrow x = 18 \text{ cm}$$

12). Two chords AB and CD of a circle intersect, internally or externally, at E then:

$$AE \times EB = CE \times ED$$



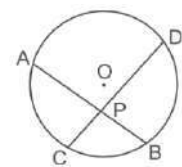
Example :- In the figure, O is the centre of the circle. Its two chords AB and CD intersect each other at the point P within the circle. If AB = 20 cm, PB = 12 cm and CP = 8 cm, then find the measure of PD.



SSC CGL 06/12/2022 (3rd Shift)

(a) 12 cm (b) 11 cm (c) 22 cm (d) 14 cm

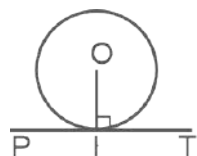
Solution :-



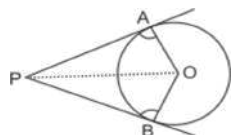
Concepts:- $PA \times PB = PC \times PD$
(intersecting chord theorem)
 $\Rightarrow 8 \times 12 = 8 \times PD \Rightarrow PD = 12 \text{ cm}$

13). The tangent to a circle at a point on its circumference is perpendicular to the radius at that point.

$$OI \perp PT$$



14). Two tangents PA and PB are drawn from an external point P on a circle with center O is equal.



- i) $\angle APO = \angle BPO$
 ii) $PA = PB$

AE^2

iii) $\angle PAO = \angle PBO = 90^\circ$

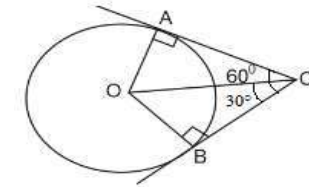
iv) $OA = OB$

Example :- If two tangents to a circle of radius 3 cm are inclined to each other at angle of 60° , then the length of each tangent is:

SSC CPO 09/11/2022 (Afternoon)

(a) $\frac{3\sqrt{3}}{4}$ cm (b) $3\sqrt{3}$ cm (c) 3 cm (d) 6 cm

Solution :-



$$\triangle AOC \cong \triangle BOC \quad (\text{by RHS})$$

In $\triangle AOC$, we have:

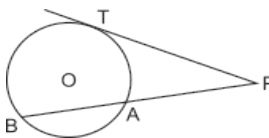
$$\tan 30^\circ = \frac{AO}{AC} = \frac{3}{AC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{3}{AC} \Rightarrow AC = 3\sqrt{3} \text{ cm}$$

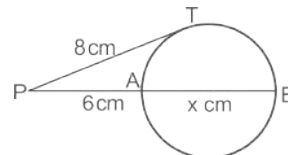
So, the length of each tangent = $3\sqrt{3}$ cm

15). For a tangent, PT, and a secant, PB, drawn to a circle from a point P,

$$PT^2 = PB \times PA$$



Example :- In the given figure, PAB is a secant and PT is a tangent to the circle from P. If PT = 8 cm. PA = 6 cm and AB = x cm. then the value of x is:



SSC CGL 14/07/2023 (1st shift)

(a) $\frac{14}{9}$ (b) $\frac{1}{3}$ (c) $\frac{14}{3}$ (d) $\frac{4}{3}$

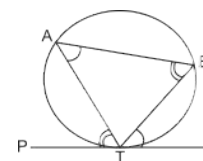
Solution :- Using tangent and secant formula :-

$$PT^2 = PA \times PB \Rightarrow 64 = 6 \times (6 + x)$$

$$\Rightarrow (6 + x) = \frac{32}{3} \Rightarrow x = \frac{14}{3} \text{ cm}$$

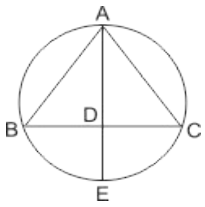
16). Angle made by a chord with a tangent is always equal to the angle made by that chord in any point of circumference.

$$\angle PTA = \angle ABT$$

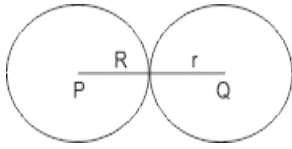


17). In the following case:

$$AB \times AC + AE \times DE =$$

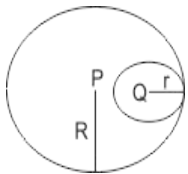


18). If two circles touch each other externally, then distance between their centres is the sum of their radius.



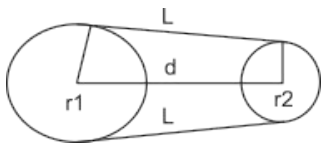
Here, $PQ = R + r$

19). If two circles touch each other internally, then distance between their centres is the difference of their radius.



Here, $PQ = R - r$

Direct common Tangents :-



Length of direct common tangent (L) = $\sqrt{d^2 - (r_1 - r_2)^2}$

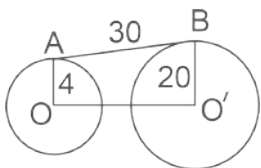
Where, d = distance between two centers and r_1, r_2 are radii of the circles.

Example :- Radii of two circles are 20 cm and 4 cm. Length of the direct common tangent is 30 cm. What is the distance between their centres ? (in cm)

SSC CHSL 17/03/2023 (2nd Shift)

(a) 36 (b) 38 (c) 34 (d) 32

Solution :-



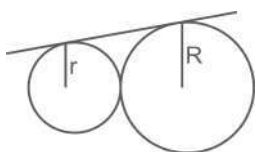
Let AB be the direct common tangent

Then, $AB =$

$$\sqrt{(\text{dis. b/w centers})^2 - (\text{diff. b/w radius of both circle})^2}$$

$$\Rightarrow (30)^2 = (OO')^2 - (20 - 4)^2$$

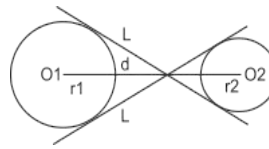
$$\Rightarrow (OO')^2 = 1156 \Rightarrow (OO') = 34 \text{ cm}$$



Length of direct common tangent (L) = $2\sqrt{Rr}$

Distance between the centre = $R + r$

Transverse common Tangents :-



Length of transverse tangent (L) = $\sqrt{d^2 - (r_1 + r_2)^2}$

Example :- If the distance between the centres of two circles is 12 cm and the radii are 5 cm and 4 cm, then the length (in cm) of the transverse common tangent is :

SSC CHSL 02/08/2023 (2nd Shift)

(a) 9 (b) $\sqrt{143}$ (c) $\sqrt{63}$ (d) 7

Solution :-

Length of transverse common tangent

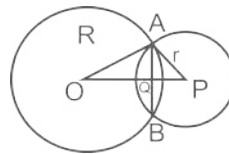
$$= \sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{12^2 - (5 + 4)^2}$$

$$= \sqrt{144 - (81)} = \sqrt{63}$$

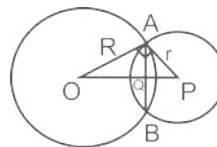
19) Two circles of radii R and r intersect and AB is the common chord, then

$$AQ = BQ = \frac{AB}{2}$$

$$\angle AQP = \angle AQO = 90^\circ$$

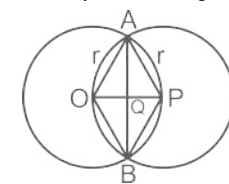


20) When $\angle OAP = 90^\circ$ i.e. The tangent of a circle at point A or B passes through the centre of the other circle.



$$OP = \sqrt{R^2 + r^2}$$

21) When radii of both the circles are equal and the both the circles pass through the centres of the other circle, i.e. $R = r$



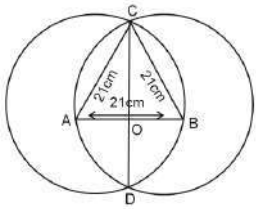
$$\Rightarrow AB = \sqrt{3}r$$

Example :- Two circles having equal radius intersect each other such that each passes through the center of the other. Sum of diameter of these two circles is 84 cm. What is the length of the common chord ? (in cm)

SSC CHSL 17/03/2023 (4th Shift)

(a) $21\sqrt{3}$ (b) $14\sqrt{3}$ (c) $28\sqrt{3}$ (d) $24\sqrt{3}$

Solution :-



In $\triangle ABC$,
 $AB = BC = CA \dots$ (radius)
 So, $\triangle ABC$ is an equilateral triangle with side equals to 21 cm.

$$\text{Height of equilateral triangle} = \frac{\sqrt{3} \times \text{side}}{2}$$

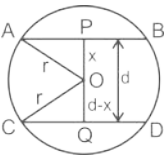
$$= \frac{21\sqrt{3}}{2} \text{ cm.}$$

$$\text{Now, Length of chord CD} = 2 \times \text{CO}$$

$$= 21\sqrt{3} \text{ cm.}$$

When two chords are drawn parallel in a circle :-

i) Opposite side of centre



$$r^2 = AP^2 + X^2 \text{ ----- (i)}$$

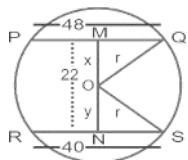
$$r^2 = CQ^2 + (d - x)^2 \text{ ----- (ii)}$$

Example :- PQ and RS are two parallel chords of a circle such that PQ is 48 cm and RS is 40 cm. If the chords are on the opposite sides of the centre and the distance between them is 22 cm, what is the radius (in cm) of the circle?

SSC CPO 09/11/2022 (Afternoon)

(a) 25 (b) 24 (c) 35 (d) 22

Solution :-



$$\text{From the given figure, } MQ = \frac{48}{2} = 24 \text{ cm}$$

$$\text{and } NS = \frac{40}{2} = 20 \text{ cm}$$

$$x + y = 22 \text{ ----- (1)}$$

In $\triangle MOQ$, we have :

$$r^2 = x^2 + 24^2 \text{ ----- (2)}$$

In $\triangle NOS$, we have :

$$r^2 = y^2 + 20^2 \text{ ----- (3)}$$

From equation (2) and (3) we have :

$$x^2 + 24^2 = y^2 + 20^2$$

$$\Rightarrow x^2 + 576 = y^2 + 400$$

$$\Rightarrow x^2 - y^2 = -176 \Rightarrow (x - y)(x + y) = -176$$

$$(x - y)22 = -176 \text{ (from eqn(1))}$$

$$x - y = -8 \text{ ----- (4)}$$

Solving eqn(1) and (4) we have :

$$x = 7 \text{ and } y = 15$$

Now putting $x = 7$ in eqn(2), we get :

$$r = \sqrt{49 + 576} = 25 \text{ cm}$$

Short Trick :-

Pythagoras triplets $\rightarrow$

(15, 20, **25**) and (7, 24, **25**)

20 and 24 are the given chords of the circle .

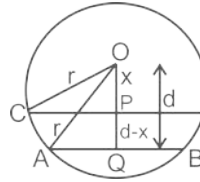
$$\text{Distance between the chord} = (15 + 7)$$

$$= 22 \text{ cm}$$

Given conditions satisfied .

Clearly , radius of the circle = 25 cm

ii) Same side of centre



$$r^2 = PC^2 + X^2 \text{ ----- (i)}$$

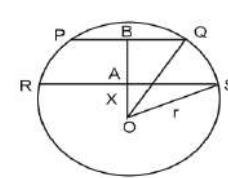
$$r^2 = AQ^2 + (d)^2 \text{ ----- (ii)}$$

Example :- PQ and RS are two parallel chords of a circle of length 10 cm and 24 cm, respectively, and lie on the same side of the centre O. If the distance between the chords is 7 cm, what is the radius (in cm) of the circle?

SSC CHSL 03/06/2022 (Afternoon)

(a) 11 (b) 9 (c) 13 (d) 7

Solution :-



$$BQ = \frac{PQ}{2} = \frac{10}{2} = 5 \text{ cm}, AS = \frac{RS}{2} = \frac{24}{2} = 12 \text{ cm}, AB = 7 \text{ cm},$$

$$OQ = OS = r$$

In $\triangle BOQ$,

$$OB^2 + BQ^2 = OQ^2$$

$$(7 + x)^2 + 5^2 = r^2 \text{ ----- (1)}$$

In $\triangle OAS$,

$$OA^2 + AS^2 = OS^2$$

$$\Rightarrow x^2 + 12^2 = r^2 \text{ ----- (2)}$$

From equation (1) and (2) We get,

$$(7 + x)^2 + 5^2 = x^2 + 12^2$$

$$49 + x^2 + 14x + 25 = x^2 + 144$$

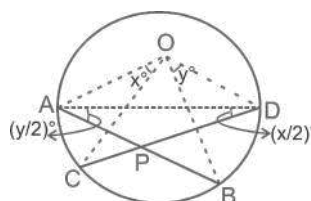
$$74 + 14x = 144$$

$$\Rightarrow 14x = 70 \Rightarrow x = 5 \text{ cm}$$

Now putting $x = 5$ in equation (2)

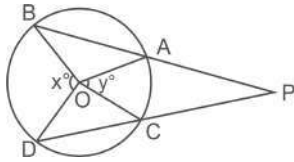
$$r^2 = 5^2 + 12^2 = 169 \Rightarrow r = 13 \text{ cm}$$

• Two Chords AB and CD of a circle with center O, intersect each other at P. If $\angle AOC = x^\circ$ and $\angle BOD = y^\circ$ then the value of $\angle BPD$ is :-



$$\angle BPD = \frac{x}{2} + \frac{y}{2}$$

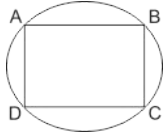
- Chords are intersecting at external points.



$$\angle BPD = \frac{x - y}{2}$$

Cyclic Quadrilateral

- i) A quadrilateral drawn inside a circle with its vertices lying on the circumference.



Sum of opposite angles = 180°

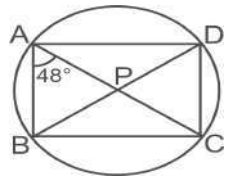
i.e. $\angle A + \angle C = \angle B + \angle D = 180^\circ$

Example :- In a circle, ABCD is a cyclic quadrilateral AC and BD intersect at each other at P. If $AB = AC$ and $\angle BAC = 48^\circ$, then the measure of $\angle ADC$ is :

SSC CGL 12/04/2022 (Evening)

(a) 104° (b) 112° (c) 132° (d) 114°

Solution:-



Here $AB = AC$, hence $\angle ABC = \angle ACB$

In

$\triangle ABC$,

$$\angle ABC + \angle ACB + \angle BAC = 180^\circ$$

$$2\angle ABC = 180^\circ - 48^\circ = 132^\circ$$

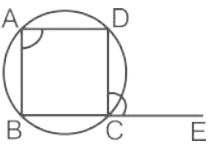
$$\angle ABC = 66^\circ$$

$$\text{So, } \angle ADC = 180^\circ - \angle ABC = 180^\circ - 66^\circ$$

$$= 114^\circ \text{ (sum of opposite angles in cyclic quadrilateral is } 180^\circ)$$

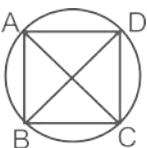
- ii) Exterior angle of the cyclic quadrilateral is equal to opposite interior angle.

$$\angle A = \angle DCE$$



- iii) **Ptolemy's Theorem:** If ABCD is a cyclic quadrilateral, then

$$AB \times DC + BC \times AD = AC \times BD$$



If the diagonal of cyclic quadrilateral bisects other diagonal then :-



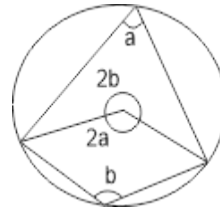
- i) When AC cuts BD ;

$$AB \times BC = CD \times AD$$

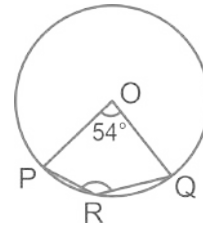
- ii) When BD cuts AC ;

$$AB \times AD = BC \times CD$$

Important result :- In the following type of cyclic quadrilateral remember the relationship between the angles, as shown:



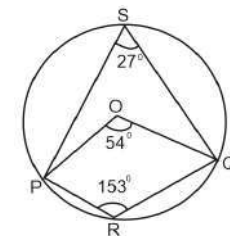
Example:- In the given figure, O is the centre of the circle. $\angle POQ = 54^\circ$. What is the measure (in degree) of $\angle PRQ$?



SSC CGL 13/04/2022 (Afternoon)

(a) 235 (b) 137 (c) 207 (d) 153

Solution:-



Let S be any point on the circumference.

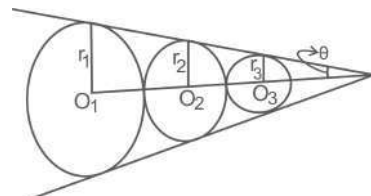
$$\angle PSQ = \frac{1}{2} \times \angle POQ = 27^\circ$$

$$\angle PSQ + \angle PRQ$$

$$= 180^\circ \text{ (cyclic quadrilateral)}$$

$$\angle PRQ = 180^\circ - 27^\circ = 153^\circ$$

When the radius are in GP :-



$$r^2 = \sqrt{r_1 \times r_3}$$

and

$$\frac{r_2}{r_1} = \frac{r_3}{r_2} = \frac{1 + \sin \theta}{1 - \sin \theta}$$

	Property	Rectangle	Square	Parallelogram	Rhombus	Trapezium
Sides	All sides are equal	✗	✓	✗	✓	✗
	Opposite sides are equal	✓	✓	✓	✓	✗
	Opposite sides are parallel	✓	✓	✓	✓	✗
Angles	All angles are equal	✓	✓	✗	✗	✗
	Opposite angles are equal	✓	✓	✓	✓	✗
	Sum of two adjacent angles is 180	✓	✓	✓	✓	✗
Diagonals	Bisect each other	✓	✓	✓	✓	✗
	Bisect perpendicular	✗	✓	✗	✓	✗
Formulas	Area	$l \times b$	a^2	$l \times h$	$\frac{1}{2} \times d_1 \times d_2$	$\frac{1}{2} \times \text{Sum of parallel Sides} \times \text{Height}$
	Perimeter	$2 \times (l + b)$	$4a$	$2 \times (l + b)$	$4a$	Sum of all Sides

Mensuration (2D)

[Surface Area and Perimeter]

2- D : Under 2 dimensions we will study

- Triangles
- Quadrilateral
- Polygons
- Circle

Note:- For 2D figures, area and perimeter are calculated.

Triangle

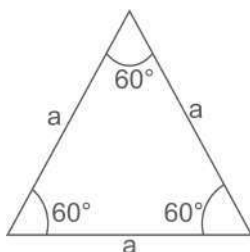
For a triangle with height, h, and base, b;

$$\text{Area} = \frac{1}{2} \times b \times h$$

Perimeter = Sum of all sides

Special Cases

1) Equilateral Triangle - All sides are equal and each angle is 60°



- Area = $\frac{\sqrt{3}}{4} a^2 = \frac{h^2}{\sqrt{3}}$
- Height = $\frac{\sqrt{3}}{2} a$
- Perimeter = $3a$
- Inradius(r) = $\frac{a}{2\sqrt{3}} = \frac{h}{3}$
- Circumradius(R) = $\frac{a}{\sqrt{3}} = \frac{2h}{3}$

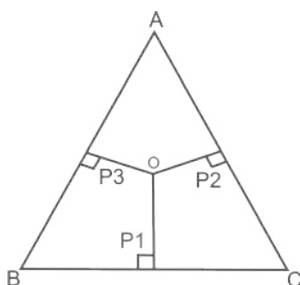
Example:- If the area of an equilateral triangle is $25\sqrt{3} \text{ cm}^2$, then the length of each side of the triangle is :

RRC Group D 23/08/2022 (Afternoon)

- (a) 12 cm (b) 5 cm (c) 8 cm (d) 10 cm

Solution :- Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2$

$$\Rightarrow \frac{\sqrt{3}}{4} \times (\text{side})^2 = 25\sqrt{3} \Rightarrow \text{side} = \sqrt{\frac{25\sqrt{3} \times 4}{\sqrt{3}}} = 10 \text{ cm}$$



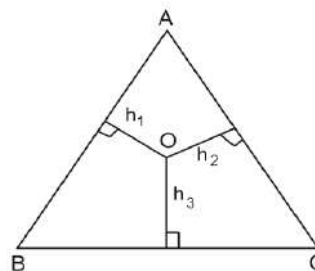
If P1, P2 and P3 are perpendicular to a side from a point inside the triangle, then

- Height of triangle = $P1 + P2 + P3$
- side of triangle = $\frac{2}{\sqrt{3}} (P1 + P2 + P3)$
- Area of triangle = $\frac{(P1 + P2 + P3)^2}{\sqrt{3}}$

Example :- 'O' is a point in the interior of an equilateral triangle. The perpendicular distance from 'O' to the sides are $\sqrt{3} \text{ cm}$, $2\sqrt{3} \text{ cm}$, $5\sqrt{3} \text{ cm}$. The perimeter of the triangle is: SSC CGL 13/12/2022 (4th Shift)

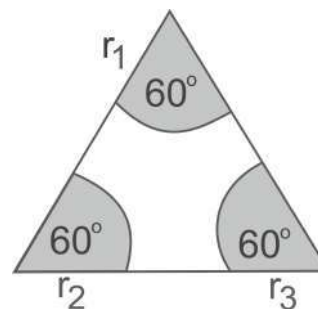
- (a) 48 cm (b) 32 cm (c) 24 cm (d) 64 cm

Solution :-



$$\begin{aligned} \text{Side (a)} &= \frac{2}{\sqrt{3}} (h_1 + h_2 + h_3) \\ &= \frac{2}{\sqrt{3}} (\sqrt{3} + 2\sqrt{3} + 5\sqrt{3}) = 16 \text{ cm} \end{aligned}$$

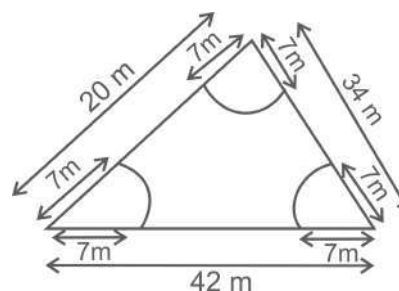
Perimeter of equilateral triangle = $3 \times a = 48 \text{ cm}$



- Area of shaded part = $\frac{\pi}{6} (r_1^2 + r_2^2 + r_3^2)$
- Area of unshaded part = $\frac{\sqrt{3}a^2}{4} - \frac{\pi}{6} (r_1^2 + r_2^2 + r_3^2)$

Example :- Three horses are tethered with 7 meter long ropes at the three corners of a triangular field having sides 20m, 34m and 42m. The area of the plot which can be grazed by the horses and the area of the plot which remains ungrazed

Solution :-



$$\text{Area of shaded part :- } \frac{\pi}{6} (r_1^2 + r_2^2 + r_3^2)$$

$$\frac{\pi}{6}((7)^2 + (7)^2 + (7)^2) = 77 \text{ m}^2$$

Area of unshaded part :-

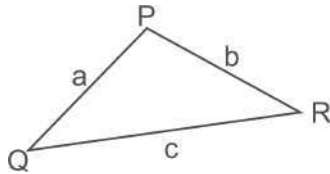
$$\text{Semiperimeter of triangle} = \frac{20 + 34 + 42}{2} = 48 \text{ m}$$

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{48(48-20)(48-34)(48-42)} = 336 \text{ m}^2$$

$$\text{So, area of unshaded part} = (336 - 77) = 259 \text{ m}^2$$

2) Scalene Triangle – Sides are of unequal lengths.



• Area = $\sqrt{s(s-a)(s-b)(s-c)}$ This formula is called Heron's formula.

Where, Semi - perimeter (s) = $\frac{a+b+c}{2}$

• Perimeter = $a + b + c$

Example :- Find the area of a triangle whose sides measure 42 cm, 36 cm and 34 cm, respectively.

RRC Group D 06/10/2022 (Morning)

(a) $112\sqrt{110} \text{ cm}^2$ (b) $56\sqrt{110} \text{ cm}^2$

(c) $14\sqrt{55} \text{ cm}^2$ (d) $28\sqrt{55} \text{ cm}^2$

Solution :- Sides = 42, 36 and 34

$$S = \frac{42 + 36 + 34}{2} = 56$$

Area of triangle =

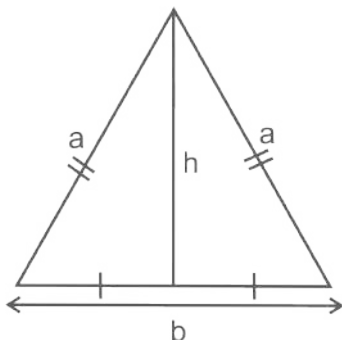
$$\sqrt{56 \times (56 - 42) \times (56 - 36) \times (56 - 34)}$$

$$\text{Area of triangle} = \sqrt{56 \times 14 \times 20 \times 22}$$

$$\text{Area of triangle} = \sqrt{4 \times 14 \times 14 \times 4 \times 5 \times 2 \times 11}$$

$$\text{Area of triangle} = 56\sqrt{110} \text{ cm}^2$$

3) Isosceles Triangle :- two sides and two angles are equal. Altitude bisects the base.



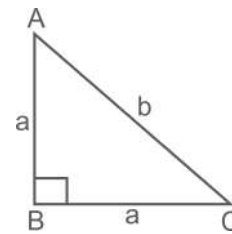
• Area = $\frac{b}{4} \sqrt{4a^2 - b^2}$

• Height = $\sqrt{a^2 - \left(\frac{b}{2}\right)^2} = \frac{1}{2} \sqrt{4a^2 - b^2}$

• Perimeter = $a + a + b = 2a + b$

• Circum radius = $\frac{a^2}{\sqrt{4a^2 - b^2}}$

If an isosceles triangle is right angle triangle then,



• Area = $\frac{1}{2}a^2$

• Hypotenuse (b) = $a\sqrt{2}$

Example :- The area of an isosceles right-angled triangle whose perimeter is 16 cm is:

Delhi Police H.C.M. 12/10/2022 (Evening)

(a) $32(5 - \sqrt{2}) \text{ cm}^2$ (b) $64(3 - 3\sqrt{3}) \text{ cm}^2$

(c) $56(3 - \sqrt{2}) \text{ cm}^2$ (d) $64(3 - 2\sqrt{2}) \text{ cm}^2$

Solution :- Let the two sides and hypotenuse of an isosceles right-angled triangle be x and y respectively. Using Pythagoras theorem, we have

$$2x^2 = y^2 \Rightarrow x^2 = \frac{y^2}{2} \Rightarrow x = \frac{y}{\sqrt{2}}$$

$$2x + y = 16 \text{ (Given)}$$

$$2 \times \frac{y}{\sqrt{2}} + y = 16,$$

$$y(\sqrt{2} + 1) = 16, y = \frac{16}{\sqrt{2} + 1}$$

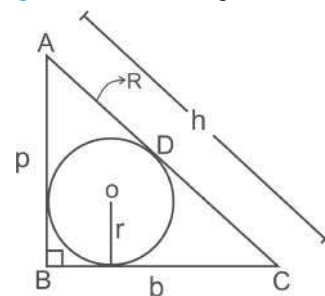
$$\text{Area of isosceles triangle} = \frac{1}{2}x^2$$

$$= \frac{1}{2} \times \frac{y^2}{2} = \frac{1}{4} \times \left(\frac{16}{\sqrt{2} + 1}\right)^2 = \frac{64}{(\sqrt{2} + 1)^2}$$

$$= \frac{64}{2 + 1 + 2\sqrt{2}} = \frac{64}{3 + 2\sqrt{2}} \text{ (on rationalization)}$$

$$= 64(3 - 2\sqrt{2}) \text{ cm}^2$$

4) Right angle Triangle :- One of the angles is 90° .



Here, p = perpendicular, b = base and, h = hypotenuse

• Area = $\frac{1}{2} \times b \times p$

• Perimeter = $p + b + h$

• Pythagoras Theorem : $h^2 = p^2 + b^2$

• Inradius (r) = $\frac{p+b-h}{2}$

• Circumradius (R) = $\frac{h}{2}$

Note :- Common Pythagoras triplets: $(2n, n^2 - 1, n^2 + 1)$, (1, 1, $\sqrt{2}$), (1, 2, $\sqrt{5}$), (3, 4, 5), (5, 12, 13), (7, 24, 25), (20, 21, 29), (9, 40, 41), (8, 15, 17), (12, 35, 37), (11, 60, 61), (65, 72, 97), (96, 110, 146). Try remembering them.

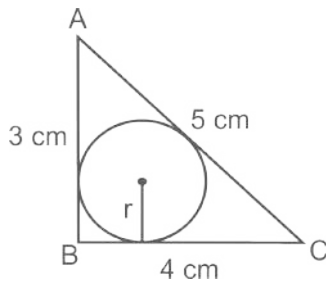
Example :- ABC is a right-angled triangle. If the lengths of

two sides containing the right angle are 4 cm and 3 cm. The radius of its incircle is :

RRB NTPC CBT - I 15/02/2021 (Morning)

- (a) 1 cm (b) 4 cm (c) 2 cm (d) 3 cm

Solution :-



$$\text{Area} = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$$

$$S = \frac{3+4+5}{2} = 6 \text{ cm}$$

$$\text{Inradius} = r = \frac{\text{area of } \Delta}{\text{semi perimeter}}$$

$$\text{Inradius} = r = \frac{6}{6} = 1 \text{ cm}$$

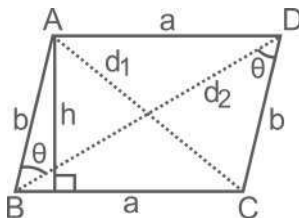
Exam hall approach:- $\text{Inradius} = \frac{a+b-c}{2} = \frac{3+4-5}{2} = 1 \text{ cm}$

Quadrilateral

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360° .

Special Cases

1) Parallelogram :- It is a quadrilateral with opposite sides parallel and equal.



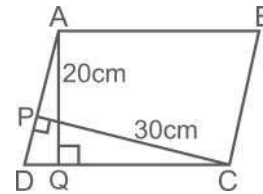
- Area = base $\times$ height
- Perimeter = $2(a + b)$
- $d_1^2 + d_2^2 = 2(a^2 + b^2)$
- In a parallelogram, opposite sides are equal, opposite angles are equal and diagonals bisect each other

Example :- The area of a parallelogram ABCD is 300 cm^2 . The distance between AB and CD is 20 cm and the distance between BC and AD is 30 cm. What is the perimeter (in cm) of the parallelogram?

SSC MTS 05/10/2021 (Evening)

- (a) 60 (b) 50 (c) 40 (d) 100

Solution :- According to question,



Area of parallelogram = Base $\times$ Altitude

For base AD,

$$300 = AD \times CP \Rightarrow 300 = AD \times 30 \Rightarrow AD = 10 \text{ cm},$$

$$BC = AD = 10 \text{ cm}$$

Similarly, for base CD,

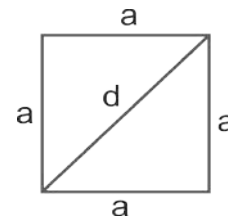
$$300 = CD \times AQ \Rightarrow 300 = CD \times 20$$

$$\Rightarrow CD = 15 \text{ cm}, AB = CD = 15 \text{ cm}$$

$$\text{So, perimeter} = AB + BC + CD + AD$$

$$= 10 + 15 + 10 + 15 = 50 \text{ cm}$$

2) Square :- It is a parallelogram with all four sides equal and each angle is equal to 90° .



- Area = $(\text{side})^2 = a^2 = \frac{d^2}{2}$

- Perimeter = $4 \times \text{side} = 4a$

- Diagonal (d) = $a\sqrt{2}$

Example :- What is the area of a square whose diagonal is $5\sqrt{2} \text{ cm}$?

SSC MTS 11/05/2023 (Morning)

- (a) 64 cm^2 (b) 25 cm^2 (c) 100 cm^2 (d) 36 cm^2

Solution :- Diagonal of square $(\sqrt{2} a) = 5\sqrt{2}$

$$\text{Side}(a) = 5 \text{ cm}$$

$$\text{Area of square } (a)^2 = 5 \times 5 = 25 \text{ cm}^2$$

3) Rectangle :- It is a parallelogram with equal opposite sides and each angle is 90°

- Area = Length $\times$ Breadth = $L \times B$

- Perimeter = $2(L + B)$

- Diagonal (d) = $\sqrt{L^2 + B^2}$

Example :- The length and the breadth of a rectangle is in the ratio of 3 : 4. If the area of the rectangle is 108 cm^2 . Find its perimeter.

SBI Clerk Pre 22/06/2019 (1st Shift)

- (a) 40 cm (b) 42 cm (c) 24 cm (d) 56 cm (e) None of these

Solution :- Let the length and breadth of rectangle be $3x$ and $4x$ respectively

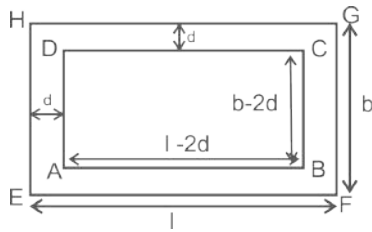
$$\text{Area of the rectangle} = 3x \times 4x = 108$$

$$\Rightarrow 12x^2 = 108 \Rightarrow x^2 = 9 \Rightarrow x = 3 \text{ cm}$$

$$\text{Length} = 3 \times 3 = 9 \text{ cm}$$

$$\text{Breadth} = 3 \times 4 = 12 \text{ cm}$$

$$\text{required perimeter} = 2(9 + 12) = 42 \text{ cm}$$

(a) Path Inside A Rectangular Field :-

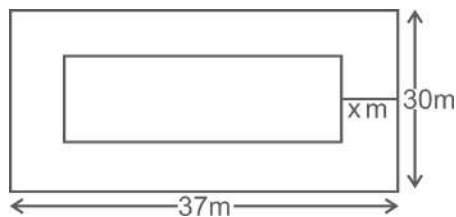
- Area of path = $2d(l + b - 2d)$
- Perimeter = $4(l + b - 2d)$

Example:- A path around the inner side of a rectangular park measuring $37\text{m} \times 30\text{m}$ occupies 570 m^2 . What is the width of the path?

RRB NTPC CBT - I 01/02/2021 (Evening)

- (a) 10 m (b) 15 m (c) 5 m (d) 28 m

Solution :-

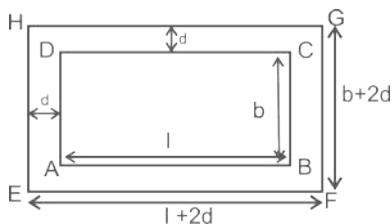


Area of path = Area of bigger rectangle - Area of smaller rectangle

$$\begin{aligned} \Rightarrow 570 &= 37 \times 30 - (37 - 2x)(30 - 2x) \\ \Rightarrow 570 &= 1110 - (1110 - 134x + 4x^2) \\ \Rightarrow 570 &= 134x - 4x^2 \Rightarrow 4x^2 - 134x + 570 = 0 \\ \Rightarrow 4x^2 - 20x - 114x - 570 &= 0 \\ \Rightarrow 4x(x - 5) - 114(x - 5) &= 0 \\ \Rightarrow (x - 5)(4x - 114) &= 0 \\ x &= 5 \text{ or } x = \frac{114}{4} \end{aligned}$$

Exam hall approach :- area of path = $2d(l + b - 2d)$

$$\begin{aligned} 570 &= 2 \times (d)[37 + 30 - 2 \times d] \\ 570 &= 2d[67 - 2d] \Rightarrow 4d^2 - 134d + 570 = 0 \\ \Rightarrow 4d^2 - 20d - 114d - 570 &= 0 \\ \Rightarrow 4d(d - 5) - 114(d - 5) &= 0 \\ \Rightarrow (d - 5)(4d - 114) &= 0 \\ \Rightarrow d &= 5 \text{ m or } d = \frac{114}{4} \text{ m} \end{aligned}$$

(b) Path Outside A Rectangular Field :-

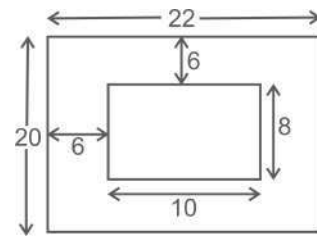
- Area of path = $2d(l + b + 2d)$
- Perimeter = $4(l + b + 2d)$

Example :- A path 6m wide runs around and outside of a rectangular plot of length 10m and breadth 8m. Find the area (in m^2) of path?

SSC MTS 07/10/2021 (Afternoon)

- (a) 440 (b) 600 (c) 80 (d) 360

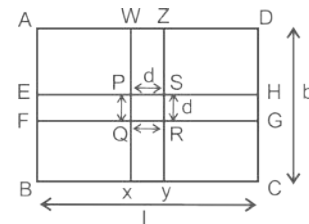
Solution :-



$$\text{Area of the path} = (22 \times 20) - (10 \times 8) = 440 - 80 = 360 \text{ m}^2$$

Exam hall approach :- Area of path = $2d(l + b + 2d)$

$$= 2 \times 6(10 + 8 + 2 \times 6) = 360 \text{ m}^2$$

(c) Path Midway of A Rectangular Field :-

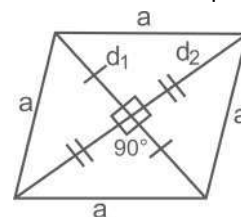
- Area of path = $d(l + b - d)$
- Perimeter = $2(l + b - 2d)$

Example :- The two cross paths of the rectangular field 40m long and 34m wide, one parallel to the breadth. The width of each path is 4m. Find the cost of gravelling the path at rupee 4.5 per m^2 .

Solution :- According to the question,

$$\begin{aligned} \text{Area of the path} &= d(l + b - d) = 4(40 + 34 - 4) = 280 \text{ m}^2 \\ \text{Required cost} &= 280 \times 4.5 = 1260 \text{ rs} \end{aligned}$$

4) Rhombus :- It is a parallelogram with all four sides equal. The opposite angles in a rhombus are equal.



Here, a = side; d_1 and d_2 are diagonals.

- Area = $\frac{1}{2} \times d_1 \times d_2$
- Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$
- Perimeter = $4a$
- $4a^2 = d_1^2 + d_2^2$
- Diagonals bisect each other at right angles.

Example:- Find the area (in cm^2) of the rhombus, lengths of whose diagonals are, respectively, 7.5 cm and 5.6 cm.

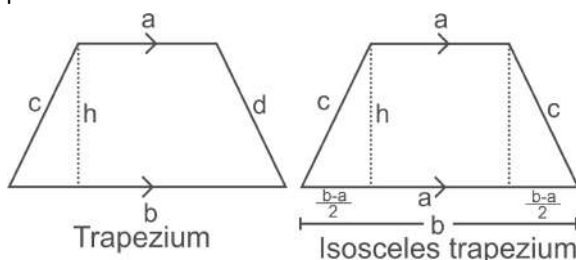
SSC MTS 11/09/2023 (3rd Shift)

- (a) 21 (b) 28 (c) 27 (d) 30

Solution :- Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$= \frac{1}{2} \times 7.5 \times 5.6 = 21 \text{ cm}^2$$

5) Trapezium :- It is a quadrilateral with one pair of opposite sides parallel.



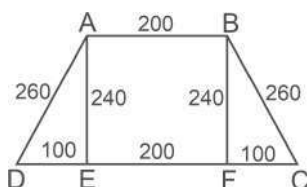
Here, a and b are parallel sides and h is the height or perpendicular distance between a and b .

- Area = $\frac{1}{2} \times \text{height} \times \text{sum of parallel side}$
- $$= \frac{1}{2} \times h \times (a + b)$$

Example:- A field is in the shape of a trapezium whose parallel sides are 200 m and 400 m long, whereas each of the other two sides is 260 m long. what is the area (in m^2) of the field ?
SSC CPO 11/12/2019 (Morning)

(a) 48000 (b) 52000 (c) 72000 (d) 60000

Solution:-



In $\triangle ADE$, by pythagoras theorem

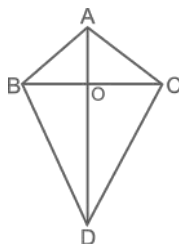
$$\Rightarrow 260^2 - 100^2 = AE^2$$

$$\Rightarrow 360 \times 160 = AE^2$$

$$\Rightarrow AE = 6 \times 4 \times 10 \Rightarrow AE = 240$$

$$\begin{aligned} \text{area of trapezium} &= \frac{1}{2} \times (200 + 400) \times 240 \\ &= 600 \times 120 = 72000 \text{ m}^2 \end{aligned}$$

6) Kite:- A kite is a quadrilateral in which two pairs of adjacent sides are equal.



- adjacent sides are equal in length. i.e, $AB = AC$, $BD = DC$
- The longer diagonal bisects the shorter diagonal.
i.e, $BO = OC$
- Perimeter = $2(a + b)$
- Area = $\frac{1}{2} \times d_1 \times d_2$

NOTE : Important Points about Quadrilaterals :-

- The diagonals of a parallelogram bisect each other.
- Diagonal of a parallelogram divides it into two triangles of equal area.

3. The diagonals of a rectangle are of equal lengths and bisect each other.

4. The diagonals of a square are equal and bisect each other at right angles.

5. A rhombus has unequal diagonals and they bisect each other at right angles.

6. A parallelogram and a rectangle have equal areas if they are on the same base and between the same parallel Lines.

Regular Polygon

In a regular polygon all sides and all interior angles are equal. A polygon is called a pentagon, hexagon, heptagon, octagon, nonagon and a decagon as they have 5, 6, 7, 8, 9, 10 sides, respectively.

If each side of a regular polygon of ' n ' sides is equal to ' a ' then:

- Area of regular pentagon = $\frac{6.9}{4} \times a^2$
- Area of regular hexagon = $6 \times \frac{\sqrt{3}}{4} a^2$
- each exterior angle = $\frac{360^\circ}{n}$
- Sum of all exterior angles = 360°
- Each interior angle = $\frac{(n-2)}{n} \times 180^\circ$
- Each interior angle = $180^\circ - \text{exterior angle}$
- Sum of all interior angles = $(n-2) \times 180$
- Number of diagonals = $\frac{n(n-3)}{2}$

Example:- What is the area of a regular hexagon the length of each of whose sides is 4 cm?

RRC Group D 27/09/2022 (Morning)

- (a) $24\sqrt{3} \text{ cm}^2$ (b) $27\sqrt{3} \text{ cm}^2$ (c) $25\sqrt{3} \text{ cm}^2$ (d) $26\sqrt{3} \text{ cm}^2$

Solution:- Area of regular hexagon = $6 \times \frac{\sqrt{3}}{4} \times a^2$

$$\Rightarrow 6 \times \frac{\sqrt{3}}{4} \times 4 \times 4 = 24\sqrt{3} \text{ cm}^2$$

Example:- If the ratio of an external angle and an internal angle of a regular polygon is 1:5, then what is the number of sides in the polygon?

SSC MTS 22/07/2022 (Afternoon)

- (a) 24 (b) 22 (c) 12 (d) 30

Solution :-

ratio of an external angle and an internal angle = 1 : 5

As per question,

$$x + 5x = 180 \Rightarrow x = 30$$

$$\text{Number of sides } (n) = \frac{360^\circ}{\text{external angle}} = \frac{360}{30} = 12$$

Example:- The number of diagonals in a 28-gon is:

Solution :- Number of diagonals in a polygon with n sides

$$= \frac{n(n-3)}{2}$$

Number of diagonals in a polygon with n sides

$$= \frac{28(28-3)}{2} = \frac{28 \times 25}{2} = 350$$

Example:- Find the measure of each interior angle of a regular polygon of 18 sides :

Solution :- The sum of interior angles of polygon of n sides
 $= (n - 2) \times 180^\circ$ (n = number of side of polygon)
 $= (18 - 2) \times 180^\circ = 2880^\circ$

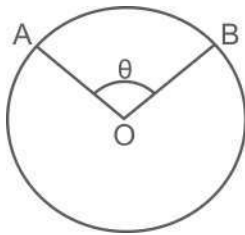
Each interior angle of the regular polygon $= \frac{2880}{18} = 160^\circ$

Exam hall approach:- Each interior angle $= \frac{(n - 2)}{n} \times 180^\circ$

Each interior angle of regular polygon $= \frac{(18 - 2)}{18} \times 180^\circ$
 $= 160^\circ$

Circle

It is a plane figure enclosed by a line on which every point is equally distant from a fixed point (centre) inside the circle.



- Area $= \pi r^2$
- Circumference (perimeter) $= 2\pi r$
- Diameter $= 2r$
- Length of Arc (AB) $= 2\pi r \times \frac{\theta}{360^\circ}$
- Area of sector AOB $= \pi r^2 \times \frac{\theta}{360^\circ}$
- Length of Arc (AB) = angle $\times$ radius, (where angle in radian)

Example :- An arc on a circle, whose length is 19.25 cm, subtends an 18° angle at the centre. What is the area of the circle? [Use $\pi = \frac{22}{7}$]

SSC CGL 21/07/2023 (4th shift)

Solution :- Length of arc $= \frac{\theta}{360} \times 2\pi r$

$$19.25 = \frac{18}{360} \times 2 \times \frac{22}{7} \times r$$

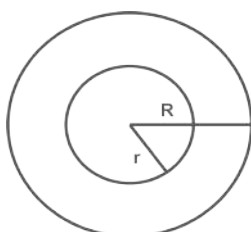
$$19.25 = \frac{1}{10} \times \frac{22}{7} \times r$$

$$r = \frac{19.25 \times 10 \times 7}{22} = 61.25$$

hence, area of the circle $= \pi r^2$

$$= \frac{22}{7} \times 61.25 \times 61.25 = 11,790.625 \text{ cm}^2$$

Circular Ring



Here, R = radius of bigger ring, r = radius of smaller ring

- Area of ring $= \pi(R^2 - r^2)$
- Difference in circumference of both the rings $= 2\pi R - 2\pi r$

Example :- The difference between the areas of two concentric circles is 264 cm^2 . What is the difference between the square of their radius? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

(a) 70 cm^2 (b) 140 cm^2 (c) 84 cm^2 (d) 64 cm^2

Solution :- Let the radius of the inner and outer circle be r and R respectively.

According to the question,

Difference between the areas of two concentric circles =

$$264 \text{ cm}^2 \Rightarrow \pi R^2 - \pi r^2 = 264 \Rightarrow \pi(R^2 - r^2) = 264$$

$$\Rightarrow \frac{22}{7} \times (R^2 - r^2) = 264 \Rightarrow R^2 - r^2 = 84 \text{ cm}^2$$

Successive Increase or Decrease

1. If the length and breadth of a rectangle are increased by a% and b%, the area of the rectangle will be increased by

$$(a + b + \frac{ab}{100})\%$$

Example :- The length and breadth of a rectangle are increased by 8% and 5%, respectively. By how much percentage will the area of the rectangle increase?

SSC CGL 01/12/2022 (1st Shift)

(a) 13.4% (b) 15.4% (c) 12.4% (d) 16.4%

Solution :- $8\% = \frac{8}{100}$, $5\% = \frac{5}{100}$

	Length	:	Breadth	=	Area
1st case	25	:	20	=	500
2nd case	27	:	21	=	567

$$\% \text{ change in area} = \frac{567 - 500}{500} \times 100 = 13.4\%$$

Exam hall approach:-

Area of rectangle = Length $\times$ Breadth

Area of rectangle is directly proportional to product of length and breadth of rectangle.

$$\% \text{ change in area} = 8 + 5 + \frac{8 \times 5}{100} = 13.4\%$$

2. If the length and breadth of a rectangle are increased and decreased by a% and b% respectively, percentage change in the area of the rectangle will be

$$(a - b - \frac{ab}{100})\%$$

Example :- The length of a rectangle is increased by 25% and breadth was decreased by 20% then what will be the percentage change in the area of the rectangle?

IBPS Clerk Pre 12/12/2020 (2nd Shift)

(a) 10 % increase (b) 0% (c) 15% increase
 (d) 25% increase (e) 20% decrease

Solution :- $25\% = \frac{1}{4}$, $20\% = \frac{1}{5}$

	Length	:	Breadth	=	Area
1st case	4	:	5	=	20

2nd case $5 : 4 = 20$

No changes in area in any case

Exam all approach:-

$$\text{Using successive method} = 25 - 20 - \frac{25 \times 20}{100}$$

= 0 % (change in area)

3. All the sides of any two dimensional figure changed by a%, then its area will change by $(2a + \frac{a^2}{100})\%$

Note:- Whenever there is a decrease, use negative value for 'a'

Example :- If each side of a rectangle is increased by 22%, then its area will increase by :

SSC CHSL 08/07/2019 (Morning)

(a) 44% (b) 50% (c) 46.65% (d) 48.84%

Solution :- %age increase in area = $22 + 22 + \frac{22 \times 22}{100}$
 $= 44 + \frac{22 \times 22}{100} = 48.84\%$

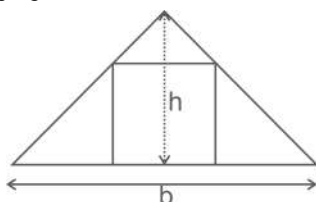
4. If all the sides of any two dimensional figure has changed (increased or decreased) by a% then its perimeter also changes by a%. In the case of a circle such changes take place because of the change in radius (or diameter).

5. Area of a square inscribed in a circle of radius 'r' is equal to $2r^2$.

6. The area of the largest triangle inscribed in a semicircle of radius r is equal to r^2 .

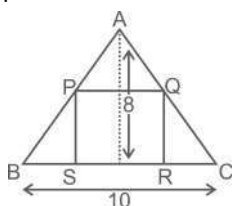
Some Direct Results

1. In the following figure:



• Side of the square = $\frac{b \times h}{b + h}$

Example :- in the given figure base and height of the triangle ABC is 10 and 8 cm respectively. PQRS is a square then the side length of the square.

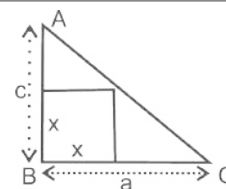


Solution :- According to the question,

$$\text{Side of the square} = \frac{b \times h}{b + h}$$

$$\text{Side of the square} = \frac{10 \times 8}{10 + 8} = \frac{40}{9} \text{ cm}$$

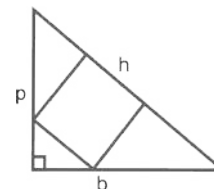
2. In the following figure:



• side of square, $x = \frac{a \times c}{a + c}$

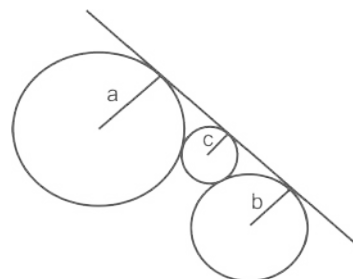
• area of inscribed square = $(\frac{a \times c}{a + c})^2$

3. In the following figure:



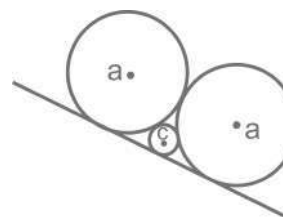
• Side of square = $\frac{p \times b \times h}{p^2 + b^2 + pb}$

4. If a, b and c are radii of three circles and all three circles have a common tangent as shown, then



• $\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$

Example :- In the given figure All three circles are tangents to the same line and also to each other. Circles a and a have equal radii. Find the radius of a, if the radius of c is equal to 20 cm.

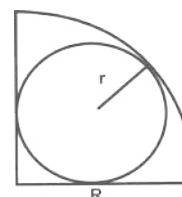


Solution :- According to the question, Let the radius of the circle a is x cm.

then, formula :- $\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{a}}$

$$\Rightarrow \frac{1}{\sqrt{20}} = \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{20}} = \frac{1+1}{\sqrt{x}} \Rightarrow \sqrt{x} = \sqrt{80}, x = 80 \text{ cm}$$

5. In the following figure

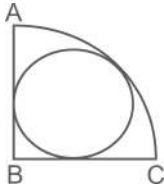


• $r = (\sqrt{2} - 1)R$

Example :- if ABC is a quarter circle and a circle is inscribed

in it and if $AB = 24$ cm, then find the radius of the smaller circle.

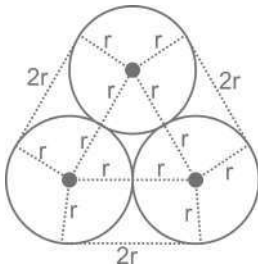
Solution :-



$$\text{Radius of small circle} = (\sqrt{2} - 1)R$$

$$\Rightarrow (\sqrt{2} - 1)24$$

6. In the following figure

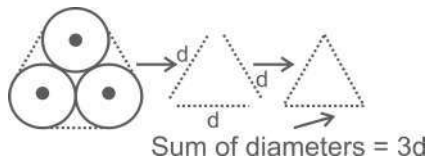


$$\bullet \text{ Length of string} = 2\pi r + 3 \times 2r = 2r(\pi + 3)$$

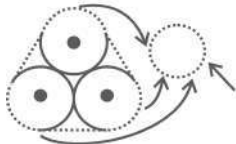
Example :- Three circles of radius 6 cm are kept touching each other. The string is tightly tied around these three circles. What is the length of the string?

SSC CGL 03/12/2022 (2nd Shift)

Solution :-



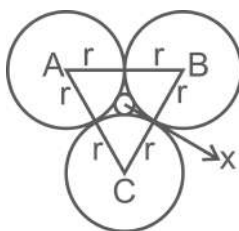
$$\text{Circumference of Circle} = 2\pi r$$



$$\text{the length of string} = 2r(\pi + 3)$$

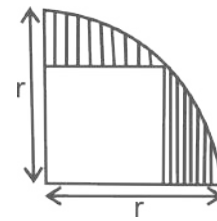
$$\text{the length of string} = 2 \times 6 + (\pi + 3) = 36 + 12\pi$$

7. If 'r' be the radius of 3 identical circles then, Radius(x) of the circle touching all the 3 circles is as shown:



$$\bullet x = \frac{2 - \sqrt{3}}{\sqrt{3}} \times r$$

8. In the given figure



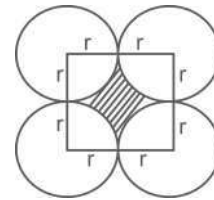
$$\bullet \text{ Side of square} = \frac{r}{\sqrt{2}}$$

$$\bullet \text{ Area of square} = \left(\frac{r}{\sqrt{2}}\right)^2$$

$$\bullet \text{ Area of shaded region} = \text{area of quadrant} - \text{area of square}$$

$$= \frac{1}{4}\pi r^2 - \left(\frac{r}{\sqrt{2}}\right)^2$$

9. In the following figure:



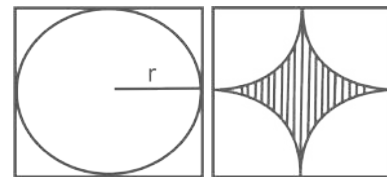
$$\bullet \text{ Area of shaded region} = \text{Area of square} - \text{Area of one circle}$$

$$= (2r)^2 - \pi r^2 = 4r^2 - \frac{22}{7}r^2 = r^2\left(4 - \frac{22}{7}\right) = \frac{6}{7}r^2$$

$$\text{Area of shaded region} = \frac{6}{7}r^2$$

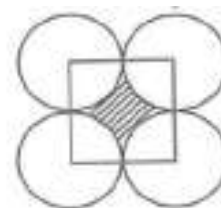
Where r = radius of each circle

10. In the following figure:



$$\bullet \text{ Area of shaded region} = \text{area of square} - \text{area of one complete circle.}$$

Example :- in the given figure if the radius of the circle is 28 cm, then find the area of the shaded region.



Solution :-

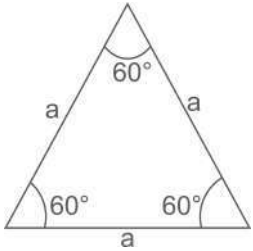
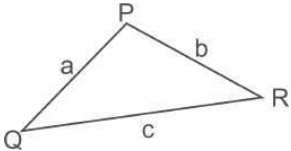
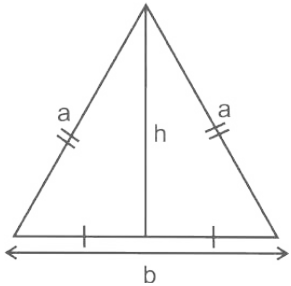
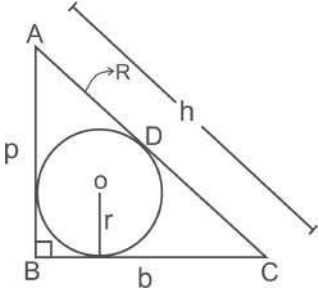
$$\text{Area of shaded region} = \text{area of square} - \text{area of one complete}$$

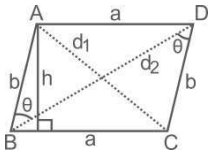
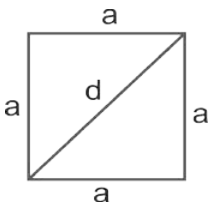
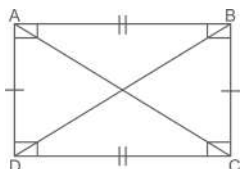
$$\text{circle} \Rightarrow 56^2 - \frac{22}{7} \times 28 \times 28 = 672 \text{ cm}^2$$

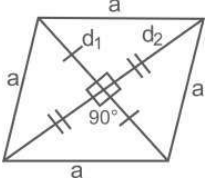
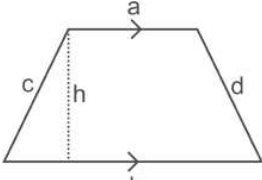
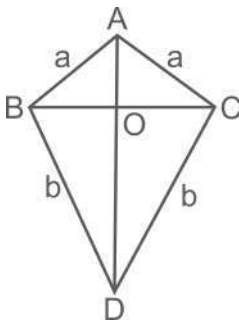
Exam hall approach:-

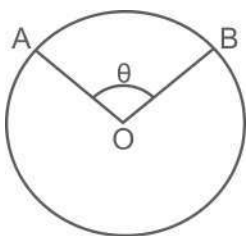
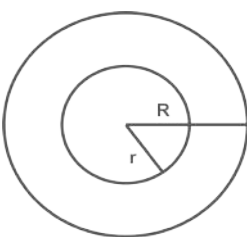
$$\text{Area of shaded region} = \frac{6}{7}r^2 = \frac{6}{7} \times 28 \times 28 = 672 \text{ cm}^2$$

Basic Formulae Table

Triangle	Shapes	Area	Perimeter
Equilateral Triangle		$\frac{\sqrt{3}}{4} a^2 = \frac{h^2}{\sqrt{3}}$	$3a$
Scalene Triangle		$\sqrt{s(s-a)(s-b)(s-c)}$ Where, Semi - perimeter (s) = $\frac{a+b+c}{2}$	$a + b + c$
Isosceles Triangle		$\frac{b}{4} \sqrt{4a^2 - b^2}$	$a + a + b = (2a + b)$
Right angle Triangle		$\frac{1}{2} \times b \times p$	$p + b + h$

Quadrilateral	Shapes	Area	Perimeter	Diagonal
Parallelogram		base $\times$ height	$2(a + b)$	$d_1^2 + d_2^2 = 2(a^2 + b^2)$
Square		$(side)^2 = a^2 = \frac{d^2}{2}$	$4 \times \text{side} = 4a$	$a\sqrt{2}$
Rectangle		Length $\times$ Breadth	$2(L + B)$	$\sqrt{L^2 + B^2}$

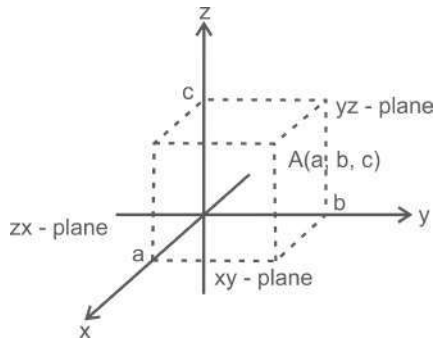
Rhombus	 $\text{Side (a)} = \frac{1}{2} \sqrt{d_1^2 + d_2^2}$	$\frac{1}{2} \times d_1 \times d_2$	4a	$4a^2 = d_1^2 + d_2^2$
Trapezium	 Trapezium	$\frac{1}{2} \times \text{height} \times \text{sum of parallel side}$ Or $\frac{1}{2} \times h \times (a + b)$	Sum of all sides	
Kite		$\frac{1}{2} \times d_1 \times d_2$	2 (a + b)	

Circle	Shapes	Area	Circumference
Circle		$= \pi r^2$	$2\pi r$
Circular Ring		$\pi(R^2 - r^2)$	Difference in circumference $2\pi R - 2\pi r$

Mensuration (3D)

[Volume]

Objects or shape that has three dimensions i.e. x, y, z is called a 3D object or shape



Important terms :-

Perimeter	Perimeter is the total distance around the shape or the length of the boundary of any closed shape.
Area	Area is the amount of space occupied by a two - dimensional figure. It is expressed in square units.
Surface Area	Surface Area is the total area occupied by the surfaces of a 3D object. They are classified into two parts - Curved or Lateral Surface Area and Total Surface Area.
Volume	Volume is the amount of space occupied by a 3D shape. It is expressed in cubic meter.
Lateral surface area	Area of 4 walls.

Important Unit Conversion

- $1m^3 = 1000$ litres
- 1 litre = 1000 cm^3
- 1 Hectare = 10000 m^2
- 1 Acre = $4046.86 \approx 4047\text{ m}^2$

Example :- A cuboidal water tank is 12 m long, 9 m wide and 7.5 m deep. How many litres of water can it hold?

SSC CHSL 08/08/2023 (1st Shift)

(a) 840000 (b) 805000 (c) 780000 (d) 810000

Solution :- $1m^3 = 1000$ litre

Volume of cuboid = $l \times b \times h = 12 \times 9 \times 7.5$

$= 810m^3 = 810000$ litre

Some Other Important Unit Conversion

Kilometer	Kilolitre	Kilogram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Hectometer	Hectolitre	Hectogram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Decameter	Decalitre	Decagram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Meter	Litre	Gram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Decimeter	Decilitre	Decigram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Centimeter	Centilitre	Centigram
$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$	$\times 10 \downarrow \uparrow \div 10$
Millimeter	Millilitre	Milligram

Example 1 :- Convert 10 km into m, cm and mm ?

Solution :- $10\text{ km} = 10 \times 1000 = 10000\text{ m}$

$10\text{ km} = 10 \times 100000 = 10 \times 10^5\text{ cm}$

$10\text{ km} = 10 \times 10^6\text{ mm}$

Example 2 :- Convert 10 mm into km, cm and m ?

Solution :- $10\text{ mm} = \frac{10}{10^6} = 10^{-5}\text{ km}$

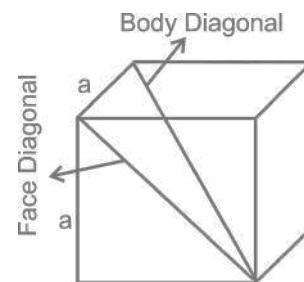
$10\text{ mm} = \frac{10}{10} = 1\text{ cm}$

$10\text{ mm} = \frac{10}{10^3} = 10^{-2}\text{ m}$

Three Dimensional Figures & Formula

Cube

A cube is a three-dimensional figure with all sides equal and has 6 faces, 8 vertices and 12 edges.



- Volume = a^3
 - Area of each Face = a^2
 - Lateral surface area = $4a^2$
 - Total surface area = $6a^2$
 - Face Diagonal = $a\sqrt{2}$
 - Body Diagonal = $a\sqrt{3}$
 - Sum of all edges = $12a$
- Here, a = length of the side

Example :- The volume of a cube is 2.197 cm^3 . Find the side (cm) of the cube

SSC MTS 19/05/2023 (Morning)

(a) 1.3 (b) 1.4 (c) 1.1 (d) 1.2

Solution :- Let side of cube = a

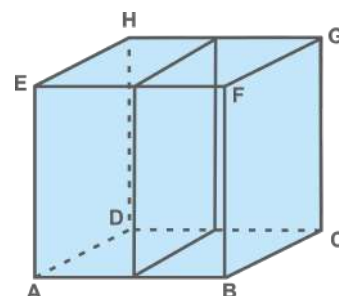
According to the question,

Volume of cube (a^3) = 2.197

$a = \sqrt[3]{2.197} = 1.3\text{ cm}$

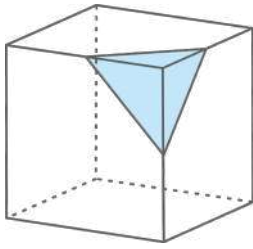
Cutting of Cube :-

1). Cutting Cube horizontally or Vertically to one of the side, then one cut adds two new surfaces

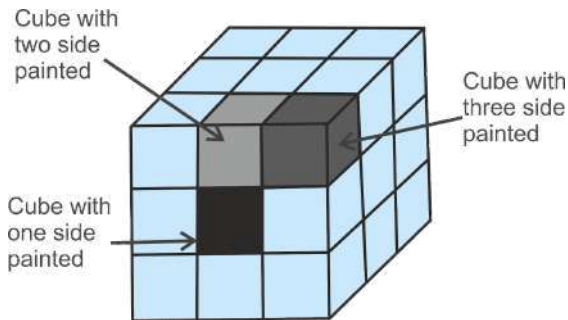


1 cut increases area by $2a^2$

2). Cutting cube at corner adds 2 new Triangular surface area (Pyramid)



Find the number of cubes of painted faces formula :-



Number of cubes with 0 side painted = $(n - 2)^3$

Number of cubes with 1 sides painted = $(n - 2)^2 \times 6$

Number of cubes with 2 sides painted = $(n - 2) \times 12$

Number of cubes with 3 sides painted = 8 (always)

Where $n = \frac{\text{length of large cube}}{\text{length of each small cube}}$

Example :- A cube of side 125cm is painted red on all the faces and then cut into smaller cubes of side 25 cm each. Find the number of smaller cubes having at least two faces painted.

SSC CGL 12/04/2022 (Evening)

(a) 48 (b) 36 (c) 44 (d) 52

Solution :- Given : Side of the large cube = 125 cm

Side of each smaller cubes = 25 cm

$$n = \frac{125}{25} = 5$$

We know that, number of cubes with 2 sides painted

$$= 12 (n - 2) = 12 (5 - 2) = 36$$

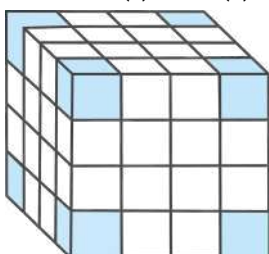
and number of cubes with 3 sides painted = 8 (always)

So number of such smaller cubes having at least two faces painted = $36 + 8 = 44$

Example :- A cube of side 80cm is painted yellow on all the faces and then cut into smaller cubes of sides 8cm each. Find the number of smaller cubes having all the three faces painted.

SSC CGL 12/04/2022 (Afternoon)

(a) 32 (b) 28 (c) 8 (d) 64

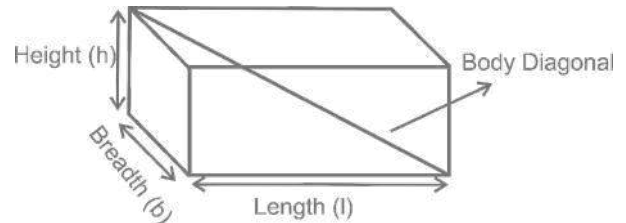


A cube painted by all the faces and then cut into n number of cubes, will always have 8 smaller cubes painted along three faces.

Number of cubes with 3 faces painted = 8 (always)

Cuboid

A rectangular body having 3D rectangular shape, is called a cuboid.



- Volume = $l \times b \times h$
- Total surface area = $2(lb + bh + lh)$
- Lateral Surface Area = $2(l + b) \times h$
- Face diagonal = $\sqrt{l^2 + b^2}$ or $\sqrt{l^2 + h^2}$
- Body Diagonal = $\sqrt{l^2 + b^2 + h^2}$
- Volume of cuboid = $\sqrt{a \times b \times c}$
(Here a, b and c are the area of three faces)

Example :- The areas of the three adjacent faces of a cuboid are 32 cm^2 , 24 cm^2 and 48 cm^2 . What is the volume (in cm^3) of the cuboid?

SSC CGL 04/06/2019 (Evening)

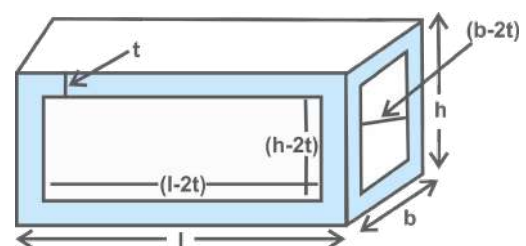
(a) 192 (b) 256 (c) 288 (d) 128

Solution :- Volume of cuboid = $\sqrt{a \times b \times c}$
(Here a, b and c are the area of three faces)

$$\text{Volume of cuboid} = \sqrt{32 \times 24 \times 48} = 192 \text{ cm}^3$$

Box

A box has its shape like a cube or cuboid. The amount that a box can hold or contain is called the capacity of the box. Capacity means internal volume.



- Surface area of an open box
 $= 2(\text{length} + \text{breadth}) \times \text{height} + \text{length} \times \text{breadth}$
 $= 2(l + b) \times h + l \times b$
- Capacity of box
 $= (l - 2t)(b - 2t)(h - 2t);$
 where, t = thickness of box
- Volume of the material of the box
 $= \text{External volume} - \text{Internal volume (or capacity)}$
 $= lbh - [(l - 2t)(b - 2t)(h - 2t)]$

Example :- A swimming pool is 40 m in length, 30 m in

breadth and 2.2 m in depth. The cost of cementing its floor and the four sides at Rs. $25/m^2$ is:

SSC CPO 16/03/2019 (Afternoon)

(a) Rs.43,980 (b) Rs.37540 (c) Rs.34260 (d) Rs.37700

Solution :- Total area to be painted

$$= 40 \times 30 + 2(40 \times 2.2 + 30 \times 2.2) = 1508 m^2$$

$$\text{Total cost} = 1508 \times 25 = \text{Rs. } 37700$$

Room

A rectangular room has four walls (surfaces) and opposite walls have equal area.

- Total Area of walls = $2(l + b) \times h$
- Total volume of the room = $l \times b \times h$
- Area of floor or roof = $l \times b$

Example :- The length, breadth, and height of a room are 10 m, 8 m and 6 m respectively. Find the cost of white washing the walls of the room and the ceiling at the rate of ₹7.50 per m^2 .

SSC CGL 24/07/2023 (2nd shift)

(a) Rs.2220 (b) Rs.1850 (c) Rs.2150 (d) Rs.2000

Solution :- Given, length = 10m, breadth = 8m, height = 6m

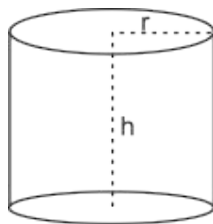
$$\text{Area of four walls} = 2(l + b) \times h$$

$$= 2(10 + 8) \times 6 = 216 m^2$$

$$\text{Area of the ceiling} = \text{length} \times \text{breadth} = 10 \times 8 = 80 m^2$$

$$\text{Cost of white washing the walls and ceiling} = (216 + 80) \times 7.5 = \text{Rs. } 2,220$$

Cylinder



- Volume of cylinder = area of base $\times$ height = $\pi r^2 h$
- Lateral/Curved surface area = Perimeter of base $\times$ height = $2\pi r h$
- Total surface Area = curved surface area + area of both the circles = $2\pi r^2 + 2\pi r h = 2\pi r(r + h)$
- When the rectangular sheet is folded along its length, then the length becomes the circumference of the base of the cylinder and breadth becomes the height of the cylinder.

Example :- The lateral area of a cylinder is $3168 cm^2$ and its height is 48 cm. Find the volume.

SSC CGL 19/07/2023 (2nd shift)

(a) $5244 cm^3$ (b) $5544 cm^3$ (c) $5644 cm^3$ (d) $16632 cm^3$

Solution :- Lateral surface area of the cylinder = $2\pi r h$

According to the question,

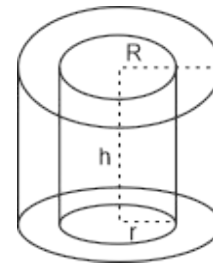
$$2 \times \frac{22}{7} \times r \times 48 = 3168$$

$$r = \frac{3168 \times 7}{2 \times 22 \times 48} = \frac{21}{2} \text{ cm}$$

$$\text{Now, volume of the cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times 48 = 16632 cm^3$$

Hollow Cylinder



- Volume of hollow cylinder = $\pi(R^2 - r^2)h$
- Curved surface area = $2\pi(R + r)h$
- Total surface area = $2\pi(R + r)h + 2\pi(R^2 - r^2)$
 $= 2\pi(R + r)\{h + R - r\}$

Where, R = External radius of cylinder, r = internal radius of cylinder, h = height

Example :- Volume of material used in constructing a hollow cylinder of height 24 cm and thickness 1 cm is 96π . What is the sum of the internal and external radius?

RRB JE 24/05/2019 (Evening)

(a) 7 cm (b) 3 cm (c) 4 cm (d) 5 cm

Solution :- Given :- h = 24 cm,

thickness (R - r) = 1 cm

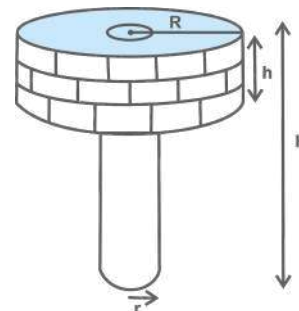
We know that volume of hollow cylinder = $\pi(R^2 - r^2) \times h$

$$\Rightarrow \pi[(R + r)(R - r)] \times h = 96\pi$$

$$\therefore \{a^2 - b^2 = (a + b)(a - b)\}$$

$$\Rightarrow (R + r) \times 1 \times 24 = 96 \Rightarrow (R + r) = \frac{96}{24} = 4 \text{ cm}$$

Embankment



Volume of the cylindrical well = Volume of the hollow cylindrical embankment

$$\Rightarrow \pi r^2 H = \pi[R^2 - r^2] \times h \Rightarrow r^2 H = [R^2 - r^2] \times h$$

Example :- A well of radius 2 m is dug 16.8 m deep. The soil taken out of it is spread evenly all around the well to form a 48 cm high embankment. What is the width (in m) of the embankment?

NTPC CBT II (13/06/2022) 2nd Shift

(a) 8 (b) 10 (c) 9.6 (d) 8.4

Solution :- Radius(r) = 2 m

Height = 16.8 m

Let the radius of embankment = R

Width of embankment = R - r

As per question,

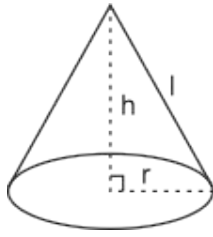
$$\frac{22}{7} \times 2 \times 2 \times 16.8 = \frac{22}{7} \times (R^2 - 4) \times 0.48$$

$$R^2 = 144 \Rightarrow R = 12$$

$$\text{Width of embankment} = R - r = 12 - 2 = 10 \text{ m}$$

Cone

A solid and round body with a round base and pointed peak.



- Volume = $\frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \pi r^2 h$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Curved surface area = $\pi r l = \pi r \sqrt{r^2 + h^2}$
- Total surface area = $\pi r l + \pi r^2 = \pi r(l + r)$
- Cone formed by rotating right angled triangle about its height:

Example :- The area of the base of a cone is 186.34 cm^2 . If the height of the cone is nine seventh of its radius, then what is its volume (in cm^3) ?

DDA JE (Civil) 01/04/2023 (Morning)

(a) 614.922 (b) 611.657 (c) 612.754 cm^3 (d) 613.841

Solution :- Area of base of cone = $\pi r^2 = 186.34 \text{ cm}^2$,

$$r^2 = \frac{186.34 \times 7}{22} = 59.29 \text{ cm}^2 = r = 7.7 \text{ cm},$$

$$\text{Height of cone} = \frac{9 \times 7.7}{7} = 9.9 \text{ cm},$$

$$\begin{aligned} \text{Volume of cone} &= \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 7.7 \times 7.7 \times 9.9 \\ &= 614.922 \text{ cm}^3 \end{aligned}$$

Example :- The circumference of base of a right circular cone is 88 cm. If the height of the cone is 28 cm, then what is the curved surface area of the cone ?

SSC CHSL 09/03/2023 (2nd Shift)

(a) $670\sqrt{3} \text{ cm}^2$ (b) $616\sqrt{5} \text{ cm}^2$ (c) $627\sqrt{3} \text{ cm}^2$ (d) $661\sqrt{5} \text{ cm}^2$

Solution :- According to the question,
Circumference of the base of the cone
($2\pi r$) = 88cm

$$\text{Then, } r = 88 \times \frac{7}{22} \times \frac{1}{2} = 14 \text{ cm}$$

Slant height of the cone (l)

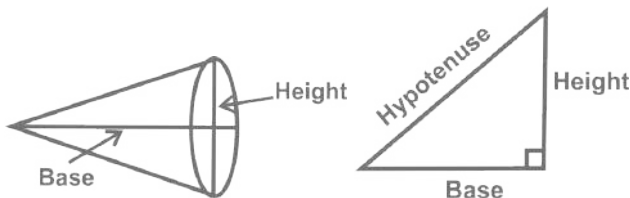
$$= \sqrt{(r)^2 + (h)^2} = \sqrt{(14)^2 + (28)^2} = \sqrt{980} = 14\sqrt{5}$$

Therefore, curved surface area of the cone ($\pi r l$)

$$= \frac{22}{7} \times 14 \times 14\sqrt{5} = 616\sqrt{5} \text{ cm}^2$$

Formation Of Cones By Rotating Triangles

1). Rotation of Right Angle triangle along Base :-



Radius of Cone = Height of Triangle

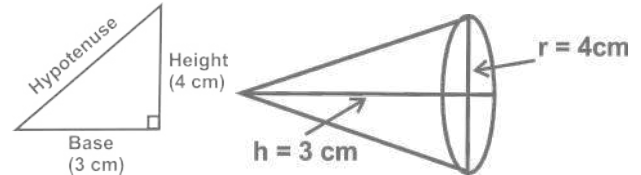
Height of Cone = Base of Triangle

Where : radius = Height , Height = Base and
slant height = Hypotenuse

Hint: to form a cone, the side through which the triangle is rotated, always remains constant.

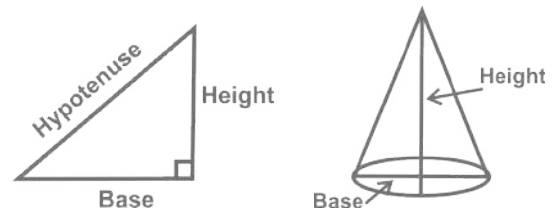
Example :- A right angled triangle of height 4 cm and base 3 cm is rotated around the height of the triangle to form a cone. Find the volume of cone formed. (Take $\pi = \frac{22}{7}$)

Solution :- According to the question,



$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times (4)^2 \times 3 = \frac{352}{7} \text{ cm}^3$$

2). Rotation of Right Angle triangle along Height :-



Radius of Cone = Base of Triangle

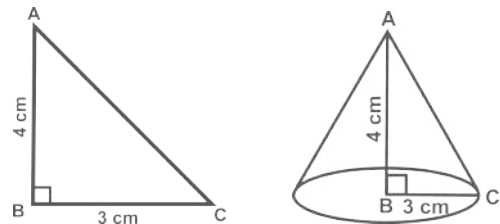
Height of cone = Height of Triangle

Where : radius = Base, Height(cone) = Height(triangle) and slant height = Hypotenuse

Example :- A right angled triangle of height 4cm and base 3 cm is rotated around the height of the triangle to form a cone. Find the volume and surface area of cone formed.

(Take $\pi = \frac{22}{7}$)

Solution :-



According to the question,

The height of the triangle is rotated to form a cone.

Then, Height of the cone = 4 cm and radius of a cone = 3 cm

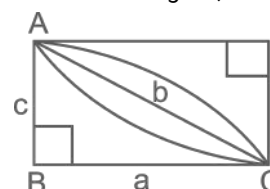
$$\Rightarrow \text{Volume of a cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times (3)^2 \times 4 = \frac{264}{7} \text{ cm}^3$$

Using Triplet (3, 4, 5) = slant height of a cone = 5 cm

$$\Rightarrow \text{Surface area of a cone} = \pi r l = \frac{22}{7} \times 3 \times 4 = \frac{264}{7} \text{ cm}^2$$

3). Rotation of Right Angle triangle along Hypotenuse :-

When given triangle is rotated along AC,

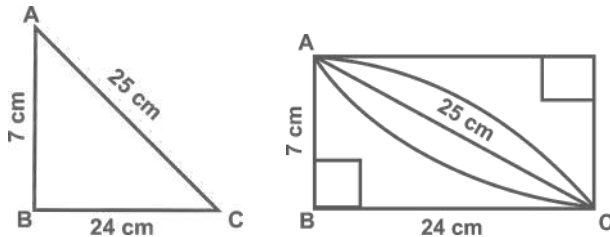


Here, diameter of the cone = 'b', and 2 different slant heights = 'a' and 'c'.

$$\text{Sum of volume of 2 cones} = \frac{1}{3} \pi \frac{(ac)^2}{b}$$

Example :- A right angled triangle having two sides of 7 cm and 24 cm respectively is rotated along its hypotenuse. Find the total volume of the shape formed.

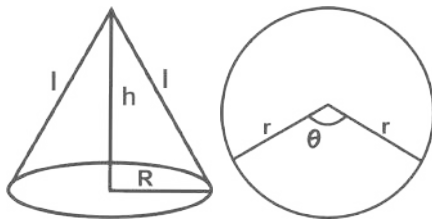
Solution :- According to the question,



By using triplet (7, 24, 25) = diameter of a cone = 25 cm and two different slant height 7 cm and 24 cm

$$\begin{aligned} \text{Total volume of 2 cones} &= \frac{1}{3} \pi \frac{(ac)^2}{b} \\ &= \frac{1}{3} \times \frac{22}{7} \times \frac{(7 \times 24)^2}{25} \\ &= \frac{1}{3} \times \frac{22}{7} \times \frac{(49 \times 576)}{25} = 1182.72 \text{ cm}^3 \end{aligned}$$

Rotation of sector to form a cone :-

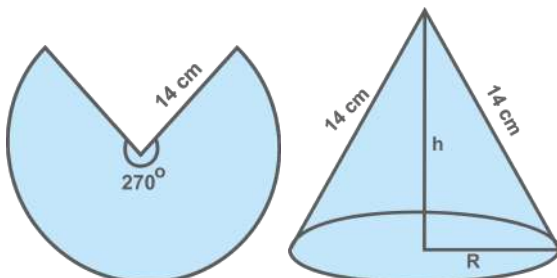


$$R = \frac{\theta}{360} r$$

Slant height of cone (L) = Radius of sector (r)
Curved Surface Area of cone = Area of Sector

Example :- The diagram below shows a sector of a circle of radius 14 cm. The angle of the sector is 270°. The sector is folded to form a cone. Find the radius of the cone.

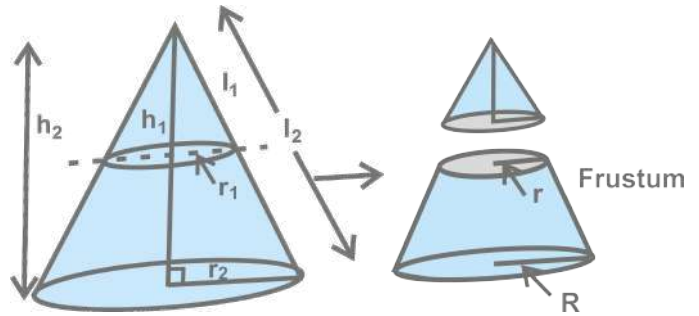
Solution :-



According to the question,

$$\begin{aligned} \text{Radius of the cone (R)} &= \frac{\theta}{360} \times r (\text{radius of a sector}) \\ &= \frac{270}{360} \times 14 = 10.5 \text{ cm} \end{aligned}$$

Results Of Cutting Of Cone



Using the property of similarity, we get the following results:

$$\frac{r_1}{r_2} = \frac{h_1}{h_2} = \frac{l_1}{l_2}$$

Considering the two cones i.e., smaller and bigger, we have:

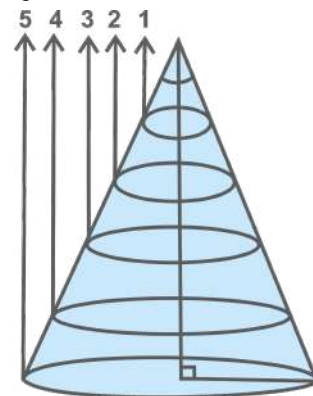
Ratio of CSA of smaller and bigger cone =

$$\frac{\text{smaller cone CSA}}{\text{bigger cone CSA}} = \frac{\pi r_1 l_1}{\pi r_2 l_2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{h_1}{h_2}\right)^2 = \left(\frac{l_1}{l_2}\right)^2$$

Ratio of volume of 2 cones

$$\frac{\text{smaller cone volume}}{\text{bigger cone volume}} = \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2}$$

For the following figure :



The ratio of CSA of 5 parts is as:

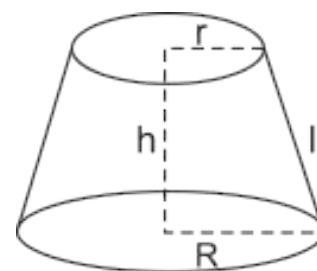
$$1^2 : 2^2 - 1^2 : 3^2 - 2^2 : 4^2 - 3^2 : 5^2 - 4^2 = 1 : 3 : 5 : 7 : 9$$

The ratio of Volume of 5 parts is as:

$$1^3 : 2^3 - 1^3 : 3^3 - 2^3 : 4^3 - 3^3 : 5^3 - 4^3 = 1 : 7 : 19 : 37 : 61$$

Frustum Of Cone

If a cone is cut by a plane parallel to its base, so as to divide the cone into two parts: upper part and lower part, then the lower part is called frustum.



- Slant height (l) = $\sqrt{h^2 + (R - r)^2}$
- Curved Surface Area = $\pi(R + r)l$
- Total surface area = $\pi(R + r)l + \pi R^2 + \pi r^2$
 $= \pi\{(R + r)l + R^2 + r^2\}$
- Volume = $\frac{1}{3} \pi h (r^2 + R^2 + rR)$

Example :- The radii of the ends of a frustum of a cone 7 cm high are 5 cm and 3 cm. Find its volume correct to one decimal place. (Use $\pi = \frac{22}{7}$)

SSC CGL 12/12/2022 (2nd Shift)

(a) 345.6 cm³ (b) 359.3 cm³ (c) 379.3 cm³ (d) 369.3 cm³

Solution: Volume of a frustum = $\frac{1}{3} \times \pi \times h(R^2 + r^2 + R \times r)$

$$= \frac{1}{3} \times \frac{22}{7} \times 7(5^2 + 3^2 + 5 \times 3)$$

$$= \frac{22}{3}(25 + 9 + 15) = \frac{22 \times 49}{3} = \frac{1078}{3} = 359.3 \text{ cm}^3$$

Example :- A solid frustum of a cone is of height 8 cm. If the radii of its lower and upper ends are 3 cm and 9 cm respectively, then its slant height is:

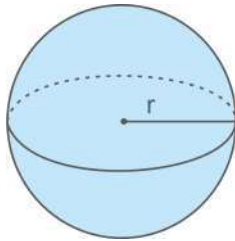
RRB NTPC CBT - I 11/03/2021 (Morning)

(a) 9cm (b) 12 cm (c) 10 cm (d) 15 cm

Solution:- Slant height (L^2) = $H^2 + (R - r)^2$

$$\Rightarrow L^2 = 8^2 + (9 - 3)^2 \Rightarrow L^2 = 64 + 36 \Rightarrow L^2 = 100 \Rightarrow L = 10 \text{ cm}$$

Sphere



- Volume of sphere = $\frac{4}{3} \pi r^3$

- Curved Surface area = Total surface area = $4\pi r^2$

Example :- If the volume of a sphere is 24,416.64 cm³, find its surface area (take $\pi = 3.14$) correct to two places of decimal. SSC CGL 03/12/2022 (1st Shift)

(a) 3069.55 cm² (b) 4069.44 cm²
(c) 5069.66 cm² (d) 6069.67 cm²

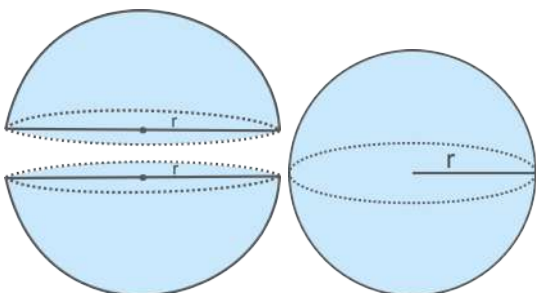
Solution :- volume of a sphere = $\frac{4}{3} \times \pi r^3$

$$\text{Now, } 24416.64 = \frac{4}{3} \times 3.14 \times r^3$$

$$r^3 = \frac{7776 \times 3}{4} = 5832 \Rightarrow r = 18 \text{ cm}$$

$$\text{Now, surface area of sphere} = 4\pi r^2 \\ = 4 \times 3.14 \times 18 \times 18 \Rightarrow 4069.44 \text{ cm}^2$$

Cutting of sphere symmetrically from center :-

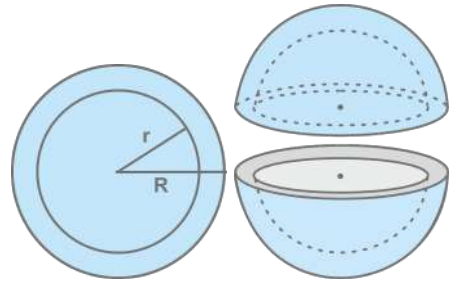


- 1 cut increases area = $2\pi r^2$

$$\text{Total surface area} = 4\pi r^2 + 2\pi r^2 = 6\pi r^2$$

- N cut increases area = $2\pi r^2 N$

Hollow Sphere Or Spherical Shell



- Volume of hollow sphere = $\frac{4}{3} \pi (R^3 - r^3)$

- Internal surface area = $4\pi r^2$

- External surface area = $4\pi R^2$

- Total surface area of a hollow hemisphere

$$= 2\pi(R^2 + r^2) + \pi(R^2 - r^2)$$

Here, R = external radius and r = internal radius

Example :- Find the total surface area (in square cm) and volume (in cube cm) of a hollow hemisphere whose outer radius is 10 cm and inner radius is 5 cm.

RRC Group D 06/10/2022 (Afternoon)

Solution :- Total surface area of a hollow hemisphere

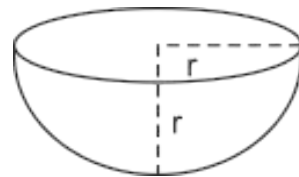
$$= 2\pi(R^2 + r^2) + \pi(R^2 - r^2)$$

$$= 2\pi(10^2 + 5^2) + \pi(10^2 - 5^2) = 250\pi + 75\pi = 325\pi \text{ cm}^2$$

$$\text{Volume of hollow hemisphere} = \frac{4}{3} \pi (R^3 - r^3)$$

$$= \frac{4}{3} \pi (10^3 - 5^3) = \frac{4 \times 825}{3} \pi = 1100\pi \text{ cm}^3$$

Hemisphere



- Volume of the hemisphere = $\frac{2}{3} \pi r^3$

- Total surface area = $3\pi r^2$

- Curved surface area = $2\pi r^2$

Where, r = radius

Example :- The total surface area of a solid hemisphere is 4158 cm². Find its volume (in cm³).

SSC CPO 05/10/2023 (1st Shift)

(a) 9702 (b) 19404 (c) 1848 (d) 462

Solution :- T.S.A of hemisphere = $3\pi r^2$

$$\Rightarrow 3\pi r^2 = 4158 \Rightarrow r^2 = \frac{4158 \times 7}{3 \times 22} \Rightarrow r = 21 \text{ cm}$$

Volume of hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21 = 19404 \text{ cm}^3$$

Example :- The total surface area of a hemisphere is 64.57 cm^2 . Find its curved surface area (cm^2).

SSC CHSL 08/06/2022 (Morning)

(a) 64.57 (b) 53.44 (c) 43.04 (d) 33.04

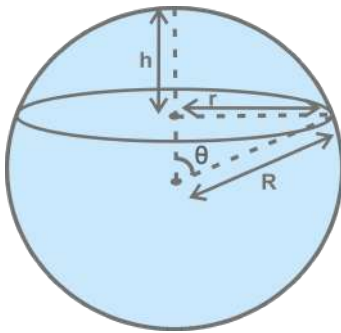
Solution :-

The total surface area of a hemisphere = 64.57 cm^2

$$\Rightarrow 3\pi r^2 = 64.57 \Rightarrow 3 \times \frac{22}{7} \times r^2 = 64.57 \Rightarrow r^2 = 6.848$$

$$\text{Curved surface area} = 2 \times \frac{22}{7} \times 6.848 = 43.04 \text{ cm}^2$$

Spherical Cap



Where : r = radius of base of the cap, R = radius of the sphere, h = height of the spherical cap and θ = angle between the rays from the centre of the sphere to the apex of the cap

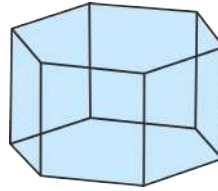
	Using R and h	Using r and h	Using R and θ
Volume	$\frac{\pi h^2}{3}(3R - h)$	$\frac{\pi h}{6} \times (3r^2 + h^2)$	$\frac{\pi R^3}{3} \times (2 + \cos\theta) \times (1 - \cos\theta)^2$
Area	$2\pi Rh$	$\pi(r^2 + h^2)$	$2\pi R^2(1 - \cos\theta)$

Prism

(Prism based on different shapes)

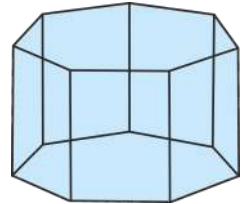
Triangular	Rectangular
→ 5 faces(2 triangular and 3 rectangular) → 9 edges → 6 vertices	→ 6 faces (all rectangular) → 12 edges → 8 vertices
Pentagonal	Square
→ 7 faces(2 pentagonal and 5 rectangular) → 15 edges → 10 vertices	→ 6 faces(2 square and 4 rectangular) → 12 edges → 8 vertices

Hexagonal



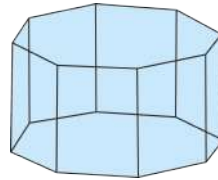
→ 8 faces(2 Hexagonal and 6 rectangular)
→ 18 edges
→ 12 vertices

Heptagonal



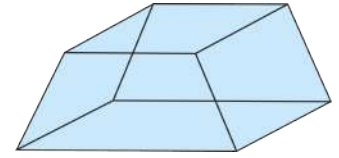
→ 9 faces(2 Hexagonal and 7 rectangular)
→ 19 edges
→ 14 vertices

Octagonal



→ 10 faces(2 Octagonal and 8 rectangular)
→ 24 edges
→ 16 vertices

Trapezoidal



→ 6 faces(2 trapezoidal and 4 rectangular)
→ 12 edges
→ 8 vertices

- Volume of prism = area of base $\times$ height
- Lateral surface area = Perimeter of base $\times$ height
- Total surface Area = Lateral surface area + area of base and top surface

Example:- The base of a right prism is an equilateral triangle whose side is 10 cm. If height of this prism is $10\sqrt{3}$ cm, then what is the total surface area of prism?

SSC CGL Tier II (08/08/2022)

(a) $125\sqrt{3}$ (b) $325\sqrt{3}$ (c) $150\sqrt{3}$ (d) $350\sqrt{3}$

Solution :- Total surface area of prism

$$= [\text{Perimeter of base} \times \text{height}] + [2 \times \text{area of base}]$$

$$= [(10 + 10 + 10) \times 10\sqrt{3}] + [2 \times \frac{\sqrt{3}}{4} \times 10^2]$$

$$= [30 \times 10\sqrt{3}] + 50\sqrt{3} = 300\sqrt{3} + 50\sqrt{3} = 350\sqrt{3} \text{ cm}^2$$

Example:- The base of a right prism is a regular hexagon of side 5 cm. If its height is $12\sqrt{3}$ cm, then its volume (in cm^3)

SSC CGL Tier II (15/11/2020)

(a) 900 (b) 1800 (c) 1350 (d) 675

Solution :- Regular hexagon consists of 6 equilateral triangles.

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2 = \frac{25\sqrt{3}}{4}$$

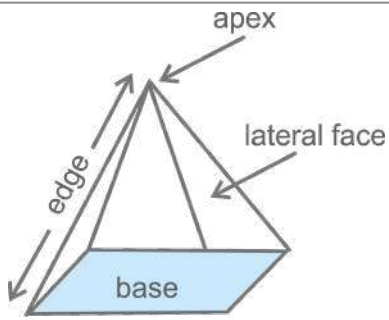
$$\text{Area of base} = 6 \times \text{Area of triangle} = \frac{150\sqrt{3}}{4}$$

$$\text{Volume of prism} = \text{area of base} \times \text{height}$$

$$= \frac{150\sqrt{3}}{4} \times 12\sqrt{3} = 1350 \text{ cm}^3$$

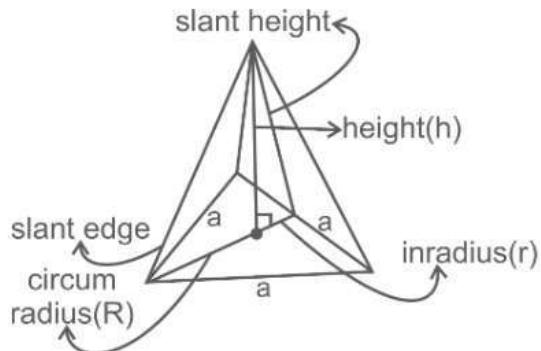
Pyramid

Note :- If the base is not round, it will be called a pyramid. A pyramid can have various shapes of the base example: square, rectangular, triangular etc.



- Pyramid means a structure with regular polygon as its base and sloping sides that meet in a point at the top.
- In Pyramid, with n sided regular polygon at its base, total number of vertices = $n + 1$
- Volume = $\frac{1}{3} \times \text{base area} \times \text{height}$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Lateral surface area = $\frac{\text{Perimeter} \times \text{slant height}}{2}$
- Total Surface area = Lateral surface area + Area of base

(1) Equilateral Triangular Pyramid :-



- Volume = $\frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 \times h$
- Curved Surface area = $\frac{1}{2} \times 3a \times l$
- Slant Height (l) = $\sqrt{r^2 + h^2} = \sqrt{\left(\frac{a}{2\sqrt{3}}\right)^2 + h^2}$
- Slant Edge = $\sqrt{R^2 + h^2} = \sqrt{\left(\frac{a}{\sqrt{3}}\right)^2 + h^2}$
- Total Surface Area = $\frac{1}{2} \times 3al + \frac{\sqrt{3}}{4} a^2$

Example :- The base of a right pyramid is an equilateral triangle with area $16\sqrt{3} \text{ cm}^2$. If the area of one of its lateral faces is 30 cm^2 , then its height (in cm) is:
SSC CGL Tier II (13/09/2019)

(a) $\sqrt{\frac{209}{12}}$ (b) $\sqrt{\frac{739}{12}}$ (c) $\sqrt{\frac{643}{12}}$ (d) $\sqrt{\frac{611}{12}}$

Solution :- Area of an equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

$$\Rightarrow 16\sqrt{3} = \frac{\sqrt{3}}{4} a^2 \Rightarrow a = 8$$

Radius of the pyramid = inradius of the triangle

$$= \frac{a}{2\sqrt{3}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

And Area of its lateral face = $\frac{1}{2} \times \text{base} \times \text{slant height}$

$$= \frac{1}{2} \times 8 \times \text{slant height} = 30 \Rightarrow \text{slant height} = \frac{15}{2}$$

Height of the pyramid = $\sqrt{l^2 - r^2}$

$$= \sqrt{\left(\frac{15}{2}\right)^2 - \left(\frac{4}{\sqrt{3}}\right)^2} = \sqrt{\frac{611}{12}} \text{ cm}$$

Example :- A pyramid has an equilateral triangle as its base, of which each side is 8 cm. Its slant edge is 24 cm. The whole surface area of the pyramid (in cm^2) is:
SSC CGL Tier II (06/03/2023)

- (a) $(16\sqrt{3} + 24\sqrt{35})$ (b) $(12\sqrt{3} + 24\sqrt{35})$
(c) $(24\sqrt{3} + 36\sqrt{35})$ (d) $(16\sqrt{3} + 48\sqrt{35})$

Solution :- LSA = $\frac{1}{2} \times \text{Perimeter of Base} \times \text{Slant height}$

TSA of pyramid = LSA + Area of Base

$$(\text{Slant Edge})^2 = (\text{Slant height})^2 + \left(\frac{\text{Side}}{\sqrt{3}}\right)^2$$

$$(24)^2 = (\text{Slant height})^2 + \left(\frac{8}{\sqrt{3}}\right)^2$$

$$(24)^2 - \left(\frac{8}{\sqrt{3}}\right)^2 = \text{Slant height}^2$$

$$\frac{576 \times 3 - 64}{3} = \text{Slant height}^2$$

$$\sqrt{\frac{1664}{3}} = \text{Slant height}$$

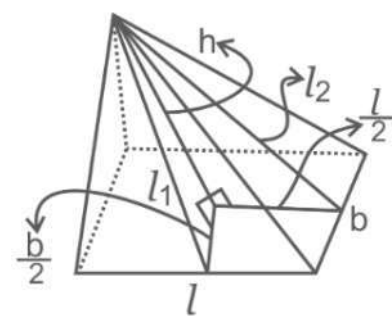
$$\sqrt{\frac{16 \times 104}{3}} = \text{Slant height}$$

$$\text{Slant height} = 4\sqrt{35} \text{ cm}$$

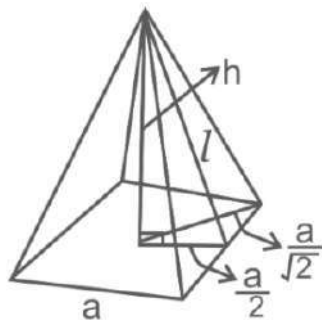
$$\text{LSA} = 48\sqrt{35} \text{ cm}$$

$$\text{TSA} = 16\sqrt{3} + 48\sqrt{35} \text{ cm}^2$$

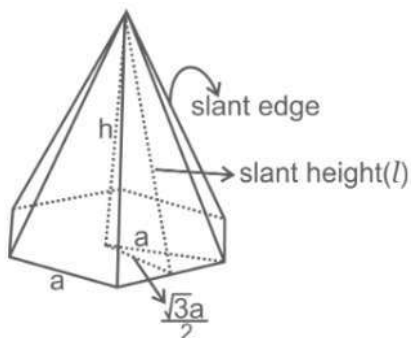
(2) Rectangular Pyramid :-



- Curved Surface area = $2 \times \frac{1}{2} l \times l_1 + 2 \times \frac{1}{2} b \times l_2$
- Total Surface Area = C.S.A + lb
- Volume = $\frac{1}{3} \times lb \times h$
- There are two slant height
- First slant height (l_1) = $\sqrt{\left(\frac{b}{2}\right)^2 + h^2}$
- Second slant height (l_2) = $\sqrt{\left(\frac{l}{2}\right)^2 + h^2}$

(3) Square Pyramid :-

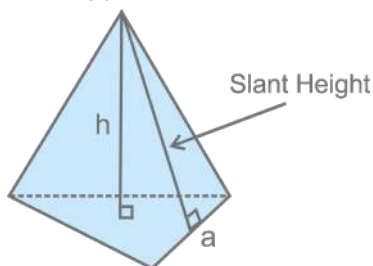
- Curved Surface area = $\frac{1}{2} \times 4a \times l$
- Total Surface Area = $\frac{1}{2} \times 4a \times l + a^2$
- Volume = $\frac{1}{3} \times a^2 \times h$
- Slant Height (l) = $\sqrt{\left(\frac{a}{2}\right)^2 + h^2}$
- Slant Edge = $\sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 + h^2}$

(4) Hexagonal Pyramid :-

- Curved Surface area = $\frac{1}{2} \times 6a \times l$
- Total Surface Area = $\frac{1}{2} \times 6a \times l + 6 \times \frac{\sqrt{3}}{4} a^2$
- Volume = $\frac{1}{3} \times 6 \times \frac{\sqrt{3}}{4} a^2 \times h$
- Slant Height (l) = $\sqrt{\left(\frac{\sqrt{3}}{2} a\right)^2 + h^2}$
- Slant Edge = $\sqrt{a^2 + h^2}$

Tetrahedron

It is a 3D figure made by joining four equilateral triangles.



- Height of Tetrahedron (h) = $\frac{\sqrt{2}}{\sqrt{3}} a$
- Slant height = $\frac{\sqrt{3}}{2} a$

- Volume (V) = $\frac{1}{3} \times \text{area of base} \times \text{height}$

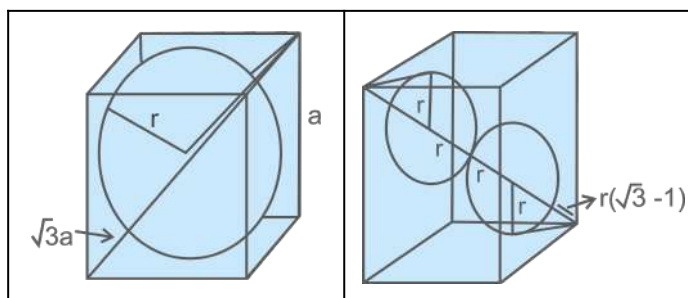
$$= \frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 \times \frac{\sqrt{2}}{\sqrt{3}} a = \frac{\sqrt{2}}{12} a^3$$
- Lateral surface area = $3 \times \frac{\sqrt{3}}{4} a^2$
- Total surface area = $\frac{\sqrt{3}}{4} a^2 \times 4 = \sqrt{3} a^2$

Example :- The box is in the form of tetrahedron of each side $3\sqrt{2}$ cm. What is the volume of the given box ?

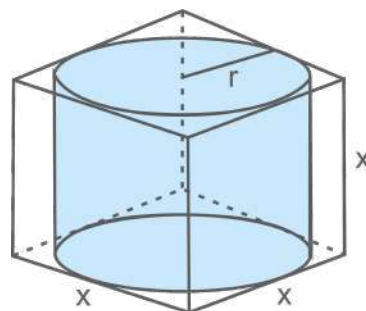
Solution :- Given that :

Side of a tetrahedron (a) = $3\sqrt{2}$ then,

$$\begin{aligned} \text{Volume of a tetrahedron} &= \frac{\sqrt{2}}{12} (3\sqrt{2})^3 = \frac{\sqrt{2} \times 54\sqrt{2}}{12} \text{ cm}^3 \\ &= \frac{54}{6} \text{ cm}^3 = 9 \text{ cm}^3 \end{aligned}$$

Some Combination of 3D Objects

- Shortest distance from vertex of cube to sphere = $\frac{a(\sqrt{3} - 1)}{2}$
(when 1 sphere in a cube)
- When two equal sphere in a cube, then
 $\Rightarrow \sqrt{3}a = 2r(1 + \sqrt{3})$

Maximum Cylinder inside the Cube :-

$$\text{Radius of cylinder} = \frac{\text{Edge of the cube}}{2} = \frac{x}{2}$$

Height of cylinder = Edge of the cube

Then,

Volume of cube : Volume of cylinder

$$\Rightarrow x^3 : \pi \left(\frac{x}{2}\right)^2 x$$

$$\Rightarrow 1 : \frac{22}{7} \times \frac{1}{4}$$

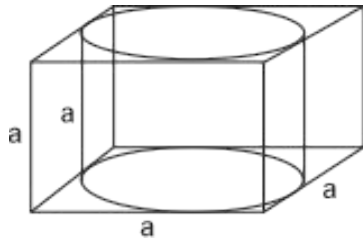
$$\frac{\text{Volume of Cube}}{\text{Volume of Cylinder}} = \frac{14}{11}$$

Example :- A right circular cylinder of maximum possible size is cut out from a solid wooden cube. The remaining material of the cube is what percentage of the original cube?

(take $\pi = 3.14$)

SSC CPO 13/12/2019 (Morning)

(a) 22.4 (b) 21.5 (c) 22.8 (d) 21.8

Solution :-Diameter of the cylinder = a

$$r = \frac{a}{2} \text{ and } h = a$$

The percentage of remaining material of the cube to the

$$\text{original cube} = \frac{a^3 - \frac{\pi a^3}{4}}{a^3} \times 100$$

$$= \frac{4 - \pi}{4} \times 100 = 0.86 \times 25 = 21.42 \approx 21.5 \%$$

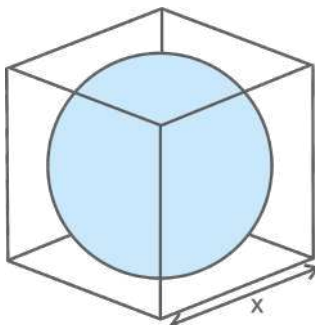
Exam hall approach :- We know that,

$$\Rightarrow \frac{\text{Volume of Cube}}{\text{Volume of Cylinder}} = \frac{14}{11}$$

According to the question,

Percentage of remaining material of the cube to the original

$$\text{cube} = \frac{3}{14} \times 100 = 21.42\% \approx 21.5 \%$$

Maximum Sphere inside the Cube :-Diameter of Sphere ($2r$) = edge of cube = x

Volume of cube : volume of sphere

$$\Rightarrow x^3 : \frac{4}{3}\pi\left(\frac{x}{2}\right)^3$$

$$\frac{\text{Volume of Cube}}{\text{Volume of sphere}} = \frac{21}{11}$$

Example :- What is the volume of the largest sphere that can be made in a cube of side 3.5 m ?(Take $\pi = \frac{22}{7}$)**Solution :-** According to the question,

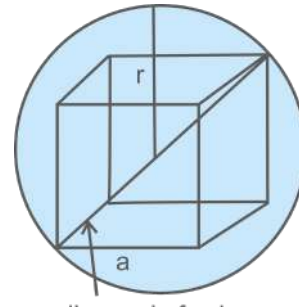
$$\text{Volume of the cube} = (3.5)^3 = 42.875 \text{ m}^3$$

We know that,

$$\frac{\text{Volume of Cube}}{\text{Volume of sphere}} = \frac{21}{11}$$

So, 21 unit = 42.875

$$\Rightarrow 11 \text{ unit} = \frac{42.875}{21} \times 11 = 22.45 \text{ m}^3$$

Maximum cube inside the Sphere :-

diagonal of cube

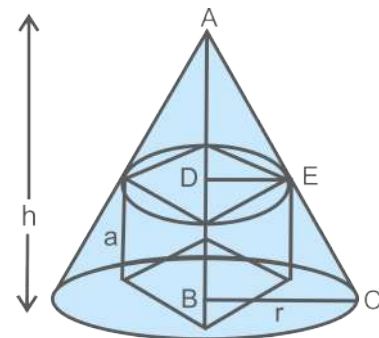
Diagonal of cube = Diameter of sphere

$$\Rightarrow \sqrt{3}a = 2r \Rightarrow a = \frac{2r}{\sqrt{3}}$$

$$\text{Volume of sphere : Volume of cube} \Rightarrow \frac{4}{3}\pi r^3 : \left(\frac{2r}{\sqrt{3}}\right)^3$$

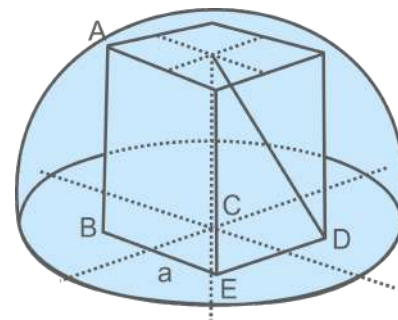
$$\Rightarrow \frac{4}{3} \times \frac{22}{7} : \frac{8}{3\sqrt{3}}$$

$$\frac{\text{Volume of Sphere}}{\text{Volume of Cube}} = \frac{11\sqrt{3}}{7}$$

Maximum cube inside the Cone :-

$$\triangle ABC \sim \triangle ADE$$

$$\Rightarrow \frac{AB}{AD} = \frac{BC}{DE} \Rightarrow \frac{h}{h-a} = \frac{r}{\frac{a}{\sqrt{2}}} \Rightarrow a = \frac{\sqrt{2}rh}{\sqrt{2}r+h}$$

Maximum cube inside the Hemisphere :-Let R be the radius of the hemisphere and ' a ' be the side of the cube. C is the Centre of the hemisphere.

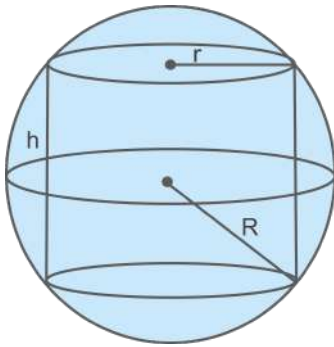
$$BD = \sqrt{2}a$$

$$\text{So, } BC = \frac{a}{\sqrt{2}}$$

$$\text{In } \triangle ABC, AC^2 = AB^2 + BC^2$$

$$\text{Now, } R^2 = \frac{3a^2}{2}$$

$$a = \sqrt{\frac{2}{3}} R$$

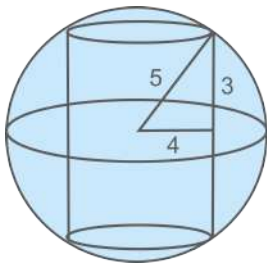
Maximum Cylinder inside Sphere :-

$$\frac{\text{Volume of Cylinder}}{\text{Volume of sphere}} = \frac{1}{\sqrt{3}}$$

Example :- A cylinder is inscribed in a sphere. The radius of the sphere is $\frac{5}{6}$ times the height of the cylinder. What is the ratio of the volume of the cylinder to the volume of the sphere?
NTPC CBT II (10/05/2022) 1st Shift

- (a) 27 : 50 (b) 72 : 125 (c) 6 : 5 (d) 36 : 25

Solution :-



Given :- $R = \frac{5}{6} h \Rightarrow \frac{R}{h} = \frac{5}{6}$

Total height = 6 units

So half height = 3 units

Radius of sphere (R) = 5 units

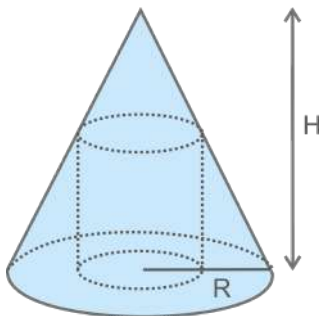
By pythagoras theorem

Radius of cylinder (r) = $\sqrt{R^2 - h^2}$

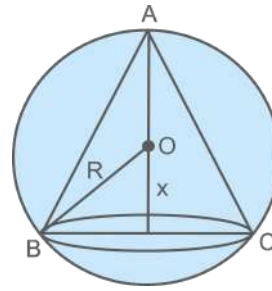
= $\sqrt{25 - 9} = \sqrt{16} = 4$ units

Now required ratio

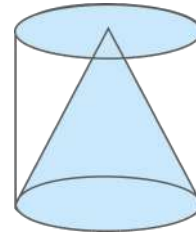
$$\frac{\text{Volume of cylinder}}{\text{Volume of Sphere}} = \frac{\pi \times 4 \times 4 \times 6}{\frac{4}{3} \pi \times 5 \times 5 \times 5} = \frac{72}{125}$$

Maximum Cylinder inside Cone :-

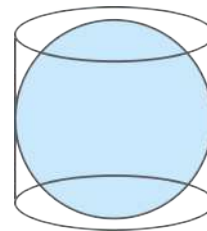
$$\frac{\text{Volume of Cylinder}}{\text{Volume of Cone}} = \frac{4}{9}$$

Maximum Cone inside Sphere :-

$$\frac{\text{Volume of Cone}}{\text{Volume of Sphere}} = \frac{8}{27}$$

Maximum Cone inside Cylinder :-

$$\frac{\text{Volume of Cylinder}}{\text{Volume of Cone}} = \frac{3}{1}$$

Maximum Sphere inside Cylinder :-

$$\frac{\text{Volume of Cylinder}}{\text{Volume of Sphere}} = \frac{3}{2}$$

Example :- A cylinder has some water in it at a height of 16 cm. If a sphere of radius 9 cm is put into it, then find the rise in the height of the water if the radius of the cylinder is 12 cm.
SSC CPO 05/10/2023 (1st Shift)

- (a) 1.6 cm (b) 8.75 cm (c) 8 cm (d) 6.75 cm

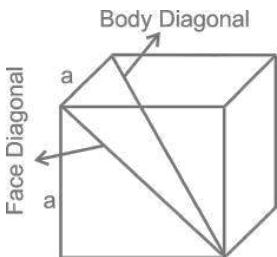
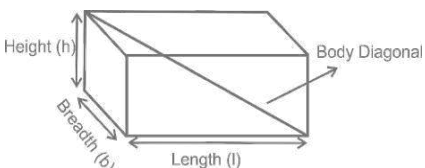
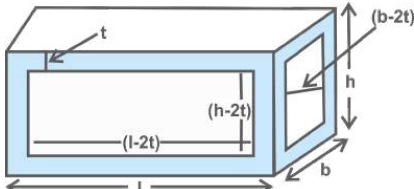
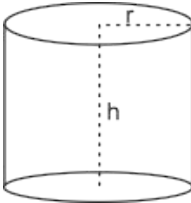
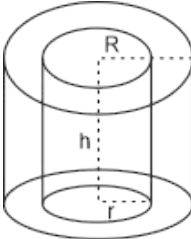
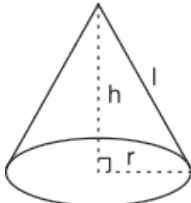
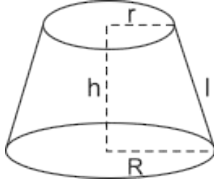
Solution :- According to the question,

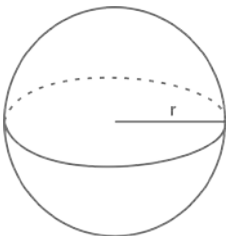
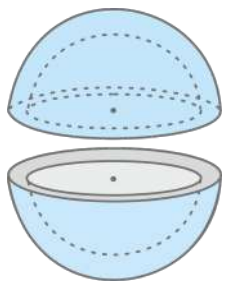
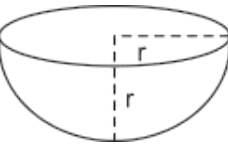
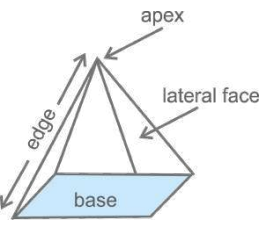
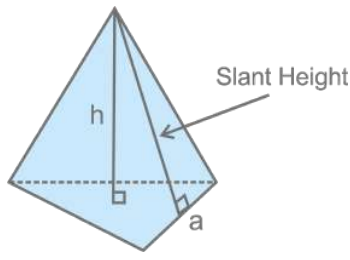
Volume of cylinder = Volume of sphere

$$\pi r^2 h = \frac{4}{3} \pi r^3 \Rightarrow 12 \times 12 \times h$$

$$= \frac{4}{3} \times 9 \times 9 \times 9 \Rightarrow h = \frac{4 \times 9 \times 9 \times 9}{3 \times 12 \times 12} \Rightarrow h = 6.75 \text{ cm}$$

Basic Formulae Table

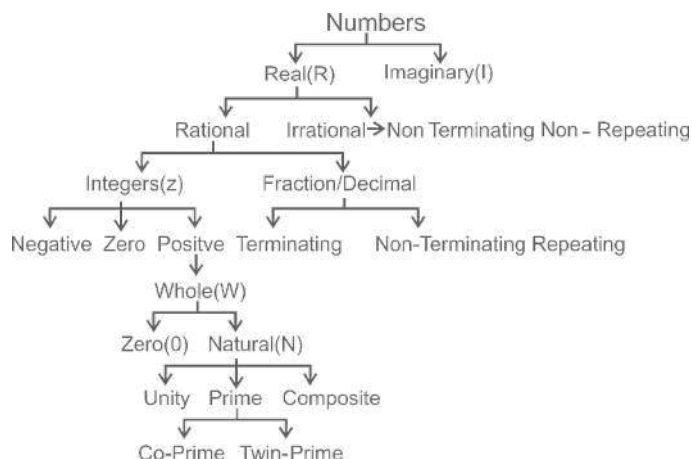
Three dimensional figure	Shape	Volume	Lateral surface area	Total surface area
Cube		a^3	$4a^2$	$6a^2$
Cuboid		$l \times b \times h$	$2(l + b) \times h$	$2(lb + bh + lh)$
Box		$(l - 2t)(b - 2t)(h - 2t)$ where, t = thickness of box	—	—
Cylinder		area of base $\times$ height = $\pi r^2 h$	$2\pi r h$	$2\pi r^2 + 2\pi r h = 2\pi r(r + h)$
Hollow Cylinder		$\pi(R^2 - r^2)h$	$2\pi(R + r)h$	$2\pi(R + r)h + 2\pi(R^2 - r^2)$
Cone		$\frac{1}{3} \times \text{area of base} \times \text{height} = \frac{1}{3}\pi r^2 h$	$\pi r l = \pi r \sqrt{r^2 + h^2}$	$\pi r l + \pi r^2 = \pi r(l + r)$
Frustum		$\frac{1}{3}\pi h(r^2 + R^2 + rR)$	$\pi(R + r)l$	$\pi\{(R + r)l + R^2 + r^2\}$

Sphere		$\frac{4}{3} \pi r^3$	$4\pi r^2$	—
Hollow Sphere		$\frac{4}{3} \pi (R^3 - r^3)$	—	$2\pi(R^2 + r^2) + \pi(R^2 - r^2)$
Hemisphere		$\frac{2}{3} \pi r^3$	$2\pi r^2$	$3\pi r^2$
Prism	—	area of base $\times$ height	Perimeter of base $\times$ height	Lateral surface area + area of base and top surface
Pyramid		$\frac{1}{3} \times \text{area of base} \times \text{height}$	$\frac{\text{Perimeter} \times \text{slant height}}{2}$	Lateral surface area + Area of base
Tetrahedron		$\frac{1}{3} \times \text{area of base} \times \text{height} = \frac{\sqrt{2}}{12} a^3$	$3 \times \frac{\sqrt{3}}{4} a^2$	$\frac{\sqrt{3}}{4} a^2 \times 4 = \sqrt{3} a^2$

Number System

Number System : - A number system is defined as a system of writing to express numbers. It is the mathematical notation for representing numbers of a given set by using digits or other symbols in a consistent manner.

Classification of Numbers



(1). Real Numbers (R) : Real numbers are numbers that include both rational and irrational numbers. In general all the arithmetic operations can be performed on these numbers and they can be represented in the number line also.

Examples : 0, +8, - 14, $\frac{4}{9}$, 1.8 etc

(2). Imaginary Numbers (i) : Imaginary Numbers are defined as the square root of the negative number where it doesn't have a definite value. It is mostly written in the form of real number multiplied by imaginary unit called "i".

Examples : $\sqrt{-3}$, $\sqrt{7}$

Note :- $i = \sqrt{-1}$, $i^2 = -1$, $i^4 = 1$

Complex number: A complex number is the sum of a real number and an imaginary number. A complex number is of the form $a + ib$ and is usually represented by z .

Standard form :-

$$Z = a + ib$$

↙
↘

 Real part Imaginary part

Imaginary number

Types of Real Number

(a) Rational Numbers : Numbers whose exact value can be determined. Also a number which can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$ is called a rational number. For example,

Examples : $\frac{3}{4} = 0.75$, $\frac{4}{5} = 0.8$, $\frac{9}{-5}$, $\frac{22}{7}$ etc.

(b) Irrational Numbers : Numbers whose exact value cannot be determined. A number whose decimal expansion is

non-terminating non - recurring is irrational.

Example : $\pi = 3.142857142857 \dots$

Types of Rational Number

(a) Integer (Z): Integers are numbers that can be positive, negative or zero, but cannot be a fraction.

$I = \dots, -4, -3, -2, -1, 0, 1, 2, 3, 4 \dots$

(b) Fraction / Decimal :- Fractions are written in the form of $\frac{p}{q}$, where $q \neq 0$, while in decimals, the whole number part and fractional part are connected through a decimal point, for example, 0.5.

Types of Integer (Z)

(a) Zero (0) : Zero is neither a positive nor a negative integer. It is a neutral number i.e. zero has no sign (+ or -).

(b) Positive Integers (Z^+) : Positive numbers are called positive integers. These integers lie on the right side of 0 on a number line.

$(Z^+) = 1, 2, 3, 4, 5 \dots$

(c) Negative Integers (Z^-) : Negative numbers are called negative integers. These integers lie on the left side of 0 on a number line.

$(Z^-) = -1, -2, -3, -4, -5 \dots$

Types of Positive Integers(z)

Whole Numbers (W) :- Any positive number without a fractional or decimal part is referred to as a whole number.

Examples: 0, 1, 2, 3, 4... etc.

Natural Numbers (N) :- All positive integers from 1 to infinite are called natural numbers.

Examples :- 1, 2, 3, 4 ... etc.

Types of Natural Number

Prime number : A number which is divisible by 1 and itself. 2 is only an even prime number.

Example :- 2, 3, 5, 7, 11, etc.

Note :-

Total prime no. between 1 - 50 → 15

Total prime no. between 50 - 100 → 10

Total prime no. between 1 - 100 → 25

Total prime no. between 1 - 500 → 95

Total prime no. between 1 - 1000 → 168

Note :- Every prime number can be written in $(6x \pm 1)$ form, But every $(6x \pm 1)$ may not be a prime number.

Example :- $6 \times 3 - 1 = 17$ (Prime No),

$6 \times 4 + 1 = 25$ (Not Prime No)

Unity Number :- A number representing a single or the only entity. For example, a line segment of unit length is a line segment of length 1.

Composite number : A number which has two or more than 2

factors are called Composite numbers.

Example :- 4, 6, 8, 9, 10, 12 etc.

Note :- 1 is neither prime nor composite.

Smallest composite number $\rightarrow$ 4

Smallest odd composite number $\rightarrow$ 9

If x and y are any two odd prime numbers then, $x^2 + y^2$ and $x^2 - y^2$ is a composite number.

Types of Prime Number

(a) Co-prime number :- Two numbers are called Co-prime numbers if their HCF is 1.

Example :- (2 and 3), (6 and 11).

Note :- Two prime numbers are always co-prime numbers to each other. Any two consecutive integers are always co-prime number to each other.

(b) Twin prime :- Two consecutive prime numbers with difference 2 are called Twin-prime numbers.

Example : (3, 5) (5, 7) (11, 13) (17, 19) etc.

3, 5, 7 is the only pair of prime numbers with a gap of 2.

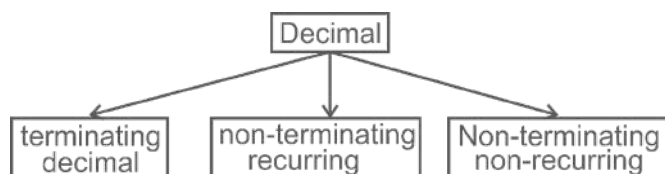
Prime Number Test

For finding whether any number is a prime number or not, we need to find the nearest square root of given number, then we need to find out whether the given number is divisible by any prime number less than the obtained number or not. If it is divisible then it is not a prime number and if not divisible then it is a prime number.

Example :- Find whether 177 is a prime number or not.

Solution :- Nearest square root of 177 is 13. Now we need to check whether 177 is divisible by prime numbers less than 13. On checking we find that 177 is divisible by 3. Hence, 177 is not a prime number.

Decimal Number



Terminating decimal :- Can be written in $\frac{p}{q}$ form where $q \neq 0$ and have a finite number of digits after the decimal point. They are also called rational numbers i.e., $0.4 \rightarrow \frac{2}{5}$

Non-terminating recurring :- Can be written in $\frac{p}{q}$ form where $q \neq 0$ but they never end, they repeat one or more numbers after the decimal point i.e., $0.\overline{5} \rightarrow \frac{5}{9}$

Non terminating Non repeating :- Can't be written in $\frac{p}{q}$ form And they continue endlessly, with no group of digits repeating. Also called irrational numbers. i.e., $\pi, \sqrt{2}, \sqrt{3}$

Perfect Number :- A perfect number is a positive integer that is equal to the sum of its factors except for the number itself.

Example: 6, 28, 496 etc.

$6 \rightarrow$ factors (1, 2, 3, 6) $\rightarrow 1 + 2 + 3 = 6$

$28 \rightarrow$ factors (1, 2, 4, 7, 14, 28) $\rightarrow 1 + 2 + 4 + 7 + 14 = 28$

Note :- 6 is the smallest Perfect number.

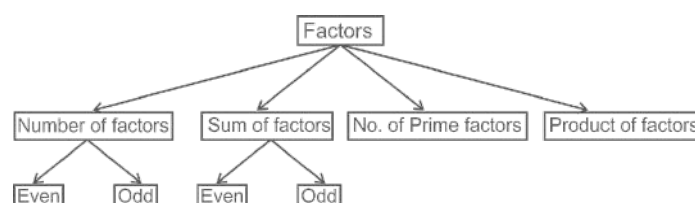
Basics of Number System

(1). Face Value :- It is nothing but the number itself about which it has been asked.

Eg: In the number 23576, face value of 5 is 5 and face value of 7 is 7.

(2). Place Value :- The place value of a number depends on its position in the number. Each position has a value 10^n , the places to its right.

Example:- In the number 23576, place value of 5 is 500 and place value of 3 is 3000.



The factors of a number are the numbers that divide it completely without leaving any remainder.

Example:- 24 can be completely divided by 1, 2, 3, 4, 6, 8, 12 and 24, so these numbers are factors of 24.

Prime factorisation of a number :- When a number is written in the form of multiplication of its prime factors, it's called prime factorisation.

i.e. Prime factorisation of 24.

$$\begin{array}{r|l} 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline & 3 \end{array}$$

$$24 \rightarrow 2 \times 2 \times 2 \times 3 \text{ or } 2^3 \times 3^1$$

Find total Number of factors :- To find the number of factors we write the number in the form of prime factors and then add +1 to the exponent of prime factors and multiply them.

$$\text{If } N = a^p \times b^q \times c^r \times \dots$$

$$\text{Then the total number of factors} = (p+1) \times (q+1) \times (r+1) \dots$$

$$\text{i.e. } 24 = 2^3 \times 3^1$$

$$\text{Number of factors of } 24 \rightarrow (3+1) \times (1+1) = 4 \times 2 = 8.$$

Find sum of factors :- If $N = a^p \times b^q \times c^r \times \dots$

$$\text{Then Sum of factors} = (a^0 + a^1 + a^2 + \dots + a^p) \times (b^0 + b^1 + b^2 + \dots + b^q) \times (c^0 + c^1 + c^2 + \dots + c^r) \dots$$

OR

$$\frac{a^{p+1}-1}{a-1} \times \frac{b^{q+1}-1}{b-1} \times \frac{c^{r+1}-1}{c-1} \times \dots$$

Example:- $24 = 2^3 \times 3^1$,

Solution :- Sum of all factors

$$= (2^0 + 2^1 + 2^2 + 2^3) \times (3^0 + 3^1) = 15 \times 4 = 60.$$

OR

$$= \frac{2^{3+1} - 1}{2-1} \times \frac{3^{1+1} - 1}{3-1} = \frac{2^4 - 1}{1} \times \frac{3^2 - 1}{2} = 15 \times 4 = 60$$

Number of even factors of a number :- To find the number of even factors of a number, we add +1 to the exponents of prime numbers except 2.

Note :- If a number doesn't have 2 as its factor it will have 0 even factors)

$$\text{If } N = a^p \times b^q \times c^r \times \dots$$

$$\text{Then total number of even factors} = p \times (q + 1) \times (r + 1) \times \dots$$

Example :- Find the number of even factors of 120.

Solution :- $120 = 2^3 \times 3^1 \times 5^1$

$$\text{Number of even factors} = 3 \times (1 + 1) \times (1 + 1) = 3 \times 2 \times 2 = 12$$

Note :- Even factors = Total factors - odd factors

Sum of even factors : we shall ignore 2^0 ,

Example :- Find the sum of even factors of 120.

Solution :- Sum of even factors $= (2^1 + 2^2 + 2^3)(3^0 + 3^1)(5^0 + 5^1) = 14 \times 4 \times 6 = 336.$

OR

$$\begin{aligned} &= \frac{2^{3+1} - 2}{2-1} \times \frac{3^{1+1} - 1}{3-1} \times \frac{5^{1+1} - 1}{5-1} \\ &= \frac{2^4 - 2}{1} \times \frac{3^2 - 1}{2} \times \frac{5^2 - 1}{4} = 14 \times 4 \times 6 = 336 \end{aligned}$$

Number of odd factors of a number :- To find the number and sum of odd factors of a number, we have to ignore the exponents of 2.

$$\text{If } N = a^p \times b^q \times c^r \times \dots$$

Where $a = 2$

$$\text{Then total number of odd factors} = (q + 1) \times (r + 1) \times \dots$$

Example :- Find the number of odd factors of 120.

Solution :- $120 = 2^3 \times 3^1 \times 5^1$

$$\text{Required number} = (1 + 1)(1 + 1) = 4$$

The exponent of 2 is completely ignored.

Sum of odd factors of a number :-

$$\text{Sum of odd factors of } 120 = (3^0 + 3^1)(5^0 + 5^1) = 4 \times 6 = 24$$

OR

$$= \frac{3^{1+1} - 1}{3-1} \times \frac{5^{1+1} - 1}{5-1} = \frac{3^2 - 1}{2} \times \frac{5^2 - 1}{4} = 4 \times 6 = 24.$$

Sum of reciprocal of all factors :- $\frac{\text{Sum of factors}}{\text{Number}}$

Example :- Find the sum of reciprocal of all factors of 120.

Solution :- $120 = 2^3 \times 3^1 \times 5^1$

$$\text{Sum of factors} = (2^0 + 2^1 + 2^2 + 2^3)(3^0 + 3^1)(5^0 + 5^1) = 360$$

$$\text{Sum of reciprocal of all factors} = \frac{\text{Sum of factors}}{\text{Number}} = \frac{360}{120} = 3$$

Product of factors = (Number) ^{$\frac{\text{Total number of factors}}{2}$}

Example :- Find the product of factors of 20.

Solution :- $20 = 1, 2, 4, 5, 10, 20.$

$$\text{Product of factors} = (20)^{\frac{6}{2}} = (20)^3 = 8000$$

Average of Factors = $\frac{\text{Sum of factors}}{\text{Number of factors}}$

Total number of Prime factors :- If $N = a^p \times b^q \times c^r \times \dots$ here a, b, c are prime factor so total number of prime factor in given number $= p + q + r$

Number of factors which are co-prime to each other :-

If $N = a^p \times b^q$ where a, b are prime factor

Then Number of coprime factor will be $= [(p + 1)(q + 1) + pq]$

If $N = a^p \times b^q \times c^r$

Then Number of coprime factor will be $= [(p + 1)(q + 1)(r + 1) + pq + qr + rp + 3pqr]$

Some Important Results of Factors :-

$$* 1001 = 7 \times 11 \times 13$$

$$1001 \times abc = abcabc$$

$$1001 \times 234 = 234234$$

Example :- Which of the following is a factor of 531531 ?

(a) 15 (b) 13 (c) 11 (d) both b and c

Solution :- $531531 = 1001 \times 531 = 7 \times 11 \times 13 \times 531$

So, both 11 and 13 are factors of 531531.

$$* 111 = 37 \times 3, 1001 \times 111 = 111111,$$

When a single digit is written 6 times, 3, 7, 11, 13, and 37 are factors of it.

Example :- Which of the following is a factor of 222222 ?

(a) 17 (b) 57 (c) 68 (d) 74

Solution :- $222222 = 2 \times 111111$

$$= 2 \times 3 \times 7 \times 11 \times 13 \times 37$$

Clearly, $2 \times 37 = 74$ is one of the factors.

Recurring Decimal

Recurring decimals are referred to as numbers that are uniformly repeated after the decimal. Some rational numbers produce recurring decimals after converting them into decimal numbers, but all irrational numbers produce recurring decimals after converting them into decimal form.

i.e.

$$(1). \frac{1}{3} = 0.333333 \dots = 0.\bar{3}$$

$$(2). 0.\bar{9} = \frac{9}{9} = 1$$

$$(3). 0.5\bar{327} = \frac{5327 - 53}{9900} = \frac{5274}{9900}$$

$$(4). 2.5327 = 2 + \frac{5327 - 53}{9900} = 2 \frac{5274}{9900}$$

Divisibility Rule

By 2 :- When the last digit is 0 or an even number. **i.e.** 520, 588

By 3 :- Sum of digits of a number is divisible by 3
i.e. 1971, 1974

By 4 :- When last two digits are divisible by 4 or, they are zeros
i.e. 1528, 1700

By 5 :- When last digit of a number is 0 or 5
i.e. 1725, 1790

By 6 :- When the number is divisible by 2 and 3 both. **i.e.** 36, 72

By 7 :- Subtract twice the last digit from the number formed by the remaining digits.

i.e. 651 divisible by 7

$65 - (1 \times 2) = 63$. Since 63 is divisible by 7, so is 651.

Exam Hall Approach for divisibility of 7 :-

Example :- What 16807 divisible by 7 ?

Step 1 : Starting from the right side , divide the number into triplet, i.e 016/807

Step 2 : Putting the sign '+' and '-' alternately on the triplet from the right side i.e $016^-/807^+$

Step 3 : After solving this , divide the resultant number 7

$$\frac{807 - 16}{7} = \frac{791}{7} = 113$$

Hence 16807 is divisible by 7

By 8 :- When the last three digits are divisible by 8. **i.e.** 2256

By 9 :- When sum of digit is divisible by 9 **i.e.** 9216

By 10 :- When the last digit is 0. **i.e.** 452600

By 11:- When the difference between the sum of odd and even place digits is equal to 0 or multiple of 11 .

i.e. 217382

Sum of odd place digits = $2 + 7 + 8 = 17$

Sum of even place digits = $1 + 3 + 2 = 6$

$17 - 6 = 11$, hence 217382 is divisible by 11.

Exam Hall Approach :-

Example :- How 217382 divisible by 11 ?

Step 1 : Divide the given number into pair from any same side , then find the difference between them

$$\begin{array}{ccc} 2 & 1 & 7 & 3 & 8 & 2 \\ \boxed{+1} & & \boxed{+4} & & \boxed{+6} & \end{array}$$

Step 2 : After solving this , the resultant number is divide by

$$11 \Rightarrow \frac{1 + 4 + 6}{11} = \frac{11}{11} = 1$$

If the resultant is divisible by 11 , then 217382 is also divisible by 11.

By 13 :- - If adding four times the last digit to the number formed by the remaining digits is divisible by 13, then the number is divisible by 13.

i.e. 1326 is divisible by 13

$132 + (6 \times 4) = 156$. Repeat the same process for 156 .

$15 + (6 \times 4) = 39$.so 39 is divisible by 13

BY 17 :-The divisibility rule of 17 states, "If five times the last digit is subtracted from a number made up of the remaining digits and the remainder is either 0 or a multiple of 17, then the number is divisible by 17"

i.e. 221: $22 - 1 \times 5 = 17$

Important Formulas

1. Sum of first n natural number = $\frac{n(n+1)}{2}$

2. Sum of first n odd numbers = n^2

3. Sum of first n even numbers = $n(n+1)$

4. Sum of square of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$

5. Sum of cubes of first n natural number = $\left(\frac{n(n+1)}{2}\right)^2$

6. $(x^m - a^m)$ is divisible by $(x - a)$ for every natural number m.

7. $(x^m - a^m)$ is divisible by $(x + a)$ and $(x - a)$ for even values of m.

8. $(x^m + a^m)$ is divisible by $(x + a)$ for odd values of m.

9. Number of prime factors of a^p, b^q, c^r, d^s is $p + q + r + s$ when a, b, c, d are all prime numbers.

10. HCF of $(a^m - 1)$ and $(a^n - 1) = [(a^{HCF(m,n)} - 1)]$

Number of Zeros in an Expression

We shall understand this concept with the help of an example.

Let's find the number of zeros in the following expression:

$$24 \times 32 \times 17 \times 23 \times 19 = (2^3 \times 3^1) \times 2^5 \times 17 \times 23 \times 19$$

Notice that a zero is made only when there is a combination of 2 and 5. Since there is no '5' here there will be no zero in the above expression.

i.e. $8 \times 15 \times 23 \times 17 \times 25 \times 22 =$

$$2^3 \times (3^1 \times 5^1) \times 23 \times 17 \times 5^2 \times 2^1 \times 11$$

In this expression there are 4 twos and 3 fives. From this 3 pairs of 5×2 can be formed. Therefore, there will be 3 zeros in the final product.

Example :- Find the number of zeros in the value of:

$$2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20}$$

Solution :- $2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20}$

$$= 2^2 \times 5^4 \times 2^{12} \times 2^8 \times 5^8 \times 2^{10} \times 3^{10} \times 3^{12} \times 5^{12} \times 2^{42} \times 2^{32} \times 5^{16} \times 2^{18} \times 5^{18} \times 5^{40}$$

Zeros are possible with a combination of 2×5 Here the number of 5's are less so the number of zeros will be limited to the number of 5's.

In this expression number of fives are:

$$5^4 \times 5^8 \times 5^{12} \times 5^{16} \times 5^{18} \times 5^{40};$$

i.e. $4 + 8 + 12 + 16 + 18 + 40 = 98$

The Number of Zeros in n!

To find the number of zeros in n!, we divide "n" by 5 until we get a number less than 5, and then we add all the quotients so

obtained.

i.e. Find the number of zeros in 36!

$$\begin{array}{r|l} 5 & 36 \\ \hline 5 & 7 \quad (1) \\ \hline & 1 \quad (2) \end{array}$$

The number of zeros = 7 + 1 = 8.

Remainder Theorem

The basic remainder formula is :-

$$\begin{array}{c} \text{Divisor} \leftarrow d \overline{) n} \rightarrow \text{Dividend} \\ \begin{array}{l} \xrightarrow{q} \text{Quotient} \\ \xrightarrow{r} \text{Remainder} \end{array} \end{array}$$

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Some Important Properties

1. Remainders are additive :- Let us assume that when $a_1, a_2, a_3, \dots, a_n$ are individually divided by d , the respective remainders are $r_1, r_2, r_3, \dots, r_n$.

Now, if we divide $(a_1 + a_2 + a_3 + \dots + a_n)$ by d , we get the same remainder as when we get by dividing $(r_1 + r_2 + r_3 + \dots + r_n)$ by d .

Example :- Find the remainder when $55 + 45 + 68$ is divided by 16.

Solution :- When we divide 55, 45 and 68 by 16 we obtain the respective remainders as 7, 13 and 4. Now we can obtain the required remainder by dividing the sum of remainders (i.e., $7 + 13 + 4 = 24$) by 16, which is 8 instead of dividing the sum of actual given numbers (i.e., $55 + 45 + 68 = 168$) by 16, which also gives us the same remainder 8.

2. Remainders are Multiplicative :- When $a_1, a_2, a_3, \dots, a_n$ are divided by a divisor d the respective remainders obtained are $r_1, r_2, r_3, \dots, r_n$ then the remainder when $(a_1 \times a_2 \times a_3 \times \dots \times a_n)$ is divided by 'd' can be obtained by dividing $(r_1 \times r_2 \times r_3 \times \dots \times r_n)$ by d .

Example :- What will be the remainder when $201 \times 198 \times 80$ is divided by 35?

Solution :- When $201 \times 198 \times 80$ is divided by 35 the remainders can be obtained by dividing the product of the remainders which are obtained by dividing 201, 198, 80 individually. Now since the remainders are 26, 23, 10 so the required remainder = Remainder of $(26 \times 23 \times 10)$ when divided by 35.

$$\begin{aligned} \text{Thus the required remainder} &= \frac{26 \times 23 \times 10}{35} \\ &= \frac{26 \times 230}{35} = \frac{26 \times 20}{35} \end{aligned}$$

Thus 30 is the remainder.

Example :- What will be the remainder when 17×23 is divided by 12?

Solution :- We can express this as:

$$\begin{aligned} 17 \times 23 &= (12 + 5) \times (12 + 11) \\ &= 12 \times 12 + 12 \times 11 + 5 \times 12 + 5 \times 11 \end{aligned}$$

In the above expression we will find that remainder will depend on the last term i.e. 5×11

$$\text{Now, } \text{rem.} \left(\frac{5 \times 11}{12} \right) = 7.$$

$$\text{So, } \frac{12 \times 12 + 12 \times 11 + 5 \times 12 + 5 \times 11}{12} \text{ and } \frac{5 \times 11}{12}$$

remainder is same in both cases which is 7.

Negative Remainder

Taking a negative remainder will make our calculation easier.

Examples :

$$(i). \text{rem} \left(\frac{7 \times 8}{9} \right) = \text{rem} \left(\frac{-2 \times -1}{9} \right) = -2 \times -1 = 2$$

$$(ii). \text{rem} \left(\frac{55 \times 56}{57} \right) = \text{rem} \left(\frac{-2 \times -1}{57} \right) = -2 \times -1 = 2$$

$$(iii). \text{rem} \left(\frac{7 \times 10}{9} \right) = \text{rem} \left(\frac{-2 \times 1}{9} \right) = -2 \times 1 = -2 \text{ or, } 7(9-2)$$

Large Power Concepts

Look at the following examples :

$$(i). \text{rem} \left(\frac{28^{12345}}{9} \right) = \text{rem} \left(\frac{(27+1)^{12345}}{9} \right) \\ = \text{rem} \left(\frac{1^{12345}}{9} \right) = 1^{12345} = 1$$

$$(ii). \text{rem} \left(\frac{26^{12345}}{9} \right) = \text{rem} \left(\frac{(27-1)^{12345}}{9} \right) \\ = \text{rem} \left(\frac{-1^{12345}}{9} \right) = -1^{12345} = -1 \text{ or } 8(9-1)$$

Some important results :-

If $\frac{N}{D}$ gives remainder R

then, $\frac{2N}{D} \rightarrow$ Remainder is $2R$

then, $\frac{3N}{D} \rightarrow$ Remainder is $3R$

then, $\frac{N^2}{D} \rightarrow$ Remainder is R^2

then, $\frac{3N+2}{D} \rightarrow$ Remainder is $3R + 2$

Note :- Remainder < Divisor (Always)

Application of Remainder Theorem

Example :- Find the last two digits of the expression?

$$22 \times 31 \times 44 \times 27 \times 37 \times 43 ?$$

Solution :- If we divide the above expression by 100, we will get the last two digits as remainder.

$$\Rightarrow \text{rem} \left(\frac{22 \times 31 \times 44 \times 27 \times 37 \times 43}{100} \right)$$

dividing by 4 to make it simple

$$= \text{rem} \left(\frac{22 \times 31 \times 11 \times 27 \times 37 \times 43}{25} \right)$$

$$= \text{rem} \left(\frac{132 \times 22 \times 216}{25} \right)$$

$$= \text{rem} \left(\frac{7 \times 22 \times 16}{25} \right) \Rightarrow \text{rem} \left(\frac{4 \times 16}{25} \right) = \text{rem} \left(\frac{14}{25} \right) = 14$$

Since we had divided by 4 initially now to get the correct answer, we need to multiply the remainder by 4.
So remainder will be $14 \times 4 = 56$, which will also be the last two digits of the expression.

Fermat's Theorem :- If 'p' and 'q' are two co-prime numbers then the remainder obtained when p^{q-1} is divided by 'q' is always 1.

i.e. $\text{rem } \frac{p^{q-1}}{q} = 1$

Example:- What will be the remainder when 3^4 is divided by 5?

Solution :- As 3 and 5 are co-prime.

So, $\text{rem } \left(\frac{3^4}{5}\right) = 1$

Euler Number :- The Euler number of a number x means the number of natural numbers which are less than x are co-prime to x.

i.e. the Euler number of 6 will be 2 as the natural numbers 1 & 5 are the only two numbers which are less than 6 and are also co-prime to 6.

Mathematically, the Euler number of a number z denoted by the symbol E(z) is calculated as explained below.

$$E(z) = z \times \left[1 - \frac{1}{a}\right] \times \left[1 - \frac{1}{b}\right] \times \left[1 - \frac{1}{c}\right] \dots$$

where a, b and c are the different prime factors of z.

Example :- What is the Euler number of 20?

Solution :- The prime factorization of 20 is $2^2 \times 5$.

So, the Euler number of 20 will be

$$E(20) = 20 \times \left[1 - \frac{1}{2}\right] \times \left[1 - \frac{1}{5}\right] = 20 \times \frac{1}{2} \times \frac{4}{5} = 8$$

Hence, the Euler number of 20 is 8.

Important Note : For all prime numbers the Euler number will be the number one less than the given number.

Euler's Theorem : If 'p' and 'q' are two co-prime number then,

$$\text{Remainder } \left[\frac{p^E}{q} \right] = 1 \quad \text{Euler's number}$$

Example :- What is the remainder when 13^{18} is divided by 19?

Solution :- 13 and 19 both are co-prime

So Euler number of 19 is **18** (The Euler number of a prime number is always 1 less than the number)

$$\text{Rem. } \left(\frac{13^{18}}{19}\right) = 1$$

Example :- What is the remainder when 19^{32} is divided by 15?

Solution :- Factors of 15 is 3×5 .

$$E(15) = 15 \times \left[1 - \frac{1}{3}\right] \times \left[1 - \frac{1}{5}\right] = 15 \times \frac{2}{3} \times \frac{4}{5} = 8$$

$$\text{Now Rem. } \left(\frac{19^8}{15}\right) = 1$$

Note :- Euler's theorem is applicable only for co-prime numbers.

Wilson's Theorem : If 'p' is a prime number then it divides

$$(p-1)! + 1.$$

i.e. 5 divides $(5-1)! + 1 = 4! + 1 = 25$

It also means that when $(p-1)!$ is divided by 'p' the remainder is -1 or 'p-1'.

Example :- Find the remainder when $60!$ is divided by 61?

Solution :- By using Wilson's theorem

$$\text{Rem. } \left[\frac{(61-1)!}{61}\right] = -1 \text{ or } 60.$$

Cyclicity and Unit Digit

Cyclicity Table :-

Number	Cyclicity	Power Cycle
1	1	1
2	4	2, 4, 8, 6
3	4	3, 9, 7, 1
4	2	4, 6
5	1	5
6	1	6
7	4	7, 9, 3, 1
8	4	8, 4, 2, 6
9	2	9, 1
10	1	0

Any power in number having unit digit 0, 1, 5, 6 $\rightarrow$ will have same Unit Digit

$$4 \rightarrow (4)^{\text{ODD}} = \text{Unit digit} \rightarrow 4$$

$$(4)^{\text{even}} = \text{Unit digit} \rightarrow 6$$

$$9 \rightarrow (9)^{\text{ODD}} = \text{Unit digit} \rightarrow 9$$

$$(9)^{\text{even}} = \text{Unit digit} \rightarrow 1$$

Example :- Find the unit digit of 512×216 ?

Solution :-

$$512 \times 216 = 12 \text{ Unit digit } 2$$

Example :- Find the unit digit of $512 + 216$?

Solution :-

$$512 + 216 = 8 \text{ Unit digit}$$

Example :- Find the unit digit of $512 - 216$?

Solution :-

$$512 - 216 = 12 - 6 = 6 \text{ Unit digit}$$

Example :- Find the unit digit of $(912)^{19821}$?

Solution :- Cyclicity of 2 is 4.

Therefore,

$$\text{Unit digit of } (2)^{19821} = 2^{4 \times 4955} \times 2^1 = 2$$

Note :- Factorial greater than 3 is always divisible by 4.

Example :- Find Unit digit of $2^{9!}$.

Solution :- $2^{9!} \rightarrow 2^4 \rightarrow 16 \rightarrow 6$ (unit digit)

Example: Find Unit digit of $493^{512!} \times 217^{396!}$.

Solution:- $493^{512!} \times 217^{396!} \rightarrow (3)^{512!} \times (7)^{396!}$
 $\rightarrow (3)^4 \times (7)^4 \rightarrow 1 \times 1 \rightarrow 1$ (unit digit)

Counting of Numbers

Example :- How many 5 will come from 350 to 600.

Solution :- frequency of zero will be equal to frequency of 5.

frequency of 5 from 0 to 600 = $\left[\frac{600}{5}\right] + \left[\frac{600}{5^2}\right] + \left[\frac{600}{5^3}\right]$

$= \left[\frac{600}{5}\right] + \left[\frac{600}{25}\right] + \left[\frac{600}{125}\right] = 120 + 24 + 4 = 148$

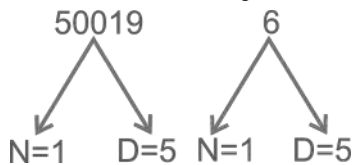
And frequency of 5 from 0 to 350 = $\left[\frac{350}{5}\right] + \left[\frac{350}{5^2}\right] + \left[\frac{350}{5^3}\right]$

$= \left[\frac{350}{5}\right] + \left[\frac{350}{25}\right] + \left[\frac{350}{125}\right] = 70 + 14 + 2 = 86$

Frequency of 5 from 350 to 600 = $148 - 86 = 62$

Digits Required to Write Counting

Concept :- N = Number and D = digit



No of digit 1 - 9 = 9D

No of digit 1 - 99 = 189D

No of digit 1 - 999 = 2889D

Example :- How many digits are required to write counting from 1 to 300 ?

Solution :- according to questions,

No of digit 1 - 9 $\Rightarrow (9 \times 1) = 9D$

No of digit 10 - 99 $\Rightarrow (90 \times 2) = 180D$

No of digit 100 - 300 $\Rightarrow (201 \times 3) = 603D$

Total required digits = $(9 + 180 + 603) = 792D$

Concept of Successive Division

If a number is divided by a and b respectively, the remainder remain r and R respectively. In such a situation that number can be known.

Example :- When a number is successively divided by 3, 4 and 7, the remainders obtained are 2, 3 and 5, respectively. What will be the remainder when 84 divides the same number?

Solution :-

Formulae used : Dividend = Divisor $\times$ Quotient + Remainder

Method 1 : Starting from the last Let last divisor = 0

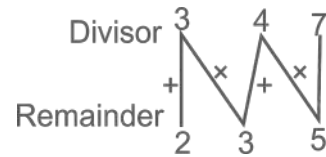
The number gives a remainder 5 when it is divisible by 7
 $\Rightarrow 7 \times 0 + 5 = 5$

The number gives a remainder 3 when it is divisible by 4
 $\Rightarrow 4 \times 5 + 3 = 23$

The number gives a remainder 2 when it is divisible by 3
 $\Rightarrow 3 \times 23 + 2 = 71$

So when 71 is divided by 84 remainder will be 71

Method 2 :



Step 1 : Multiply $5 \times 4 = 20$

Step 2 : add 3 to the 20 = 23

Step 3 : Multiply the result of step 2 by 3 = 69

Step 4 : add 2 to the 69 = 71 thus the value of N = 71

Therefore remainder will be 71

When Digits are Reversed

Example :- If the digits of a two digit number is reversed, then the number is decreased by 36. Which of the following is correct regarding the number ?

I. The difference of the digits is 4.

II. The value of the number can be 84.

III. Number is always a composite number.

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(a) I, II and III (b) II and III (c) I and III (d) I and II

Solution :-

I. Let the two digit number be $10x + y$

According to question,

$(10x + y) - (10y + x) = 36$

$9x - 9y = 36 \Rightarrow x - y = 4$

II. The value of no can be 84 as $84 - 48 = 36$

III. the number formed in the given case may or may not be composite no. i.e. 84 which is a composite no and 73 which is a prime no.

Clearly, we can see that option (d) is the correct one.

HCF and LCM

LCM (Least common multiple) of two or more given numbers is the least number that is exactly divisible by each of them.

HCF (Highest common factor) of two or more numbers is the greatest number that divides each of them exactly. HCF is also known as the 'Highest common Divisor' (HCD) and the 'Greatest Common Divisor' (GCD).

The concept of multiples and factors :-

If X , Y , and Z are three natural numbers and $X \times Y = Z$, then

- (i) X and Y are called the factors of Z .
- (ii) Z is said to be divisible by X and Y .
- (iii) Z is said to be a multiple of X and Y .

Example: The set of positive Integers which are factors of 18 is (1, 2, 3, 6, 9, 18).

Method of Finding HCF

To find the HCF of the given numbers :-

1. Break the given numbers into their prime factors.
2. The HCF will be the product of all the prime factors common to all the numbers.

Let us learn the process of finding HCF with the help of some solved examples.

HCF by Factorisation Method

Example :- HCF of 6, 12, 18

Solution :- Firstly find out the factors of 6, 12, 18 and then multiply the common factors.

$$\begin{array}{r|l} 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array} \quad \begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array} \quad \begin{array}{r|l} 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$6 = 2 \times 3, 12 = 2 \times 2 \times 3, 18 = 2 \times 3 \times 3$$

$$\text{HCF} = 2 \times 3 = 6$$

Try to find HCF and LCM of 3, 6, 9, 12 yourself.

HCF of 3, 6, 9, 12 can also be found by Division method. It is useful when the numbers are bigger.

HCF by Division Method

Example :- HCF of 24, 48, 72, and 100.

Solution :- To start the division method select the smallest two numbers.

$$\begin{array}{r} 24 \overline{)48} (2 \\ \underline{-48} \\ \text{XX} \end{array} \quad \begin{array}{r} 48 \overline{)72} (1 \\ \underline{-48} \\ 24 \end{array} \quad \begin{array}{r} 24 \overline{)100} (4 \\ \underline{-96} \\ 4 \end{array}$$

$$\text{HCF of 24 and 48} = 24$$

$$\text{HCF of 24, 48 and 72} = 24$$

$$\text{HCF of 24, 48, 72, and 100} = 4.$$

Note :-

- (i). HCF of two prime numbers is always 1.
- (ii). HCF of co-prime numbers is always 1.

Method of Finding LCM

The Least Common Multiple of two or more numbers is the smallest number which is exactly divisible by all of them. In other words, it is the product of the highest powers of all the prime factors of the given numbers.

To find the LCM of given numbers :-

1. Break the given numbers into their prime factors.
2. The LCM will be the product of the highest power of all the factors that occur in the given numbers.

LCM by Factorisation Method

Example :- Find the LCM of 96, 36 and 18.

Solution :- $96 = 2 \times 2 \times 2 \times 2 \times 3$

$$= 2^5 \times 3^1; 36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$

$$18 = 2 \times 3 \times 3 = 2^1 \times 3^2$$

$$\text{LCM} = 2^5 \times 3^2 = 288$$

Example :- LCM of 6, 12, 8 ?

Solution :-

$$\begin{array}{r|l} 2 & 6, 12, 8 \\ \hline 2 & 3, 6, 4 \\ \hline 3 & 3, 3, 2 \\ \hline 2 & 1, 1, 2 \\ \hline & 1, 1, 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 3 \times 2 = 24$$

Relation Between HCF and LCM

(1). $\text{LCM} \times \text{HCF} = 1\text{st number} \times 2\text{nd number}$

Example :- For numbers 8 and 12,

Solution :- $\text{LCM} = 24$ and $\text{HCF} = 4$

$$\text{Now, } \text{LCM} \times \text{HCF} = 24 \times 4 = 96$$

$$\text{also, } 8 \times 12 = 96$$

(2). HCF of some numbers is always a factor of LCM of the numbers.

Note :- If the HCF of two numbers is 1 then they are called co-prime numbers.

$$(3). \frac{\text{LCM}}{\text{HCF}} = \text{Product};$$

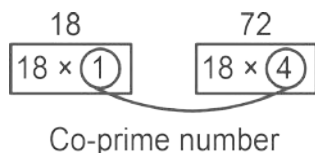
where LCM and HCF are of two numbers

N_1 and N_2 . If we find two co-prime factors, F_1 and F_2 , of the Product as obtained above.

(4).

1st number 2nd number
 $H \times X$ $H \times Y$

 X, Y are co-prime numbers
 Where, $\text{HCF} = H$

Example :-

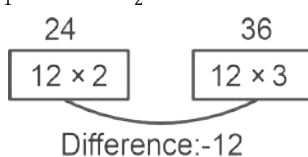
(5). If 1st number = N_1 and 2nd number = N_2

HCF of N_1 = Hx

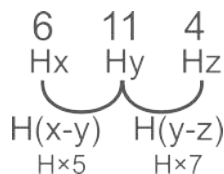
HCF of N_2 = Hy

And HCF of N_1 and N_2 = H

So, Difference between N_1 and N_2 = $Hx - Hy = H(x - y)$

Example :- $N_1 = 24$ and $N_2 = 36$ 

Note :- HCF is always either the difference of two numbers or factors of difference of two numbers.



(6). When the second divisor is a factor of the first divisor, then the second remainder is obtained by dividing the first remainder by the second divisor.

Example :- When 29 is divided by 8, the remainder obtained is 5, then what will be the remainder when the same number is divided by 4?

Solution :- Here, the second divisor that is 4 is a factor of the first divisor that is 8. So, on dividing the first remainder that is 5 by the second divisor that is 4, we get our second remainder which is 1. So, the required answer is 1

LCM and HCF of Fraction

$$\text{LCM of Fraction} = \frac{\text{LCM of numerator}}{\text{HCF of denominator}}$$

$$\text{HCF of Fraction} = \frac{\text{HCF of numerator}}{\text{LCM of denominator}}$$

Example :- LCM and HCF of $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ **Solution :-**

$$\text{LCM} = \frac{\text{LCM of numerator}}{\text{HCF of denominator}} = \frac{\text{LCM of } 1, 2, 3}{\text{HCF of } 2, 3, 4} = \frac{6}{1}$$

$$\text{HCF} = \frac{\text{HCF of numerator}}{\text{LCM of denominator}} = \frac{\text{HCF of } 1, 2, 3}{\text{LCM of } 2, 3, 4} = \frac{1}{12}$$

LCM and HCF of Decimal

Example :- Find the HCF and LCM of 0.63, 1.05 and 2.1

SBI PO Pre 19/12/2022 (2nd Shift)

(a) 0.21 and 6.30 (b) 1.05 and 6.30

(c) 2.1 and 0.63 (d) 0.63 and 1.05

Solution :- Numbers be 0.63, 1.05 and 2.1

Multiplying each number by 100

Then, numbers = 63, 105, 210

 $63 = 3 \times 3 \times 7$, $105 = 3 \times 5 \times 7$, $210 = 2 \times 3 \times 5 \times 7$

Now, LCM(63, 105, 210)

 $= 2 \times 3 \times 3 \times 5 \times 7 = 630$

Then, LCM (0.63, 1.05, 2.10) = 6.30

And, HCF (63, 105, 210) = $3 \times 7 = 21$

Then, HCF (0.63, 1.05, 2.10) = 0.21

Hence, HCF and LCM be 0.21 and 6.30 respectively

LCM of Indices

(1). When the base of the given number is the same, then the number with the highest power will be the LCM of the given number.

Example :- Find the LCM of $3^2, 3^4, 3^8, 3^9$?**Solution :-** Here base are same so the LCM of given number will be the highest power of 3 i.e 3^9

(2). When the base is not the same, and there are no common factors in the base, then the product of the given number will be the LCM

Example :- Find the LCM of $3^2, 2^4$?**Solution :-** LCM = $3^2 \times 2^4 = 144$

HCF of Indices

(1). When the base of the given number are same, then the number with the smallest power will be the HCF of the given number.

Example :- Find the HCF of $3^2, 3^4, 3^8, 3^9$?**Solution :-** HCF = 3^2

(2). When the base is not the same, and there are no common factors in the base, then the HCF of the given number will be 1.

Example :- Find the Hcf of $3^2, 5^4, 7^8$ **Solution :-** HCF = 1

Some Application of LCM :-

The Smallest number which is exactly divisible by x, y, z = LCM of (x, y, z) The smallest number which when divided by x, y, z and leaves the remainder r in each case = (LCM of x, y, z) + r

The smallest number which when divided by x, y, z leaves remainder a, b, c respectively = LCM (x, y, z) - k

Where $k = (x - a) = (y - b) = (z - c)$

Some Application of HCF :-

The largest number which is exactly divisible by x, y, z = HCF of (x, y, z)

The largest number which when divided by x, y, z and leaves the remainder r in each case

$$= \text{HCF} [(x - r), (y - r), (z - r)] \text{ or } \text{HCF} [(x - y), (y - z), (z - x)]$$

The largest number which when divided by x, y, z and leaves the remainder a, b, c respectively = $\text{HCF} [(x - a), (y - b), (z - c)]$

LCM and HCF of Polynomials

$$\text{LCM} \times \text{HCF} = \text{First Polynomial} \times \text{Second Polynomial}$$

$$\text{LCM} = \frac{\text{First Polynomial} \times \text{Second Polynomial}}{\text{HCF}}$$

$$\text{HCF} = \frac{\text{First Polynomial} \times \text{Second Polynomial}}{\text{LCM}}$$

Note :- LCM of two or more polynomials is the common factor of least degree.

HCF of two or more polynomials is the common factor of Highest degree.

$$\text{HCF of } (4^{315} - 1)(4^{25} - 1) = 4^5 - 1 = 1023$$

Example :- The LCM of $x^2 - 8x + 15$ and $x^2 - 5x + 6$ is:
SSC CPO 05/10/2023 (2nd Shift)

- (a) $(x + 5)(x + 2)(x + 3)$ (b) $(x - 5)(x - 2)(x - 3)$
(c) $(x + 5)(x - 2)(x - 3)$ (d) $(x - 2)(x - 3)^2(x - 5)$

Solution :- According to the question,

$$x^2 - 8x + 15$$

$$x^2 - (5 + 3)x + 15$$

$$x(x - 5) - 3(x - 5)$$

$$(x - 5), (x - 3)$$

$$\text{And } x^2 - 5x + 6$$

$$x^2 - (3 + 2)x + 6$$

$$x(x - 3) - 2(x - 3)$$

$$(x - 3), (x - 2)$$

So the LCM will be $(x - 5), (x - 2), (x - 3)$

Example :- What is the HCF of the polynomials $(x^3 - 8)$, $(x^3 - 6x^2 + 12x - 8)$ and $(x^3 - 4x^2 + 4x)$?
SSC MTS 12/09/2023 (3rd Shift)

(a) $(x - 1)$ (b) $(x - 2)$ (c) $(x - 8)$ (d) $(x - 4)$

Solution :- $(x^3 - 8) = (x - 2)(x^2 + 2x + 4)$

$$(x^3 - 6x^2 + 12x - 8) = (x - 2)^3$$

$$\text{and } (x^3 - 4x^2 + 4x) = x(x - 2)^2$$

HCF of $(x^3 - 8)$, $(x^3 - 6x^2 + 12x - 8)$ and

$$(x^3 - 4x^2 + 4x) = (x - 2)$$

$$> \text{HCF of } [a^x \pm 1, a^y \pm 1] = a^{\text{HCF}(x,y)} \pm 1$$

Example :- Find the HCF of $(4^{315} - 1)$ and $(4^{25} - 1)$.

SSC MTS 26/07/2022 (Morning)

- (a) 1 (b) $(4^{25} - 1)$ (c) 1024 (d) 1023

Solution :- $(4^{315} - 1)(4^{25} - 1)$

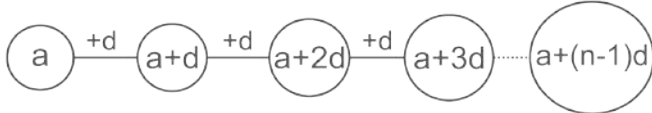
HCF of 315 and 25 = 5

A.P. and G.P.

Arithmetic Progression (A.P.)

A progression in which the difference between every two successive terms is the same. It is possible to obtain a formula for the n th term from an arithmetic progression.

1st Term 2nd Term 3rd Term 4th Term..... n th Term



d = Common Difference

Arithmetic Progression (A.P.) Formula

Series : $a_1, a_2, a_3, a_4, \dots, a_n$

Common difference of an arithmetic progression $d = a_2 - a_1$

a) n th term of an A. P.

$a_n = a_1 + (n - 1)d$... $\{a_1 = \text{first term}, n = \text{number of terms}\}$

Or $a_n = \text{last term} - (n - 1)d$

Example:- Find the ninth term of an arithmetic progression with the first term as 5 and the common difference as 4.

RRB NTPC CBT - I 08/02/2021 (Evening)

(a) 35 (b) 39 (c) 37 (d) 41

Solution:- 1st term (a) = 5,

common difference (d) = 4

9th term of AP = $a + (n - 1)d$

$= 5 + (9 - 1) \times 4 = 5 + 32 = 37$

b) Sum of n terms of an A.P. $\rightarrow S_n = \frac{n}{2} [2a_1 + (n-1)d]$

$S_n = \frac{n}{2} [a_1 + a_1 + (n-1)d]$

$S_n = \frac{n}{2} [a_1 + a_n]$

Or $S_n = \frac{n}{2} [\text{First term} + \text{Last term}]$

Example:- Find the sum upto 151 term of the sequence 243, 256, 269,

(a) 183917 (b) 183916 (c) 183915 (d) 183918

Solution:- Given A.P. = 243, 256, 269, 151th term

First term (a) = 243, common difference (d) = 13

$S_{151} = \frac{n}{2} [2 \times a + (n - 1)d]$

$= \frac{151}{2} [2 \times 243 + 150 \times 13]$

$S_{151} = 151 \times 1218 = 183918$

Example:- Which term of the arithmetic progression 88, 90, 92, 94... is 142?

(a) 30^{th} (b) 29^{th} (c) 28^{th} (d) 27^{th}

Solution :- $a_n = a + (n - 1)d$

Where $d = 90 - 88 = 92 - 90 = 2$

$a_n = a_1 + (n - 1)d$

$142 = 88 + (n - 1) \times 2$

$$2n = 142 - 88 \Rightarrow n = \frac{56}{2} = 28^{th}$$

Special Series

$$1). \frac{1}{a \times b} + \frac{1}{b \times c} + \frac{1}{c \times d} + \frac{1}{d \times e} + \frac{1}{e \times f} + \dots +$$

$\frac{1}{(n-1) \times n}$ then its value will be

$$= \frac{1}{\text{common difference}} \left[\frac{1}{a} - \frac{1}{n} \right]$$

Example:- Find the value of $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5}$

$$+ \frac{1}{5 \times 6} + \dots + \frac{1}{9 \times 10}$$

RRB JE 24/05/2019 (Evening)

(a) $\frac{1}{10}$ (b) $\frac{9}{10}$ (c) $\frac{5}{11}$ (d) $\frac{2}{5}$

Solution:- If series are in the form of $\frac{1}{a \times b} + \frac{1}{b \times c} + \frac{1}{c \times d}$

$$+ \frac{1}{d \times e} + \frac{1}{e \times f} + \dots + \frac{1}{(n-1) \times n}$$

then its value will be $= \left[\frac{1}{a} - \frac{1}{n} \right]$

$$\text{Given series } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \dots$$

$$+ \frac{1}{9 \times 10} \text{ Hence its value will be } \left[1 - \frac{1}{10} \right] = \frac{9}{10}$$

$$2). \frac{1}{1 \times 4 \times 7} + \frac{1}{4 \times 7 \times 10} + \frac{1}{7 \times 10 \times 13} + \frac{1}{10 \times 13 \times 16} +$$

$\frac{1}{13 \times 16 \times 19}$ then its value will be $\left(\frac{1}{\text{diff of 1st and 3rd number}} \right)$

$$\left[\frac{1}{\text{product of 1st two numbers}} - \frac{1}{\text{product of last two numbers}} \right]$$

$$= \left(\frac{1}{7-1} \right) \left[\frac{1}{1 \times 4} - \frac{1}{16 \times 19} \right]$$

$$3). \frac{1}{1 \times 3 \times 5} + \frac{1}{1 \times 4} + \frac{1}{3 \times 5 \times 7} + \frac{1}{4 \times 7} + \frac{1}{5 \times 7 \times 9} +$$

$\frac{1}{7 \times 10} + \dots 20 \text{ times}$

$$= \left(\frac{1}{1 \times 3 \times 5} + \frac{1}{3 \times 5 \times 7} + \dots + \frac{1}{19 \times 21 \times 23} \right) + \left(\frac{1}{1 \times 4} + \right.$$

$\frac{1}{4 \times 7} + \dots + \frac{1}{28 \times 31} \left. \right)$

$$= \frac{1}{4} \left(\frac{1}{1 \times 3} - \frac{1}{21 \times 23} \right) + \frac{1}{3} \left(\frac{1}{1 \times 1} - \frac{1}{31} \right)$$

$$= \frac{1}{4} \times \frac{160}{21 \times 23} - \frac{10}{21} = \frac{6070}{14973}$$

$$\text{Note :- } \frac{1}{a \times b} = \frac{1}{b-a} \left(\frac{1}{b} - \frac{1}{a} \right)$$

$$\frac{1}{a \times b \times c} = \frac{1}{c-a} \left(\frac{1}{ab} - \frac{1}{bc} \right)$$

$$\frac{1}{a \times b \times c \times d} = \frac{1}{d-a} \left(\frac{1}{abc} - \frac{1}{bcd} \right)$$

4). Sum of 1st n natural numbers

$$1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2}$$

5). Sum of 1st n even natural number

$$2 + 4 + 6 + 8 + \dots + 2n = n(n+1)$$

6). Sum of 1st n odd natural number

$$1 + 3 + 5 + 7 + 9 + \dots (2n - 1) = n^2$$

7). Sum of squares of 1st n natural numbers

$$1^2 + 2^2 + 3^2 + \dots n^2 = \frac{n(n+1)(2n+1)}{6}$$

8). Sum of Cube of 1st n natural numbers

$$1^3 + 2^3 + 3^3 + \dots n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

9). Sum of first 'n' terms of the series of alternate positive and negative numbers :

Case 1 :- when 'n' is even

$$1^2 - 2^2 + 3^2 - 4^2 + \dots = \frac{-n(n+1)}{2}$$

Case 2 :- when 'n' is odd

$$1^2 - 2^2 + 3^2 - 4^2 + \dots = \frac{n(n+1)}{2}$$

Arithmetic mean :- $\frac{\text{Sum of all observations}}{\text{Total no of observations}}$

1). If A be the AM(arithmetic mean) between x and y then ,

$$A = \frac{x+y}{2}$$

2). In general , If $a_1, a_2, a_3, a_4, \dots, a_n$ terms are in AP, So their

AM(arithmetic mean) will be

$$A.M. = \frac{\text{sum of } a_1, a_2, a_3, a_4, \dots, a_n \text{ terms}}{n}$$

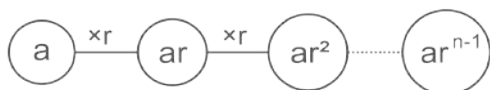
3). AP in odd Consecutive term :- $a - d, a, a + d$

4). AP in even Consecutive term :- $a - 3d, a - d, a + d, a + 3d$

Geometric Progression (G.P.)

A geometric progression is a sequence in which each term has a fixed ratio, known as a common ratio. G.P. stands for geometric progression. The geometric sequence is generally written as $a, ar, ar^2, \dots$, where a is the first term, and r is the sequence's common ratio. Both negative and positive numbers are possible for the common ratio.

1st Term 2nd Term 3rd Term.....nth Term



$r = \text{Common Ratio}$

Geometric Progression (G.P) Formula

The common ratio of a geometric progression $(r) = \frac{\text{Second term}}{\text{first term}}$

a). The formula for the nth term of a geometric progression with the first term (a) and common ratio (r) $\rightarrow (a_n) = ar^{n-1}$

b). The formula for the sum of geometric progression formula,

$$\text{If } |r| < 1, S_n = \frac{a(1-r^n)}{1-r}$$

$$\text{If } |r| > 1, S_n = \frac{a(r^n - 1)}{r - 1}$$

The sum of infinite geometric progression formula,

$$S_\infty = \frac{a}{1-r}, \text{ where } r < 1$$

Example:- How many terms are there in G.P 3, 18, 108, ..., 23328 ?

(a) 4 (b) 8 (c) 6 (d) 10

Solution :- G.P :- 3, 18, 108, ..., 23328

First term $\rightarrow a = 3$, common ratio $(r) = \frac{\text{Second term}}{\text{first term}} = 6$

nth term of the G.P. $= T_n = ar^{n-1}$

$$\Rightarrow 23328 = 3 \times (6)^{n-1} \Rightarrow (6)^5 = (6)^{n-1}$$

number of term $(n) = 6$

Example:- Find the sum of the G.P:

$\frac{8}{5}, \frac{8}{25}, \frac{8}{125}, \frac{8}{625}, \dots$ to n terms.

(a) $2(1 - (\frac{1}{5})^n)$ (b) $4(1 - (\frac{1}{5})^n)$

(c) $\frac{2}{5}(1 - (\frac{1}{5})^n)$ (d) $\frac{5}{4}(1 - (\frac{1}{5})^n)$

Solution:- According to question,

$$\text{Common ratio } (r) = \frac{\text{2nd term}}{\text{1st term}} = \frac{\frac{8}{25}}{\frac{8}{5}} = \frac{1}{5}$$

$$S_n = \frac{a\{1 - (r)^n\}}{(1 - r)} = \frac{\frac{8}{5}\{1 - (\frac{1}{5})^n\}}{(1 - \frac{1}{5})} = \frac{\frac{8}{5}\{1 - (\frac{1}{5})^n\}}{\frac{4}{5}} = 2\{1 - (\frac{1}{5})^n\}$$

Geometric Mean :- The (G.M) of series containing n observations is the nth root of the product of the values.

(1). If G be the GM between x and y then , $G = \sqrt{xy}$

(2). In general , If a , b , c , dn terms are in GP ,

So their GM $= \sqrt[n]{a \cdot b \cdot c \dots n \text{ terms}}$

(3). GP in odd Consecutive term :- $\frac{a}{r}, a, ar$

(4). GP in even Consecutive term :- $\frac{a}{r^2}, \frac{a}{r}, ar, ar^2$

Relation Between A.M. And G.M.

Let A and G be the AM(arithmetic mean) and GM(Geometric mean) of two positive real number x and y

$$A = \frac{x+y}{2}, \quad G = \sqrt{xy}$$

$$A - G = \frac{(\sqrt{x} - \sqrt{y})^2}{2} \geq 0 \dots (i)$$

From equation(i) we obtain the relationship

$$A.P \geq G.P$$

Example:- If the arithmetic mean and geometric mean of two observations are 10 and 5, respectively, then find the sum of the squares of the observations.

RRB NTPC CBT - I 11/03/2021 (Evening)

(a) 295 (b) 275 (c) 225 (d) 350

Solution:- Arithmetic mean = 10

Let the numbers be a and b

According to question ,

$$a + b = 20$$

$$\text{And, } \sqrt{ab} = 5 \Rightarrow ab = 25$$

$$\begin{aligned}
 \text{Now, } (a + b)^2 &= a^2 + b^2 + 2ab \\
 &= 20^2 = a^2 + b^2 + 2 \times 25 \\
 &= 400 = a^2 + b^2 + 50 \Rightarrow a^2 + b^2 = 350
 \end{aligned}$$

Harmonic Progression (HP)

The Harmonic Progression is the reciprocal of the arithmetic progression .

$$HP = \frac{1}{AP}$$

Harmonic mean(HM) :-

(1). If H be the HM between x and y then ,

$$H = \frac{2xy}{x + y}$$

(2). In general , If $a_1, a_2, a_3, a_4, \dots, a_n$ terms are in HP , So their HM will be

$$HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$$

Relation between AM, GM, HM of two positive numbers a and b, then

i). If $AM = \frac{a+b}{2}$, $GM = \sqrt{ab}$, and $HM = \frac{2ab}{a+b}$

ii). $AM \geq GM \geq HM$.

iii). $G^2 = A \times H$ and therefore A, G, H are in G.P.

Simplification

Conversion of complex arithmetic expressions into simple ones is called Simplification.

VBODMAS Rule

V- Vinculum or bar ('Bar bracket') '—'

* First solve vinculum i.e. bar.

Eg: $(7 - 5 - 4) = ?$

First solve $5 - 4$ i.e. 1 then $7 - 1 = 6$.

B (Brackets)- Brackets are to be solved in the following sequence -

- Small brackets ('Circular bracket') '()''
- Middle brackets ('Curly bracket') '{}'
- Square bracket ('Big bracket') '[]'

* For modulus e.g. $|-3|$ we write magnitude only not sign i.e. 3 in this case.

O (of) - Operation of 'of' is simplified.

D (Division) - Operation of division is done.

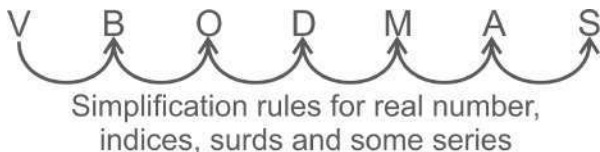
M (Multiplication) - Operation of multiplication is done.

A (Addition) - Addition operation is done.

S (Subtraction) - Subtraction operation is done.

Note :- We solve the questions in this order (**VBODMAS**).

Order of the above operations is same as the order given and applied from left to right -



Example :- Simplify the expression

$$35 + 25 - [15 - \{13 \times (2 + 1)\}]$$

Solution :- $35 + 25 - [15 - \{13 \times (2 + 1)\}]$

$$35 + 25 - [15 - \{13 \times 3\}]$$

$$35 + 25 - [15 - 39]$$

$$35 + 25 - [-24]$$

$$35 + 25 + 24 = 84$$

KEY POINTS :-

1). If $\sqrt{x} = x$, then the required number will be (?) = x^2

$$2). \sqrt{a^4 \times b^4 \times c^4} = a^2 b^2 c^2$$

$$3). \sqrt{a^n \times b^m} = a^{\frac{n}{2}} \times b^{\frac{m}{2}}$$

$$4). \frac{\sqrt{x}}{\sqrt{y}} = \sqrt{\frac{x}{y}}$$

$$5). \sqrt{x} \times \sqrt{y} = \sqrt{xy}$$

6). If in a given number, the total number of digits are n and if n is even, the digits in the square root of that number will have $\frac{n}{2}$

digits and if n is odd, then the number of digits will be $\frac{n+1}{2}$.

Example :- Square root of 25 and 225 respectively has how many digits ?

Solution :- Total number of digits in 25 = 2 (even)

$$\text{Digits in the square root of } 25 = \frac{n}{2} = \frac{2}{2} = 1 \text{ (i.e. 5)}$$

Total number of digits in 225 = 3 (odd)

$$\text{Digits in the square root of } 225 = \frac{3+1}{2} = 2 \text{ (i.e. 15)}$$

$$7). \sqrt{x \sqrt{x \sqrt{x \dots n}}} = x^{\frac{2^n - 1}{2^n}}$$

Example :- Find the value of $\sqrt{2 \sqrt{2 \sqrt{2 \sqrt{2 \sqrt{2}}}}}$

Solution :- Here, $n = 5$,

$$\therefore \sqrt{2 \sqrt{2 \sqrt{2 \sqrt{2 \sqrt{2}}}}} = 2^{\frac{2^5 - 1}{2^5}} = 2^{\frac{31}{32}}$$

$$8). \sqrt{x \sqrt{x \sqrt{x \dots \infty}}} = x;$$

Example :- $\sqrt{7 \sqrt{7 \sqrt{7 \dots \infty}}} = 7$

9). To find the value of $\sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$, find the factors of x , such that the difference between the factors is 1, then the larger factor will be the result.

10). To find the value of $\sqrt{x - \sqrt{x - \sqrt{x - \dots \infty}}}$, find the factors of x , such that the difference between the factors is 1, then the smaller factor will be the result.

Example :- Find the value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}$

Solution:- Let $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}} = x$

$$\Rightarrow \sqrt{12 + x} = x$$

$$\Rightarrow 12 + x = x^2 \Rightarrow x^2 - x - 12 = 0$$

$$\Rightarrow x^2 - 4x + 3x - 12 = 0$$

$$\Rightarrow (x - 4)(x + 3) = 0$$

$$\Rightarrow x = 4 \text{ and } x = -3 \therefore x = 4$$

Exam hall approach:- The factors of 12 with a difference of one are 4, 3. Here, 4 is the larger number.

$\therefore$ Value will be 4.

Example :- Find the value of $\sqrt{30 - \sqrt{30 - \sqrt{30 - \dots \infty}}}$

Solution :- The factors of 30 with a difference of 1 are 5, 6. Here, 5 is the smaller number. $\therefore$ Value will be 5.

11). If any number is in $\frac{1}{\sqrt{a} \pm \sqrt{b}}$ form, then we multiply by its rationalization factor $\sqrt{a} \pm \sqrt{b}$ in both numerator and denominator.

Example :- Find the value of $\frac{1}{\sqrt{9} - \sqrt{8}}$.

Solution :- $\frac{1}{\sqrt{9}-\sqrt{8}} = \frac{1}{\sqrt{9}-\sqrt{8}} \times \frac{\sqrt{9}+\sqrt{8}}{\sqrt{9}+\sqrt{8}}$
 $= \frac{\sqrt{9}+\sqrt{8}}{(\sqrt{9})^2 - (\sqrt{8})^2} = \frac{3+2\sqrt{2}}{1} = 3 + 2\sqrt{2}$

Cube And Cube Roots

Important points :-

1). If a number is multiplied by itself twice, the result of this multiplication is called the cube of that number.

i.e. :- $2 \times 2 \times 2 = 2^3$

Cube root of $2^3 = \sqrt[3]{2^3} = 2^{3 \times \frac{1}{3}} = 2^1 = 2$

2). Algebraic method to calculate cube:

$$(a + b)^3 = a^3 + 3ab(a + b) + b^3$$

$$(a - b)^3 = a^3 - 3ab(a - b) - b^3$$

Important Formulas

1). $(a + b)^2 = a^2 + 2ab + b^2$

2). $(a - b)^2 = a^2 - 2ab + b^2$

3). $(a^2 - b^2) = (a + b)(a - b)$

4). $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

5). $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

6). $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$

7). $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$

8). $a^3 + b^3 + c^3 - 3abc =$

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

Here, if $(a + b + c) = 0$, then $a^3 + b^3 + c^3 - 3abc = 0$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

9). $(a + b + c)^2 = (a^2 + b^2 + c^2) + 2(ab + bc + ca)$

Bar based concept :-

$0.\overline{x} = \frac{x}{9}$	$0.\overline{xyz} = \frac{xyz - x}{990}$
$0.\overline{xy} = \frac{xy}{99}$	$0.\overline{xyz} = \frac{xyz - xy}{900}$

Example :- The value of $0.\overline{23} + 0.1\overline{43} - 0.\overline{2}$

Solution :- $\frac{23}{99} + \left(\frac{143 - 1}{990}\right) - \frac{2}{9}$
 $= \frac{230 + 142 - 220}{990} = \frac{152}{990} = 0.15\overline{2}$

Continuous Fraction

Example :- $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{3}{5}}}} = ?$

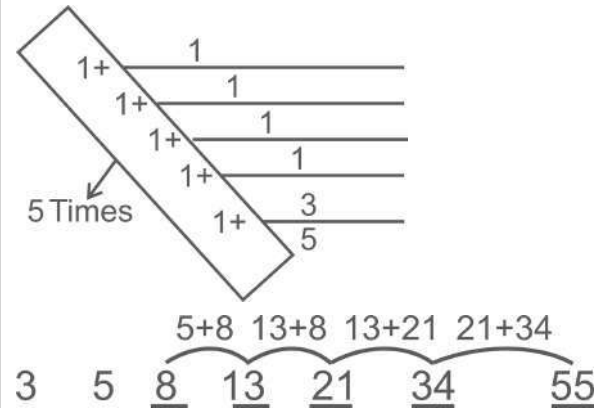
Solution :-

Step 1 : First write the last term $\left(\frac{3}{5}\right)$

Step 2 : Write first '3' and then write '5'

Step 3 : As many times as '1' is given in the question, further numbers will appear, and to find the next number, immediately add the previous number to that number.

i.e. $5 + 3 = 8, 8 + 5 = 13$



Step 4 : And finally , write the last number as a fraction with the number immediately preceding it .

Required number = $\frac{55}{34}$

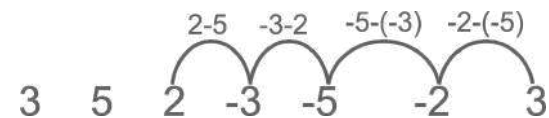
Example :- $1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{1 - \frac{3}{5}}}} = ?$

Solution:-

Step 1 : First write the last term $\left(\frac{3}{5}\right)$

Step 2 : Write '3' first and then write '5'

Step 3 : As Many times as '1' is given in the question Then , immediately subtract the number from the previous one to the next one .



Step 4 : And finally , write the last number as a fraction with the number immediately preceding it Required number = $-\frac{3}{2}$

Example :- $1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}} = ?$

Solution:-

Step 1 : First write the last term $\left(\frac{1}{4}\right)$

Step 2 : Write '1' first and then write '4'

Step 3 :

$$1 \xrightarrow{+2} 4 \xrightarrow{+5} 9 \xrightarrow{+21} 31 \xrightarrow{+9} 40$$

Step 4 : And finally , write the last number as a fraction with the number immediately preceding it Required Number = $\frac{40}{31}$

Example :- $5 - \frac{1}{3 - \frac{1}{2 - \frac{1}{5}}} = ?$

Solution :-

Step 1 : First write the last term ($\frac{1}{5}$)

Step 2 : Write '1' first and then write '5'

Step 3 :

$$1 \xrightarrow{-5} 5 \xrightarrow{-9} 9 \xrightarrow{-22} 22 \xrightarrow{-101} 101$$

Step 4 : And finally , write the last number as a fraction with the number immediately preceding it Required number : $\frac{101}{22}$

Example : $\frac{1}{a + \frac{1}{b + \frac{1}{c}}} = \frac{13}{29}$,

Find the value of $a + b - c = ?$

Solution :- Here

$$\begin{array}{r} 13 \overline{) 29} \text{ (2) } a \\ \underline{26} \\ 3 \overline{) 13} \text{ (4) } b \\ \underline{12} \\ 1 \overline{) 3} \text{ (3) } c \\ \underline{3} \\ \hline \end{array}$$

$$\Rightarrow a = 2, b = 4, c = 3$$

$$\text{So } a + b - c = 2 + 4 - 3 = 3$$

Mixed Number ($p\frac{q}{r}$) Based Question

Type - 1

Series based :-

Example :- $999\frac{1}{5} + 999\frac{2}{5} + \dots + 999\frac{4}{5}$

$$999 \times 4 + \frac{1+2+3+4}{5} = 3996 + \frac{4 \times 5}{2} = 3996 + 2 = 3998$$

Type - 2

Example :

$$\begin{aligned} 55\frac{2}{3} + 45\frac{4}{5} + 41\frac{11}{15} &= 55 + \frac{2}{3} + 45 + \frac{4}{5} + 41 + \frac{11}{15} \\ &= 141 + \left(\frac{2}{3} + \frac{4}{5} + \frac{11}{15}\right) = 141 + \left(\frac{10+12+11}{15}\right) \end{aligned}$$

$$= 141 + \frac{33}{15} = 141 + \frac{11}{5} = 141\frac{11}{5}$$

$$= 141 + 2 + \frac{1}{5} = 143\frac{1}{5}$$

Type - 3

If difference between numerator and denominator is 1

Example : $\frac{1}{7} + \left(9999\frac{692}{693}\right) \times 99$

Here $693 - 692 = 1$ so we can write $\frac{692}{693} = 1 - \frac{1}{693}$

$$= \frac{1}{7} + \left(9999 + 1 - \frac{1}{693}\right) \times 99$$

$$= \frac{1}{7} + \left(10000 - \frac{1}{693}\right) \times 99$$

$$= \frac{1}{7} + 990000 - \frac{99}{693} = 990000$$

Type - 4

If denominator of a number same as multiplier

Example : $45\frac{92}{99} \times 99$

Add 7 in 92 so it becomes 99 the same as the denominator.

$$45\frac{(92+7)}{99} \times 99 = (45 + 1) \times 99 = 4554$$

Now subtract 7 from the resultant = $4554 - 7 = 4547$

Comparison of Fraction Based Questions

Example :- Write in ascending order

$$\frac{2}{3}, \frac{8}{6}, \frac{5}{4}, \frac{3}{2}$$

Solution :- L.C.M of (3, 6, 4, 2) = 12

$$\frac{2}{3} \times 12, \frac{8}{6} \times 12, \frac{5}{4} \times 12, \frac{3}{2} \times 12$$

$$8, 16, 15, 18$$

It is clear that $8 < 15 < 16 < 18$

$$\text{So the ascending order} = \frac{2}{3}, \frac{5}{4}, \frac{8}{6}, \frac{3}{2}$$

Algebra

Square Formulas

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $a^2 - b^2 = (a + b)(a - b)$

Example :- What is the value of

$$(k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$$

SSC CHSL 15/03/2023 (2nd Shift)

(a) $k^{64} - \frac{1}{k^{64}}$ (b) $\frac{k^{32} - \frac{1}{k^{32}}}{k - \frac{1}{k}}$ (c) $(k^{32} - \frac{1}{k^{32}})$ (d) $\frac{k^{32} - \frac{1}{k^{32}}}{k + \frac{1}{k}}$

Solution :-

Note - make given equation in terms of $(a + b)(a - b)$

On Multiplying and dividing by $(k + \frac{1}{k})$

$$\begin{aligned} &= \frac{(k + \frac{1}{k})}{(k + \frac{1}{k})} \times (k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}}) \\ &= \frac{(k^2 - \frac{1}{k^2})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})} \\ &= \frac{(k^4 - \frac{1}{k^4})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})} \\ &= \frac{(k^8 - \frac{1}{k^8})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})} \\ &= \frac{(k^{16} - \frac{1}{k^{16}})(k^{16} + \frac{1}{k^{16}})}{(k + \frac{1}{k})} = \frac{(k^{32} - \frac{1}{k^{32}})}{(k + \frac{1}{k})} \end{aligned}$$

- $a^4 - b^4 = (a^2 + b^2)(a^2 - b^2) = (a^2 + b^2)(a + b)(a - b)$
- $a^2 + \frac{1}{a^2} = (a + \frac{1}{a})^2 - 2$
- $a^2 + \frac{1}{a^2} = (a - \frac{1}{a})^2 + 2$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca)$
- $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$

If $a = b = c = k$ then, $a^2 + b^2 + c^2 = ab + bc + ca$

Example:- What is the possible value of $(a + b + c) - 3$, if $a^2 + b^2 + c^2 = 9$ and $ab + bc + ca = 8$?
SSC CGL 03/12/2022 (4th Shift)

- (a) 5 (b) 3 (c) 9 (d) 2

Solution:- $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$

$$(a + b + c)^2 = 9 + 2 \times 8$$

$$a + b + c = 5$$

$$\text{Now, } (a + b + c) - 3 = 5 - 3 = 2$$

Cube Formula

- $(a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$
- $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$
- $a^3 + \frac{1}{a^3} = (a + \frac{1}{a})^3 - 3(a + \frac{1}{a})$
- $a^3 - \frac{1}{a^3} = (a - \frac{1}{a})^3 + 3(a - \frac{1}{a})$

Most Repeated Formulae in Exams

$$1). a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Example :- If $a + b + c = 5$, $a^3 + b^3 + c^3 = 85$ and $abc = 25$, then find the value of $a^2 + b^2 + c^2 - ab - bc - ca$.

SSC CGL 13/12/2022 (4th Shift)

- (a) 2 (b) 4 (c) 6 (d) 8

Solution :- Formula : $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 85 - 3 \times 25 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 85 - 75 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 10 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = \frac{10}{5} = 2$$

$$2). a^3 + b^3 + c^3 - 3abc =$$

$$\frac{1}{2}(a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$$

Example :- If $x = 222$, $y = 223$ and $z = 224$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CGL 12/12/2022 (4th Shift)

- (a) 2007 (b) 2004 (c) 2006 (d) 2005

Solution :- Formula used :

$$x^3 + y^3 + z^3 - 3xyz =$$

$$\frac{1}{2}(x + y + z)[(x - y)^2 + (y - z)^2 + (z - x)^2]$$

$$= \frac{(222 + 223 + 224)}{2}[(222 - 223)^2$$

$$+ (223 - 224)^2 + (224 - 222)^2]$$

$$= \frac{669}{2}[(-1)^2 + (-1)^2 + (2)^2]$$

$$= \frac{669}{2} \times 6 = 2007$$

3). $a^3 + b^3 + c^3 - 3abc = (a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$

Example :- If $a + b + c = 5$ and $ab + bc + ca = 7$, then the value of $a^3 + b^3 + c^3 - 3abc$ is:

SSC CPO 05/10/2023 (1st Shift)

(a) 20 (b) 25 (c) 15 (d) 30

Solution :-

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$= (5)[(5)^2 - 3(7) = 5(4) = 20$$

4) $a^3 + b^3 + c^3 - 3abc = \frac{1}{2} (a + b + c) [3(a^2 + b^2 + c^2) - (a + b + c)^2]$

Example:- If $x + y + z = 8$, and $x^2 + y^2 + z^2 = 20$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is _____.

SSC CPO 09/11/2022 (Evening)

(a) 16 (b) 10 (c) 15 (d) -16

Solution:- Formula used :

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2} (a + b + c) [3(a^2 + b^2 + c^2) - (a + b + c)^2]$$

$$= \frac{1}{2} (8) [3(20) - (8)^2] = 4 [60 - 64] = -16$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= a^3 + b^3 + c^3 - 3abc$$

$$\text{When } (a + b + c) = 0 \text{ or } (a^2 + b^2 + c^2 - ab - bc - ca) = 0$$

• when, $(a + b + c) = 0$; $a \neq b \neq c$

$$(a^2 + b^2 + c^2 - ab - bc - ca) = 0; a = b = c$$

• $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(a + c)$

• $a^4 + a^2 b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$

• If $(x^2 + y^2 + z^2) = 0$, then $x = 0, y = 0, z = 0$

For Increasing Power

1) If $x + \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 - 2$

Example:- If $(a + \frac{1}{a}) = 6$, then what is the value of $\frac{3}{4}(a^2 + \frac{1}{a^2})$?

SSC CHSL 02/08/2023 (2nd Shift)

(a) 22.5 (b) 34 (c) 25.5 (d) 36

Solution:- According to question,

$$a + \frac{1}{a} = 6 \Rightarrow a^2 + \frac{1}{a^2} = (6)^2 - 2 = 34$$

$$\text{Now, } \frac{3}{4} (a^2 + \frac{1}{a^2}) = \frac{3}{4} \times 34 = \frac{51}{2} = 25.5$$

2) If $x - \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 + 2$

Example:- If $(x - \frac{1}{x}) = 0.4$, and $x > 0$, what is the value of

$$(x^2 + \frac{1}{x^2})?$$

SSC CHSL 07/08/2023 (2nd Shift)

(a) 4.16 (b) 3.32 (c) 1.84 (d) 2.16

Solution:- $x - \frac{1}{x} = 0.4$

$$x^2 + \frac{1}{x^2} = (0.4)^2 + 2 = 2.16$$

3) If $x + \frac{1}{x} = a$, then $x^3 + \frac{1}{x^3} = a^3 - 3a$

Example:- If $2x + \frac{2}{x} = 5$, then the value of $(x^3 + \frac{1}{x^3} + 2)$

will be :

SSC CHSL 02/08/2023 (1st Shift)

(a) $\frac{81}{11}$ (b) $\frac{81}{7}$ (c) $\frac{71}{8}$ (d) $\frac{81}{8}$

Solution:- $2x + \frac{2}{x} = 5 \Rightarrow x + \frac{1}{x} = \frac{5}{2}$

$$x^3 + \frac{1}{x^3} = (\frac{5}{2})^3 - 3 \times (\frac{5}{2}) = \frac{65}{8}$$

$$x^3 + \frac{1}{x^3} + 2 = \frac{65}{8} + 2 = \frac{81}{8}$$

4) If $x - \frac{1}{x} = a$, then $x^3 - \frac{1}{x^3} = a^3 + 3a$

Example:- If $(y - \frac{1}{y}) = 9$, find the value of $(y^3 - \frac{1}{y^3})$

SSC CHSL 04/08/2023 (3rd Shift)

(a) 766 (b) 756 (c) 702 (d) 729

Solution:- $y - \frac{1}{y} = 9$

$$y^3 - \frac{1}{y^3} = (9)^3 + 3 \times 9 = 756$$

5) If $x + \frac{1}{x} = a$, then $x^4 + \frac{1}{x^4} = (a^2 - 2)^2 - 2$

Example:- If $\frac{x^2 + 1}{x} = 5$, then find the value

$$\text{of } x^4 + \frac{1}{x^4} - 36.$$

SSC CHSL 09/08/2023 (2nd Shift)

(a) 491 (b) 149 (c) 419 (d) 194

Solution:- $x + \frac{1}{x} = 5$

$$x^2 + \frac{1}{x^2} = 5^2 - 2 = 23$$

$$x^4 + \frac{1}{x^4} = 23^2 - 2 = 527$$

$$\text{Now, } x^4 + \frac{1}{x^4} - 36 = 527 - 36 = 491$$

6) If $x - \frac{1}{x} = a$, then $x^4 + \frac{1}{x^4} = (a^2 + 2)^2 - 2$

Example:- If $(x - \frac{1}{x}) = 10$, what is the value of $(x^4 + \frac{1}{x^4})$?

SSC CGL 26/07/2023 (3rd shift)

(a) 10404 (b) 10402 (c) 10406 (d) 10400

Solution:- Given, $x - \frac{1}{x} = 10$

$$x^2 + \frac{1}{x^2} = (10)^2 + 2 = 102$$

$$\text{Now, } x^4 + \frac{1}{x^4} = 102 \times 102 - 2 = 10402$$

7) If $x + \frac{1}{x} = a$, then $x^5 + \frac{1}{x^5} = (a^2 - 2)(a^3 - 3a) - (a)$

or

$$(x^5 + \frac{1}{x^5}) = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

Example:- If $(x + \frac{1}{x}) = 5$, then the value of $(x^5 + \frac{1}{x^5})$ is:

SSC CHSL 07/08/2023 (2nd Shift)

(a) 2535 (b) 2540 (c) 2515 (d) 2525

Solution:- $x + \frac{1}{x} = 5 \Rightarrow x^2 + \frac{1}{x^2} = 23$

$$\Rightarrow x^3 + \frac{1}{x^3} = 110$$

Now,

$$x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

$$\Rightarrow 23 \times 110 - 5 = 2530 - 5 = 2525$$

8) If $x - \frac{1}{x} = a$, then $x^5 - \frac{1}{x^5} = (a^2 + 2)(a^3 + 3a) - (a)$

or

$$x^5 - \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$$

Example:- If $(y - \frac{1}{y}) = -9$, what will be the value of

$$(y^5 - \frac{1}{y^5})?$$

SSC CHSL 08/08/2023 (4th Shift)

(a) -62757 (b) -62748

(c) -62739 (d) -59049

Solution:-

$$(y - \frac{1}{y}) = -9 \Rightarrow (y^2 + \frac{1}{y^2}) = 83$$

$$\text{and } (y^3 - \frac{1}{y^3}) = (-9)^3 + 3 \times (-9)$$

$$= -756$$

$$y^5 - \frac{1}{y^5} = \{(y^2 + \frac{1}{y^2})(y^3 - \frac{1}{y^3})\} - (y - \frac{1}{y})$$

$$= \{83 \times (-756)\} - (-9)$$

$$= \{(-62748)\} - (-9) = (-62739)$$

9) If $x + \frac{1}{x} = a$, then $x^6 + \frac{1}{x^6} = (a^3 - 3a)^2 - (2)$

Example:- If $(x + \frac{1}{x}) = 2\sqrt{2}$, and $x > 1$, what is the value of

$$(x^6 - \frac{1}{x^6})?$$

SSC CGL 19/07/2023 (1st shift)

(a) $140\sqrt{2}$ (b) $116\sqrt{2}$ (c) $144\sqrt{2}$ (d) $128\sqrt{2}$

Solution:- $x + \frac{1}{x} = 2\sqrt{2}$

$$x - \frac{1}{x} = \sqrt{(2\sqrt{2})^2 - 4} = 2$$

$$x^3 - \frac{1}{x^3} = (x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$$

$$= (2)^3 + 3 \times 2 = 14$$

$$x^3 + \frac{1}{x^3} = (x + \frac{1}{x})^3 - 3(x + \frac{1}{x})$$

$$= (2\sqrt{2})^3 - 3 \times 2\sqrt{2} = 16\sqrt{2} - 6\sqrt{2} = 10\sqrt{2}$$

Now,

$$x^6 - \frac{1}{x^6} = (x^3 - \frac{1}{x^3})(x^3 + \frac{1}{x^3})$$

$$\Rightarrow x^6 - \frac{1}{x^6} = 14 \times 10\sqrt{2} = 140\sqrt{2}$$

10) If $x + \frac{1}{x} = a$, then $x^6 - \frac{1}{x^6} = (a^3 - 3a)(a^3 + 3a)$

Example:- If $(x + \frac{1}{x}) = 5\sqrt{2}$, and $x > 1$, what is the value of

$$(x^6 - \frac{1}{x^6})?$$

SSC CHSL 02/08/2023 (2nd Shift)

(a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$

(c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

Solution:- $x + \frac{1}{x} = 5\sqrt{2}$

$$x^3 + \frac{1}{x^3} = (5\sqrt{2})^3 - 3(5\sqrt{2})$$

$$= 250\sqrt{2} - 15\sqrt{2} = 235\sqrt{2}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \sqrt{\left(x^3 + \frac{1}{x^3}\right)^2 - 4}$$

$$= \sqrt{(235\sqrt{2})^2 - 4}$$

$$= \sqrt{110450 - 4} = \sqrt{110446} = 49\sqrt{46}$$

$$\text{Now, } x^6 - \frac{1}{x^6} = (x^3 + \frac{1}{x^3})(x^3 - \frac{1}{x^3})$$

$$= 235\sqrt{2} \times 49\sqrt{46}$$

$$= 235 \times 49 \times 2 \times \sqrt{23} = 23030\sqrt{23}$$

For Parallel Power

1.	If $x + \frac{1}{x} = a$, then $x - \frac{1}{x} = \sqrt{a^2 - 4}$
2.	If $x - \frac{1}{x} = a$, then $x + \frac{1}{x} = \sqrt{a^2 + 4}$

3.	If $x + \frac{1}{x} = a$, then $x^2 - \frac{1}{x^2} = a\sqrt{a^2 - 4}$
4.	If $x - \frac{1}{x} = a$, then $x^2 - \frac{1}{x^2} = a\sqrt{a^2 + 4}$

Some special cases :-

1) If $x + \frac{1}{x} = 2$, then $x = 1$

if $x + \frac{1}{x} = -2$, then $x = -1$

Example :- If $x + \frac{1}{x} = 2$, then what is the value of $x^{64} + x^{121}$?

- (a) 2 (b) -2 (c) 1 (d) 3

Solution :- $x + \frac{1}{x} = 2$; $x = 1$
 $= 1^{64} + 1^{121} = 2$

2) If $x + \frac{1}{x} = \pm\sqrt{3}$, then $x^6 = -1$

Example :- If $x + \frac{1}{x} = \sqrt{3}$ then the value of $x^{12} + x^9 + x^6 + x^3 + 1$ is:

- (a) 1 (b) -1 (c) 0 (d) -2

Solution :- $x^{12} + x^9 + x^6 + x^3 + 1$
 $= x^6(x^6 + 1) + x^3(x^3 + 1) + 1 = x^6(-1 + 1) + x^3(-1 + 1) + 1 = 1$

Short trick :- If the exponential difference between two terms is 6 then their value becomes zero.
 $x^{12} + x^9 = 0$

3) If $x + \frac{1}{x} = 1$, then $x^3 = -1$

If $x + \frac{1}{x} = -1$, then $x^3 = 1$

Example :- If $x + \frac{1}{x} = 1$, then the value of $x^{12} + x^9 + x^6 + x^3 + 1$ is:

- SSC CGL 06/12/2022 (1st Shift)
 (a) 1 (b) -1 (c) 0 (d) -2

Solution :- $x^{12} + x^9 + x^6 + x^3 + 1$
 $= x^9(x^3 + 1) + x^3(x^3 + 1) + 1 = x^9(-1 + 1) + x^3(-1 + 1) + 1 = 1$

Short trick :- If the exponential difference between two terms is 3 then their value becomes zero.
 $x^{12} + x^9 = 0$

Some more Special Case :-

►► if $a^2 - ab + b^2 = 0$ then $a^3 + b^3 = 0$

►► if $\frac{1}{a} - \frac{1}{b} = \frac{1}{a-b}$ then $a^3 + b^3 = 0$

►► if $\frac{1}{b} + \frac{1}{a} = \frac{1}{a+b}$ then $a^3 - b^3 = 0$

►► if $\frac{a}{b} + \frac{b}{a} = -1$ then $a^3 - b^3 = 0$

►► if $\frac{a}{b} + \frac{b}{a} = 1$ then $a^3 + b^3 = 0$

►► if $a^2 + a + 1 = 0$ then $a^3 - 1 = 0$ or $a^3 = 1$

►► if $ab(a+b) = 1$ then $\frac{1}{a^3 b^3} - a^3 - b^3 = 3$

Componendo - Dividendo Rule

If $\frac{a}{b} = \frac{c}{d}$ then, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

If $\frac{a+b}{a-b} = \frac{a}{b}$ then $\frac{c}{d} = \frac{c+d}{c-d}$

• If $(x-a)^2 + (y-b)^2 + (z-c)^2 = 0$ then by putting $x = a, y = b, z = c$ in an expression we can find the value of the expression.

• If $x + y = a$, then xy will be maximum when $x = y$.

• If $xy = a$, then $(x + y)$ will be minimum when $x = y = \sqrt{a}$

Rules for Making a Perfect Square Polynomial :-

1. Middle term = $\pm \sqrt{\text{First term} \times \text{Third term}} \times 2$

2. First term = $\frac{(\text{Middle term})^2}{4 \times \text{Third term}}$

3. Third term = $\frac{(\text{Middle term})^2}{4 \times \text{First term}}$

Example:- Check $9x^2 + 12xy + 4y^2$ polynomial is perfect square or not?

Solution:-

Step 1 :- We know that

Middle term = $\pm \sqrt{\text{First term} \times \text{Third term}} \times 2$

$12 = \pm \sqrt{9 \times 4} \times 2 \Rightarrow 12 = 6 \times 2 \Rightarrow 12 = 12$

Step 2 :- First term = $\frac{(\text{Middle term})^2}{4 \times \text{Third term}}$

$9 = \frac{(12)^2}{4 \times 4} \Rightarrow 9 = 9$

Step 3 :- Third term = $\frac{(\text{Middle term})^2}{4 \times \text{First term}}$

$4 = \frac{(12)^2}{4 \times 9} \Rightarrow 4 = 4$

Therefore the given polynomial is a perfect square.

Exam hall approach :-

$9x^2 + 12xy + 4y^2$

$\Rightarrow (3x)^2 + 2 \times 3x \times 2y + (2y)^2 \Rightarrow (3x + 2y)^2$

Therefore the given polynomial is a perfect square.

Algebra Formula for Irrational Number

1.	$\sqrt{ab} = \sqrt{a}\sqrt{b}$
2.	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
3.	$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$
4.	$(\sqrt{a} + \sqrt{b})^2 = a + 2\sqrt{ab} + b$
5.	$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$

Algebra Formula

Laws of Exponents

Multiplication rule	$a^x \times a^y = a^{x+y}$
Division rule	$a^x \div a^y = a^{x-y}$
Power of power rule	$(a^x)^y = a^{xy}$
Power of product rule	$(ab)^x = a^x b^x$
Power of fraction rule	$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
Zero exponent	$a^0 = 1$
Negative exponent	$a^{-x} = \frac{1}{a^x}$
Fractional exponent	$a^{\frac{x}{y}} = \sqrt[y]{a^x}$

Factor Formula

- $(x+a)(x+b) = x^2 + x(a+b) + ab$
- $(x-a)(x-b) = x^2 - x(a+b) + ab$
- $(x-a)(x+b) = x^2 - x(a-b) - ab$
- $(x+a)(x-b) = x^2 + x(a-b) - ab$

We can solve this type of identities by value putting :-

- If $x + y = 2z$, then the value of $\frac{z}{x-z} + \frac{y}{y-z} = 1$ ($x \neq y \neq z$)
- If $x + y = 2z$, then $\frac{x}{x-z} + \frac{y}{y-z} = 2$ ($x \neq y \neq z$)
- If $x = a + \frac{1}{a}$ and $y = a - \frac{1}{a}$ then $\sqrt{x^4 + y^4 - 2x^2y^2} = 4$

Higher Power Formula

- $x^4 - \frac{1}{x^4} = (x^2 + \frac{1}{x^2})(x + \frac{1}{x})(x - \frac{1}{x})$
- $x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$
- $x^5 - \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$
- $x^7 + \frac{1}{x^7} = (x^4 + \frac{1}{x^4})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$
- $x^7 - \frac{1}{x^7} = (x^4 + \frac{1}{x^4})(x^3 - \frac{1}{x^3}) + (x - \frac{1}{x})$
- $x^8 - \frac{1}{x^8} = (x^4 + \frac{1}{x^4})(x^2 + \frac{1}{x^2})(x + \frac{1}{x})(x - \frac{1}{x})$

Some Other Results

- $\frac{(a+b)^2 - (a-b)^2}{ab} = 4$
- $\frac{(a+b)^2 + (a-b)^2}{a^2 + b^2} = 2$
- If $a + b + c = abc$, then $\frac{(1-a^2)(1-b^2)}{ab} + \frac{(1-b^2)(1-c^2)}{bc} + \frac{(1-c^2)(1-a^2)}{ca} = 4$

4.) If $bc + ca + ab = abc$ then,

$$\frac{(a+b)}{ab(c-1)} + \frac{(b+c)}{bc(a-1)} + \frac{(c+a)}{ca(b-1)} = 1$$

5.) $ab(a-b) + bc(b-c) + ca(c-a) = (b-a)(b-c)(c-a)$

6.) If $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$ (Where $a \neq b \neq c$) then abc is equal to ± 1

- If $x + y = 0$
 $x = -y, x = 0, y = 0$
- If $x^2 + y^2 = 0$
 $x^2 = -y^2, x^2 = 0, y^2 = 0$

Example :- If $x^2 + y^2 + z^2 = 2(x + z - 1)$, then the value of $x^3 + y^3 + z^3$ is equal to :

SSC CHSL 11/08/2023 (3rd Shift)

(a) 6 (b) 1 (c) 2 (d) 8

Solution :- given : $x^2 + y^2 + z^2 = 2(x + z - 1)$

$$x^2 + y^2 + z^2 = 2x + 2z - 2$$

$$x^2 - 2x + z^2 - 2z + 2 + y^2 = 0$$

$$x^2 - 2x + z^2 - 2z + y^2 + 2 = 0$$

$$x^2 - 2x + 1 + z^2 - 2z + 1 + y^2 = 0$$

$$(x-1)^2 + (z-1)^2 + y^2 = 0$$

$$x = 1, z = 1, y = 0$$

Now, $x^3 + y^3 + z^3 = 1 + 0 + 1 = 2$

Short trick :- for the value of any variable ($x, y, z, \dots$)

Divide coefficient by 2 and write with their sign.

Addition /subtraction in given equation :-

Example :- If $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$ then the value of $\frac{1}{1-a}$

$$+ \frac{1}{1-b} + \frac{1}{1-c} = ?$$

Solution :- on adding 3 both sides of the equation.

$$\frac{a}{1-a} + 1 + \frac{b}{1-b} + 1 + \frac{c}{1-c} + 1 = 1 + 3$$

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 4$$

Example :- If $x = 16$, then the value of $(x^4 - 17x^3 + 17x^2 - 17x + 17)$.

(a) 1 (b) 0 (c) 3 (d) 2

Solution :- $x^4 - 16x^3 - x^3 + 16x^2 + x^2 - 16x - x + 17$

$$= x^4 - x.x^3 - x^3 + x.x^2 + x^2 - x.x - x + 17$$

$$= x^4 - x.x^3 - x^3 + x.x^2 + x^2 - x.x - x + 17 = -x + 17 = -16 + 17 = 1$$

Surds and Indices

Indices

Indices are numeric values that act as power or exponent to a particular number.

i.e. a^n

Here 'a' is any number and is called the base of a^n .

'n' is any natural number and is called index or exponent of a^n .

Law of Indices

$$\begin{aligned} (1). a^m \times a^n &= a^{m+n} & (2). \frac{a^m}{a^n} &= a^{m-n} & (3). (a^m)^n &= a^{mn} \\ (4). \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n} & (5). a^0 &= 1 & (6). a^{\frac{1}{n}} &= \sqrt[n]{a} \\ (7). p^{-n} &= \frac{1}{p^n} & (8). (ab)^n &= a^n \times b^n & (9). \frac{1}{a^n} &= a^{-n} \end{aligned}$$

If 'a' is negative number and 'n' is an odd number then

$$(-a)^n = -a^n$$

Some Important Result :-

1. If $a^m = k$, then $a = (k)^{\frac{1}{m}}$
2. If $a^{\frac{1}{m}} = k$, then $a = k^m$
3. If $a^m = b^n$, then $a = (b)^{\frac{n}{m}}$ and $b = (a)^{\frac{m}{n}}$
4. If $a^m = a^n$, then $m = n$
5. The HCF of $(a^m - 1)$ and $(a^n - 1)$ is equal to the $(a^{\text{hcf of } m, n} - 1)$

Surds

When a root of rational number cannot be obtained exactly, then that root is called a surd.

i.e. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$ etc.

$\sqrt[3]{2}$, $\sqrt[3]{3}$, $\sqrt[3]{4}$, $\sqrt[3]{5}$, $\sqrt[3]{6}$ etc.

but $\sqrt{4} = 2$, $\sqrt[3]{27} = 3$, $\sqrt[5]{32} = 2$ are not the surds.

So, we can say that "all surds are irrational numbers but all irrational numbers are not the surds."

In surds $\sqrt[n]{a}$, the value of n is called the order and a is called the radicand.

Types of surds :-

Simple Surds :- Simple Surds have a single number, also called as monomial.

i.e. $\sqrt{2}$.

Similar Surds :- Similar surds have a common root factor.

i.e. $3\sqrt{2}$ and $7\sqrt{2}$.

Mixed Surds :- Mixed Surds can be defined as a product of an irrational number with a rational number.

i.e. $3\sqrt{7}$ and $2\sqrt{6}$.

Compound Surds :- A compound surd has algebraic addition or

subtraction of two or more than two surds.

i.e. $(\sqrt{6} + \sqrt{3})$

Conjugate or complementary surds :-

For every surd of the form $\sqrt{a} + \sqrt{b}$ there exists another surd $\sqrt{a} - \sqrt{b}$ which is called the conjugate surd of previous surd and vice-versa.

i.e. $(2\sqrt{3} + 5\sqrt{3})$ and $(2\sqrt{3} - 5\sqrt{3})$ are the pair of conjugate surds.

Laws of surds

$$\begin{aligned} (1). \sqrt[n]{a} &= a^{\frac{1}{n}} & (2). \sqrt[n]{a} \times \sqrt[n]{b} &= \sqrt[n]{ab} \\ (3). \frac{\sqrt[n]{a}}{\sqrt[n]{b}} &= \sqrt[n]{\frac{a}{b}} & (4). \sqrt[m]{\sqrt[n]{a}} &= \sqrt[mn]{a} \\ (5). (\sqrt[n]{a})^m &= \sqrt[n]{a^m} & (6). a^{\frac{m}{n}} &= \sqrt[n]{a^m} \end{aligned}$$

Example :- If $3\sqrt{5}$ is given, find out the rationalizing factor.

Solution :- $3\sqrt{5} \times \sqrt{5} = 3 \times 5 = 15$, so the rationalizing factor of $3\sqrt{5}$ is $\sqrt{5}$.

Rationalisation :- Making two or more surds a rational number by multiplication, is called the rationalisation.

➤ If any number is in $\frac{1}{\sqrt{a} \pm \sqrt{b}}$ form, then we multiply by its rationalization factor $\sqrt{a} \mp \sqrt{b}$ in both numerator and denominator.

Example :- Find the value of $\frac{1}{\sqrt{9} - \sqrt{8}}$

Solution :- On rationalising $\frac{1}{\sqrt{9} - \sqrt{8}}$

$$= \frac{1}{\sqrt{9} - \sqrt{8}} \times \frac{\sqrt{9} + \sqrt{8}}{\sqrt{9} + \sqrt{8}} = \frac{\sqrt{9} + \sqrt{8}}{\sqrt{9}^2 - \sqrt{8}^2} = \frac{3 + 2\sqrt{2}}{1} = 3 + 2\sqrt{2}$$

Some Important Result :-

$$(1). \text{ If } y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$$

$$\text{Then, } y = \frac{1 + \sqrt{1 + 4x}}{2}$$

Note :- To find the value of $\sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$, find the factors of x, such that the difference between the factors is 1, then the larger factor will be the result.

Example :- $y = \sqrt{72 + \sqrt{72 + \sqrt{72 + \dots \infty}}}$

Solution :- We know that,

$$\begin{aligned} \Rightarrow y &= \frac{1 + \sqrt{1 + 4x}}{2} \Rightarrow y = \frac{1 + \sqrt{1 + 4 \times 72}}{2} = \frac{1 + \sqrt{1 + 288}}{2} \\ &= \frac{1 + \sqrt{289}}{2} = \frac{1 + 17}{2} = \frac{18}{2} = 9 \end{aligned}$$

Exam Hall Approach :-

Factorise 72 into two successive integers i.e. 8×9

In addition series, Greater number will be the answer i.e. 9

$$(2). \text{ If } y = \sqrt{x - \sqrt{x - \sqrt{x - \dots \infty}}}$$

Then $y = \frac{-1 + \sqrt{1+4x}}{2}$

Note :- To find the value of $\sqrt{x - \sqrt{x - \sqrt{x - \dots \infty}}}$, find the factors of x , such that the difference between the factors is 1, then the smaller factor will be the result.

Example :- $y = \sqrt{56 - \sqrt{56 - \sqrt{56 - \dots \infty}}}$

Solution :- We know that,

$$\Rightarrow y = \frac{-1 + \sqrt{1+4x}}{2}$$

$$\Rightarrow y = \frac{-1 + \sqrt{1+4 \times 56}}{2} = \frac{-1 + \sqrt{225}}{2} = \frac{-1 + 15}{2} = \frac{14}{2} = 7$$

Exam Hall Approach :-

Factorise 56 into two successive integers i.e. 8×7

$$\sqrt{56 - 1\sqrt{56 - 1\sqrt{56 - \dots \infty}}}$$

In Subtraction series, a smaller number will be the answer i.e. 7

(3). If $y = \sqrt{x \sqrt{x \sqrt{x \dots \infty}}}$
Then, $y = x$

(4). If $y = \sqrt{x \sqrt{x \sqrt{x \dots \infty}}}$
Then, $y = x^{\frac{2^n - 1}{2^n}}$

(5). $\sqrt[n]{a \times \sqrt[n]{a \times \sqrt[n]{a \times \dots \infty}}} = \sqrt[n-1]{a}$

(6). $\sqrt[n]{a \div \sqrt[n]{a \div \sqrt[n]{a \div \dots \infty}}} = \sqrt[n+1]{a}$

(7). $\sqrt{a + b\sqrt{a + b\sqrt{a + b\sqrt{a + \dots \infty}}}} = \frac{\sqrt{4a+b^2} + b}{2}$

(8). $\sqrt{a - b\sqrt{a - b\sqrt{a - b\sqrt{a - \dots \infty}}}} = \frac{\sqrt{4a+b^2} - b}{2}$

Example :- $y = \sqrt{63 - 2\sqrt{63 - 2\sqrt{63 - 2\sqrt{63 - \dots \infty}}}}$

Solution :- We know that,

$$\Rightarrow y = \frac{\sqrt{4a+b^2} - b}{2} = \frac{\sqrt{4 \times 63 + 2^2} - 2}{2} = \frac{\sqrt{252 + 4} - 2}{2} = \frac{\sqrt{256} - 2}{2} = \frac{16 - 2}{2} = \frac{14}{2} = 7$$

Exam hall approach :-

Here

$$\sqrt{63 - 2\sqrt{63 - 2\sqrt{63 - 2\sqrt{63 - \dots \infty}}}}$$

In Subtraction series, a smaller number will be the answer i.e. 7

(9). $\sqrt{a + b\sqrt{a - b\sqrt{a + b\sqrt{a - b\sqrt{a + \dots \infty}}}}} = \frac{\sqrt{4a-3b^2} + b}{2}$

(10). $\sqrt{a - b\sqrt{a + b\sqrt{a - b\sqrt{a + \dots \infty}}}} = \frac{\sqrt{4a-3b^2} - b}{2}$

(11). $\sqrt{a + \sqrt{a - \sqrt{a + \sqrt{a - \dots \infty}}}} = \frac{\sqrt{4a-3} + 1}{2}$

(12). $\sqrt{a - \sqrt{a + \sqrt{a - \sqrt{a + \dots \infty}}}} = \frac{\sqrt{4a-3} - 1}{2}$

(13). If $y = \left(\sqrt[p]{\sqrt[q]{\sqrt[r]{a^x}}} \right)^n$ then $y = a^{\left(\frac{x \times z \times n}{p \times q} \right)}$

(14). $\frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} - \sqrt{y}} + \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} = 2 \times \left(\frac{x+y}{x-y} \right)$

Example :- If $x = \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$, $y = \frac{\sqrt{7} - \sqrt{5}}{\sqrt{7} + \sqrt{5}}$ then $(x+y) = ?$

Solution :- $x+y = 2 \times \left(\frac{7+5}{7-5} \right) = 12$

(15). $\frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} - \sqrt{y}} - \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \left(\frac{4xy}{x-y} \right)$

(16). $(\sqrt{x} + \sqrt{y})^2 - (\sqrt{x} - \sqrt{y})^2 = 4\sqrt{xy}$

(17).

Example :- If $m = \sqrt{13 + \sqrt{13 + \sqrt{13 + \sqrt{13 + \dots \infty}}}}$ and

$n = \sqrt{13 - \sqrt{13 - \sqrt{13 - \sqrt{13 - \dots \infty}}}}$ Find $m - n = ?$

Solution :-

$$m = \sqrt{13 + \sqrt{13 + \sqrt{13 + \sqrt{13 + \dots \infty}}}}$$

$$n = \sqrt{13 - \sqrt{13 - \sqrt{13 - \sqrt{13 - \dots \infty}}}}$$

$$m - n = \sqrt{13 + \sqrt{13 + \sqrt{13 + \sqrt{13 + \dots \infty}}}} - \sqrt{13 - \sqrt{13 - \sqrt{13 - \sqrt{13 - \dots \infty}}}}$$

$$m - n = \sqrt{13 + \sqrt{13 + \sqrt{13 + \sqrt{13 + \dots \infty}}}} - \sqrt{13 - \sqrt{13 - \sqrt{13 - \sqrt{13 - \dots \infty}}}}$$

Here $m = a+1$ and $n = a$

$$m - n = a+1 - a \Rightarrow m - n = 1$$

(18).

Example :- $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}} = ?$

Solution :- $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}$

$$= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}$$

$$= \sqrt{10 + \sqrt{25 + 11}} = \sqrt{10 + 6} = \sqrt{16} = 4$$

Square Root Expression Inside the Square Root

Example :- Solve $\sqrt{7 - 2\sqrt{6}}$?

Solution :-

$$\sqrt{7 - 2\sqrt{6}} = \sqrt{(\sqrt{6})^2 + (\sqrt{1})^2 - 2(\sqrt{6}) \times (\sqrt{1})}$$

$$= \sqrt{(\sqrt{6} - \sqrt{1})^2} = (\sqrt{6} - \sqrt{1})$$

Note :- $(a - b)^2 = a^2 + b^2 - 2ab$

➤ If $x + y = \text{constant}$, then

$(xy)_{\max}$ = Difference of x and y should be minimum.

$(xy)_{\min}$ = Difference of x and y should be maximum.

Example :- The smallest in ,

$$(\sqrt{8} + \sqrt{5}), (\sqrt{7} + \sqrt{6}), (\sqrt{10} + \sqrt{3}), (\sqrt{11} + \sqrt{2})$$

Solution :- Here Sum is constant = 13

$$\Rightarrow \text{Smallest} = (\sqrt{11} + \sqrt{2}) \text{ (Diff. Max)}$$

$$\text{Largest} = (\sqrt{7} + \sqrt{6}) \text{ (Diff. Min)}$$

Example :- Find the smallest of among

$$(\sqrt{23} + 2\sqrt{3}), (\sqrt{31} + 2), (\sqrt{24} + \sqrt{11}), (\sqrt{29} + \sqrt{6})$$

Solution:- $(\sqrt{23} + \sqrt{12}), (\sqrt{31} + \sqrt{4}), (\sqrt{24} + \sqrt{11}),$

$$(\sqrt{29} + \sqrt{6})$$

Here Sum is constant = 35

$$\Rightarrow \text{Smallest} = (\sqrt{31} + \sqrt{4}) \text{ (Diff. Max)}$$

$$\text{Largest} = (\sqrt{23} + \sqrt{12}) \text{ (Diff. Min)}$$

➤ If $x - y = \text{constant}$, then

$(xy)_{\max}$ = Sum of x and y should be minimum.

$(xy)_{\min}$ = Sum of x and y should be maximum .

Example :- Find the greatest and smallest among ?

$$(\sqrt{201} - \sqrt{199}), (\sqrt{101} - \sqrt{99}), (\sqrt{301} - \sqrt{299}), (\sqrt{401} - \sqrt{399})$$

Solution :- Here difference is constant = 2

$$\text{Maximum Sum} = 401 + 399 = 800$$

$$\text{Smallest} = (\sqrt{401} - \sqrt{399})$$

$$\text{Minimum Sum} = 101 + 99 = 200$$

$$\text{Greatest} = (\sqrt{101} - \sqrt{99})$$

Example :- Find the greatest and smallest value among these

$$\sqrt{2}, \sqrt[4]{3}, \sqrt[3]{4}, \sqrt[6]{6} ?$$

Solution :- Here LCM of indices 2, 3, 4, 6 is 12

$$(2^{\frac{1}{2}})^{12}, (3^{\frac{1}{4}})^{12}, (4^{\frac{1}{3}})^{12}, (6^{\frac{1}{6}})^{12}$$

$$2^6, 3^3, 4^4, 6^2$$

$$64, 27, 256, 36$$

$$\text{Greatest value} = \sqrt[3]{4}$$

$$\text{Smallest value} = \sqrt[4]{3}$$

Polynomial and Quadratic Equation

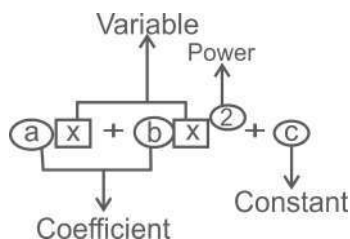
Polynomials are mathematical expressions made up of variables and constants by using arithmetic operations like addition, subtraction, and multiplication.

General form : $a_0 + a_1x + a_2x^2 + a_3x^3 \dots + a_nx^n$

Where $n \in \mathbb{N}$

It is denoted by - $p(x), q(x), g(x), f(x)$

Some Terms Related To Polynomials



Degree of Polynomials

The highest or greatest exponent of the variable in a polynomial is known as the degree of a polynomial.

Some Important Types of Polynomial

1). Zero Polynomial :-

It is a polynomial whose degree is equal to zero.

i.e. 3, 5, 7

They are also called constant polynomials

2). Linear polynomial :

It is a polynomial with degree 1. i.e. $ax + c$

3). quadratic polynomial :-

It is a polynomial with degree 2. i.e. $ax^2 - c$.

4). Cubic Polynomial :-

It is a polynomial with degree 3. i.e. $ax^3 + b^2 - cx - d$.

5). Biquadratic polynomial :-

It is polynomial with degree 4. i.e. $ax^4 + bx^3 - cx + x - d$.

Properties of Polynomials

1). Polynomial of n-th degree has exactly 'n' complex/real roots.

2). If a polynomial P is divisible by a polynomial Q, then every zero of Q is also a zero of P.

3). **Bezout's Theorem** :- Polynomial P(x) is divisible by binomial $x - a$, if and only if $P(a) = 0$.

This is also known as the factor theorem.

4). If a polynomial P is divisible by two coprime polynomials Q and R, then it is divisible by $Q \times R$.

5). **Remainder Theorem** :- The remainder when a polynomial $f(x)$ is divided by $(x - a)$ is $f(a)$.

$$\text{i.e. } \frac{f(x)}{(x-a)} = f(a)$$

6). **Rational Root Theorem** :- A rational root of a polynomial functional $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_2x^2 + a_1x + a_0$ is of the form p/q , where p is a factor of a_0 and q is a factor of a_n . This theorem is very helpful in finding the rational zeros of a polynomial.

Linear Equations in Two Variables

The linear equations in two variables are the equations in which each of the two variables is of the highest order (exponent) of 1 and may have one, none, or infinitely many solutions.

The standard form :- $ax + by + c = 0$, where x and y are the two variables.

The solutions can also be written in ordered pairs like (x, y).

i.e. $10x + 5y = 6$ and $4x + 6y = 8$ are linear equations in two variables.

System of linear equations :- There are basically three types of systems of linear equations and three types of solutions.

⇒ An independent system has exactly one solution pair (x, y).

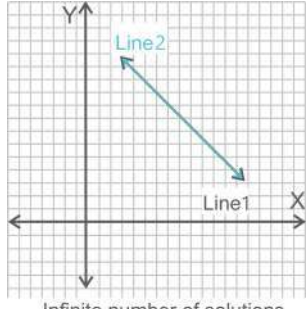
The point where the two lines intersect is the only solution.

⇒ An inconsistent system has no solution. Notice that the two lines are parallel and will never intersect.

⇒ A dependent system has infinitely many solutions. The lines are coincident. They are the same line, so every coordinate pair on the line is a solution to both equations.

Types of solution and their conditions :-

Types of solutions	Conditions	Graphical representation
Unique Solutions	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Unique Solutions Consistent and independent
No solutions	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	No Solution (Inconsistent and independent)

Infinite number of solutions	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	 Infinite number of solutions (consistent and dependent)
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Example :- Solve the following system of linear equation

$$2x + 3y = 11$$

$$3x + 2y = 9$$

Solution :- $2x + 3y = 11$ (i)

$$3x + 2y = 9$$
 (ii)

Multiply equation (i) by 2 and equation (ii) by 3 we get

$$4x + 6y = 22$$
 (a)

$$9x + 6y = 27$$
 (b)

Subtracting equation (a) from equation (b) we get

$$5x = 5 \Rightarrow x = 1$$

Putting the value of x in equation (i) we get

$$2 + 3y = 11 \Rightarrow 3y = 9 \Rightarrow y = 3$$

Example :- For what value of p does the system of equations $18x + 36y + 45 = 0$ and $px - 54y + 67 = 0$ have no solution?

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(a) 54 (b) -27 (c) -36 (d) 27

Solution :- (b) Given,

$$18x + 36y + 45 = 0 \text{ (1)}$$

$$px - 54y + 67 = 0 \text{ (2)}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (no solution)}$$

$$\frac{18}{p} = \frac{36}{(-54)} \neq \frac{45}{67}$$

$$p = \frac{18 \times (-54)}{36} = -27$$

Quadratic Equation

An algebraic statement of the second degree in x is called a quadratic equation.

Standard Form :- $ax^2 + bx + c = 0$

$$\begin{array}{c} \text{Coefficient of } x \\ \uparrow \\ \textcircled{a}x^2 + \textcircled{b}x + \textcircled{c} = 0 \\ \swarrow \quad \searrow \\ \text{Coefficient of } x^2 \quad \text{Constant} \end{array}$$

One fundamental rule of the quadratic equation is that the value of first constant can never be zero.

i.e. $2x^2 + 3x + 2 = 0$

Roots of Quadratic Equation :-

For quadratic equation : $ax^2 + bx + c = 0$, then

- Sum of roots $(\alpha + \beta) = -\frac{b}{a} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$
- Product of roots $(\alpha\beta) = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$

where, α and β are roots of a given quadratic equation.

There are two methods of finding the roots of quadratic equation.

1. Standard method :-

Using this method you can solve any quadratic equation.

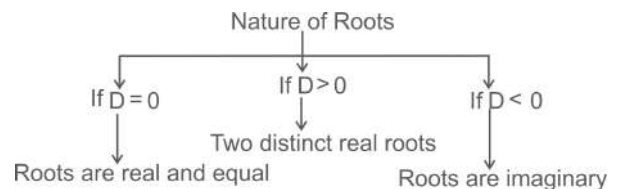
Suppose the equation is $ax^2 + bx + c = 0$.

$$\text{The value of } x \text{ will be } = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

It is known as **Sridharacharya formula**.

Here, $\sqrt{b^2 - 4ac} = D$ or Discriminant

Nature of roots :-



If $D > 0$ there will be two distinct real roots,

If $D < 0$, there will be no real roots

If $D = 0$, there will be two real and equal roots.

Example :- find the nature of roots of equation

$$2x^2 - 4x - 2 = 0$$

Solution :- According to question,

$$a = 2, b = -4, c = -2$$

$$D = b^2 - 4ac = (-4)^2 - 4 \times 2 \times (-2)$$

$$D = 16 + 16$$

$$D = 32 \Rightarrow D > 0 \text{ So, roots are real and distinct}$$

After applying the standard method we will get two value one is negative and other is positive.

Example :- $x^2 - 3x - 4 = 0$

Solution :- $a = 1, b = -3, c = -4$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times (-4)}}{2 \times 1} \\ &= \frac{3 \pm \sqrt{9 + 16}}{2 \times 1} \\ &= \frac{3 + 5}{2}, \frac{3 - 5}{2} \Rightarrow 4, -1 \end{aligned}$$

2. Factorisation method :-

Factorisation is a method of breaking the algebraic expressions into the product of their factors. If we multiply the factors again, then they will result in the original expression.

Example :- $x^2 + 5x + 6 = 0$

Solution :- $x^2 + 3x + 2x + 6 = 0$

$$(x + 3)(x + 2) = 0$$

$$x = -3, x = -2$$

Another method of solving quadratic equation :-

This method is useful when two quadratic equations are given and we have to compare the roots of both equations.

1. $12x^2 - 28x + 15 = 0$

2. $4y^2 - 20y + 21 = 0$

Step 1 :- Similar to factorization, split the middle term into products of their factors.

$$\text{For } 12x^2 - 28x + 15 = 0$$

Sum = - 28 and product = 180.

Hence the split term are - 10 and - 18

Step 2 :- Now change the sign of split numbers, i.e. they become +10 and +18.

Step 3 :- Divide each of these numbers by the coefficient of x^2 .

In this case it will become $\frac{5}{6}, \frac{3}{2}$

Repeat the same process to find the value of y and compare them with x.

Here the value of y will be $\frac{3}{2}, \frac{7}{2}$

Here $x \leq y$.

Exam Hall Oriented Approach :-

In exam two quadratic equations will be given one in the form of x or p other in the form of y or q. We have to find the value of these variables and then compare these values to arrive at the conclusion mentioned below.

- (a) $x > y$ (b) $x < y$
 (c) $x \geq y$ (d) $x \leq y$
 (e) $x = y$ or relation can't be established.

Method To Choose Correct Options

If	If	If	If	If	If
$x_1 > y_1$	$x_1 < y_1$	$x_1 > y_1$	$x_1 < y_1$	$x_1 > y_1$	$x_1 > y_1$
$x_1 > y_2$	$x_1 < y_2$	$x_1 = y_2$	$x_1 < y_2$	$x_1 > y_2$	$x_1 > y_2$
$x_2 > y_1$	$x_2 < y_1$	$x_2 > y_1$	$x_2 = y_1$	$x_2 < y_1$	$x_2 < y_1$
$x_2 > y_2$	$x_2 < y_2$	$x_2 > y_2$	$x_2 < y_2$	$x_2 > y_2$	$x_2 < y_2$
Then $x > y$	Then $x < y$	Then $x \geq y$	Then $x \leq y$	Then relation Can't be established	Then Relation Can't be established

Note :- It means the symbols '>' and '<' can't come together. If they are together then a relationship can't be established will be the correct answer.

Example :- In the given question, two equations numbered I and II are given. Solve both the equations and mark the appropriate answer,

$$\text{I. } x^2 - 20x + 91 = 0 \quad \text{II. } y^2 - 11y + 30 = 0$$

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- (a) $x > y$ (b) $x < y$ (c) $x \geq y$ (d) $x \leq y$
 (e) $x = y$ or relationship can't be established

Solution :- From equation I:

$$x^2 - 20x + 91 = 0$$

$$x^2 - 13x - 7x + 91 = 0$$

$$x(x - 13) - 7(x - 13) = 0$$

$$(x - 7)(x - 13) = 0$$

$$x = 7, x = 13$$

From equation II:

$$y^2 - 11y + 30 = 0$$

$$y^2 - 6y - 5y + 30 = 0$$

$$y(y - 6) - 5(y - 6) = 0$$

$$(y - 5)(y - 6) = 0$$

$$y = 5, y = 6$$

Value of x	Value of y	Relation
7	5	$x > y$
7	6	$x > y$
13	5	$x > y$
13	6	$x > y$

Hence, $x > y$.

Quick way to solve the quadratic equations :-

Sign visible in x / y's question equation	Sign of actual value of x or y. Larger value sign, smaller value sign
+, +	-, -
-, +	+, +
+, -	-, +
-, -	+, -

Example :-

$$x^2 + 7x + 12 = 0$$

$$y^2 - 5y + 6 = 0$$

Solution :- The signs of x's equations are + and +, which means their solution is - and -. Both negative values.

The signs of y's equations are - and +, which means their solution values are + and +. Both positive values.

Obviously, x's possible values are both negative... And y's possible values are both positive. Hence, the solution is $x < y$.

Example :-

$$x^2 - 12x + 32 = 0$$

$$y^2 - 7y + 12 = 0$$

Solution :-

The question's symbol are -, + which means both values will be positive for each variable of x and y.

$$x^2 - 4x - 8x + 32 = 0$$

$$x(x - 4) - 8(x - 4) = 0$$

$$x = 4, x = 8$$

$$y^2 - 7y + 12 = 0$$

$$y^2 - 3y - 4y + 12 = 0$$

$$y(y - 3) - 4(y - 3) = 0 \Rightarrow y = 3, y = 4$$

Hence, $x \geq y$.

Maximum and Minimum Value of Quadratic Equation

For Quadratic Equation : $P(x) = ax^2 + bx + c$

When $a > 0$ then **Minimum value** at $x = -\frac{b}{2a}$

When $a < 0$ then **Maximum value** at $x = -\frac{b}{2a}$

Cubic Equation

An algebraic statement of the third degree in x is called a Cubic equation.

Standard Form :- $ax^3 + bx^2 + cx + d = 0$

It is a polynomial with degree 3.

Diagram illustrating the identification of coefficients in the equation $ax^3 + bx^2 + cx + d = 0$:

- a is the Coefficient of x^3 .
- b is the Coefficient of x^2 .
- c is the Coefficient of x .
- d is the Constant.

One fundamental rule of the quadratic equation is that the value of first constant can never be zero.

$$\text{i.e. } 2x^3 + 3x^2 + 1x + 2 = 0$$

Roots of Cubic Equation :-

- For Cubic equation : $ax^3 + bx^2 + cx + d = 0$, then

$$\text{Sum of its roots } (\alpha + \beta + \gamma) = -\frac{b}{a} = -\frac{\text{Coefficient of } x^2}{\text{Coefficient of } x^3}$$

$$\text{Sum of Product of its roots } (\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a} = \frac{\text{Coefficient of } x}{\text{Coefficient of } x^3}$$

$$\text{Product of its roots } (\alpha\beta\gamma) = \frac{d}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^3}$$

where, α and β are roots of a given quadratic equation.

Some Important Properties

- 1). The values of 'x' that satisfy the equation $p(x) = 0$ are called roots of the cubic polynomial $p(x)$.
- 2). Since the degree of $p(x)$, the cubic equation $p(x) = 0$ can have a maximum of 3 roots.
- 3). The roots of a cubic equation can be found by using synthetic division, factoring, or rational root theorem.

Formation of Polynomial from given roots :-

1). Quadratic Equation :-

If α and β are the roots of a quadratic equation then the equation is

$$ax^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$$

2). Cubic Equation :-

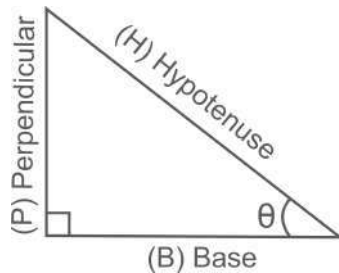
If α , β and γ are the roots of a cubic equation then the equation is

$$ax^3 - (\text{sum of roots})x^2 + (\text{sum of product of roots})x + (\text{product of roots}) = 0$$

Trigonometry

It is that branch of Mathematics which deals with the measurements of the sides and angles of a triangle and also the problems associated with angles.

Trigonometric Ratio :- In a right angled triangle, as shown below



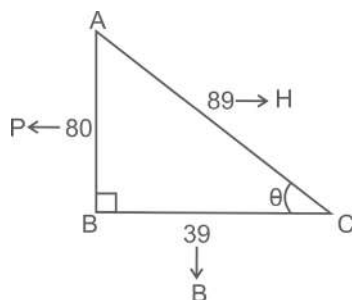
$\sin \theta = \frac{P}{H}$	$\cos \theta = \frac{B}{H}$	$\tan \theta = \frac{P}{B}$
$\operatorname{cosec} \theta = \frac{H}{P}$	$\sec \theta = \frac{H}{B}$	$\cot \theta = \frac{B}{P}$

Example :- If θ is an acute angle and $\cos \theta = \frac{39}{89}$ what is the value of $\sin \theta$

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- (a) $\frac{49}{89}$ (b) $\frac{79}{89}$ (c) $\frac{80}{89}$ (d) $\frac{50}{89}$

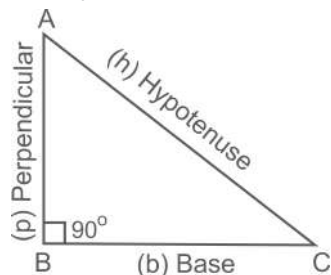
Solution :-



According to questions,

$$\sin \theta = \frac{P}{H} = \frac{80}{89}$$

Pythagoras theorem :- the sum of the squares of the perpendicular and base of a right triangle is equal to the square of its hypotenuse.



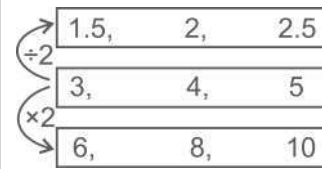
If b , p and h be three sides of right angle triangle then,

$$h^2 = b^2 + p^2$$

Note :- if a , b , and c are three numbers in the form of triplet, if we multiply or divide with a constant term then these numbers will also be in the form of triplet.

Example :- 3, 4, 5 are triplet

Then,



are also triplet

Some Important Triplet

1.	3, 4, 5	10.	16, 63, 65
2.	5, 12, 13	11.	15, 112, 113
3.	8, 15, 17	12.	13, 84, 85
4.	7, 24, 25	13.	28, 45, 53
5.	12, 35, 37	14.	33, 56, 65
6.	9, 40, 41	15.	69, 260, 269
7.	11, 60, 61	16.	60, 221, 229
8.	20, 21, 29	17.	105, 208, 233
9.	39, 80, 89	18.	120, 209, 269

Other triplet can be formed as :- Let x is even number then triplet will be $(2x, x^2 - 1, x^2 + 1)$ If x is odd number then triplet will be $(x, \frac{x^2}{2} - 0.5, \frac{x^2}{2} + 0.5)$

Concept :- If $a \sin \theta + b \cos \theta = c$ where a, b and c are triplet then:-

$$\sin \theta = \frac{a}{c}, \cos \theta = \frac{b}{c}, \tan \theta = \frac{a}{b}$$

Example :- $3 \sin \theta + 4 \cos \theta = 5$ (1)

$$\text{so, } \sin \theta = \frac{3}{5} \text{ and } \cos \theta = \frac{4}{5}$$

On dividing eq 1 by 5

$$\frac{3}{5} \sin \theta + \frac{4}{5} \cos \theta = \frac{5}{5} \text{(1)}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

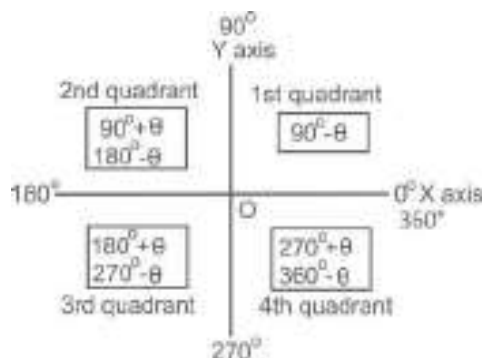
Basic Identities based on above ratio :-

$\sin \theta = \frac{1}{\operatorname{cosec} \theta}$	$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$
$\therefore \sin \theta \times \operatorname{cosec} \theta = 1$	

$\cos \theta = \frac{1}{\sec \theta}$	$\sec \theta = \frac{1}{\cos \theta}$
$\therefore \cos \theta \times \sec \theta = 1$	

$\tan \theta = \frac{1}{\cot \theta} = \frac{\sin \theta}{\cos \theta}$	$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$
$\therefore \tan \theta \times \cot \theta = 1$	

Trigonometric ratios of allied angles



Type - I ($90^\circ - \theta$)

$$\begin{aligned}\sin(90^\circ - \theta) &= \cos \theta, & \cos(90^\circ - \theta) &= \sin \theta \\ \cot(90^\circ - \theta) &= \tan \theta, & \tan(90^\circ - \theta) &= \cot \theta \\ \sec(90^\circ - \theta) &= \operatorname{cosec} \theta, & \operatorname{cosec}(90^\circ - \theta) &= \sec \theta\end{aligned}$$

Example :- find the value of $\frac{\sin 37^\circ}{\cos 53^\circ}$

Solution :- we know that, $\sin(90^\circ - \theta) = \cos \theta$

$$\text{So, } \frac{\sin 37^\circ}{\cos 53^\circ} = \frac{\sin 37^\circ}{\cos(90^\circ - 37^\circ)} = \frac{\sin 37^\circ}{\sin 37^\circ} = 1$$

Type - II ($90^\circ + \theta$)

$$\begin{aligned}\sin(90^\circ + \theta) &= \cos \theta, & \cos(90^\circ + \theta) &= -\sin \theta \\ \cot(90^\circ + \theta) &= -\tan \theta, & \tan(90^\circ + \theta) &= -\cot \theta \\ \sec(90^\circ + \theta) &= -\operatorname{cosec} \theta, & \operatorname{cosec}(90^\circ + \theta) &= \sec \theta\end{aligned}$$

Example :- find the value of $\sin 120^\circ$

Solution :- We know that, $\sin(90^\circ + \theta) = \cos \theta$

$$\sin 120^\circ = \sin(90^\circ + 30^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Type - III ($180^\circ - \theta$)

$$\begin{aligned}\sin(180^\circ - \theta) &= \sin \theta, & \cos(180^\circ - \theta) &= -\cos \theta \\ \cot(180^\circ - \theta) &= -\cot \theta, & \tan(180^\circ - \theta) &= -\tan \theta \\ \sec(180^\circ - \theta) &= -\sec \theta, & \operatorname{cosec}(180^\circ - \theta) &= \operatorname{cosec} \theta\end{aligned}$$

Type - IV ($180^\circ + \theta$)

$$\begin{aligned}\sin(180^\circ + \theta) &= -\sin \theta, & \cos(180^\circ + \theta) &= -\cos \theta \\ \cot(180^\circ + \theta) &= \cot \theta, & \tan(180^\circ + \theta) &= \tan \theta \\ \sec(180^\circ + \theta) &= -\sec \theta, & \operatorname{cosec}(180^\circ + \theta) &= \operatorname{cosec} \theta\end{aligned}$$

Type - V ($270^\circ - \theta$)

$$\begin{aligned}\sin(270^\circ - \theta) &= -\cos \theta, & \cos(270^\circ - \theta) &= -\sin \theta \\ \cot(270^\circ - \theta) &= \tan \theta, & \tan(270^\circ - \theta) &= \cot \theta \\ \sec(270^\circ - \theta) &= -\operatorname{cosec} \theta, & \operatorname{cosec}(270^\circ - \theta) &= -\sec \theta\end{aligned}$$

Example :- find the value of $\cot 240^\circ$.

Solution :- We know that, $\cot(270^\circ - \theta) = \tan \theta$

$$\cot 270^\circ = \cot(270^\circ - 30^\circ) = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Type - VI ($270^\circ + \theta$)

$$\begin{aligned}\sin(270^\circ + \theta) &= -\cos \theta, & \cos(270^\circ + \theta) &= \sin \theta \\ \cot(270^\circ + \theta) &= -\tan \theta, & \tan(270^\circ + \theta) &= -\cot \theta \\ \sec(270^\circ + \theta) &= \operatorname{cosec} \theta, & \operatorname{cosec}(270^\circ + \theta) &= -\sec \theta\end{aligned}$$

Example :- find the value of $\cot 300^\circ$.

Solution :- We know that, $\cot(270^\circ + \theta) = -\tan \theta$

$$\cot 300^\circ = \cot(270^\circ + 30^\circ) = -\tan 30^\circ = -\frac{1}{\sqrt{3}}$$

Type - VII ($n \times 360^\circ + \theta$)

$$\begin{aligned}\sin(n \times 360^\circ + \theta) &= \sin \theta, & \cos(n \times 360^\circ + \theta) &= \cos \theta \\ \cot(n \times 360^\circ + \theta) &= \cot \theta, & \tan(n \times 360^\circ + \theta) &= \tan \theta \\ \sec(n \times 360^\circ + \theta) &= \sec \theta, & \operatorname{cosec}(n \times 360^\circ + \theta) &= \operatorname{cosec} \theta\end{aligned}$$

Example :- find the value of $\cos 1020^\circ$.

Solution :- We know that, $\cos(n \times 360^\circ + \theta) = \cos \theta$

$$\cos 1020^\circ = \cos(1080^\circ - 60^\circ)$$

1080 multiple of 360°, so no change in trigonometry function.

$$\cos(3 \times 360^\circ - 60^\circ) = \cos 60^\circ = \frac{1}{2}$$

Example :- Simplify the following:

$7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$
SSC CHSL 09/08/2023 (3rd Shift)

(a) 7 (b) -7 (c) -1 (d) 1

Solution :-

We know that, $\tan \theta = \cot(90^\circ - \theta)$ and $\tan \theta \times \cot \theta = 1$ according to question,

$$\begin{aligned}& 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ) \\ &= 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ) \\ &= 7(\tan 79^\circ \tan 77^\circ \tan 71^\circ \tan 61^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ) \\ &= 7(\tan 79^\circ \cot 79^\circ \tan 77^\circ \cot 77^\circ \tan 71^\circ \cot 71^\circ \tan 61^\circ \cot 61^\circ) \\ &= 7(1 \times 1 \times 1 \times 1) = 7\end{aligned}$$

Alternate method :- If $A + B = 90^\circ$, then $\cot A \times \cot B = 1$

$$\begin{aligned}& 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ) \\ &= 7(\cot 11^\circ \times \cot 79^\circ)(\cot 13^\circ \times \cot 77^\circ)(\cot 19^\circ \times \cot 71^\circ) \\ &\quad (\cot 29^\circ \times \cot 61^\circ) \\ &= 7(1 \times 1 \times 1 \times 1) = 7\end{aligned}$$

Conversion of trigonometric ratios :-

- $\sin(-x) = -\sin x$
- $\operatorname{cosec}(-x) = -\operatorname{cosec} x$
- $\cos(-x) = \cos x$
- $\sec(-x) = \sec x$
- $\tan(-x) = -\tan x$
- $\cot(-x) = -\cot x$

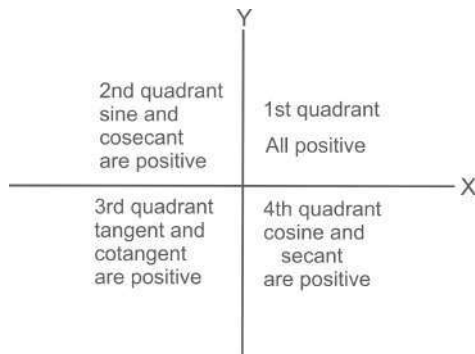
Example :- What is the value of $\sin(-405^\circ)$?

SSC CGL 20/07/2023 (3rd shift)

Solution :- we know that, $\sin(-x) = -\sin x$

$$\sin(-405^\circ) = -\sin(360 + 45^\circ) = -\sin 45^\circ = \frac{-1}{\sqrt{2}}$$

Signs of the trigonometric ratios in different quadrants :-



• $90^\circ, 270^\circ$ (odd multiple of 90°) will be changed

Changes will be in

$\sin\theta \leftrightarrow \cos\theta, \tan\theta \leftrightarrow \cot\theta, \operatorname{cosec}\theta \leftrightarrow \sec\theta$

• $180^\circ, 360^\circ$ (even multiple of 90°) will not be changed

• x axis $\pm \theta \rightarrow$ no change

• y axis $\pm \theta \rightarrow$ changes like

Example :- $\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ = ?$
SSC CGL 20/07/2023 (1st shift)

Solution :- $\operatorname{cosec} 2910^\circ = \operatorname{cosec}(360^\circ \times 8 + 30^\circ)$ {•x axis $\pm \theta \rightarrow$ no change}

$$= \operatorname{cosec} 30^\circ = 2$$

$$\sec 4260^\circ = \sec(360^\circ \times 11 + 300^\circ)$$

$$= \sec 300^\circ = \sec(360^\circ - 60^\circ) = \sec 60^\circ = 2$$

$$\tan 2565^\circ = \tan(360^\circ \times 7 + 45^\circ) = \tan(45^\circ) = 1$$

$$\cot 1755^\circ = \cot(360^\circ \times 4 + 315^\circ)$$

$$= \cot(315^\circ) = \cot(270^\circ + 45^\circ) \text{}\{ \bullet \text{ y axis } \pm \theta \rightarrow \text{changes like } (\tan\theta \leftrightarrow \cot\theta) \text{ } \theta \text{ lies in 4th quadrant so } \tan\theta \text{ is negative} \}$$

$$= -\tan 45^\circ = -1$$

$$\text{Now, } \operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ$$

$$\Rightarrow 2 + 2 + 1 - 1 = 4$$

Circular Measure of Angles

(1). sexagesimal system :-

$$1 \text{ degree} = \frac{1}{90} \text{ right angle}$$

$$1 \text{ minute} = \frac{1}{60} \text{ degree}$$

$$1 \text{ second} = \frac{1}{60} \text{ minute}$$

(2). centesimal or french system :-

$$1 \text{ right angle} = 100 \text{ grade}$$

$$1 \text{ grade} = 100 \text{ minute}$$

$$1 \text{ minute} = 100 \text{ second}$$

(3). circular system :-

$$1^\circ = 60' = 60 \text{ minute}$$

$$1' = 60'' = 60 \text{ second}$$

Relation Between Radian And Degree

$$1 \text{ radian} = \frac{180^\circ}{\pi} = 57^\circ 16' \text{ approx.}$$

$$1^\circ = \frac{\pi}{180} \text{ radian} = 0.01746 \text{ radian approx}$$

Example :- change the $-47^\circ 30'$ degree into the radian

Solution :- $-47^\circ 30'$ we know that $1^\circ = 60'$

$$\text{So, } -47^\circ 30' = -47\frac{1}{2}^\circ = \frac{-95^\circ}{2}$$

Here $180^\circ = \pi$ radian

According to question,

$$= \frac{\pi}{180^\circ} \times \left(\frac{-95^\circ}{2}\right) \text{ radian} = \frac{-19}{72} \pi \text{ radian}$$

Degree	30°	45°	60°	90°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Example :- $16 \cos^3 \frac{\pi}{6} - 12 \cos \frac{\pi}{6} = \underline{\hspace{2cm}}$

SSC CGL 01/12/2022 (4th Shift)

(a) 0 (b) 2 (c) 1 (d) -1

Solution :- $16 \cos^3 \frac{\pi}{6} - 12 \cos \frac{\pi}{6}$

$$= 16 \cos^3 30^\circ - 12 \cos 30^\circ = 16 \left(\frac{\sqrt{3}}{2}\right)^3 - 12 \times \frac{\sqrt{3}}{2}$$

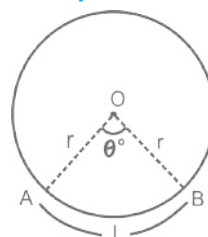
$$= 16 \left(\frac{3\sqrt{3}}{8}\right) - 6\sqrt{3} \Rightarrow 0$$

Alternate method :-

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

$$16\cos^3 \frac{\pi}{6} - 12\cos \frac{\pi}{6} = 4\cos 3 \times \frac{\pi}{6} \Rightarrow 4\cos \frac{\pi}{2} = 0$$

Concept :-



If r is the radius of circle, an arc of length l subtends an angle of θ radian, then

$$\bullet \quad \overline{AB} = l = 2\pi r \times \frac{\theta^\circ}{360^\circ}$$

$$\bullet \quad l = r \times \frac{\pi \theta^\circ}{180^\circ}$$

$$\bullet \quad l = r \times \theta \quad (\text{where, } \theta = \theta^\circ \times \frac{\pi}{180^\circ})$$

Example :- If the angle of the sector with radius 4 units is 45° , then find the length of the sector.

Solution :- according to question,

$$l = r \times \theta \Rightarrow l = 4 \times 45^\circ \times \frac{\pi}{180^\circ} \Rightarrow l = 4 \times 45^\circ \times \frac{\pi}{180^\circ}$$

$$\Rightarrow l = \frac{22}{7}$$

Important Identities

1). $\sin^2 \theta + \cos^2 \theta = 1$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

2). $\sec^2 \theta - \tan^2 \theta = 1$

$$* \sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

3). $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

$$* \operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

$$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

$$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

We can derive all other identities from these three basic identities.

Type - 1

Example :- Find the value of

$$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$$

RRC Group D 06/09/2022 (Morning)

(a) $5 + \operatorname{cosec}^2 A - \sec^2 A$ (b) $5 + \operatorname{cosec}^2 A + \sec^2 A$

(c) $5 + \sec A + \operatorname{cosec} A$ (d) $5 - \operatorname{cosec}^2 A + \sec^2 A$

Solution :- According to question,

$$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$$

$$\Rightarrow \sin^2 A + \operatorname{cosec}^2 A + 2 + \cos^2 A + \sec^2 A + 2$$

$$\Rightarrow 5 + \operatorname{cosec}^2 A + \sec^2 A \quad \text{using identity } [\sin^2 A + \cos^2 A = 1]$$

Type - 2

Example :- $\sec^2 \theta + (\operatorname{cosec}^2 \theta - 1) - (1 + \tan^2 \theta) - \cot^2 \theta = \underline{\hspace{2cm}}$.

RRC Group D 16/09/2022 (Morning)

(a) 1 (b) 0 (c) $\sec^2 \theta$ (d) $\cot^2 \theta$

Solution :- according to question,

$$\sec^2 \theta + (\operatorname{cosec}^2 \theta - 1) - (1 + \tan^2 \theta) - \cot^2 \theta$$

$$= \sec^2 \theta + \cot^2 \theta - \sec^2 \theta - \cot^2 \theta = 0$$

$$\{\text{using identity } \sec^2 \theta = 1 + \tan^2 \theta, \cot^2 \theta = \operatorname{cosec}^2 \theta - 1\}$$

Type - 3

Example :- If $\operatorname{cosec} A - \cot A = 3$ then $\operatorname{cosec} A + \cot A = ?$

RRB NTPC CBT - I 14/03/2021 (Morning)

(a) $\frac{1}{3}$ (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{5}$

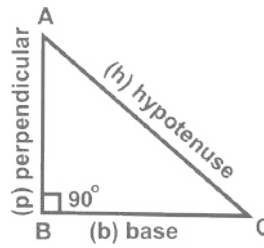
Solution :- We know that

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$\Rightarrow (\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A) = 1$$

$$\Rightarrow \operatorname{cosec} A - \cot A = \frac{1}{3}$$

Some Other knowledge About Identities



We know that, $b^2 + p^2 = h^2$ ----- (1)

• In equation (1), Dividing the both side by h^2 we get,

$$\frac{b^2}{h^2} + \frac{p^2}{h^2} = 1 \Rightarrow \sin^2 \theta + \cos^2 \theta = 1$$

• In equation (1), Dividing the both side by p^2 we get,

$$\frac{h^2}{p^2} = \frac{b^2}{p^2} + 1 \Rightarrow \frac{h^2}{p^2} - \frac{b^2}{p^2} = 1 \Rightarrow \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

• In equation (1), Dividing the both side by b^2 we get,

$$\frac{h^2}{b^2} = \frac{p^2}{b^2} + 1 \Rightarrow \frac{h^2}{b^2} - \frac{p^2}{b^2} = 1 \Rightarrow \sec^2 \theta - \tan^2 \theta = 1$$

Example :- $8\sin^4 \beta + 9\cos^4 \beta = 6$ then find the value of $9\operatorname{cosec}^6 \beta + 4\sec^6 \beta$.

Solution :- we know that, $\sin^2 \theta + \cos^2 \theta = 1$

$$8\sin^4 \beta + 9\cos^4 \beta = 6 \quad \{\text{Divide the both side by 6}\}$$

$$\left(\frac{8}{6}\sin^2 \beta\right) \cdot \sin^2 \beta + \left(\frac{9}{6}\cos^2 \beta\right) \cdot \cos^2 \beta = \frac{6}{6}$$

$$\sin^2 \beta = \frac{6}{8} = \frac{3}{4}, \cos^2 \beta = \frac{6}{9} = \frac{2}{3}$$

According to question,

$$9\operatorname{cosec}^6 \beta + 4\sec^6 \beta \Rightarrow 9\left(\frac{4}{3}\right)^3 + 4\left(\frac{3}{2}\right)^3 = 64 \times 3 + 27 \times 2 \Rightarrow 246$$

Values of Trigonometric functions for different values of θ :-

θ	0°	30°	45°	60°	90°	180°	270°	360°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not Defined	0	Not Defined	0
$\cot \theta$	Not Defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	Not Defined	0	Not Defined
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not Defined	-1	Not Defined	1
$\operatorname{cosec} \theta$	Not Defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	Not Defined	-1	Not Defined

Example :- If $k(\tan 45^\circ \sin 60^\circ) = \cos 60^\circ \cot 30^\circ$, then the value of k is :

SSC CPO 04/10/2023 (1st Shift)

(a) 1 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) 2

Solution :- $k(\tan 45^\circ \sin 60^\circ) = \cos 60^\circ \cot 30^\circ$
 $= k(1 \times \frac{\sqrt{3}}{2}) = \frac{1}{2} \times \sqrt{3} \Rightarrow k = 1$

Example :- What is the value of $\frac{1 + \sin^2 45^\circ}{1 - \cos^2 45^\circ}$?

SSC CHSL 15/03/2023 (3rd Shift)

(a) 2 (b) 3 (c) $\frac{3}{2}$ (d) $\frac{1}{2}$

Solution :- $\frac{1 + \sin^2 45^\circ}{1 - \cos^2 45^\circ} = \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = 3$

Concept :-

If A and B are complement angle $(A + B) = 90^\circ$ then :-

$\sin A = \cos B$, $\tan A = \cot B$, $\sec A = \csc B$

$\tan A \cdot \tan B = 1$, $\cot A \cdot \cot B = 1$

$\sin A \cdot \sec B = 1$, $\cos A \cdot \csc B = 1$

$\sin^2 A + \cos^2 B = 1$

Example :- find the value of $\tan 10^\circ \cdot \tan 25^\circ \cdot \tan 65^\circ \cdot \tan 80^\circ$

Solution :- we know that $(A + B) = 90^\circ$ then $\tan A \cdot \tan B = 1$
 So, $\tan 10^\circ \cdot \tan 25^\circ \cdot \tan 65^\circ \cdot \tan 80^\circ$

$\tan 10^\circ \cdot \tan 25^\circ \cdot \tan 65^\circ \cdot \tan 80^\circ$
 $= 1$

Example :- if $\cot 2\theta \cdot \cot 3\theta = 1$ find the value of

$\sin \frac{5\theta}{2} \cdot \cos \frac{5\theta}{2}$

Solution :- we know that $(A + B) = 90^\circ$ then $\cot A \cdot \cot B = 1$

So, $\cot 2\theta \cdot \cot 3\theta = 1 \Rightarrow 5\theta = 90^\circ$

Now, $\sin \frac{5\theta}{2} \cdot \cos \frac{5\theta}{2} = \sin \frac{90^\circ}{2} \cdot \cos \frac{90^\circ}{2}$
 $= \sin 45^\circ \cos 45^\circ = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2}$

Range of All Trigonometric Functions

Trigonometric functions	Minimum	Maximum
$\sin x, \cos x$ (odd powers)	-1	+1
$\sin x, \cos x$ (even powers)	0	+1
$\tan x, \cot x$ (odd powers)	$-\infty$	$+\infty$
$\tan x, \cot x$ (even powers)	0	$+\infty$
$\sec x, \csc x$ (odd powers)	$-\infty$	$+\infty$
$\sec x, \csc x$ (even powers)	+1	$+\infty$

Maximum and Minimum value of the trigonometric expressions of the different form :-

• $a \sin x \pm b \cos x \Rightarrow$ Maximum value $= \sqrt{a^2 + b^2}$
 Minimum value $= -\sqrt{a^2 + b^2}$

• $a \sin^2 x + b \cos^2 x$

If, $a > b \Rightarrow$ Maximum value $= a$, Minimum value $= b$

If $a < b \Rightarrow$ Maximum value $= b$, Minimum value $= a$

• $a \operatorname{cosec}^2 x + b \sec^2 x \Rightarrow$ Maximum value $= \infty$,

Minimum value $= (\sqrt{a} + \sqrt{b})^2$

• $a \sin^2 x + b \operatorname{cosec}^2 x$ } Maximum $\rightarrow \infty$

• $a \tan^2 x + b \cot^2 x$ } Minimum $\rightarrow 2\sqrt{ab}$

• $a \cos^2 x + b \sec^2 x$ }

• $\sin^{2n} x + \cos^{2m} x \Rightarrow$ Maximum value $= 1$

Minimum value $=$ (use $x = 45^\circ$)

Example :- Find the minimum and maximum value of $\sin^2 x + \cos^4 x$?

Solution :- Maximum value of $\sin^2 x + \cos^4 x = 1$

Minimum value $= \sin^2 45^\circ + \cos^4 45^\circ = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

• $(\sin x \cos x)^n \Rightarrow$ Maximum value $= \infty$

$\Rightarrow$ Minimum value $= \left(\frac{1}{2}\right)^n$

Example :- Find the minimum value of $\sin^4 x \cos^4 x$?

Solution :- Minimum value $= \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

Compound Formulae or Sum and Difference Identities

• $\sin(A + B) = \sin A \cdot \cos B + \cos A \cdot \sin B$

• $\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$

• $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

• $\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

• $\tan(A + B) = \frac{\tan A + \tan B}{1 - (\tan A \times \tan B)}$

• $\tan(45^\circ + B) = \frac{1 + \tan B}{1 - \tan B}$

• $\tan(A - B) = \frac{\tan A - \tan B}{1 + (\tan A \times \tan B)}$

• $\tan(45^\circ - B) = \frac{1 - \tan B}{1 + \tan B}$

• $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$

• $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

• $\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - (\tan A \tan B - \tan B \tan C - \tan C \tan A)}$

Example :- Find the value of $\sin 15^\circ$.

RRB NTPC CBT - I 18/01/2021 (Morning)

(a) $\frac{\sqrt{3} + 1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3} - 1}{2\sqrt{2}}$ (c) $\frac{\sqrt{3} + 1}{\sqrt{2}}$ (d) $\frac{\sqrt{3} - 1}{\sqrt{2}}$

Solution :- $\sin 15^\circ = \sin(45^\circ - 30^\circ)$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

Example :- find the value of $\frac{\sin 59^\circ \cos 31^\circ + \cos 59^\circ \sin 31^\circ}{\cos 20^\circ \cos 25^\circ - \sin 20^\circ \sin 25^\circ}$

Solution :- $\frac{\sin 59^\circ \cos 31^\circ + \cos 59^\circ \sin 31^\circ}{\cos 20^\circ \cos 25^\circ - \sin 20^\circ \sin 25^\circ}$

$$= \frac{\sin(59^\circ + 31^\circ)}{\cos(20^\circ + 25^\circ)} \Rightarrow \frac{\sin 90^\circ}{\cos(45^\circ)} = \sqrt{2}$$

Example :- Using trigonometric formulas, find the value of

$$\left[\frac{\sin(x-y)}{\sin(x+y)} \right] \left[\frac{\tan x + \tan y}{\tan x - \tan y} \right]$$

SSC CGL 18/07/2023 (1st shift)

(a) -2 (b) 2 (c) 0 (d) 1

Solution :- $\left(\frac{\sin(x-y)}{\sin(x+y)} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$

$$= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$$

$$= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$$

$$= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$$

$$= \left(\frac{\tan x - \tan y}{\tan x + \tan y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right) = 1$$

Example :- The value of $\tan 75^\circ - \cot 75^\circ$ is equal to :

RRB NTPC CBT - I 24/07/2021 (Evening)

(a) $2 - \sqrt{3}$ (b) $1 + 2\sqrt{3}$ (c) $2\sqrt{3}$ (d) $2 + \sqrt{3}$

Solution :-

$$\tan 75^\circ - \cot 75^\circ = \tan 75^\circ - \tan(90^\circ - 75^\circ) = \tan 75^\circ - \tan 15^\circ$$

$$\text{We know that, } \tan(A - B) = \frac{\tan A - \tan B}{1 + (\tan A \times \tan B)}$$

$$\tan 75^\circ - \tan 15^\circ = \tan(75^\circ - 15^\circ) \{1 + (\tan 75^\circ \times \tan 15^\circ)\}$$

$$= \tan 60^\circ \times 2 = 2\sqrt{3}$$

Some important derived identities :

- $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$
- $(\sec A + \tan A - 1)(\sec A + \tan A + 1) = 2 \tan A$
- $\tan^2 A - \sin^2 A = \tan^2 A \cdot \sin^2 A$
- $\sin^6 A + \cos^6 A + 3 \sin^2 A \cdot \cos^2 A = 1$

Example :- The value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$ is :

SSC CPO 09/11/2022 (Afternoon)

(a) 1 (b) 2 (c) 3 (d) 0

Solution :-

$$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$$

Putting $A = 45^\circ$, we have :

$$(1 + \cot 45^\circ - \operatorname{cosec} 45^\circ)(1 + \tan 45^\circ + \sec 45^\circ) - 1$$

$$= (2 - \sqrt{2})(2 + \sqrt{2}) - 1 \Rightarrow 4 - 2 - 1 = 1$$

Exam hall approach :-

$$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1 = 2 - 1 = 1$$

Double Angle Identities :-

- $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$

- $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A} = 2 \cos^2 A - 1$$

- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

- $\cot 2A = \frac{\cot^2 A - 1}{2 \cot A}$

Example :- If $\tan A = \frac{3}{4}$, find the values of $\sin 2A$, $\cos 2A$, and $\tan 2A$.

Solution :- we know that, $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$

$$\sin 2A = \frac{2 \times \frac{3}{4}}{1 + \left(\frac{3}{4}\right)^2} = \frac{\frac{3}{2}}{1 + \frac{9}{16}} = \frac{3 \times 16}{2 \times 25} = \frac{24}{25}$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A} = \frac{1 - \left(\frac{3}{4}\right)^2}{1 + \left(\frac{3}{4}\right)^2}$$

$$= \frac{7 \times 16}{16 \times 25} = \frac{7}{25}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} = \frac{2 \times \frac{3}{4}}{1 - \left(\frac{3}{4}\right)^2}$$

$$= \frac{\frac{3}{2}}{1 - \frac{9}{16}} = \frac{3 \times 16}{2 \times 7} = \frac{24}{7}$$

Triple Angle Identities :-

- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$
- $\cot 3A = \frac{\cot^3 A - 3 \cot A}{3 \cot^2 A - 1}$

Example :- find the value of $3 \sin 20^\circ - 4 \sin^3 20^\circ$

Solution :- we know that, $\sin 3A = 3 \sin A - 4 \sin^3 A$

$$\text{So, } 3 \sin 20^\circ - 4 \sin^3 20^\circ$$

$$= \sin 3 \times 20^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

Example :- find the value of $4 \cos^3 10^\circ - 3 \cos 10^\circ$

Solution :- we know that, $4 \cos^3 A - 3 \cos A$

$$\text{So, } 4 \cos^3 10^\circ - 3 \cos 10^\circ$$

$$= \cos 3 \times 10^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Some important result :-

- $\sin \theta \cdot \sin 2\theta \cdot \sin 4\theta = \frac{1}{4} \sin 3\theta$
- $\sin(60^\circ - \theta) \cdot \sin \theta \cdot \sin(60^\circ + \theta) = \frac{1}{4} \sin 3\theta$
- $\sin^2(60^\circ - \theta) + \sin^2 \theta + \sin^2(60^\circ + \theta) = \frac{3}{2}$

- $\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta = \frac{1}{4} \cos 3\theta$
- $\cos(60^\circ - \theta) \cdot \cos \theta \cdot \cos(60^\circ + \theta) = \frac{1}{4} \cos 3\theta$
- $\cos^2(60^\circ - \theta) + \cos^2(\theta) + \cos^2(60^\circ + \theta) = \frac{3}{2}$
- $\tan \theta \cdot \tan 2\theta \cdot \tan 4\theta = \tan 3\theta$
- $\tan(60^\circ - \theta) \cdot \tan \theta \cdot \tan(60^\circ + \theta) = \tan 3\theta$

Example :- find the value of $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80^\circ$

Solution :- we know that, $\sin \theta \cdot \sin 2\theta \cdot \sin 4\theta = \frac{1}{4} \sin 3\theta$

$$\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80^\circ = \frac{1}{4} \sin 60^\circ = \frac{1}{4} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{8}$$

Example :- find the value of

$$\tan 10^\circ \cdot \tan 50^\circ \cdot \tan 60^\circ \cdot \tan 70^\circ$$

Solution :- we know,

$$\tan(60^\circ - \theta) \cdot \tan \theta \cdot \tan(60^\circ + \theta) = \tan 3\theta$$

$$\text{So, } \tan 10^\circ \cdot \tan 50^\circ \cdot \tan 60^\circ \cdot \tan 70^\circ$$

$$\tan 30^\circ \cdot \sin 60^\circ = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

Transformations of sums or differences into products :-

- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$
- $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

Example :- What is $\sin \alpha - \sin \beta$?

SSC CGL 01/12/2022 (2nd Shift)

- (a) $2 \cos \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}$ (b) $2 \sin \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}$
 (c) $2 \cos \frac{\alpha-\beta}{2} \sin \frac{\alpha+\beta}{2}$ (d) $2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}$

Solution :- Formula :-

$$\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}$$

Example :- What is $\cos 6\theta - \cos 8\theta$?

Solution :- Formula :-

$$\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$\cos 6\theta - \cos 8\theta = -2 \sin \frac{6\theta+8\theta}{2} \sin \frac{6\theta-8\theta}{2} = 2 \sin 7\theta \sin \theta$$

Important Concept :-

- $a \sin x + b \cos x = m$, $a \cos x - b \sin x = n$

$$\text{Then, } a^2 + b^2 = m^2 + n^2$$

Example :- if $3 \sin \theta + 5 \cos \theta = 5$. Find the value of $3 \cos \theta - 5 \sin \theta$?

Solution :- $3 \sin \theta + 5 \cos \theta = 5$. and $3 \cos \theta - 5 \sin \theta = n$

$$\text{Then use } a^2 + b^2 = m^2 + n^2$$

$$3^2 + 5^2 = 5^2 + n^2$$

- $a \sec x + b \tan x = m$

$$a \tan x - b \sec x = n$$

$$\text{Then, } a^2 - b^2 = m^2 - n^2$$

- If $\sec x$ and $\tan x$ are replaced by $\operatorname{cosec} x$ and $\cot x$ respectively. Then the outcome will be the same.

Example :- If $29 \sec \theta - 21 \tan \theta = 20$ Find the value of $29 \tan \theta - 21 \sec \theta$?

$$\text{Solution :- } 29 \tan \theta - 21 \sec \theta = \sqrt{20^2 - 29^2 + 21^2} = 0$$

Some Important results :-

(i). If $A + B + C = 180^\circ$. then,

- $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
- $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$
- $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$
- $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$
- $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

Example :- find the value of $\tan 34^\circ + \tan 76^\circ + \tan 70^\circ$ is equal

Solution :- If $A + B + C = 180^\circ$. then,

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

According to question,

$$34^\circ + 76^\circ + 70^\circ = 180^\circ$$

$$\tan 34^\circ + \tan 76^\circ + \tan 70^\circ = \tan 34^\circ \tan 76^\circ \tan 70^\circ$$

(ii). If $A + B + C = 90^\circ$. then,

- $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$
- $\cot A + \cot B + \cot C = \cot A \cot B \cot C$

(iii). If $A + B = 45^\circ$ or 225° . then

- $(1 + \tan A)(1 + \tan B) = 2$
- $(1 - \cot A)(1 - \cot B) = 2$

Example :- find the value of $(1 + \tan 5^\circ)(1 + \tan 40^\circ)$.

Solution :- we know that, If $A + B = 45^\circ$. then

$$(1 + \tan A)(1 + \tan B) = 2$$

$$\text{Hence, } (1 + \tan 5^\circ)(1 + \tan 40^\circ) = 2$$

Example :- if $A + B = 45^\circ$ then find $\frac{\tan A}{1 - \tan A} \cdot \frac{\tan B}{1 - \tan B}$?

Solution :- according to questions,

$$\frac{\tan A}{1 - \tan A} \cdot \frac{\tan B}{1 - \tan B} = \frac{\frac{1}{\cot A}}{1 - \frac{1}{\cot A}} \cdot \frac{\frac{1}{\cot B}}{1 - \frac{1}{\cot B}}$$

$$= \frac{1}{\cot A - 1} \cdot \frac{1}{\cot B - 1}$$

$$\text{If } A + B = 45^\circ. \text{ then } (1 - \cot A)(1 - \cot B) = 2$$

$$\text{Hence, } = \frac{1}{\cot A - 1} \cdot \frac{1}{\cot B - 1} = \frac{1}{2}$$

Transformations of products into sums or differences :-

- $2\cos C \cdot \cos D = \cos(C + D) + \cos(C - D)$
- $-2\sin C \cdot \sin D = \cos(C + D) - \cos(C - D)$
- $2\sin C \cdot \cos D = \sin(C + D) + \sin(C - D)$
- $2\cos C \cdot \sin D = \sin(C + D) - \sin(C - D)$

$$\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$\sec \theta - \tan \theta = p$$

then,

$$\sec \theta = \frac{p^2 + 1}{2p}, \tan \theta = \frac{1 - p^2}{2p}$$

$$\sin \theta = \frac{1 - p^2}{p^2 + 1}$$

Example :- If $\sec \theta - \tan \theta = m$, then $2 \tan \theta = ?$

SSC CHSL 02/08/2023 (3rd Shift)

(a) $\frac{1-m}{m}$ (b) $\frac{1-m^2}{m}$ (c) $\frac{1+m^2}{m}$ (d) $\frac{m^2-1}{m}$

Solution :- according to question,

$$\tan \theta = \frac{1-p^2}{2p} = \frac{1-m^2}{2m}$$

$$\text{So, } 2 \tan \theta = \frac{1-m^2}{2m} \times 2 = \frac{1-m^2}{m}$$

- $\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$

$$\operatorname{cosec} \theta - \cot \theta = p \text{ then;}$$

$$\operatorname{cosec} \theta = \frac{p^2 + 1}{2p}, \cot \theta = \frac{1 - p^2}{2p}$$

$$\cos \theta = \frac{1 - p^2}{p^2 + 1}$$

Example :- If $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = x$, then what is

$$\frac{1}{\operatorname{cosec} \theta + \cot \theta} - \frac{1}{\sin \theta} \text{ equal to, where } 0 < \theta < \frac{\pi}{2} ?$$

UPSC CDS - II (04/09/2022)

(a) $-x$ (b) x (c) $\frac{1}{x}$ (d) $-\frac{1}{x}$

Solution :- We know that, $\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$

According to question,

$$\text{If } \frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = x$$

$$\Rightarrow \operatorname{cosec} \theta + \cot \theta - \operatorname{cosec} \theta = x \Rightarrow \cot \theta = x$$

$$\text{So, } \frac{1}{\operatorname{cosec} \theta + \cot \theta} - \frac{1}{\sin \theta} \Rightarrow \operatorname{cosec} \theta - \cot \theta - \operatorname{cosec} \theta$$

$$= -\cot \theta$$

$$\Rightarrow \frac{1}{\operatorname{cosec} \theta + \cot \theta} - \frac{1}{\sin \theta} = -x$$

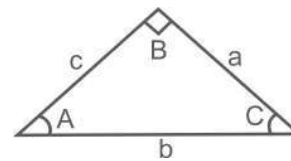
Value of some special trigonometry angle :-

Serial.No.	Angle
1.	$\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$
2.	$\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$
3.	$\sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$

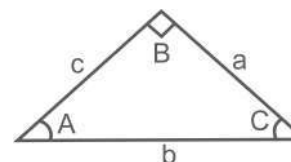
4.	$\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \sin 72^\circ$
5.	$\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$
6.	$\sin 54^\circ = \frac{\sqrt{5}+1}{4} = \cos 36^\circ$
7.	$\sin 22\frac{1}{2}^\circ = \frac{\sqrt{2}-\sqrt{2}}{2}$
8.	$\cos 22\frac{1}{2}^\circ = \frac{\sqrt{2}+\sqrt{2}}{2}$

Example :- Find the value of $\sin 75^\circ$.

Solution :- $\sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Laws of sine formula :-

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Laws of cosines :-

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Half angle formula :-

$$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$$

- $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2} = \frac{2 \tan \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}$

- $\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$

- $\cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 1 - 2 \sin^2 \frac{A}{2}$

$$= \frac{1 - \tan^2 \frac{A}{2}}{1 + \tan^2 \frac{A}{2}} = 2 \cos^2 \frac{A}{2} - 1$$

- $\tan^2 \frac{A}{2} = \frac{1 - \cos A}{1 + \cos A}$

- $\tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$

- $\cot^2 \frac{A}{2} = \frac{1 + \cos A}{1 - \cos A}$

- $\cot A = \frac{\cot^2 \frac{A}{2} - 1}{2 \cot \frac{A}{2}}$

Example :- find the value of $\frac{1 - \tan^2 22\frac{1}{2}^\circ}{1 + \tan^2 22\frac{1}{2}^\circ}$.

Solution :- we know that, $\cos A = \frac{1 - \tan^2 \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}$

$$\frac{1 - \tan^2 22\frac{1}{2}^\circ}{1 + \tan^2 22\frac{1}{2}^\circ} = \frac{1 - \tan^2 \frac{45^\circ}{2}}{1 + \tan^2 \frac{45^\circ}{2}} = \cos 45^\circ \Rightarrow \frac{1}{\sqrt{2}}$$

Example :- find the value of $\frac{\sin x}{\sin \frac{x}{4}}$.

Solution :- $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$

according to question,

$$\frac{\sin x}{\sin \frac{x}{4}} = \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{\sin \frac{x}{4}} = \frac{4 \sin \frac{x}{4} \cdot \cos \frac{x}{4} \cdot \cos \frac{x}{2}}{\sin \frac{x}{4}} = 4 \cos \frac{x}{4} \cos \frac{x}{2}$$

General solution of trigonometric function :-

$$\sin x = 0 \Rightarrow x = n\pi$$

$$\cos x = 0 \Rightarrow x = (2n + 1) \frac{\pi}{2}$$

$$\tan x = 0 \Rightarrow x = n\pi$$

$$\sin x = \sin y \Rightarrow x = n\pi + (-1)^n y$$

$$\cos x = \cos y \Rightarrow x = 2n\pi \pm y$$

Some useful facts :-

$$\left(\frac{1 - \tan \theta}{1 - \cot \theta}\right)^2 = \tan^2 \theta$$

$$\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \frac{1 - \sin \theta}{\cos \theta} = \sec \theta - \tan \theta$$

$$\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{1 - \cos \theta}{\sin \theta} = \operatorname{cosec} \theta - \cot \theta$$

$$\sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} = \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$$

$$\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$$

$$(\tan \theta - \cot \theta)(\tan^2 \theta + \cot^2 \theta + 1) = \tan^3 \theta - \cot^3 \theta$$

Some other concept :-

1). If $A + B = 180^\circ$

$$\tan A + \tan B = 0$$

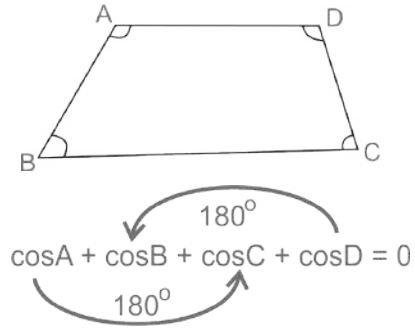
$$\cos A + \cos B = 0$$

$$\sec A + \sec B = 0$$

$$\cot A + \cot B = 0$$

Example :- find the value of $\cos A + \cos B + \cos C + \cos D$ if ABCD is a cyclic quadrilateral

Solution :-



2). If $\cot \theta + \tan \theta = x$ and $\sec \theta - \cos \theta = y$ then

$$(x^2 y)^{\frac{2}{3}} - (xy^2)^{\frac{2}{3}} = 1$$

3). If $\sin \theta + \sin^2 \theta = 1$ then $\sin \theta = \cos^2 \theta$ and $\cos^2 \theta + \cos^4 \theta = 1$ and $\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta + 2\cos^4 \theta + 2\cos^2 \theta - 2 = 1$

4). If $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$ and $\frac{x}{a} \sin \theta - \frac{y}{b} \cos \theta = 1$ then

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 2$$

5). (i) (a) $\sin 1^\circ \cdot \sin 2^\circ \cdot \sin 3^\circ \dots \sin 180^\circ = 0$

(b) $\sin 1^\circ \cdot \sin 2^\circ \cdot \sin 3^\circ \dots$ to greater than $\sin 180^\circ = 0$

(ii) (a) $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 90^\circ = 0$

(b) $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots$ to greater than $\cos 90^\circ = 1$

(iii) $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ = 1$

(iv) $\sin^2 \theta + \sin^2 (x + \theta) + \dots + \sin^2 [90 - (x + \theta)] +$

$$\sin^2 (90 - \theta) = \frac{\text{Total terms}}{2}$$

Example :- find the value of $\sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ$.

Solution :- we know that, $\sin^2 \theta + \sin^2 (x + \theta) + \dots + \sin^2 [90 - (x + \theta)] + \sin^2 (90 - \theta) = \frac{\text{Total terms}}{2}$

According to question,

$$\text{Total number of term} = t_n = a + (n - 1)d$$

$$a = 5, d = 1$$

$$85 = 5 + (n - 1)1 \Rightarrow n = 81$$

$$\text{So, } \sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ$$

$$= \frac{\text{Total terms}}{2} = \frac{81}{2} = 40\frac{1}{2}$$

Use of Componendo and Dividendo to solve some other trigonometric function :-

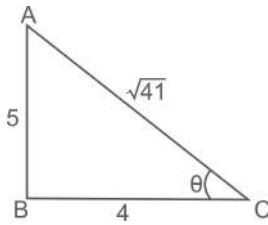
If $\frac{x}{y} = \frac{a}{b}$, then $\frac{x+y}{x-y} = \frac{a+b}{a-b}$

Example :- if $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 9$ find the value of $\tan \theta$ and $\cos \theta$.

Solution :- apply C & D $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 9$

$$\frac{(\sin \theta + \cos \theta) + (\sin \theta - \cos \theta)}{(\sin \theta + \cos \theta) - (\sin \theta - \cos \theta)} = \frac{9 + 1}{9 - 1}$$

$$\frac{2\sin \theta}{2\cos \theta} = \frac{10}{8} \Rightarrow \tan \theta = \frac{5}{4} \Rightarrow \tan \theta = \frac{\text{perpendicular}}{\text{base}} = \frac{5}{4}$$



$$\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{4}{\sqrt{41}}$$

Example :- if $2 \cot \theta = 3$ find the value of $\frac{2 \cos \theta - \sin \theta}{2 \cos \theta + \sin \theta}$.

Solution :- $2 \cot \theta = 3 \Rightarrow \cot \theta = \frac{3}{2}$

$$\frac{\cos \theta}{\sin \theta} = \frac{3}{2} \Rightarrow \frac{2 \cos \theta}{\sin \theta} = \frac{3}{1} \quad \{ \because \text{multiply both side by 2} \}$$

apply C & D

$$\frac{2 \cos \theta + \sin \theta}{2 \cos \theta - \sin \theta} = \frac{3 + 1}{3 - 1} \Rightarrow \frac{2 \cos \theta + \sin \theta}{2 \cos \theta - \sin \theta} = \frac{4}{2}$$

$$\frac{2 \cos \theta - \sin \theta}{2 \cos \theta + \sin \theta} = \frac{1}{2}$$

Important Value Putting Method in Trigonometry

Type - I

Note :-

Step.1 :- some trigonometric cases to solve the questions first we calculate value of angle from the given equation

Step.2 :- put the value in other equations

Example :- if $\sin \alpha \sin \beta - \cos \alpha \cos \beta + 1 = 0$ then find the value of $1 + \cot \alpha \tan \beta = ?$

Solution :- according to questions,

$$\sin \alpha \sin \beta - \cos \alpha \cos \beta + 1 = 0$$

$$\cos \alpha \cos \beta - \sin \alpha \sin \beta = 1$$

We know that, $\cos (A + B) = \cos A \cos B - \sin A \sin B$

$$\cos \alpha \cos \beta - \sin \alpha \sin \beta = 1$$

$$\cos (\alpha + \beta) = \cos 0^\circ \Rightarrow (\alpha + \beta) = 0^\circ \Rightarrow \beta = -\alpha$$

$$\text{Hence, } 1 + \cot \alpha \tan \beta = 1 + \cot \alpha \tan (-\alpha) = 0$$

Type - II

Note :- when, in the trigonometric function given such as:-

$$(\sec \theta - \operatorname{cosec} \theta = 0),$$

$$(\sin \theta - \cos \theta = 0),$$

$$(\tan \theta - \cot \theta = 0)$$

then put $\theta = 45^\circ$ to solve the question.

Example :- if $\sec \theta - \operatorname{cosec} \theta = 0$ and θ is an acute angle, then what is the value of $\sec^2 \theta + \operatorname{cosec}^2 \theta$?

Solution :- according to questions,

$$\sec \theta - \operatorname{cosec} \theta = 0 \quad \{ \text{put } \theta = 45^\circ \}$$

$$\sec 45^\circ - \operatorname{cosec} 45^\circ \Rightarrow \sqrt{2} - \sqrt{2} = 0$$

$$0 = 0 \quad \{ \text{hence L.H.S. = R.H.S. satisfied} \}$$

$$\text{So, } \sec^2 \theta + \operatorname{cosec}^2 \theta \Rightarrow \sec^2 45^\circ + \operatorname{cosec}^2 45^\circ$$

$$(\sqrt{2})^2 + (\sqrt{2})^2 = 4$$

Type - III

Note :- when, in the trigonometric function given such as:-

$$(\sec \theta + \frac{1}{\cos \theta} = 2), (\sec \theta + \cos \theta = 2),$$

then put $\theta = 0^\circ$ to solve the question.

Example :- If $\sec \theta + \frac{1}{\cos \theta} = 2$, find the value of $\sec^{55} \theta + \frac{1}{\sec^{55} \theta}$?

Solution :- $\sec \theta + \frac{1}{\cos \theta} = 2$

$$\sec \theta + \sec \theta = 2$$

$$2 \sec \theta = 2 \Rightarrow \sec \theta = 1 = \sec 0^\circ \Rightarrow \theta = 0^\circ$$

$$\text{So, } \sec^{55} \theta + \frac{1}{\sec^{55} \theta} = \sec^{55} 0^\circ + \frac{1}{\sec^{55} 0^\circ} = 1 + 1 = 2$$

Exam Hall approach :- we can put $\theta = 0^\circ$ in this type of question

$$\sec \theta + \frac{1}{\cos \theta} = 2 \Rightarrow \sec 0^\circ + \frac{1}{\cos 0^\circ} = 2 \Rightarrow 2$$

$$\sec^{55} 0^\circ + \frac{1}{\sec^{55} 0^\circ} = 2$$

Type - IV

Example :- If $\tan^2 \theta + 3 \sec \theta - 9 = 0$, where $0 < \theta < 90^\circ$, then what is the value of $12 \cot^2 \theta + 3 \operatorname{cosec} \theta$?

UPSC CDS - I (10/04/2022)

(a) $(\sqrt{3} + 1)^2$ (b) $(\sqrt{3} + 2)^2$ (c) $(2\sqrt{3} + 1)^2$ (d) $(3\sqrt{3} + 1)^2$

Solution :- According to the question,

Put $\theta = 60^\circ$

$$\tan^2 60^\circ + 3 \sec 60^\circ - 9 = 0$$

$$(\sqrt{3})^2 + 3 \times 2 - 9 = 0 \Rightarrow 9 - 9 = 0$$

Now, $12 \cot^2 \theta + 3 \operatorname{cosec} \theta$

$$12 \cot^2 60^\circ + 3 \operatorname{cosec} 60^\circ$$

$$12 \times \left(\frac{1}{\sqrt{3}}\right)^2 + 3 \times \frac{2}{\sqrt{3}}$$

$$4 + 2\sqrt{3} = (3 + 1 + 2 \times 1 \times \sqrt{3})$$

$$(\sqrt{3} + 1)^2$$

Type - V

Note :- some special cases to solve the questions we can put the value of $\theta = 30^\circ, 60^\circ$ both in questions and options.

Example :- What is the values of $\tan \frac{\theta}{2}$?

SSC CGL 26/07/2023 (4th shift)

(a) $\frac{\cos \theta}{1 - \sin \theta}$ (b) $\frac{\sin \theta}{1 - \cos \theta}$ (c) $\frac{\cos \theta}{1 + \cos \theta}$ (d) $\frac{\sin \theta}{1 + \cos \theta}$

Solution :- put $\theta = 60^\circ$

$$\tan \frac{\theta}{2} = \tan \frac{60^\circ}{2} = \frac{1}{\sqrt{3}}$$

put $\theta = 60^\circ$ in the options one by one to satisfy the value

Then, options d satisfy.

$$\frac{\sin \theta}{1 + \cos \theta} \Rightarrow \frac{\sin 60^\circ}{1 + \cos 60^\circ} \Rightarrow \frac{\frac{\sqrt{3}}{2} \times 2}{2 \times 3} = \frac{1}{\sqrt{3}}$$

Alternate method :-

$$\sin 2\theta = 2 \sin\theta \cos\theta$$

$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\left(\frac{\theta}{2}\right) \Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1+\cos\theta}$$

Type - VI

Note :-

(i) some most special cases to solve the questions we do not need to put the value of $\theta = 30^\circ, 60^\circ \dots$ etc in questions.

(ii) if the variable are given then put the value of any variable equal to 0.

Example :- if $a\sin^2\theta + b\cos^2\theta = c$ then find the value of $\tan\theta = ?$

(a) $\sqrt{\frac{c-b}{a-c}}$ (b) $\sqrt{\frac{b-c}{a-c}}$ (c) $\sqrt{\frac{c-b}{c-a}}$ (d) $\sqrt{\frac{c+b}{a+c}}$

Solution :- $a\sin^2\theta + b\cos^2\theta = c$ {put $c = 0$ }

$$a\sin^2\theta = -b\cos^2\theta \Rightarrow \frac{\sin^2\theta}{\cos^2\theta} = \frac{-b}{a} \Rightarrow \tan^2\theta = \frac{-b}{a}$$

put $c = 0$ in the options one by one to satisfy the value

Then, options a satisfy.

$$\sqrt{\frac{c-b}{a-c}} = \sqrt{\frac{0-b}{a-0}} = \frac{-b}{a}$$

Type - VII

Note :-

(i) in general trigonometric function to solve the question we put the value of $\theta = 30^\circ, 60^\circ, 90^\circ$ in the cosine, sine, sec and cosec function

(ii) in the tan, and cot function put the value of $\theta = 45^\circ$

Example :- The value of $\sqrt{\frac{1+\cos 2A}{1-\cos 2A}} = ?$ (Note: A is non-zero)

RRB NTPC CBT - I 16/01/2021 (Evening)

(a) $\tan A$ (b) $\sin A$ (c) $\cos A$ (d) $\cot A$

Solution :- $\sqrt{\frac{1+\cos 2A}{1-\cos 2A}}$

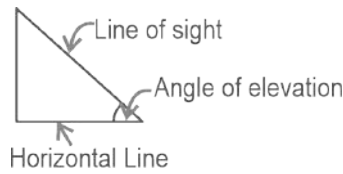
Short trick : put $A = 30^\circ$

$$\sqrt{\frac{1+\cos 60}{1-\cos 60}} = \sqrt{\frac{1+\frac{1}{2}}{1-\frac{1}{2}}} = \sqrt{3} = \cot 30^\circ = \cot A$$

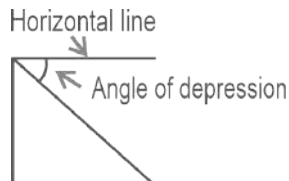
Height and Distance

In this chapter following concepts / terminologies will be used.

Angle Of Elevation :- The angle made by the line of sight with the horizontal line is called the angle of elevation.



Angle Of Depression :- The angle made by the line of sight with the horizontal line is called the angle of depression.



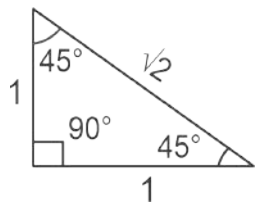
Note: When we come near object, the angle of elevation increases and when we go away from the object, the angle of elevation decreases.

Note: Angle of elevation and angle of depression are always equal.

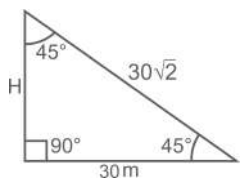
Some Important Relation

S.No.	Angle	Ratios of sides
1	$45^\circ : 45^\circ : 90^\circ$	$1 : 1 : \sqrt{2}$
2	$30^\circ : 60^\circ : 90^\circ$	$1 : \sqrt{3} : 2$
3	$15^\circ : 75^\circ : 90^\circ$	$(\sqrt{3} - 1) : (\sqrt{3} + 1) : 2\sqrt{2}$
4	$30^\circ : 30^\circ : 120^\circ$	$1 : 1 : \sqrt{3}$

Case (1)



Example:- The angle of elevation of the top of a tower from a point on the ground, 45° which is 30 m from the foot of the tower. Find the height of the tower?



Solution :- Let the height of the tower be 'H'

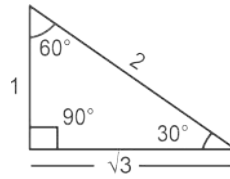
$$\tan 45^\circ = \frac{H}{30}$$

$$1 = \frac{H}{30} \Rightarrow H = 30 \text{ m}$$

Exam hall approach:- 1 unit = 30 m

$$1 \text{ unit} = ? \Rightarrow ? = 30 \text{ m}$$

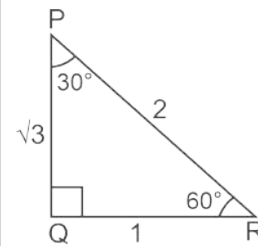
Case (2)



Example:- The angle of elevation of the top of a building at a distance of 70 m from its foot on a horizontal plane is found to be 60° . Find the height of the building
SSC CGL Tier II (03/03/2023)

- (a) $70\sqrt{3}$ m (b) $60\sqrt{3}$ m (c) $50\sqrt{3}$ m (d) $70\sqrt{2}$ m

Solution:-



$$\text{Angle} \rightarrow 30^\circ : 60^\circ : 90^\circ$$

$$\text{Side} \rightarrow 1 : \sqrt{3} : 2$$

According to the questions ,

$$1 \text{ unit} = 70 \text{ m}$$

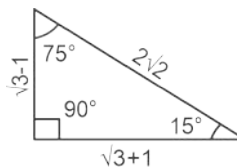
$$\text{Then , height of the building } (\sqrt{3} \text{ units}) = 70\sqrt{3} \text{ m.}$$

Exam hall approach:-

$$1 \text{ unit} = 70 \text{ m}$$

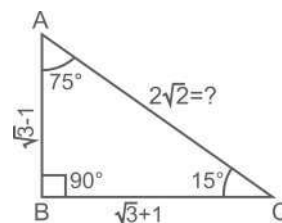
$$\sqrt{3} \text{ unit} = 70\sqrt{3} \text{ m.}$$

Case (3)



Example:- The angle of elevation of a ladder leaning against a wall is 15° and the foot of the ladder is 4 m away from the wall. The length of the ladder is :

Solution:-



Let AB be the wall and AC be the ladder.

Then, $\angle ACB = 15^\circ$ and $BC = 4 \text{ m}$.

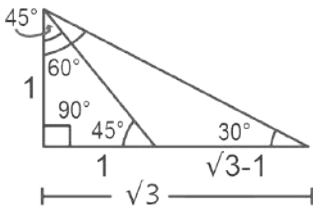
$$\sqrt{3} + 1 \text{ unit} = 4 \text{ m}$$

$$2\sqrt{2} \text{ unit} = ? \Rightarrow ? = \frac{8\sqrt{2}}{\sqrt{3} + 1}$$

When Angle Changes

Case (1)

When the angle changes from 30° to 45°

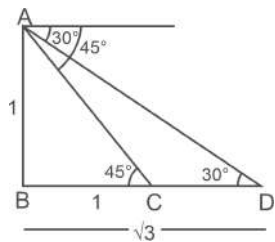


Example :- As observed from the top of a lighthouse, 42 m high above sea level, The angle of depression of a ship sailing directly towards it changes from 30° to 45° . The distance travelled by the ship during the period of observation is:

SSC CHSL 04/08/2021 (Evening)

- (a) $42(1 - \sqrt{3})$ (b) $42(\sqrt{3} + 1)$ (c) $42(\sqrt{3} - 1)$ (d) 42

Solution :-



From the given figure, we have;

$$CD = BD - BC = (\sqrt{3} - 1) \text{ unit}$$

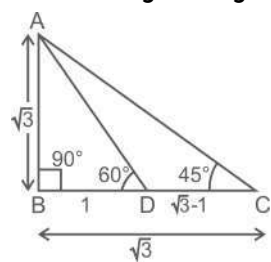
Here, 1 unit ----- 42 m

Then, $(\sqrt{3} - 1) \text{ unit}$ ----- $42(\sqrt{3} - 1) \text{ m}$

So, the distance travelled by the ship during the period of observation = $42(\sqrt{3} - 1) \text{ m}$

Case (2)

When the angle changes from 45° to 60°

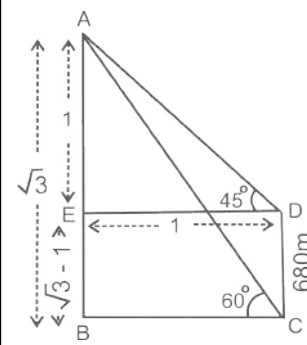


Example :- The angle of elevation of the top of a tower from the top of a building whose height is 680 m is 45° and the angle of elevation of the top of the same tower from the foot of the same building is 60° . What is the height (in m) of the tower?

SSC CGL Tier II (07/03/2023)

- (a) $340(3 + \sqrt{3})$ (b) $310(3 - \sqrt{3})$
(c) $310(3 + \sqrt{3})$ (d) $340(3 - \sqrt{3})$

Solution :-



Let $AB = \text{Height of tower} = \sqrt{3} \text{ unit}$

Building height = $BE = CD = 680 \text{ m}$

ATQ,

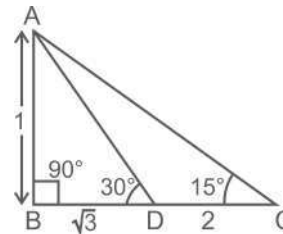
$$\Rightarrow (\sqrt{3} - 1) \text{ unit} = 680$$

$$\Rightarrow \sqrt{3} \text{ unit} = \frac{680}{\sqrt{3} - 1} \times \sqrt{3}$$

$$= \frac{680\sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = 340(3 + \sqrt{3}) \text{ m}$$

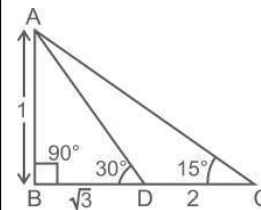
Case (3)

When the angle changes from 15° to 30°



Example :- A car is moving at uniform speed towards a tower. It takes 15 minutes for the angle of depression from the top of tower to the car to change from 15° to 30° . What time after this, the car will reach the base of the tower?

Solution :-

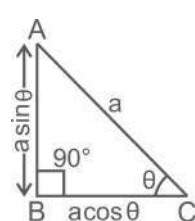
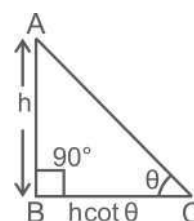


According to the question,

$$\Rightarrow 2 \text{ unit} = 15 \text{ min}$$

$$\Rightarrow (\sqrt{3}) \text{ unit} = \frac{15\sqrt{3}}{2} \text{ min}$$

Case (4)



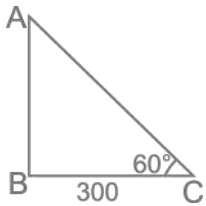
Example:- 300 m from the foot of a cliff on a level ground, the angle of elevation of the top of a cliff is 60° . Find the height of

the cliff.

RRB NTPC CBT - I 03/04/2021 (Evening)

(a) $200\sqrt{3}$ m (b) $150\sqrt{3}$ m (c) $300\sqrt{3}$ m (d) $250\sqrt{3}$ m

Solution:-



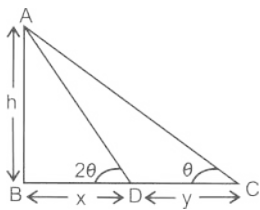
In triangle ABC

$BC = 300$ m

$$\tan \theta = \frac{AB}{BC} \Rightarrow \tan 60^\circ = \frac{AB}{300} \Rightarrow \sqrt{3} = \frac{AB}{300} \Rightarrow AB = 300\sqrt{3}$$

Case (5)

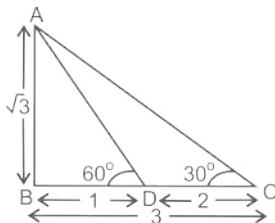
When the angle change from θ to 2θ



$$h^2 = y^2 - x^2$$

Example :- The angle of elevation of a point A on the ground from the top of a tower is 30° . On going 10 m towards the foot of the tower, the angle of elevation changes to 60° . What will be the height of the tower ?

Solution :-



According to the question,

$\Rightarrow 2 \text{ unit} = 10 \text{ m}$ (given that)

$\Rightarrow 1 \text{ unit} = 5 \text{ m}$

Now, height of the tower $(\sqrt{3}) = 5\sqrt{3} \text{ m}$

Exam Hall approach

Using identity,

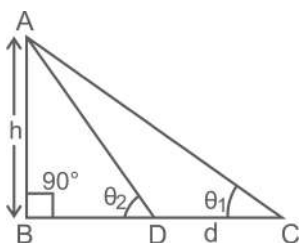
$$h^2 = y^2 - x^2$$

Where $DC(y) = 10 \text{ m}$, $BD(x) = 5 \text{ m}$ and $AB(h) = ?$

So,

$$\Rightarrow h^2 = 10^2 - 5^2 = 75 \Rightarrow h = \sqrt{75} \Rightarrow h = 5\sqrt{3} \text{ m}$$

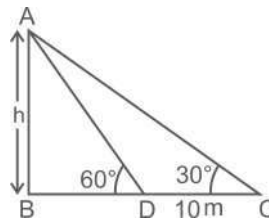
Case (6)



$$d = h(\cot \theta_1 - \cot \theta_2)$$

Example:- The angle of elevation of a point A on the ground from the top of a tower is 30° . On going 10 m towards the foot of the tower, the angle of elevation changes to 60° . What will be the height of the tower ?

Solution:-



Using identity,

$$d = h(\cot \theta_1 - \cot \theta_2)$$

$$\Rightarrow 10 = h(\cot 30^\circ - \cot 60^\circ)$$

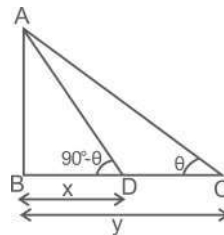
$$\Rightarrow 10 = h(\sqrt{3} - \frac{1}{\sqrt{3}}) \Rightarrow 10 = h(\frac{3-1}{\sqrt{3}})$$

$$\Rightarrow 10 = h \frac{2}{\sqrt{3}} = \frac{10\sqrt{3}}{2} = 5\sqrt{3}$$

Other Important Result

when angle are complementary

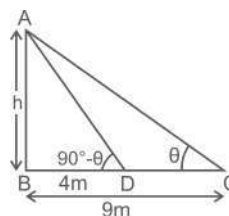
(a)



$$h = \sqrt{xy}$$

Example :- The angles of elevation of the top of a tower from two points at a distance of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Find the height of the tower.

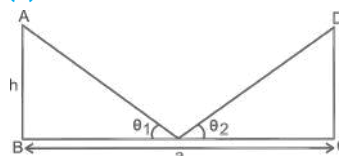
Solution :-



Using identity,

$$\Rightarrow h = \sqrt{xy} \Rightarrow h = \sqrt{4 \times 9} \Rightarrow h = 6 \text{ m}$$

(b)



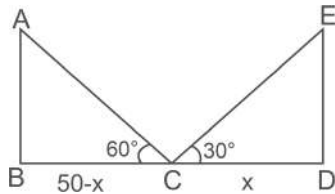
$$a = h(\cot\theta_1 + \cot\theta_2)$$

Example:- Two vertical lamp-posts of equal height are on either side of a road 50 m wide. At a point on the road between the lamp - posts, the elevations of the tops of lamp - posts are 60° and 30° . Find the height of the lamp - post.

RRB JE 27/05/2019 (Evening)

- (a) 22.50 m (b) 23.75 m
(c) 18.65 m (d) 21.65 m

Solution:-



In $\triangle ABC$

$$\tan 60^\circ = \frac{AB}{(50-x)}$$

$$\Rightarrow AB = (50-x)\sqrt{3} \dots\dots\dots \text{eq. (1)}$$

In $\triangle ECD$

$$\tan 30^\circ = \frac{ED}{x} = \frac{AB}{x} \dots\dots\dots \text{eq. (2)}$$

Solve eq. (1) and eq. (2)

$$\frac{x}{\sqrt{3}} = (50-x)\sqrt{3} \Rightarrow x = 150 - 3x$$

$$\Rightarrow 4x = 150 \Rightarrow x = \frac{75}{2}$$

So length of lamp-posts = $ED = AB$

$$= \frac{75}{2 \times \sqrt{3}} = \frac{25 \times \sqrt{3}}{2} = 12.5 \sqrt{3}$$

$$= 12.5 \times 1.73 = 21.65 \text{ m}$$

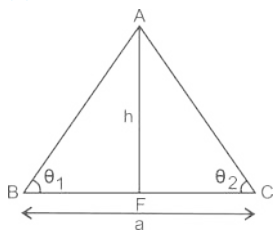
Exam hall approach:-

$$50 = h(\cot\theta_1 + \cot\theta_2)$$

$$50 = h(\cot 60^\circ + \cot 30^\circ) \Rightarrow h = \frac{25\sqrt{3}}{2} = 12.5\sqrt{3}$$

$$= 12.5 \times 1.73 = 21.65 \text{ m}$$

(c)



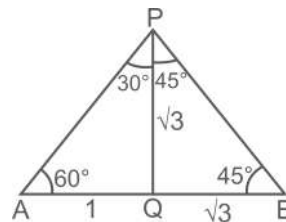
$$a = h(\cot\theta_1 + \cot\theta_2)$$

Example :- Two points A and B are on the ground and on opposite sides of a tower. A is closer to foot of tower by 42 m than B. If the angles of elevation of the top of the tower from A and B are 60° and 45° respectively, then the height of the tower is closest to :

SSC CPO 13/12/2019 (Evening)

- (a) 94.6 m (b) 99.4 m (c) 88.2 m (d) 87.6 m

Solution:- Let PQ be the height of the tower.



We will use this identity when we are given the direct value of AB.

$$\Rightarrow a = h(\cot\theta_1 + \cot\theta_2)$$

According to the question,

$$\Rightarrow 1 \text{ unit} = \frac{42}{\sqrt{3}-1}$$

$$\Rightarrow (\sqrt{3} + 1) \text{ unit} = \frac{42(\sqrt{3} + 1)}{\sqrt{3}-1} = AB$$

So,

$$\Rightarrow \frac{42(\sqrt{3} + 1)}{\sqrt{3}-1} = h(\cot 60^\circ + \cot 45^\circ)$$

$$\Rightarrow \frac{42(\sqrt{3} + 1)}{\sqrt{3}-1} = h\left(\frac{1}{\sqrt{3}} + 1\right) \Rightarrow \frac{42(\sqrt{3} + 1)}{\sqrt{3}-1} = h\left(\frac{1 + \sqrt{3}}{\sqrt{3}}\right)$$

$$\Rightarrow h = \frac{42\sqrt{3}}{\sqrt{3}-1} = \frac{42 \times 1.73}{0.73} \approx 99.5 \text{ m}$$

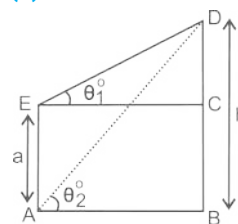
Exam Hall approach :

ATQ,

$$(\sqrt{3} - 1) \text{ unit} = 42 \text{ m}$$

$$\text{Then, } \sqrt{3} \text{ unit} = \frac{42\sqrt{3}}{\sqrt{3}-1} = \frac{42 \times 1.73}{0.73} \approx 99.5 \text{ m}$$

(d)



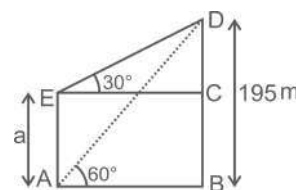
$$h = \frac{a \cot\theta_1}{\cot\theta_1 - \cot\theta_2}$$

Example:- From the top of a 195 m high cliff, the angles of depression of the top and bottom of the tower are 30° and 60° respectively. Find the height of the tower (in m).

SSC CGL 18/04/2022 (Evening)

- (a) $195\sqrt{3}$ (b) 195 (c) 130 (d) 65

Solution:-



Let the height of tower (EA) = a

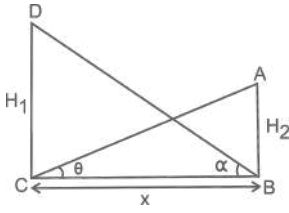
Using the identity,

$$\Rightarrow h = \frac{a \cot\theta_1}{\cot\theta_1 - \cot\theta_2} \Rightarrow 195 = \frac{a \cot 30^\circ}{\cot 30^\circ - \cot 60^\circ}$$

$$\Rightarrow 195 = \frac{a \times \sqrt{3}}{\sqrt{3} - \frac{1}{\sqrt{3}}} \Rightarrow 195 = \frac{a \times \sqrt{3}}{\frac{3-1}{\sqrt{3}}} \Rightarrow 195 = \frac{a \times 3}{2}$$

$$\Rightarrow a = 130 \text{ m}$$

(e)



$$\theta + \alpha = 90^\circ \text{ then,}$$

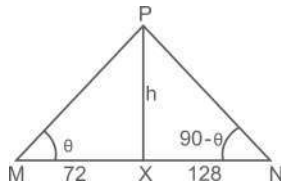
$$x = \sqrt{H_1 H_2}$$

Example:- The angle of elevation of the top of a tall building from the points M and N at the distances of 72m and 128m, respectively, from the base of the building and in the same straight line with it, are complementary. The height of the building (in m) is ?

SSC CGL 12/04/2022(Afternoon)

(a) 84 (b) 96 (c) 80 (d) 90

Solution:- Let the height of building be h

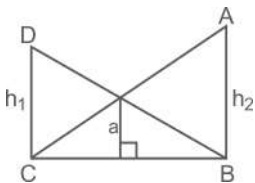


when angles are complementary

$h^2 = \text{Product of the distance of two points from the building}$

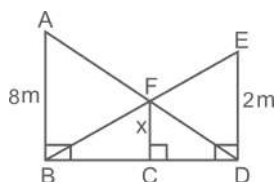
$$h = \sqrt{72 \times 128} = \sqrt{9216} = 96 \text{ m}$$

(f)



$$\frac{1}{a} = \frac{1}{h_1} + \frac{1}{h_2}$$

Example:- In the following figure, AB = 8 metre, FC = x metre and ED = 2 metre are each perpendicular to BD then find the value of x.



Solution:- We know that,

$$\Rightarrow \frac{1}{a} = \frac{1}{h_1} + \frac{1}{h_2}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{8} + \frac{1}{2} \Rightarrow \frac{1}{x} = \frac{1+4}{8} = \frac{5}{8} \Rightarrow x = \frac{8}{5}$$

When object changes its Position from one point to another point

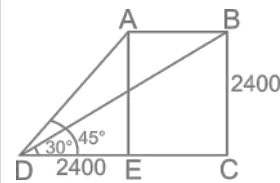
Example:- The angle of elevation of an aeroplane from a point on the ground is 45° . After a flight of 12 seconds horizontally, the angle of elevation changes to 30° . If the aeroplane is flying at height of 2400 m, Find the speed of the aeroplane (in km/h).

NTPC CBT II (15/06/2022) 3rd Shift

(a) $72(1+\sqrt{3})$ (b) $720(\sqrt{3}+1)$

(c) $720(\sqrt{3}-1)$ (d) $72(\sqrt{3}-1)$

Solution :-



$$\tan 45^\circ = \frac{AE}{DE} = \frac{2400}{DE} \Rightarrow DE = 2400 \text{ m}$$

$$\text{Now, } \tan 30^\circ = \frac{BC}{DE + EC}$$

$$\frac{1}{\sqrt{3}} = \frac{2400}{2400 + EC}$$

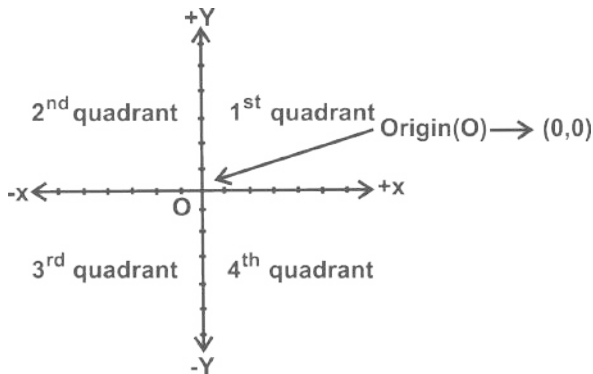
$$EC = 2400(\sqrt{3} - 1)$$

$$\text{Speed of the aeroplane} = \frac{2400(\sqrt{3} - 1)}{12} \times \frac{18}{5}$$

$$= 720(\sqrt{3} - 1) \text{ km/h}$$

Coordinate Geometry

Coordinate geometry is a system of geometry where the position of points on the plane is described by using an ordered pair of numbers. The x- coordinate and the y- coordinate of a point taken together is known as coordinates of the point



Key points :-

- **Coordinate Axes :-** The perpendicular lines in the cartesian system are x axis and y axis . Both of them together are known as coordinate axes.
- **Origin :-** The point of intersection of coordinate axes is origin. The coordinates of origin are (0 , 0).
- **Abscissa :-** The x - coordinate is called abscissa. It is the perpendicular distance of a point from y axis.
- **Ordinate :-** The y-coordinate is called ordinate . It is the perpendicular distance of a point from x-axis.
- If $y = a$, where a is constant then $y = a$ denotes a straight line parallel to x-axis.
- If $x = a$, where a is constant then $x = a$ denotes a straight line parallel to y-axis.
- For passing a line through the origin, one of the coordinates of the given equation must be (0, 0).

Important Formula

Distance Formula

If points $P(x_1, y_1)$, $Q(x_2, y_2)$ lie on the plane xy then,

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example :- If the distance between two points $(x, 7)$ and $(1, 15)$ is 10 units, then the possible values of $x = ?$
RRB NTPC CBT - I 08/03/2021 (Evening)

Solution :- Distance formula $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$10 \text{ unit} = \sqrt{(1 - x)^2 + (15 - 7)^2}$$

$$10 = \sqrt{(x^2 + 1 - 2x) + (8)^2}$$

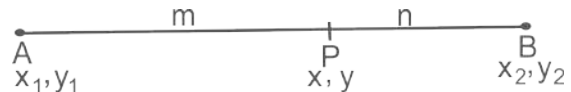
$$100 = x^2 + 1 - 2x + 64$$

$$100 - 65 = x^2 - 2x \Rightarrow x^2 - 2x - 35 = 0$$

$$\text{Where, } x = 7, -5$$

Section Formula

(a) Internal division :



$$\frac{PA}{PB} = \frac{m}{n} \Rightarrow P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Example :- The coordinates of the point that divides the line segment joining the points $(-5, 5)$ and $(7, -3)$ internally in the ratio $3 : 1$ are given by:

RRB ALP Tier - I (29/08/2018) Morning

(a) $(-2, 3)$ (b) $(\frac{5}{2}, 0)$ (c) $(1, 1)$ (d) $(4, -1)$

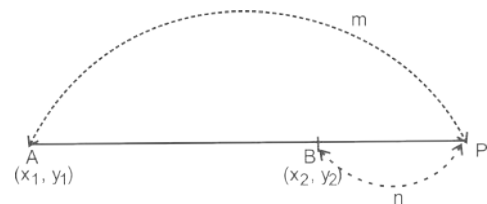
Solution :- As we know, The coordinates of the point that divides the line segment joining the points (x_1, y_1) and (x_2, y_2) internally in the ratio $m : n$ are given by ;

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

So, the required coordinates of the point

$$= \left(\frac{3 \times 7 + 1 \times -5}{3+1}, \frac{3 \times -3 + 1 \times 5}{3+1} \right) = \left(\frac{16}{4}, \frac{-4}{4} \right) = (4, -1)$$

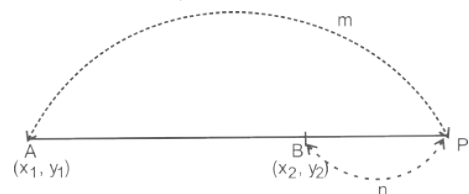
(b) External division :



$$\frac{PA}{PB} = \frac{m}{n} \Rightarrow P(x, y) = \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right)$$

Example :- Using the section formula, determine the coordinates of the point C that externally divides the points A $(-3, -1)$, and B $(-1, 0)$ in a ratio of $2 : 3$.

Solution :- We know that,



$$P(x, y) = \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right)$$

According to the question,

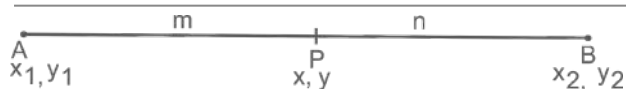
Required coordinates of the point

$$= \left(\frac{(2 \times -1) - (3 \times -3)}{2-3}, \frac{(2 \times 0) - 3 \times -1}{2-3} \right)$$

$$= \left(\frac{-2+9}{-1}, \frac{0+3}{-1} \right) = (-7, -3)$$

(c) Midpoint Division :

This is derived from the division formula. The coordinates of a point that divides line segments in the ratio $1 : 1$ means $(m : n) = (1 : 1)$



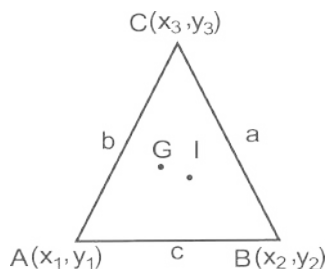
$$\frac{PA}{PB} = \frac{m}{n} = \frac{1}{1} \Rightarrow P(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Example :- Find the coordinates of the points which divide the line segment joining A(-2, 2) and B(2, 8) into two equal parts.

Solution :- Let the point = P

$$\Rightarrow P(x, y) = \left(\frac{-2 + 2}{2}, \frac{2 + 8}{2} \right) \Rightarrow P(x, y) = (0, 5)$$

Centroid (G) And Incentre (I) Formula



$$\text{Centroid (G)} = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

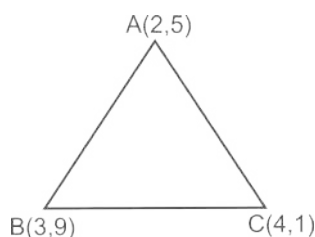
$$\text{Incentre (I)} = \left(\frac{x_1 a + x_2 b + x_3 c}{a + b + c}, \frac{y_1 a + y_2 b + y_3 c}{a + b + c} \right)$$

Example :- Find the centroid of a triangle formed by the points (2, 5), (3, 9) and (4, 1)

RRB NTPC CBT - I 11/03/2021 (Evening)

(a) (6, 6) (b) (9, 15) (c) (3, 9) (d) (3, 5)

Solution :-



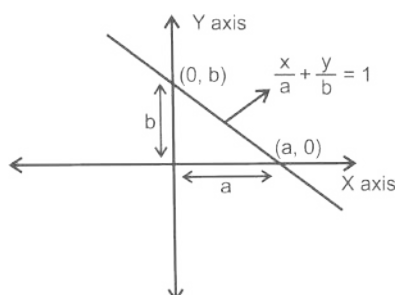
Let, Centroid of a triangle = (X, Y)

$$\text{Now, } X = \frac{x_1 + x_2 + x_3}{3} = \frac{2 + 3 + 4}{3} = \frac{9}{3} = 3$$

$$\text{And } Y = \frac{y_1 + y_2 + y_3}{3} = \frac{5 + 9 + 1}{3} = \frac{15}{3} = 5$$

Hence, Centroid of a triangle = (X, Y) $\Rightarrow$ (3, 5)

Intercept form of a line :-



Example :- Convert the equation $12x + 3y - 24 = 0$ into intercept form. What is the x-intercept and the y-intercept of the line?

Solution :- The given equation of a line is $12x + 3y - 24 = 0$.

Then, the intercept forms the equation of a line.

$$\Rightarrow 12x + 3y = 24$$

$$\Rightarrow \frac{12x}{24} + \frac{3y}{24} = 1 \Rightarrow \frac{x}{2} + \frac{y}{8} = 1$$

Now,

x - intercept = (2, 0) and y - intercept = (0, 8)

Slope of a line :-

• If (x_1, y_1) , (x_2, y_2) are any two points on line L, then the ratio $\frac{y_2 - y_1}{x_2 - x_1}$ is called the slope of the line L.

Example :- Find the slope of the line which passes through points (7, 5) and (-3, -5).

DMRC CRA 18/02/2020 (2nd shift)

(a) 1 (b) -1 (c) $\sqrt{3}$ (d) 0

$$\text{Solution :- Slope of points } (a_1, b_1) (a_2, b_2) = \frac{b_2 - b_1}{a_2 - a_1}$$

$$\text{Then, slope of } (7, 5) \text{ and } (-3, -5) = \frac{-5 - 5}{-3 - 7} = 1$$

• Angle between two lines of slope m_1 and m_2 respectively then

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Example :- The slopes of the lines which make an angle of 45° with the line $4x - y + 5 = 0$ are.

Solution :- Given that :

$$4x - y + 5 = 0 \text{ and } \angle(\theta) = 45^\circ$$

$$\Rightarrow y = 4x + 5 \dots (i)$$

Comparing the above equation from general equation of line ($y = mx + c$)

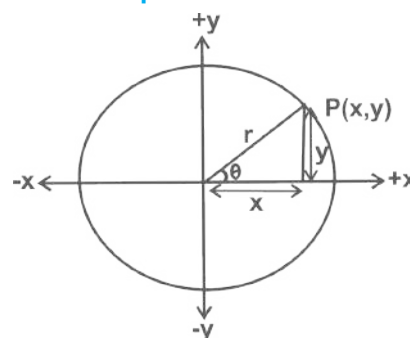
Then, $m_1 = 4$ and $m_2 = ?$

$$\Rightarrow \tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\Rightarrow \tan 45^\circ = \left| \frac{m_2 - 4}{1 + 4m_2} \right| \Rightarrow 1 = \left| \frac{m_2 - 4}{1 + 4m_2} \right| \Rightarrow 1 = \frac{m_2 - 4}{1 + 4m_2}$$

$$\Rightarrow 1 + 4m_2 = m_2 - 4 \Rightarrow m_2 = \frac{5}{3}$$

Polar coordinates of a point :-



$$X = r \cos \theta, Y = r \sin \theta$$

$$r^2 = X^2 + Y^2 \text{ and } \tan \theta = \frac{Y}{X}$$

Example :- Find the radius of the circle $x^2 + y^2 = 25$.

SSC CHSL 08/08/2023 (4th Shift)

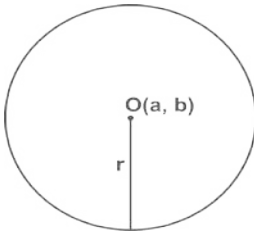
- (a) 25 units (b) 5 units
(c) 2 units (d) 12 units

Solution :- Equation of circle passing through origin :-

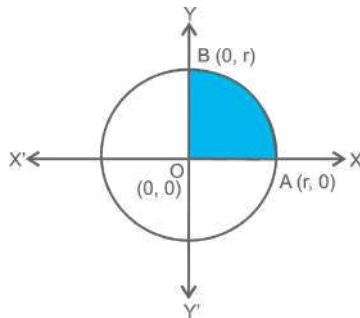
$$x^2 + y^2 = r^2 \text{}(r = \text{radius})$$

$$\text{Here, } x^2 + y^2 = 25 \Rightarrow \text{radius} = 5 \text{ units}$$

- If (a,b) is the center of circle and r is the radius then equation of circle = $(x - a)^2 + (y - b)^2 = r^2$



- if the center is origin then the equation of circle will be



i) $x^2 + y^2 = r^2$

ii) Area of enclosed circle = πr^2

Example :- if (3, 4) is the center of the circle and 5cm is the radius then find the equation of the circle .

Solution :- Equation of circle = $(x - 3)^2 + (y - 4)^2 = 5^2$

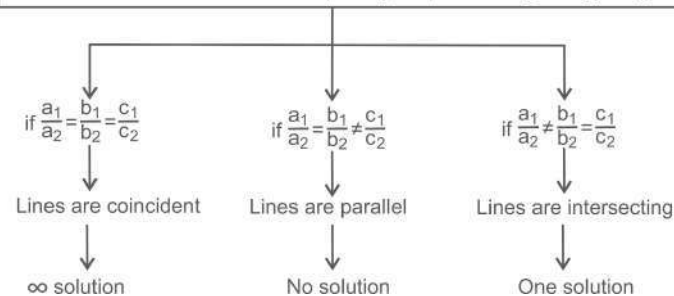
$$= x^2 + 9 - 6x + y^2 + 16 - 8y = 25$$

$$= x^2 + y^2 - 6x - 8y + 9 + 16 = 25$$

$$= x^2 + y^2 - 6x - 8y = 0$$

Concept

Points of intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$



Example :- For what value of k, the pair of equations $10x + 5y - (k - 5) = 0$, $20x + 10y - k = 0$ has an infinite solution?

Solution :-

Given that the equations $10x + 5y = (k - 5)$ and $20x + 10y = k$

$$\text{Condition of infinite solution} \rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

so,

$$\Rightarrow \frac{10}{20} = \frac{5}{10} = \frac{(k-5)}{k}$$

$$\Rightarrow \frac{5}{10} = \frac{(k-5)}{k} \Rightarrow 5k = 10k - 50 \Rightarrow 5k = 50 \Rightarrow k = 10$$

or

$$\Rightarrow \frac{10}{20} = \frac{(k-5)}{k} \Rightarrow 10k = 20k - 100 \Rightarrow 10k = 100 \Rightarrow k = 10$$

Example :- For what value of k, the pair of equations $4x - 3y = 9$, $2x + ky = 11$ has no solution?

Solution :-

Given that the equations $4x - 3y = 9$ and $2x + ky = 11$

$$\text{Condition of no solution} \rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

So,

$$\Rightarrow \frac{4}{2} = \frac{-3}{k} \neq \frac{9}{11} \Rightarrow \frac{4}{2} = \frac{-3}{k} \Rightarrow k = \frac{-3}{2}$$

Example :- For what value of k, the pair of equations $4x - 3y = 9$, $2x + ky = 11$ has one solution?

Solution :-

Given that the equations $4x - 3y = 9$ and $2x + ky = 11$

$$\text{Condition of one solution} \rightarrow \frac{a_1}{a_2} \neq \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

So,

$$\Rightarrow \frac{4}{2} \neq \frac{-3}{k} = \frac{9}{11} \Rightarrow \frac{-3}{k} = \frac{9}{11} \Rightarrow 9k = -33 \Rightarrow k = \frac{-33}{9}$$

Intercept form of equation of a line :-

- $\frac{x}{a} + \frac{y}{b} = 1$, where 'a' is x intercept of line and 'b' is y intercept of line .

- Perpendicular distance from a point (x_1, y_1) to line $ax + by + c = 0$, then

$$D = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Example :- The distance from the origin to the line $4x + 3y + 6 = 0$ is:

RRB NTPC CBT - I 04/03/2021 (Morning)

- (a) $\frac{7}{5}$ (b) $\frac{4}{5}$ (c) $\frac{6}{5}$ (d) $\frac{3}{5}$

Solution :- Length of the perpendicular from $O(0, 0)$ to the line $4x + 3y + 6 = 0$ is ,

$$D = \frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|4 \times 0 + 3 \times 0 + 6|}{\sqrt{4^2 + 3^2}} = \frac{6}{\sqrt{25}} = \frac{6}{5}$$

Distance between parallel lines :-

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0$$

$$D = \left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$$

Example :- Calculate the distance between the parallel lines $3x + 4y + 7 = 0$ and $3x + 4y - 5 = 0$.

Solution :- We know that,
 $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is

$$\Rightarrow D = \left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$$

So,

$$\Rightarrow D = \left| \frac{7 + 5}{\sqrt{3^2 + 4^2}} \right| = \left| \frac{12}{\sqrt{9 + 16}} \right| = \frac{12}{5}$$

• Equation of line :-

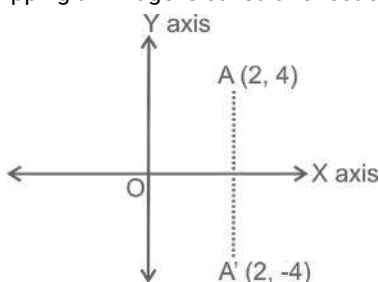
(i) The general form of a line is given as : $Ax + By + C = 0$

(ii) Slope intercept Form

Let x, y be the coordinate of a point through which a line passes, m be the slope of a line, and c be the y -intercept, then the equation of a line is given by :

$$y = mx + c$$

Reflection :- Flipping an image is called a reflection in geometry.



Reflection over

- **X-axis** $(a, b) \rightarrow (a, -b)$
- **Y-axis** $(a, b) \rightarrow (-a, b)$
- **Line Y = X** $(a, b) \rightarrow (b, a)$
- **Line Y = -X** $(a, b) \rightarrow (-b, -a)$

Example :- What is the reflection of the point $(5, -3)$ in the line $y = 3$?

SSC CGL Tier II (18/11/2020)

(a) $(5, -6)$ (b) $(-5, 3)$ (c) $(5, 9)$ (d) $(5, 3)$

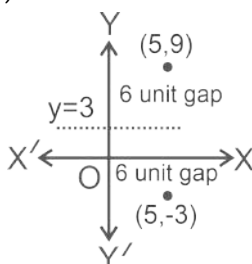
Solution :- $y = 3$ is parallel to x axis so on reflection x axis will not change

Perpendicular distance of $(5, -3)$ from $(y = 3) = 6$ units

6 units on the other side of the line

$$(y = 3) = 3 + 6 = 9$$

So the point = $(5, 9)$



• Equation of a circle :-

$x^2 + y^2 + 2gx + 2fy + c = 0$, where center of the circle is $(-g, -f)$

and radius $(r) = \sqrt{g^2 + f^2 - c}$

6 x

Example:- Find the radius of the circle

$$x^2 + y^2 + 7x + 4y + 9 = 0.$$

RRB NTPC CBT - I 04/03/2021 (Morning)

(a) $\frac{\sqrt{19}}{2}$ (b) $\frac{\sqrt{13}}{2}$ (c) $\frac{\sqrt{23}}{2}$ (d) $\frac{\sqrt{29}}{2}$

Solution :- $(x^2 + y^2 + 2gx + 2fy + c) = 0$, represents the circle with center $(-g, -f)$ and radius equal to $r^2 = g^2 + f^2 - c$

$$x^2 + y^2 + 7x + 4y + 9 = 0$$

Here, $g = -\frac{7}{2}$, $f = -2$ and $c = 9$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{\frac{49}{4} + 4 - 9} = \sqrt{\frac{49}{4} - 5} = \frac{\sqrt{29}}{2}$$

• **Area of Triangle :-** If point $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are the vertices of a triangle then area of a triangle is :

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Example :- If $(2, 7)$, $(5, 1)$, $(x, 3)$ are the vertices of a triangle whose area is 18 (sq units) then find the Possible values of x ?

RRB NTPC CBT - I 12/01/2021 (Morning)

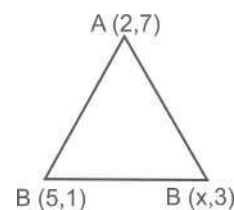
(a) -10 or -2

(b) 10 or -2

(c) 10 or 2

(d) -10 or 2

Solution :-



Area of the triangle

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$18 = \frac{1}{2} |2(1 - 3) + 5(3 - 7) + x(7 - 1)|$$

$$18 = \frac{1}{2} |-4 - 20 + 6x|$$

$$36 = |-24 + 6x|$$

In "mode", we get +ve and -ve both values

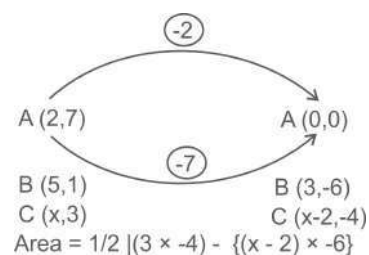
$$36 = \pm |-24 + 6x|$$

$$\text{So, } 36 = -24 + 6x \text{ and } 36 = 24 - 6x$$

$$x = 10 \text{ and } x = -2$$

Example hall approach:

Make any one of the given coordinates zero, do the same action with rest coordinates



$$18 = \frac{1}{2} |(-12) - \{-6x + 12\}|$$

$$36 = |(-12) + 6x - 12| = |6x - 24|$$

$$36 = \pm |-24 + 6x|$$

$$\text{So, } 36 = -24 + 6x \text{ and } 36 = 24 - 6x$$

Permutations and Combinations

Permutation

Definition of permutation : a permutation is an arrangement in a definite order of a number of objects taken some or all at a time.

Factorial

(I) $n! = n \times (n-1) \times \dots \times 3 \times 2 \times 1$

(II) $0! = 1$

(III) Factorials of negative numbers are not defined.

Permutation Formula :-

a) The number of permutations of 'n' different things taken 'r' at a time, where repetition is not allowed is given by

$${}_nP_r = \frac{n!}{(n-r)!} \text{ Where, } 0 \leq r \leq n$$

b) The number of permutations of 'n' different things taken 'r' at a time, where repetition is allowed is given by

$$n^r, \text{ Where, } 0 \leq r \leq n$$

c) The number of permutations of n objects taken all at a time, where p_1 objects are of the first kind, p_2 objects are of the second kind, ..., p_k objects are of the k^{th} kind and the rest if any, and if all are different is - $\frac{n!}{p_1! p_2! \dots p_k!}$

d) Circular Permutation : The number of permutations of n different objects taken r at a time, around a circle is given by

$$P_n = (n-1)!$$

Example:- A bag contains 6 red balls and 8 green balls. 2 balls are drawn at random one by one with replacement. Find the probability that both the balls are green
RRB Clerk Pre 13/09/2020 (1st Shift)

(a) $\frac{16}{49}$ (b) $\frac{25}{49}$ (c) $\frac{12}{49}$ (d) $\frac{21}{49}$ (e) None of these

Solution:- Total balls = $6 + 8 = 14$

Number of green balls = 8

2 balls are drawn at random one by one with replacement

Then, probability that both balls are green

$$= \frac{{}^8C_1}{{}^{14}C_1} \times \frac{{}^8C_1}{{}^{14}C_1} = \frac{8}{14} \times \frac{8}{14} = \frac{16}{49}$$

* In permutation, order of appearance of things is taken into account.

Example :- let there are three objects A, B & C.

When one object is taken at a time then permutation or arrangement of these object (3) - A, B, C

When two object is taken at a time then permutation or arrangement of these object (6) - (A, B), (B, A), (B, C), (C, B), (C, A), (A, C)

When three object is taken at a time then permutation or arrangement of these object (6) - (A, B, C) (A, C, B) (B, A, C) (B, C, A) (C, A, B) (C, B, A)

$$P(n, r) = {}^nP_r = \frac{n!}{(n-r)!}$$

Example :- How many three digit numbers can be formed without using the digits 0, 1, 4, 5, 7, if repetition of digits are not allowed ?

Bihar Police Cons. 14/03/2021 (2nd Shift)

(a) 60 (b) 120 (c) 504 (d) 720

Solution:- n = no. of digit that are not in 0,1,4,5,7

Hence, n = (2, 3, 6, 8, 9) = 5

r = 3

(as we need three digit number only)

$$\text{So, } {}^5P_3 = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2!}{2!} = 60$$

Example :- The number of permutations of the letters of ALLAHABAD is

Haryana CET 01/11/2021 (2nd Shift)

(a) 9! (b) $\frac{9!}{2!(4!)}$ (c) $\frac{9!}{3!}$ (d) $\frac{9!}{4!}$

Solution :- Required possible permutation

$$= \frac{\text{ALLAHABAD}}{\text{AAAA, LL}} = \frac{9!}{2!(4!)}$$

Combination

A combination is all about grouping. The number of different groups which can be formed from the available things can be found out using combinations.

a). The number of combinations of 'n' different things taken 'r' at a time, is given by

$${}_nC_r = \frac{n!}{(n-r)! r!}, 0 \leq r \leq n$$

Note :- (i) ${}_nC_1 = n$ and (ii) ${}_nC_0 = 1$

Example :- let there are three objects A, B & C.

Combination or selection of these objects taken one at a time (3) - A, B, C

Combination or selection of these objects taken two at a time (3) - (A, B), (B, C), (C, A)

Combination or selection of these objects taken three at a time (1) - (A, B, C)

$$C(n, r) = {}^nC_r = \frac{n!}{r!(n-r)!}$$

* The relationship between permutation and combination for 'r' things taken from 'n' things.

$${}_nP_r = r! \times {}^nC_r$$

Example:- If ${}^nC_x = 126$ and ${}^nP_x = 3024$ find n and x.

HCS CSAT (12/09/2021)

(a) 7,3 (b) 8,4 (c) 8,3 (d) 9,4

Solution:- ${}^nP_x = x! \cdot {}^nC_x$

$$3024 = 126 x! \Rightarrow 24 = x!$$

$$4 \times 3 \times 2 \times 1 = 4!$$

So, x = 4

Now, ${}^nC_x = {}^nC_4 = 126$

$$\frac{n!}{4!(n-4)!} = 126$$

$$\frac{n(n-1)(n-2)(n-3)(n-4)!}{4!(n-4)!} = 126$$

$$n(n-1)(n-2)(n-3) = 126 \times 4! = 6 \times 7 \times 3 \times 4 \times 3 \times 2 \times 1$$

$$n(n-1)(n-2)(n-3) = 9 \times 8 \times 7 \times 6$$

So, $n = 9$

Hence, value of $n = 9$ and $x = 4$

• Fundamental Principle of Counting (Counting without actually counting)

If an event A can occur in 'x' different ways and another event B can occur in 'y' different ways, then the total number of different ways of -

(I) simultaneous occurrence of both events is $x \times y$.

(Multiplication Principle)

(II) happening exactly one of the events is $x + y$. (Addition Principle)

Example:- A person has to go to Mumbai from Patna via Delhi. He can go to Delhi from Patna in 3 different ways and can go to Mumbai from Delhi in 4 different ways. Then, he can go to Mumbai from Patna via Delhi in $3 \times 4 = 12$ ways.

* Difference between Permutation and Combination

Permutation	Combination
are used when order/sequence of arrangement is needed.	are used when only the number of possible groups are to be found and the order/sequence of arrangement is not needed.
It is used for things of a different kind.	It is used for things of a similar kind.
The permutation of two things from three given things a,b,c is ab, ba, bc, cb, ca, ac.	The combination of two things from three given things a, b, c is ab, bc, ca.

Note :- For a given set of n and r values, the permutation answer is larger than the combination answer.

Miscellaneous

Example :- In how many different ways can the letters of the word 'POPULAR' be arranged ?

SBI Clerk Pre 29/02/2020 (1st Shift)

(a) 1650 (b) 1780 (c) 6800 (d) 2520 (e) None of these

solution:- Total number of letter in word 'POPULAR' = 7

Repeated letter = 2

So, number of different ways to arrange the word 'POPULAR'

$$= \frac{7!}{2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 2520$$

Example :- How many arrangements can be made using all the letters of the word DAUGHTER exactly once each so that the vowels always come together ?

DMRC CRA 17/02/2020 (2nd shift)

(a) 0234 (b) 3420 (c) 2340 (d) 4320

Solution:- In the word 'DAUGHTER'

Consonants = 5 letter, vowels = 3 letter

Now, all the vowels should come together so consider the bundle of vowels as one letter, then total letters will be 6.

According to the question,

So, the number of ways to arrange 6 letters = $6!$

Now, vowels also interchange their position

So, the number of ways = $3!$

Hence, required ways = $6! \times 3! = 6 \times 5 \times 4 \times 3 \times 2 \times 3 \times 2 = 4320$

Probability

Definition :- Probability can be defined as the ratio of the number of favorable outcomes to the total number of outcomes of an event.

In short Probability defines the likelihood of occurrence of an event.

Examples :- Flipping or tossing of a coin, Rolling a dice, Drawing a ball from a bag containing a number of balls, etc. all such events lead to finding the probability.

Formula for probability :

$$P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Number of total outcomes.}}$$

Important Properties of Probability

- The value of the probability of an event to happen can lie between 0 and 1 (0 can also be a probability).
- The maximum probability of an event is its sample space (sample space is the total number of possible outcomes)
- Probability of an impossible event is zero or a null set.
- There cannot be a negative probability for an event.
- If A and B are two mutually exclusive outcomes (Two events that cannot occur at the same time), then the probability of A or B occurring is the probability of A plus the probability of B.

Important Terms in Probability

1. Event :- A probability event can be defined as a set of outcomes of an experiment.
The probability of an event E is a number P(E) such that $0 \leq P(E) \leq 1$

2. Elementary event :- An event having only one outcome. Sum of all probabilities of all the elementary events of an experiment = 1.

3. Sure or certain event :- The probability of such an event is equal to 1.

4. Impossible events :- The probability of such an event is zero. For example, in an experiment of rolling a dice, the probability of getting a number greater than 6, is an impossible event.

5. Complementary events :- For any event, $P(E) + P(\bar{E}) = 1$

Where $\bar{E}$ stands for "not E"

Also, $P(\bar{E}) = 1 - P(E)$

Here E and $\bar{E}$ are called complementary events.

Example :- A card is drawn at random from a well shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen.

Solution :- $P(A) = 1 - P(\bar{A})$

$P(\bar{A}) = 1 - P(\text{red card or a queen})$

$$P(\bar{A}) = 1 - \left(\frac{26}{52} + \frac{4}{52} - \frac{2}{52} \right)$$

$$P(\bar{A}) = 1 - \frac{28}{52} = \frac{24}{52} = \frac{6}{13}$$

6. Equally likely outcomes :- All outcomes with equal probability

7. Sample space :- A sample space is a collection or a set of possible outcomes of a random experiment.

Example — Sample space for tossing of a coin once gives $S = \{H, T\}$

Sample space for tossing a coin twice gives $S = \{HH, HT, TH, TT\}$

Sample space for rolling a dice gives $S = \{1, 2, 3, 4, 5, 6\}$

8. Simple event :- If an event E has only one sample point of a sample space, it is called a simple (or elementary) event.

For example, in the experiment of tossing two coins, the sample space obtained is

$S = \{HH, HT, TH, TT\}$

There are four simple events as follows :

These are, $E_1 = \{HH\}$, $E_2 = \{HT\}$, $E_3 = \{TH\}$, $E_4 = \{TT\}$

9. Compound event :- If an event has more than one sample point, it is called a compound event.

Example — In the experiment of "tossing a coin thrice", the events :-

X — 'Exactly one head appeared'

Y — 'At Least one head appeared'

Z — 'Atmost one head appeared' etc.

are all compound events. The subsets of S associated with these events are

$X = \{HTT, THT, TTH\}$

$Y = \{HTT, THT, TTH, HHT, HTH, THH, HHH\}$

$Z = \{TTT, THT, HTT, TTH\}$

Each of the above subsets contain more than one sample point, hence they are all compound events.

Example :- A coin is tossed 3 times. The probability of getting exactly 2 heads is

UPSC CAPF (07/08/2022)

(a) $\frac{1}{3}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{5}{8}$

Solution :-

When coin is tossed three time total outcomes = $2^3 = 8$

Favourable outcomes for getting exactly two heads

= (HTH)(HHT)(THH) = 3

Hence, required probability = $\frac{3}{8}$

10. The event A and B ($A \cap B$) :-

Intersection of two sets $A \cap B$ is the set of those elements which are common to both A and B i.e., belonging to 'both A and B'

Suppose, $S = \{1, 2, 3, 4, 5, 6, 7\}$,

$A = \{2, 3, 5\}$ and $B = \{1, 3, 5\}$

Here, $A \cap B = \{3, 5\}$

11. The event 'A or B' ($A \cup B$) :-

When the sets A and B are two events associated with a sample space, then 'A or B' means The set of all values in sets A and B

Suppose, $S = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 5\}$

and $B = \{1, 3, 5\}$

Here, $A \cup B = \{1, 2, 3, 5\}$

12. Relation between above two mentioned events :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example :- For two events A and B, $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and

$$P(A \cup B) = \frac{1}{2} \text{ find } P\left(\frac{A}{B}\right) = ?$$

RRB NTPC CBT - I 08/03/2021 (Evening)

Solution :- Given that,

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{4} \text{ and } P(A \cup B) = \frac{1}{2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{1}{2} = \frac{1}{3} + \frac{1}{4} - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{2} = \frac{4+3-6}{12} = \frac{1}{12}$$

$$\text{Now, } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{1}{12} \times \frac{4}{1} = \frac{1}{3}$$

13. The event A but not B :-

Set of all those elements which are present in A but not in B.

Is denoted as : $A - B = A \cap B'$

14. The event B but not A :-

Set of all those elements which are present in B but not in A.

Is denoted as : $B - A = B \cap A'$

Take an example for clear understanding

Example :- Suppose $\rightarrow S = \{1, 2, 3, 4, 5, 6, 7\}$,

$A = \{2, 3, 5\}$ and $B = \{1, 3, 5\}$

A or B means $A \cup B = \{1, 2, 3, 5\}$

A and B means $A \cap B = \{3, 5\}$

B but not A means $B \cap A' = \{1\}$

A but not B means $A \cap B' = \{2\}$

Not A means $A' = \{1, 4, 6, 7\}$

Not B means $B' = \{2, 4, 6, 7\}$

15. Mutually exclusive events :-

Mutually exclusive events or disjoint events are those events which do not occur at the same time.

Example :-

when a coin is tossed then the result will be either head or tail, but we cannot get both the results.

16. Exhaustive event :-

Exhaustive events are those events whose union is equal to the sample space of the experiment. or

$E_1, E_2, E_3, \dots, E_n$ are said to be Exhaustive events if at least one of them necessarily happens whenever is performed.

Example :- In rolling a dice once, the possible outcomes or the sample space $\rightarrow$

Solution:- $S = \{1, 2, 3, 4, 5, 6\}$

X : a number less than 4 appears

Y : a number greater than 2 or less than 5 appears

Z : a number greater than 4 appears

Now, $X \cup Y \cup Z = \{1, 2, 3\} \cup \{3, 4\} \cup \{5, 6\} = S$

Such events are said to be exhaustive. Here, it can be clearly seen, whenever the experiment will be performed, at least one of them will necessarily occur.

17. Independent events :-

Definition 1 :- Two events are said to be independent if the occurrence of one of them is not affected by the occurrence of the other.

Example :- Consider the experiment of drawing a card from a deck of 52 playing cards, in which The elementary events are assumed to be equally likely. If E and F denote the events 'the card drawn is a spade' and 'the card drawn is an ace' respectively,

$$\text{Therefore, } P(E) = P(F) = \frac{13}{52} = \frac{1}{4}$$

That means, both are equal and are not affected by the occurrence of the other.

Definition 2 :- Two events are said to be independent if

$$P(F|E) = P(F) \text{ provided } P(E) \neq 0$$

$$P(E|F) = P(E) \text{ provided } P(F) \neq 0$$

Definition 3 :- two events are said to be independent if

$$P(E \cap F) = P(E) \times P(F)$$

Coins

General formula to represent the number of outcomes on rolling 'n' coins is 2^n .

Biased And Unbiased Coin

Unbiased coin :- A fair coin having equal probability of getting a head or a tail.

Biased coin :- an unfair coin having unequal probability of getting a head or a tail.

Such coins have more or less probability of getting a particular face.

Unbiased Coin

i) Flipping or tossing of a coin :- Suppose a coin tossed then we get two possible outcomes either a head (H) or a tail (T), there is no other possibility.

and it is impossible to predict whether the result of a toss will be a head or tail.

So, total Possible outcomes = 2

(a) Here, the probability of getting a head $P(H)$

$$\rightarrow \frac{\text{Number of times heads shows up}}{\text{number of total possible outcomes}} = \frac{1}{2}$$

(b) Here, the probability of getting a tail $P(T)$

$$\rightarrow \frac{\text{Number of times tail shows up}}{\text{number of total possible outcomes}} = \frac{1}{2}$$

ii) Flipping or tossing of two coins :- When two coins are tossed then we get a total of four (2^2) outcomes.

Sample Space = $\{(H, H), (H, T), (T, H), (T, T)\}$,

Total number of outcomes = 4

$$P(2H) = P(0T) = \frac{\text{Number of outcome with two heads}}{\text{Total Outcomes}} = \frac{1}{4}$$

$$P(1H) = P(1T) = \frac{\text{Number of outcomes with only one head}}{\text{Total Outcomes}} = \frac{2}{4} = \frac{1}{2}$$

$$P(0H) = P(2T) = \frac{\text{Number of outcome with no heads}}{\text{Total Outcomes}} = \frac{1}{4}$$

ii) Flipping or tossing of three coins :- When 3 coins are tossed then we get a total of eight ($2^3 = 8$) outcomes.

Sample Space = {(H, H, H), (H, H, T), (H, T, H), (T, H, H), (T, T, H), (T, H, T), (H, T, T), (T, T, T)}

Total number of outcomes = 8

$$P(0H) = P(3T) = \frac{\text{Number of outcomes with no heads}}{\text{Total Outcomes}} = \frac{1}{8}$$

$$P(1H) = P(2T) = \frac{\text{Number of outcomes with only one head}}{\text{Total Outcomes}} = \frac{3}{8}$$

$$P(2H) = P(1T) = \frac{\text{Number of outcome with two heads}}{\text{Total Outcomes}} = \frac{3}{8}$$

$$P(3H) = P(0T) = \frac{\text{Number of outcome with three heads}}{\text{Total Outcomes}} = \frac{1}{8}$$

Rolling Dice

* General formula to represent the number of outcomes on rolling 'n' dice is 6^n .

i) Rolling one Dice :- The total number of outcomes on rolling a die is 6, sample space is {1, 2, 3, 4, 5, 6}.

Following are a few probabilities to help in better understanding the concept of probability on rolling one dice.

$$P(\text{Even Number}) = \frac{\text{Number of even number outcomes}}{\text{Total Outcomes}} = \frac{3}{6} = \frac{1}{2}$$

$$P(\text{Odd Number}) = \frac{\text{Number of odd number outcomes}}{\text{Total Outcomes}} = \frac{3}{6} = \frac{1}{2}$$

$$P(\text{Prime Number}) = \frac{\text{Number of prime number outcomes}}{\text{Total Outcomes}} = \frac{3}{6} = \frac{1}{2}$$

ii) Rolling two Dice :- The total number of outcomes on rolling two dice is $6^2 = 36$.

The following are the sample space of 36 outcomes on rolling two dice.

1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Following are a few probabilities to help in better understanding the concept of probability on rolling two dice or rolling a dice twice.

$$P(\text{doublet}) = \frac{6}{36} = \frac{1}{6}$$

$$P(3 \text{ on at least one dice}) = \frac{11}{36}$$

$$P(\text{a sum of 7}) = \frac{6}{36} = \frac{1}{6}$$

Playing Cards

A "standard" deck of playing cards consists of 52 Cards and further categorized into 4 suits of Spade (♠), club (♣), heart (♥), and diamond (♦) constituting 13 cards each.

Each suit contains 13 cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen and King.

Key Points :- Number cards (2,3.....9,10) $\rightarrow 9 \times 4 = 36$,

Face cards (J,Q,K,) $\rightarrow 3 \times 4 = 12$

Ace cards (A) $\rightarrow 1 \times 4 = 4$

Following are few probabilities to help in better understanding the concept of probability on Playing Cards.

The probability of drawing a black card is

$$P(\text{Black card}) = \frac{26}{52} = \frac{1}{2}$$

The probability of drawing a hearts card is

$$P(\text{Hearts}) = \frac{13}{52} = \frac{1}{4}$$

The probability of drawing a face card is

$$P(\text{Face card}) = \frac{12}{52} = \frac{3}{13}$$

The probability of drawing a card numbered 4 is

$$P(4) = \frac{4}{52} = \frac{1}{13}$$

The probability of drawing a red card numbered 4 is

$$P(4 \text{ Red}) = \frac{2}{52} = \frac{1}{26}$$

Example :- Anil placed a deck of 52 cards. If you pick two cards what is the probability that both are aces?

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Solution :- Number of ace cards = 4

When 1 card is taken then, probability that it is ace = $\frac{4}{52} =$

$$\frac{1}{13}$$

remaining cards = 51

Remaining ace cards = 3

Now, out of 51 cards one card is taken, probability that it is

$$\text{ace} = \frac{3}{51} = \frac{1}{17}$$

$$\text{Hence, probability that both cards is aces} = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}$$

Conditional Probability

The probability of an event E is called conditional probability of E, given that F has already occurred and is denoted by $P(E|F)$.

Example :- Tossing of a coin thrice, or tossing three coins simultaneously gives the sample space

$S = \{HHH, THH, HTH, HHT, TTT, TTH, THT, HTT\}$

Let E be the event, two tails appear,

$$\text{therefore } E = \{TTT, HTT, THT, TTH\} \Rightarrow P(E) = \frac{4}{8} = \frac{1}{2}$$

And F be the event in which, first coin shows head, therefore,

$$H = \{HTT, HHH, HHT, HTH\} \Rightarrow P(F) = \frac{4}{8} = \frac{1}{2}$$

$$\text{Also, } E \cap F = \{\text{HTT}\} \Rightarrow P(E \cap F) = \frac{1}{4}$$

Now, suppose that if it is given that the first coin shows tail, i.e., F occurs, and then we have to find the probability of E. Then in this case, we have to find the conditional probability. The conditional probability of E, given that F has already occurred is calculated as follows

$$P(E|F) = \frac{\text{number of elementary events favorable to } E \cap F}{\text{number of elementary events favorable to } F}$$

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

$$\text{Therefore, } P(E|F) = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

Probability By Permutation and Combination

a). The number of permutations of 'n' different things taken 'r' at a time, where repetition is not allowed is given by n

$${}_nP_r = \frac{n!}{(n-r)!}$$

Where, $0 \leq r \leq n$

b). $n! = 1 \times 2 \times 3 \times \dots \times n$

c). $n! = n \times (n-1)!$

d). The number of permutations of 'n' different things taken 'r' at a time, where repetition is allowed is given by

n^r , Where, $0 \leq r \leq n$

e). The number of combinations of 'n' different things taken 'r' at a time, is given by

$${}_nC_r = \frac{n!}{(n-r)! r!}, 0 \leq r \leq n$$

Note :-

$$(i) {}_nC_1 = n \quad \text{and} \quad (ii) {}_nC_0 = 1$$

Example :- if we select 3 persons randomly from a group, consisting of 4 men, 3 women and 2 boys, then the probability that 2 men are selected is:

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Solution :-

Total persons = 9, Men = 4

Formula

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\text{Favorable outcomes} = {}^4C_2 \times {}^5C_1 {}^5C_1$$

$$= 6 \times 5 = 30$$

$$\text{Total outcomes} = {}^9C_3 = \frac{9 \times 8 \times 7}{1 \times 2 \times 3} = 84$$

$$\text{Required probability} = \frac{30}{84} = \frac{5}{14}$$

No. of ways to draw 1 red ball out of 6 = 6C_1

No. of ways to draw 2 white ball out of 4 = 4C_2

No. of ways to draw 3 balls out of 18 = ${}^{18}C_3$

$$\text{Required probability} = \frac{n(E)}{n(S)} = \frac{\text{number of favorable outcome}}{\text{total number of outcome}}$$

$$= \frac{{}^6C_1 \times {}^4C_2}{{}^{18}C_3} = \frac{3}{68}$$

Key points to note :-

a) The sum of the probability of happening of an event and not happening of an event is equal to 1. $P(A) + P(A') = 1$.

b) The probability of an impossible event or the probability of an event not happening is always equal to 0. $P(\phi) = 0$.

c) The probability of a sure event is always equal to 1. $P(A) = 1$

d) The probability of happening of any event always lies between 0 and 1. $0 < P(A) < 1$

Example :- Suppose, a bag contains '6' red, 4 white and 8 blue balls. If three balls are drawn at random, then finding the probability, of 1 red and 2 white balls as

Solution :-

Mean, Median and Mode

Mean

Mean is the average of the given numbers and is calculated by dividing the sum of given numbers by the total number of numbers. The symbol $\bar{X}$ represents the sample mean.

$$\text{Mean } (\bar{X}) = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} = \frac{\Sigma X}{n}$$

where n represents the total no of values in the sample.

Example :- Find the mean of 28, 21, 39, 33, 36, 23.

SSC MTS 05/09/2023 (3rd Shift)

(a) 30 (b) 32 (c) 26 (d) 28

Solution :-

$$\begin{aligned} \text{Mean } (\bar{X}) &= \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \\ &= \frac{28 + 21 + 39 + 33 + 36 + 23}{6} = \frac{180}{6} = 30 \end{aligned}$$

Variety of data

Mean for Raw data or Ungrouped data :-

Let. $X_1 + X_2 + X_3 + \dots + X_n$ are be n raw data

then we use find arithmetic mean

$$\text{Mean } (\bar{X}) = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$$

Example:- What is the arithmetic mean of 4, 8, 12, 16 and 20?

DDA ASO 20/09/2017 (Morning)

(a) 8 (b) 10 (c) 14 (d) 12

Solution :- Arithmetic mean = $\frac{4 + 8 + 12 + 16 + 20}{5} = 12$

Grouped data

Grouping of data plays a significant role when we have to deal with large data. This information can also be displayed using a pictograph and tabular or a bar graph form. data formed by arranging individual observations of a variable into groups, so that a frequency distribution table of these groups provides a convenient way of summarizing or analyzing the data is termed as grouped data.

Mean for discrete frequency distribution :-

$$\bar{X} = \frac{\text{Sum of all observations}}{\text{Total number of observations}} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

Example:- Find the mean of the following data:

x	19	21	23	25	27	29	31
f	13	15	16	18	16	15	13

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(a) 20 (b) 30 (c) 28 (d) 25

Solution :-

x_i	f_i	$f_i x_i$
-------	-------	-----------

19	13	247
21	15	315
23	16	368
25	18	450
27	16	432
29	15	435
31	13	403
	$f_i = 106$	$f_i x_i = 2650$

$$f_i = 106 \quad f_i x_i = 2650$$

$$\text{Mean} = \frac{2650}{106} = 25$$

Median

Median is defined as the value of middlemost observations when the data are arranged in either ascending or descending order. It is denoted by M . In other words, 50% of values of observations in the data are above median and 50% observations have value less than the median.

Median for raw data

For n observations $x_1, x_2, x_3, \dots, x_n$ median is found as follows :

$\Rightarrow$ If n is odd then use the formula

$M = \text{value of the } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$

where, n = total number of observation

Example :- if we have 15 observations, the value of the

$\left(\frac{15+1}{2}\right)^{\text{th}}$ that is 8th observation will be exactly the central value, which is called median.

$\Rightarrow$ If n is even then use the formula :

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

Example :- Find the median of the data

11, 16, 33, 15, 51, 18, 71, 75, 22, 17.

RRC Group D 17/08/2022 (Afternoon)

(a) 18 (b) 24 (c) 20 (d) 22

Solution :- Arranging in ascending order;

11, 15, 16, 17, 18, 22, 33, 51, 71, 75

$$\text{Median} = \frac{\left(\frac{10}{2}\right)^{\text{th}} \text{ term} + \left(\frac{10}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

$$\frac{(5)^{\text{th}} \text{ term} + (6)^{\text{th}} \text{ term}}{2} = \frac{18 + 22}{2} = 20$$

Median for discrete frequency distribution :- Suppose $x_1, x_2, x_3, \dots, x_k$ are the values of a variable with frequencies $f_1, f_2, f_3, \dots, f_k$ respectively.

A frequency distribution generally shows the observations arranged in ascending order. We shall use cumulative frequencies to find the median for a frequency distribution where observations are arranged in ascending order.

Median $M =$ value of the $(\frac{n+1}{2})^{\text{th}}$ observation

where, $n =$ total number of observations

Example :- In the data given below, the number of sixes and the batsmen who hit them, has been shown.

Number of Sixes	Number of batsmen
1	2
2	3
3	1
4	3
5	2

What is the median of the number of sixes ?

SSC MTS 06/08/2019 (Morning)

(a) 3 (b) 5 (c) 4 (d) 4.5

Solution :-

Number of Sixes (x_i)	Number of batsman (f_i)	Cumulative frequency (cf)
1	2	2
2	3	2 + 3 = 5
3	1	5 + 1 = 6
4	3	6 + 3 = 9
5	2	9 + 2 = 11
	$N = \frac{11}{2} = 5.5$	

Since, 5.5 lies in the 6th cumulative frequency zone. Hence, Median of sixes will be 3.

Median for continuous frequency distribution :-

A continuous frequency gives the values of the variable in the form of class intervals and they are generally arranged in ascending order. In such cases, we will use the cumulative frequencies to find the median. These cumulative frequencies will show us the class containing the median. For this, we take

Median class = class containing the $(\frac{n}{2})^{\text{th}}$ observations

where, $n =$ total number of observations

$$\text{Median } M = L + \frac{\frac{n}{2} - cf}{f} \times c$$

where, $L =$ lower boundary point of median class

$cf =$ cumulative frequency of the class prior to median class

$f =$ frequency of the median class

$c =$ length of median class

Note :- When no prior class is there, then in that case $cf = 0$

Example:- Find the median of the given frequency distribution.

Class interval	Frequency
0-10	8
10-20	7
20-30	9
30-40	12
40-50	11

RRC Group D 26/09/2022 (Evening)

(a) $\frac{215}{9}$ (b) $\frac{85}{3}$ (c) $\frac{265}{9}$ (d) $\frac{88}{3}$

Solution :-

class	Frequency	C.F
-------	-----------	-----

0 - 10	8	8
10 - 20	7	15
20 - 30	9	24
30 - 40	12	36
40 - 50	11	47

$$\frac{N}{2} = \frac{47}{2} = 23.5$$

Median class = 20 - 30

$$\text{Median} = l + \frac{\frac{N}{2} - CF}{f} \times h = 20 + \frac{23.5 - 15}{9} \times 10 = \frac{265}{9}$$

Mode

The value which gets repeated the maximum number of times or the value occurring with maximum frequency in the given data is called mode. It is denoted by M_0 .

Example :- The mode of the given data : 4, 4, 9, 9, 9, 5, 5, 5, 5, 5 is 5, as 5 is repeated the maximum number of times.

Mode for raw data and for discrete frequency distribution

:- In these cases the mode can be found simply by inspection. We can find the mode as a value among the observations which is repeated maximum number of times or the one which has maximum frequency.

Mode for continuous frequency distribution :-

The class having the maximum frequency is called as modal class of the frequency distribution.

$$\text{Mode } M_0 = L + \left\{ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right\} \times h$$

Where, $L =$ Lower limit of modal class

$f_1 =$ Frequency of modal class

$f_0 =$ Frequency of previous modal class

$f_2 =$ Frequency of next modal class

$h =$ Class size

By formula,

Note :- The above formula can be used only if the distribution has classes of equal class length. Moreover, the formula can be used only in those cases where the maximum frequency is only for one class.

Example:- Find the mode for the given distribution (rounded off to two decimal places).

Class Interval	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
Frequency	8	7	6	9	11	10

SSC CGL Tier II (02/03/2023)

(a) 35.25 (b) 40.25 (c) 30.33 (d) 28.33

Solution :-

Class Interval	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
Frequency	8	7	6	9	11	10

$$\text{Mode} = L + \left\{ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right\} \times h$$

Where, $L =$ Lower limit of modal class

$f_1 =$ Frequency of modal class

f_0 = Frequency of previous modal class

f_2 = Frequency of next modal class

h = Class size

By formula,

$$\begin{aligned}\text{Mode} &= 25 + \left\{ \frac{11 - 9}{2(11) - 9 - 10} \right\} \times 5 \\ &= 25 + \left\{ \frac{2}{22 - 19} \right\} \times 5 = 25 + \frac{10}{3} = \frac{85}{3} = 28.33\end{aligned}$$

Relation between mean, mode and Median :-

Mode = 3 (Median) - 2 (Mean)

This written in notations as : $M_0 = 3M - 2\bar{X}$

Example:- For a sample data, mean = 60 and median = 48. For this distribution, the mode is:

SSC CGL Tier II (06/03/2023)

- (a) 18 (b) 48 (c) 36 (d) 24

Solution :- Mode = 3 Median - 2 Mean

$$\text{Mode} = 3 \times 48 - 2 \times 60$$

$$\text{Mode} = 24$$

Range

The range is the difference between two extreme observations of the distributions. For eg. if A and B are the greatest and smallest values respectively of observations in a distribution, then its range is A - B.

Range = Highest observation - Lowest observation

Example :- What is the range of the given data?

11, 16, 14, 7, 11, 23, 10, 30, 20, 33, 19, 12, 17, 14

- (a) 26 (b) 25 (c) 24 (d) 19

Solution :- 11, 16, 14, 7, 11, 23, 10, 30, 20, 33, 19, 12, 17, 14

Range of the data = highest term - lowest term

$$\Rightarrow 33 - 7 = 26$$

Variance

Variance is a statistical measurement that is used to the average squared difference between data values and the mean

$$\text{Variance } (\sigma^2) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$\bar{x}$, denoted is data mean

$x_i - \bar{x}$ = calculate the difference of for each value in the data set

Example:- Find the standard deviation of the following data (rounded off to two decimal places).

5, 3, 4, 7

SSC CGL Tier II (03/03/2023)

- (a) 1.48 (b) 3.21 (c) 4.12 (d) 2.45

Solution :- Mean of the data = $\frac{5 + 3 + 4 + 7}{4} = 4.75$

Deviation about mean = 0.25, -1.75, -0.75 and 2.25

Their square = 0.0625, 3.0625, 0.5625 and 5.0625

$$\text{Now, Variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$= \frac{0.0625 + 3.0625 + 0.5625 + 5.0625}{4} = 2.1875$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{2.1875} = 1.48$$

Standard Deviation

If σ^2 is the variance, then σ , is called the standard deviation.

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Example :- If the variance of 5 values is 0.81, then what is its standard deviation ?

SSC MTS 26/10/2021 (Morning)

- (a) 0.09 (b) 0.9 (c) 2.7 (d) .027

Solution :- Standard Deviation = $\sqrt{\text{Variance}} = \sqrt{0.81} = 0.9$

Example :- Find the standard deviation for the following data 1
5, 8, 10, 11, 16

$$\text{Solution :- } \bar{x} = \frac{5 + 8 + 10 + 11 + 16}{5} = \frac{50}{5} = 10$$

evaluate $(x_i - \bar{x})$

$$10 - 5 = 5, 10 - 8 = 2, 10 - 10 = 0, 10 - 11 = -1, 10 - 16 = -6$$

Then use variance formula

$$\begin{aligned}\sigma^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \\ &= \frac{5^2 + 2^2 + 0^2 + (-1)^2 + (-6)^2}{5} = \frac{66}{5}\end{aligned}$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{\frac{66}{5}} = \sqrt{13.2}$$

Coefficient of variation

Coefficient of variation is a type of relative measure of dispersion. It is expressed as the ratio of the standard deviation to the mean. The coefficient of variation is a dimensionless quantity and is usually given as a percentage. It helps to compare two data sets on the basis of the degree of variation.

$$\text{Coefficient of variation} = \frac{\text{standard deviation}}{\text{mean}} \times 100$$

Example:- If the mean is 25 and the standard deviation is 5 then the coefficient of variation is:

RRB NTPC CBT - I 07/01/2021 (Evening)

- (a) 20% (b) 48% (c) 60% (d) 27%

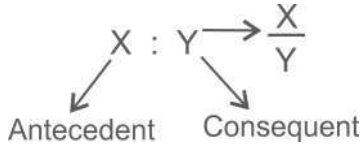
Solution :- Coefficient of Variation = $\frac{5}{25} \times 100 = 20\%$

Ratio and Proportion

Ratio

Ratio is the comparison of two quantities which is obtained by dividing the first quantity by the other.

i.e. $x : y$ can also be written as $\frac{x}{y}$ where $y \neq 0$



* Ratio always occurs between same quantities

Types of Ratio

Mixed Ratio:- Let $x : y$ and $a : b$ be two ratio then mixed ratio is $xa : yb$

Duplicate ratio (square):- Duplicate ratio of $x : y$ will be $x^2 : y^2$

Sub-duplicate Ratio (square root):- Sub-duplicate Ratio of $x : y$ will be $\sqrt{x} : \sqrt{y}$

Triplicate Ratio (cube):- Triplicate ratio of $x : y$ will be $x^3 : y^3$

Sub-triplicate Ratio (cube root) :- Sub-triplicate Ratio of $x : y$ will be $\sqrt[3]{x} : \sqrt[3]{y}$

Inverse ratio :- $x : y$ will be $y : x$

Compound Ratio :- If two or more ratios are given, then the antecedent of one is multiplied with antecedent of the other and respective consequents are also multiplied.

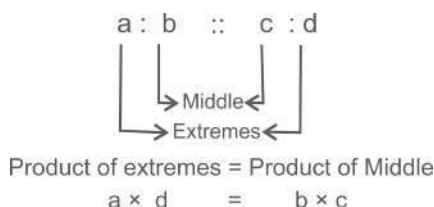
If $a : b$, $c : d$ and $e : f$ are three ratios, then their compound ratio will be $\frac{ace}{bdf}$.

Proportion

Two equivalent ratios are always in proportion.

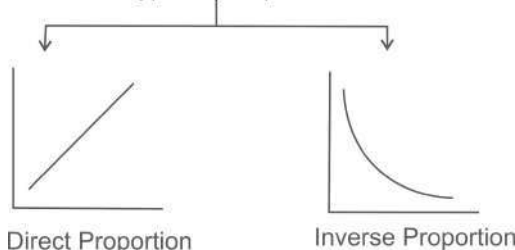
i.e. If $\frac{a}{b} = \frac{c}{d}$ then a, b, c, d are in proportion and written as

$a : b :: c : d$



Types of Proportion

Types of Proportions



Directly Proportional: If one quantity increases, the other quantity also increases and vice-versa.

i.e. If $x = ky$ and k is constant then we can say that x is directly proportional to y .

Inversely Proportional: if one quantity increases, the other quantity decreases and vice-versa.

i.e. If $x = \frac{k}{y}$ and k is constant then we can say that x is inversely proportional to y .

Example :- A varies directly as the positive square root of B, and inversely as the cube of C. If $A = 15$, when $B = 27$ and $C = 2$, then find B when $A = 9$ and $C = 2$.

SSC CGL 05/12/2022 (2nd Shift)

(a) $\frac{281}{42}$ (b) $\frac{243}{25}$ (c) $\frac{264}{37}$ (d) $\frac{275}{51}$

Solution :- $A \propto \sqrt{B} \propto \frac{1}{C^3}$

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 15 = \frac{k \times \sqrt{27}}{2^3}$$

$$k = \frac{120}{3\sqrt{3}} \dots\dots\dots \text{e.q. (1)}$$

Again ,

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 9 = \frac{120 \times \sqrt{B}}{3\sqrt{3} \times 2^3}$$

$$\sqrt{B} = \frac{9\sqrt{3} \times 8}{40} \Rightarrow B = \frac{243}{25}$$

Some results

a) If $ad = bc$, then $\frac{a}{b} = \frac{c}{d}$

b) If $ad > bc$, $\frac{a}{b} > \frac{c}{d}$

c) If $ad < bc$, $\frac{a}{b} < \frac{c}{d}$

d) If $\frac{a}{b} = \frac{c}{d}$ then $\frac{a+b}{b} = \frac{c+d}{d}$

e) If $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$ or $\frac{a}{b} - 1 = \frac{c}{d} - 1$

f) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ then $\frac{a-c}{b-d} = \frac{c-e}{d-f} = \frac{a-e}{b-f}$

g) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ then $\frac{a+c+e}{b+d+f} = \frac{a+c}{b+d} = \frac{c+e}{d+f}$

Example :- If $a : b = c : d = e : f = g : h = 1 : 3$, then what will be the value of the following :

$(pa + qc + re + sg) : (pb + qd + rf + sh)$

RRB NTPC CBT - I 05/04/2021 (Evening)

(a) 1 : 5 (b) 1 : 2 (c) 1 : 3 (d) 1 : 4

Solution :- Given ,

$a : b = c : d = e : f = g : h = 1 : 3$ Then ,

$(pa + qc + re + sg) : (pb + qd + rf + sh)$

$(p \times 1 + q \times 1 + r \times 1 + s \times 1) : (p \times 3 + q \times 3 + r \times 3 + s \times 3)$

$(p + q + r + s) : 3(p + q + r + s) \Rightarrow 1 : 3$

h) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{f}{g} = K$, where K is constant .

$$\text{Then, } K = \frac{a+c+e+f}{b+d+f+g}$$

i) For continued proportion,

$$\frac{a}{b} = \frac{c}{d} = k, a = bk, c = dk$$

j) **Mean Proportion** : Mean proportion of two numbers, a and b, is given by $\sqrt{ab}$.

Example:- The mean proportional between

$(3 + \sqrt{5})$ and $(15 - \sqrt{125})$ is _____.

Selection Post 27/06/2023 (2nd Shift)

(a) $3\sqrt{5}$ (b) $5\sqrt{5}$ (c) $9\sqrt{5}$ (d) $2\sqrt{5}$

Solution:-

$$\begin{aligned}\text{Mean proportional} &= \sqrt{(3 + \sqrt{5})(15 - \sqrt{125})} \\ &= \sqrt{(3 + \sqrt{5})5(3 - \sqrt{5})} = \sqrt{5(9 - 5)} = 2\sqrt{5}\end{aligned}$$

k) 3rd Proportional:- 3rd proportion of a and b i.e. $a : b :: b : x$

$$\Rightarrow \text{Then, } x = \frac{b^2}{a}$$

Example :- Third proportion of 15 and 120 is:

SSC CHSL 04/08/2023 (4th Shift)

(a) 880 (b) 960 (c) 900 (d) - 860

Solution :-

$$\text{Third proportional} = \frac{120 \times 120}{15} = 960$$

l) 4th Proportional: 4th proportional of a, b and c i.e. $a : b :: c : x$

$$\Rightarrow \text{Then, } x = \frac{b \times c}{a}$$

Example :- Find the fourth proportional of 12^{35} , 3^{12} and 4^{12} .

SSC MTS 14/09/2023 (3rd Shift)

(a) 12^{23} (b) 3^{24} (c) 12^{11} (d) 4^{24}

Solution :-

$$\begin{aligned}\text{Fourth proportional} &= \frac{3^{12} \times 4^{12}}{12^{35}} \\ \Rightarrow \frac{3^{12} \times 4^{12}}{3^{35} \times 4^{35}} &= 3^{-23} \times 4^{-23} = (3 \times 4)^{-23} = 12^{-23}\end{aligned}$$

m) **Componendo and Dividendo rule:**

$$\text{If } a : b :: c : d, \text{ Then } \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Some important results

1) If the numerator and denominator of a ratio is multiplied and divided by the same number, the ratio remains unchanged

$$\text{i.e. } \frac{a}{b} = \frac{a \times k}{b \times k} = \frac{\frac{a}{k}}{\frac{b}{k}}$$

Ratio is independent of change in scale only and not of origin.

2) If there is 'n' number of unequal fractions such as $\frac{a_1}{b_1}, \frac{a_2}{b_2},$

$\frac{a_3}{b_3}, \frac{a_4}{b_4}, \dots, \frac{a_n}{b_n}$ then the ratio of $(a_1+a_2+a_3+\dots+a_n)$ and $(b_1+b_2+b_3+\dots+b_n)$ lies between lowest and highest values of these fractions.

Example :- Three unequal fractions are $\frac{1}{2}, \frac{2}{3}, \frac{4}{7}$. So, the ratio of $(1 + 2 + 4) : (2 + 3 + 7) = 7:12$. Among these fractions, lowest fraction is $\frac{1}{2}$ and the highest fraction is $\frac{4}{7}$. So, the required value lies between $\frac{1}{2}$ and $\frac{4}{7}$.

3) If an amount A is to be divided between x and y in the ratio m : n then

$$x = \frac{m}{m+n} \times A \text{ and } y = \frac{n}{m+n} \times A$$

Example :- Two numbers are in the ratio 5 : 7

If the sum of the numbers is 108, find the smaller number.

RRC Group D 30/09/2022 (Afternoon)

(a) 42 (b) 47 (c) 36 (d) 45

Solution :-

$$\text{Smaller number} = \frac{5}{5+7} \times 108 = 45$$

4) If an ratio of amount of x and y is m : n and their difference of their share is A then

$$x = \frac{m}{m-n} \times A \text{ and } y = \frac{n}{m-n} \times A$$

Example:- Two numbers are in the ratio 5 : 3 and the difference between these two numbers is 34. Find the smaller of the two numbers.

SSC CPO 03/10/2023 (1st Shift)

(a) 68 (b) 34 (c) 51 (d) 85

Solution:-

$$\text{Smaller number} = \frac{3}{5-3} \times 34 = 51$$

5) If $aA = bB = cC$, where a,b,c are constants and A, B, C are variables. Then the ratio of $A : B : C = b \times c : c \times a : a \times b$.

Or

If $aA = bB = cC = dD$, where a,b,c,d are constants and A, B, C, D are variables. Then the ratio of $A : B : C : D = b \times c \times d : c \times a \times d : a \times b \times d : a \times b \times c$.

Example:- If $2P = 3Q = 4R = 5S$, then find $P : Q : R : S$.

Selection Post 28/06/2023 (3rd Shift)

(a) $30 : 20 : 12 : 15$ (b) $12 : 20 : 30 : 15$

(c) $20 : 30 : 15 : 12$ (d) $30 : 20 : 15 : 12$

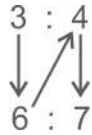
Solution:- $2P : 3Q : 4R : 5S = k$

$$P : Q : R : S = 3 \times 4 \times 5 : 4 \times 5 \times 2 : 2 \times 3 \times 5 : 2 \times 3 \times 4.$$

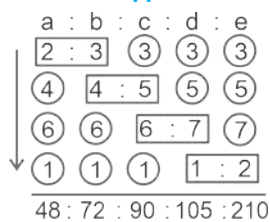
$$P : Q : R : S = 60 : 40 : 30 : 24 = 30 : 20 : 15 : 12$$

Type - 1

(a) When two or three ratios are given:

Example:- If $a : b = 3 : 4$, $b : c = 6 : 7$ then $a : b : c = ?$ **Solution:-** $a : b = 3 : 4$, $b : c = 6 : 7$.LCM of the common entities i.e $b = (4, 6) = 12$ $a : b = 3 : 4 \Rightarrow \times 3 = 9 : 12$ $b : c = 6 : 7 \Rightarrow \times 2 = 12 : 14$ So, $a : b : c = 9 : 12 : 14$ **Exam hall Approach** $3 \times 6 : 6 \times 4 : 4 \times 7 = 18 : 24 : 28$ $a : b : c = 9 : 12 : 14$

(b) When more than three ratios are given:

Example:- If $a : b = 2 : 3$, $b : c = 4 : 5$, $c : d = 6 : 7$, $d : e = 1 : 2$ then $a : b : c : d : e = ?$ **Solution:-****Exam Hall approach:-****Example:-** A sum of Rs. 12000 is divided between A, B, C and D such that the ratio of share of A and B is 8 : 9, that of B and C is 2 : 3 and that of C:D is 9 : 13. What is the difference between the shares of B and D ?

SSC CPO 13/12/2019 (Morning)

- (a) Rs. 1320 (b) Rs. 2400
(c) Rs. 2520 (d) Rs. 2760

Solution:-Ratio $\rightarrow A : B : C : D$ $8 : 9 : 9 : 9$ $2 : 2 : 3 : 3$ $9 : 9 : 9 : 13$ $16 : 18 : 27 : 39$

On adding the ratios = 100 units

100 units = 12000 $\Rightarrow$ 1 unit = 120difference between the shares of B and D = $21 \times 120 = \text{Rs } 2520$ **Example:-** If $P : Q = 3 : 4$ and $Q : R = 4 : 7$, then find the value of $(P + Q) : (Q + R) : (R + P)$.

SSC CGL Tier II (26/10/2023)

- (a) 11 : 4 : 7 (b) 10 : 3 : 11
(c) 3 : 4 : 7 (d) 7 : 11 : 10

Solution:- $P : Q : R = 3 : 4 : 7$ Hence, $(P + Q) : (Q + R) : (R + P) = (3 + 4) : (4 + 7) : (7 + 3) = 7 : 11 : 10$

Type - 2

Example:- If x is subtracted from each of 52, 47, 20 and 19, the numbers so obtained in this order are in proportion. What is the mean proportional between $(x + 13)$ and $(x - 8)$?

SSC MTS 13/10/2021 (Morning)

- (a) 12 (b) 10 (c) 15 (d) 9

Solution:- According to the question,

$$\Rightarrow 52 - x : 47 - x :: 20 - x : 19 - x$$

$$\Rightarrow \frac{(52 - x)}{(47 - x)} = \frac{(20 - x)}{(19 - x)}$$

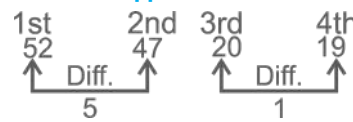
$$\Rightarrow (52 - x)(19 - x) = (47 - x)(20 - x)$$

$$\Rightarrow 988 - 71x + x^2 = 940 - 67x + x^2$$

$$\Rightarrow 48 = 4x \Rightarrow x = 12$$

Mean proportional between $(x + 13)$ and $(x - 8)$

$$= \sqrt{(12 + 13)(12 - 8)} = \sqrt{100} = 10$$

Exam hall approach :-

$$\text{Direct value of } x = \frac{5 \times (3\text{rd or } 4\text{th no.}) - 1 \times (1\text{st or } 2\text{nd no.})}{5 - 1}$$

$$\Rightarrow x = \frac{100 - 52}{4} = \frac{48}{4} = 12$$

Now,

$$\text{Mean proportion} = \sqrt{(x + 13)(x - 8)}$$

$$= \sqrt{25 \times 4} = 5 \times 2 = 10$$

Type - 3

When any sum/things distributed in two or more parts

Example:- If Rs. 825 is divided into three parts in a proportion of $\frac{1}{3} : \frac{4}{9} : \frac{3}{4}$, then the first part will be :

RRB NTPC CBT - I 08/03/2021 (Morning)

- (a) Rs.190 (b) Rs.204
(c) Rs.180 (d) Rs.210

Solution:- Rs. 825 is divided into three parts in a proportion of $\text{LCM}(3,9,4) = 36$

$$\frac{1}{3} \times 36 : \frac{4}{9} \times 36 : \frac{3}{4} \times 36$$

$$= 12 : 16 : 27 ;$$

$$\text{Then the first part will be} = 825 \times \frac{12}{55} = 180 \text{ Rs.}$$

Example:- Rs. 43,120 is divided among P, Q and R such that P's share is $\frac{4}{7}$ of total share of Q and R together and Q's share is $\frac{2}{5}$ of total share of P and R together. What is the share of R ?

SSC CHSL 21/03/2023 (3rd Shift)

- (a) Rs.15120 (b) Rs.14220
(c) Rs.16400 (d) Rs.18050

Solution:-Ratio $\rightarrow$

$$P : Q + R = 4 : 7 \Rightarrow \times 7$$

$$Q : P + R = 2 : 5 \Rightarrow 11$$

Total amount of P, Q, R = 77 units

$$\text{And R's unit} = 77 - (7 \times 4 + 11 \times 2) = 27 \text{ units}$$

$$77 \text{ units} = ₹43120$$

$$(R) 27 \text{ units} = \frac{43120}{77} \times 27 \rightarrow ₹15120$$

Type - 4

Based on Coins/Notes

When number of coins or notes asked:

Example:- In a wallet, there are coins of Rs. 1, Rs. 2 and Rs. 5 in the ratio of 2 : 5 : 3 respectively. If there is Rs. 54 in all, then how many Rs. 5 coins are there?

SSC MTS 09/05/2023 (Afternoon)

(a) 20 (b) 10 (c) 4 (d) 6

Solution:- Cost $\rightarrow ₹1 : ₹2 : ₹5$

$$\text{Ratio} \rightarrow 2x : 5x : 3x$$

$$\text{Total amount} = 2x + 10x + 15x = 27x$$

$$\Rightarrow 27x = ₹54$$

$$\text{Number of ₹5 coins (3x)} = \frac{54}{27x} \times 3x = 6$$

When Total amount asked:

Example:- In a bag there are coins of ₹5, ₹10 and ₹20 denominations. The total number of coins in the bag is 240. If the number of coins of ₹5, ₹10, ₹20 denominations are in the ratio of 2 : 3 : 5, then what is the total amount of money in the bag?

SSC MTS 12/07/2022 (Afternoon)

(a) ₹3,540 (b) ₹4,600

(c) ₹4,620 (d) ₹3,360

Solution :-

Denomination of coin: ₹5 : ₹10 : ₹20

$$\text{Ratio of coin} \rightarrow 2x : 3x : 5x$$

$$\text{Value} \rightarrow 10x : 30x : 100x$$

$$10x = 240 \Rightarrow x = 24$$

Now total amount of money in the bag

$$= 10x + 30x + 100x$$

$$= 140x = 140 \times 24 = ₹3360$$

Type - 5

Addition and subtraction

Increase / decrease in numbers:

Example:- Two numbers are in the ratio 2 : 7. If the first number is increased by 14 and the second number is decreased by 6, then the ratio becomes 4 : 3. What is the sum of the original two numbers?

RRC Group D 02/09/2022 (Morning)

(a) 45 (b) 27 (c) 36 (d) 18

Solution:- Let the numbers are $2x$ and $7x$.

$$\Rightarrow \frac{2x + 14}{7x - 6} = \frac{4}{3}$$

$$\Rightarrow 6x + 42 = 28x - 24$$

$$\Rightarrow 28x - 6x = 42 + 24$$

$$\Rightarrow 22x = 66 \Rightarrow x = 3$$

$$\text{Sum of the numbers} = 2x + 7x = 6 + 21 = 27$$

Increase/ decrease in Population:-

Example:- The population of a town increased by 10% and 20% in two successive years, but decreased by 25% in the third year. Find the ratio of the population in the third year and the population 3 years back.

RRB JE 01/06/2019 (Morning)

(a) 1 : 1 (b) 2 : 1 (c) 99 : 100 (d) 100 : 99

Solution:-

Initial : final

Population of first year $\rightarrow 10 : 11$

Population of second year $\rightarrow 5 : 6$

Population of third year $\rightarrow 4 : 3$

Final year $\rightarrow 100 : 99$

Ratio of population after 3 year and before = 99 : 100

Type - 6

Income & Expenditure

$$\text{Income} = \text{Expenditure} + \text{Saving}$$

**Expenditure $\uparrow$ then
Saving $\downarrow$**

**Expenditure $\downarrow$ then
Saving $\uparrow$**

Example:- If the ratio of Income of 2 companies is 7 : 5, while it was found that the ratio of expenditure is 5 : 3, If both the companies save Rs. 8,000 in each month. Find sum of monthly income of both the companies?

SBI Clerk Pre 19/11/2022 (2nd Shift)

(a) ₹. 50000 (b) ₹. 48000 (c) ₹. 60000

(d) ₹. 65000 (e) ₹. 72000

Solution:-

$$\begin{array}{lcl} \text{Ratio} \rightarrow & \text{Company 1} & \text{Company 2} \\ \text{Income} \rightarrow & 7 & 5 \\ \text{Expenditure} \rightarrow & 5 & 3 \\ \hline \text{saving} \rightarrow & 8000 & 8000 \end{array}$$

$$\text{Ratio (25 - 21) unit} = 40000 - 24000$$

$$\Rightarrow 4 \text{ unit} = 16000 \Rightarrow 1 \text{ unit} = 4000$$

Sum of income of both companies

$$12 \text{ unit} = 4000 \times 12 = ₹48000$$

Example:- The expenditure and savings of a person are in a ratio of 4 : 3. His savings increase by $\frac{1}{5}$ of his income and income remains the same. What is the new ratio of his expenditure and savings?

SSC CHSL 10/03/2023 (1st Shift)

- (a) 12 : 17 (b) 14 : 17
(c) 13 : 22 (d) 15 : 13

Solution:- According to the question,

Ratio → Income : expenditure : savings

Initial → 7 : 4 : 3

Change → -1.4 : +1.4

Final → 7 : 2.6 : 4.4

His savings increased $\frac{1}{5}$ th of their total income and income remains same

Therefore, the new ratio of his expenditure and savings = 2.6 : 4.4 = 13 : 22

Example:- The ratio between the monthly income and the expenditure of Vaidic is 8 : 5. If his income increases by 20% and his expenditure increases by 30%, then find the percentage increase or decrease in his monthly savings.

SSC CGL 14/07/2023 (4th shift)

- (a) $5\frac{2}{3}\%$ increase (b) $5\frac{2}{3}\%$ decrease
(c) $3\frac{1}{3}\%$ increase (d) $3\frac{1}{3}\%$ decrease

Solution:-

Ratio - Income = expenditure + savings

Initial - 80 = 50 + 30

Final - 96 = 65 + 31

Change % in savings = $\frac{1}{30} \times 100 = 3\frac{1}{3}\%$ increased

Example:- The ratio of the incomes of A and B in the last year was 4 : 3. The ratios of their individual incomes in the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income in the present year is ₹24.12 lakhs, then the sum of the income (in ₹ lakhs) of A in the last year and that of B in the present year is:

SSC CGL Tier II (29/01/2022)

- (a) 20.52 (b) 10.98 (c) 22.17 (d) 21.28

Solution:- Ratio → Last year Income

A : B = 4 : 3

Now,

Last Year : Present Year

A → $\times 4 \times 5$ (3 : 4)

B → $\times 3 \times 3$ (5 : 6)

Last Year : Present Year

A → 60 : 80

B → 45 : 54

According to the question,

(80 + 54) units = 24.12 lakh

1 unit = 0.18 lakh

Last year income of A + Present year income of B = (60 + 54) units = 114 units

114 units = 114 × 0.18 = 20.52 Lakh

Type - 7

Age

Addition of ages in given ratio of ages:-

Example:- The ratio of present ages of Manjeet and Aniruddh is 3 : 4. The ratio of their ages 27 years hence would be 9 : 11. What would be the present age of Aniruddh?

IBPS PO Pre 15/10/2022 (2nd Shift)

- (a) 36 years (b) 45 years (c) 54 years (d) 72 years
(e) 41 years

Solution:- According to the question,

Let the present age of Manjeet and Aniruddh be $3x$ and $4x$

$$\frac{3x + 27}{4x + 27} = \frac{9}{11}$$

$$33x + 297 = 36x + 243$$

$$x = \frac{297 - 243}{3} = 18$$

So the present age of Aniruddh be = $18 \times 4 = 72$ years

Exam hall approach :-

	Manjeet	Aniruddh
Present ratio	→ 3	: 4) × 2
	+27	
New ratio	→ 9	: 11) × 1

Total change ($3x$) = 27 ⇒ $x = 9$

Now, Present age of Aniruddha = $8 \times 9 = 72$ years

Subtraction of ages in given ratio of ages:-

Example:- At present, A is younger than B by 8 years. If 4 years ago, their ages were in the ratio of 1 : 2, then what is the present age of B (in years)?

SSC CGL 12/04/2022 (Morning)

- (a) 11 (b) 20 (c) 12 (d) 18

Solution:- Let the present ages of A and B be a and b years respectively.

According to the question,

$$\frac{a - 4}{b - 4} = \frac{1}{2}$$

$$2a - 8 = b - 4 \Rightarrow 2a - b = 4 \dots\dots(1)$$

$$\text{Also, } a = b - 8 \Rightarrow a - b = -8 \dots\dots(2)$$

On solving equations (1) and (2), we get

$a = 12$ and $b = 20$

So present age of B = 20 years

Note:- difference of ages between A and B is always same

	A	:	B	
	1	:	2	→ 1 unit = 8 year
-4year	↑		↑	-4year
present age:-	8 + 4	:	16 + 4	
	12	:	20	

Addition and Subtraction of ages in given ratio of ages:-

Example:- The sum of the present ages of a father and his son is 94 years. 8 years ago their respective ages were in the ratio of 2 : 1. After 10 years what will be the ratio of ages of father and son?

SSC CHSL 17/03/2023 (4th Shift)

(a) 35 : 22 (b) 14 : 13 (c) 25 : 18 (d) 16 : 15

Solution:- Let current age of father be x and son be y

$$x + y = 94 \text{ -----e.q. (1)}$$

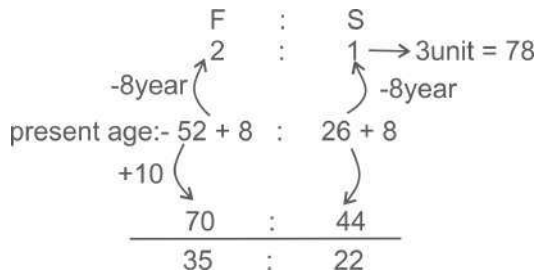
$$(8 - x) = 2(8 - y)$$

$$2y - x = 8 \text{ -----e.q. (2)}$$

On adding e.q. (1) and e.q. (2); we get

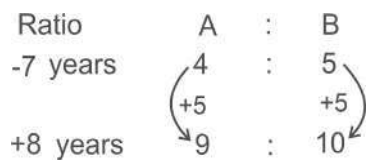
$$y = 34 \text{ and } x = 60$$

$$\text{Required ratio} = 70 : 44 = 35 : 22$$

**Example:-** Seven years ago, the ratio of the ages of A and B was 4 : 5. Eight years hence, the ratio of the ages of A and B will be 9 : 10. What is the sum of their present ages in years?

SSC CPO 23/11/2020 (Morning)

(a) 41 (b) 82 (c) 32 (d) 56

Solution:-

According to question ,

$$5 \text{ units} = 15 \text{ years} \rightarrow 1 \text{ unit} = 3 \text{ years}$$

Sum of their present age =

$$(4 + 5) \text{ units} + 14 \text{ years}$$

$$= 9 \times 3 + 14 = 41 \text{ years}$$

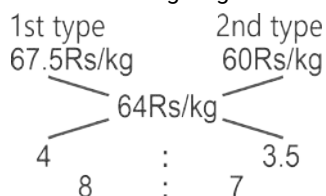
Type - 8

Average

Example:- In what ratio should rice costing Rs.67.5 per kg be mixed with rice costing Rs.60 per kg to get an average price of Rs.64 per kg?

SSC CHSL Tier II (26/06/2023)

(a) 8 : 7 (b) 5 : 6 (c) 6 : 5 (d) 7 : 8

Solution:- Using alligation method ,

Type - 9

When distribution is wrong

Example:- By mistake, instead of dividing ₹702 among Ram, Ramesh, and Naresh in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{6}$, it was divided in the ratio of 3 : 4 : 6. Who gained the most and by how much?

SSC CGL 02/12/2022 (1st Shift)

(a) Ram ₹158 (b) Ramesh ₹158

(c) Naresh ₹168 (d) Ram ₹168

Solution:-

Ram : Ramesh : Naresh

$$\frac{1}{3} : \frac{1}{4} : \frac{1}{6} \Rightarrow 4 : 3 : 2$$

Ratio → Ram : Ramesh : Naresh

Initial → 4 : 3 : 2 = 9 units × 13

Final → 3 : 4 : 6 = 13 units × 9

Make units equal as money divided is same

Ram : Ramesh : Naresh

52 : 39 : 26 = 117 units

27 : 36 : 54 = 117 units

-25u -3u +28u

According to question ,

117 units = Rs. 702

Then , 28 units = Rs 168

Clearly , Naresh got Rs.168 More

Ratio of area/volume

Example:- The radii of two cylinders are in the ratio 2 : 1 and their heights are in the ratio 5 : 2. The ratio of their volumes is:

SSC CHSL Tier II (26/06/2023)

(a) 20 : 1 (b) 15 : 1 (c) 10 : 1 (d) 5 : 1

Solution:-

$$\text{Volume of cylinder} = \pi r^2 h$$

Ratio → Initial : Final

$$(\text{radius})^2 \rightarrow (2)^2 : (1)^2$$

$$\text{height} \rightarrow 5 : 2$$

$$\text{Volume} \rightarrow 20 : 2 = 10 : 1$$

Miscellaneous

Example:- The cost of 3 cups and 5 plates is ₹1,080 and the cost of 2 cups and 4 plates is ₹840. Find the ratio of the cost of a cup to the cost of a plate.

SSC CHSL 03/08/2023 (3rd Shift)

(a) 1 : 3 (b) 2 : 3 (c) 1 : 1 (d) 1 : 2

Solution:-

$$3C + 5P = 1080 \text{ ₹ ... e.q. (1)}$$

$$2C + 4P = 840 \text{ ₹ ... e.q. (2)}$$

$$C + P = 240 \text{ ₹}$$

$$\Rightarrow 2C + 2P = 480 \text{ ₹ ...e.q. (3)}$$

On solving e.q. (3) and e.q. (2) , we get

$$2P = 360 \Rightarrow P = 180 \text{ ₹ and } C = 60 \text{ ₹}$$

$$\text{Required ratio} = 1 : 3$$

Example:- The cost of a diamond is directly proportional to the square of its weight. The cost of a 14 gm diamond is ₹2560. This diamond got broken down into two pieces in the ratio of 5 : 9. How much loss percent is incurred due to this breakage? (correct to two decimal places)

SSC CGL Tier II (08/08/2022)

- (a) 53.47 percent (b) 49.71 percent
(c) 45.92 percent (d) 55.41 percent

Solution:- It is given that, $CP \propto W^2$

$$\text{Initial weight of a diamond} = (5 + 9)^2 \\ = 14^2 = 196 \text{ unit}$$

After break down, weight of a diamond

$$= 5^2 + 9^2 = 25 + 81 = 106 \text{ unit}$$

Required loss% in weight of diamond

$$= \frac{196 - 106}{196} \times 100$$

$$= \frac{90}{196} \times 100 = 45.918 \sim 45.92\%$$

$$\frac{1}{8} = \frac{(10 + 2x) - (20 + x)}{10 + 2x}$$

$$\frac{1}{8} = \frac{x - 10}{10 + 2x} \Rightarrow 10 + 2x = 8x - 80$$

$$6x = 90 \Rightarrow x = 15$$

So, required ratio = 10 : x = 10 : 15 = 2 : 3

Example:- In a school, $\frac{5}{12}$ of the number of students are girls and the rest are boys, $\frac{4}{7}$ of the number of boys are below 14 years of age, and $\frac{2}{5}$ of the number of girls are 14 years or above 14 years of age. If the number of students below 14 years of age is 1120, then the total number of students in the school is:

SSC CPO 25/11/2020 (Evening)

- (a) 1820 (b) 1900 (c) 1920 (d) 1290

Solution:-

Ratio → Boys : Girls = 7 : 5

$$\text{Girls below 14 years of age} = 5 \times \frac{3}{5} = 3$$

$$\text{Boys below 14 years of age} = 7 \times \frac{4}{7} = 4$$

According to question ,

$$(3 + 4) = 7 \text{ units} = 1120$$

$$1 \text{ unit} = 160$$

$$\text{Total Students } (7 + 5) = 12 \times 160 = 1920$$

Example:- A man ordered 10 physics books and some chemistry books. The price of a chemistry book is twice the price of a Physics book. While preparing the bill the clerk interchanged the number of physics and chemistry books by mistake, which decreased the bill by $12\frac{1}{2}\%$. The ratio of the number of physics books to the number of chemistry books in the original order is:

SSC MTS 12/10/2021 (Evening)

- (a) 3 : 5 (b) 3 : 4 (c) 4 : 5 (d) 2 : 3

Solution:-

Let the number of chemistry books = x

let, the price of each physics book and each chemistry book be 1 unit and 2 units respectively.

$$\text{So, total correct amount} = 10 \times y + x \times 2 = 10 + 2x$$

$$\text{Total incorrect amount} = 10 \times 2 + x \times 1 = 20 + x$$

According to the question,

$$12.5 = \frac{(10 + 2x) - (20 + x)}{10 + 2x} \times 100$$

Percentage

A percentage is a number or ratio that can be expressed as a fraction of 100.

Example :- 20% mean ?

20% means $\frac{20}{100}$ number.

Also, you can see $\frac{20}{100} = \frac{1}{5} = 0.2$

i.e. 0.2 of a number is the same as 20% of that thing.

Percentages are really just another way to write a fraction.

i.e. 20% is the same as $\frac{1}{5}$.

$$\text{Percentage} = \frac{\text{Value}}{\text{Total value}} \times 100$$

Important Concept

- Percentage change = $\frac{\text{final value} - \text{initial value}}{\text{initial value}} \times 100$
- If the price of product increases or decreases by a% then the decrease or increase in us to remain expenditure same is $[\frac{a}{(100 \pm a)} \times 100] \%$. ('+' for price increase and '-' for price decrease)
- If the number is first increased by x% and then decreased by x%, then the net result will always decrease and it is equal to $(-\frac{x^2}{100})\%$. Same result will be obtained in the vice-versa case.

Successive percentage change :-

- Two successive percentage change : If the number is first increase or decrease by a% and then increase or decrease by b% then, Effective percentage change

$$2. = [\pm a \pm b + \frac{(\pm a)(\pm b)}{100}] \%$$

(+ve sign is used for increase, - ve sign is used for decrease)

- Three successive percentage change : If the number is first increased by a% then by b% and then by c%, then Effective percentage change =

$$(a + b + c + \frac{ab + bc + ca}{100} + \frac{abc}{10000}) \%$$

Note :- if a = b = c then,

$$\text{Effective percentage change} = (3a + \frac{3a^2}{100} + \frac{a^3}{10000}) \%$$

Important Fractions and their Equivalent Percentage

Fraction	Percentage	Decimal
1	100%	-
$\frac{1}{2}$	50%	0.5

$\frac{1}{3}$	$33\frac{1}{3}\%$	33.33%
$\frac{1}{4}$	25%	0.25
$\frac{1}{5}$	20%	0.2
$\frac{1}{6}$	$16\frac{2}{3}\%$	16.67%
$\frac{1}{7}$	$14\frac{2}{7}\%$	14.28%
$\frac{1}{8}$	$12\frac{1}{2}\%$	12.5%
$\frac{1}{9}$	$11\frac{1}{9}\%$	11.11%
$\frac{1}{10}$	10%	0.1
$\frac{1}{11}$	$9\frac{1}{11}\%$	9.09%
$\frac{1}{12}$	$8\frac{1}{3}\%$	8.33%
$\frac{1}{13}$	$7\frac{9}{13}\%$	7.69%
$\frac{1}{14}$	$7\frac{1}{7}\%$	7.14%
$\frac{1}{15}$	$6\frac{2}{3}\%$	6.67%
$\frac{1}{16}$	$6\frac{1}{4}\%$	6.25%
$\frac{1}{17}$	$5\frac{15}{17}\%$	5.88%
$\frac{1}{18}$	$5\frac{5}{9}\%$	5.55%
$\frac{1}{19}$	$5\frac{5}{19}\%$	5.26%
$\frac{1}{20}$	5%	0.05
$\frac{1}{21}$	$4\frac{16}{21}\%$	4.76%
$\frac{1}{22}$	$4\frac{6}{11}\%$	4.54%
$\frac{1}{23}$	$4\frac{8}{23}\%$	4.34%
$\frac{1}{24}$	$4\frac{1}{6}\%$	4.16%
$\frac{1}{25}$	4%	0.04
$\frac{1}{40}$	$2\frac{1}{2}\%$	2.5%

Some Important Derived Fraction

Fraction	Percentage	Decimal
$\frac{3}{8}$	$37\frac{1}{2}\%$	37.5%

$\frac{5}{8}$	$62\frac{1}{2}\%$	62.5%
$\frac{7}{8}$	$87\frac{1}{2}\%$	87.5%
$\frac{3}{11}$	$27\frac{3}{11}\%$	27.27%
$\frac{2}{3}$	$66\frac{2}{3}\%$	66.66%
$\frac{7}{3}$	$233\frac{1}{3}\%$	233.33%
$\frac{9}{8}$	$112\frac{1}{2}\%$	112.5%
$\frac{2}{7}$	$28\frac{4}{7}\%$	28.57%
$\frac{6}{7}$	$85\frac{5}{7}\%$	85.71%
$\frac{5}{6}$	$83\frac{1}{3}\%$	83.33%
$\frac{11}{12}$	$91\frac{2}{3}\%$	91.66%

These above fractions are very important and it will be very helpful to learn them.

Some Important Points

1. Increasing a number by 35%

35% means $\frac{35}{100}$ therefore adding 35 to the original value 100 gives $\frac{100 + 35}{100} = \frac{135}{100} = \frac{27}{20}$

Example :- A number after increasing by 36 becomes equal to its 109% then what is that number ?

RRB ALP Tier - II (23/01/2019) Morning

(a) 300 (b) 360 (c) 400 (d) 450

Solution :- increase in percentage = 9%

9% = 36

$$100\% = \frac{36}{9} \times 100 = 400$$

2. Decreasing a number by 42%

42% means $\frac{42}{100}$ therefore decreasing 42 from the original value 100 gives $\frac{100 - 42}{100} = \frac{58}{100} = \frac{29}{50}$

Example :- A number when decreased by $22\frac{1}{2}\%$ becomes

217. The number is :

RRB NTPC CBT - I 29/12/2020 (Evening)

(a) 420 (b) 280 (c) 212 (d) 315

Solution:- $22\frac{1}{2}\% = \frac{9}{40}$

Initial : Final

Number → 40 : 31

31 units = 217,

$$\text{So, the initial number (40 units)} = \frac{217}{31} \times 40 = 280$$

3. If any quantity is increased by $x\%$, then to return to the

original value, we must decrease it by $\frac{x}{100 + x} \times 100\%$

Example :- if 100 is increased by 25%, then to return to the original value, we must decrease it by 20% as shown:

Solution :-

$$\frac{25}{100 + 25} \times 100\% = \frac{25}{125} \times 100\% = 20\%$$

$$\text{Or } 100 \xrightarrow{+25\%} 125 \xrightarrow{-20\%} 100$$

4. If P is greater than Q by $y\%$, then Q is smaller than P by

$$\frac{y}{100 + y} \times 100\%$$

Example :- if P is greater than Q by 40%, then Q is smaller than P by

Solution :-

$$\frac{40}{100 + 40} \times 100\% = 28.57\%$$

5. If any quantity is decreased by $x\%$, then to return to the

original value, we must increase it by $\frac{x}{100 - x} \times 100\%$

Example :- if 100 is decreased by 20%, then to return to 100, we must

Solution :-

$$\text{increase it by } \frac{20}{100 - 20} \times 100\% = \frac{20}{80} \times 100\% = 25\%$$

$$\text{Or } 100 \xrightarrow{-20\%} 80 \xrightarrow{+25\%} 100$$

6. If P is smaller than Q by $y\%$, then Q is greater than P by

$$\frac{y}{100 - y} \times 100\%$$

Example :- if P is smaller than Q by 30%, then Q is greater than P by

Solution :-

$$\frac{30}{100 - 30} \times 100\% = \frac{3}{7} \times 100\% = 42.85\%$$

7. If a person saves 'A' rupees after spending a% on food, b% on travel, and c% on cinema of his income, then

$$\text{Monthly income} = \frac{100}{100 - [a + b + c]} \times A$$

Example :- Jay spends 50% of his monthly income on grocery and bills, 20% of his monthly income on buying clothes, 5% of his monthly income on medicines, and the remaining amount of ₹12,500 he saves. What is Jay's monthly income?

Matriculation Level 30/06/2023 (Shift - 3)

(a) ₹45,000 (b) ₹57,500 (c) ₹65,000 (d) ₹50,000

Solution :-

$$\begin{aligned} \text{Monthly income} &= \frac{100}{100 - [50 + 20 + 5]} \times 12500 \\ &= \frac{100}{25} \times 12500 = 50000 \text{ Rs.} \end{aligned}$$

Exam Hall Approach :

Jay's Savings = $100\% - (50\% + 20\% + 5\%) = 25\%$

According to the question ,

25% of total income = 12500 Rs.

Total income (100%) = $12500 \times 4 = 50000$ Rs.

8. If the side of a square or radius of a circle increases by $a\%$, then increase in area = $(2a + \frac{a^2}{100})\%$

Or

9. If side of a square or radius of a circle decreases by $a\%$, then decrease in area = $(-2a + \frac{a^2}{100})\%$

Example :- If the radius of a circle is increased by 10%. then its area would increase by:

SSC CHSL 20/03/2023 (4th Shift)

(a) 100% (b) 20% (c) 21% (d) 19%

Solution :-

$$\text{increase in area} = (2a + \frac{a^2}{100})\% = (2 \times 10 + \frac{10^2}{100})\% = 21\%$$

Exam Hall Approach :

$$\text{Fraction value of } 10\% = \frac{1}{10}$$

$$\text{Area of circle} = \pi r^2$$

Old area : new area

100 : 121

Percentage increase in area

$$= \frac{21}{100} \times 100 = 21\%$$

10. Increasing the length by $a\%$, breadth by $b\%$, and height of a cuboid by $h\%$, increases its volume by:

$$(a + b + c + \frac{ab + bc + ca}{100} + \frac{abc}{10000})\%$$

Example :- The length and breadth of a cuboid are increased by 10% and 20%, respectively, and its height is decreased by 20%. The percentage increase in the volume of the cuboid is:

SSC CHSL 14/10/2020 (Morning)

(a) $5\frac{4}{5}\%$ (b) $5\frac{1}{5}\%$ (c) $5\frac{2}{5}\%$ (d) $5\frac{3}{5}\%$

Solution :- increases its volume

$$= (10 + 20 - 20 + \frac{10 \times 20 - 20 \times 20 - 20 \times 10}{100} - \frac{10 \times 20 \times 20}{10000})\%$$

$$= (10 - \frac{400}{100} - \frac{4}{10}) = 6 - \frac{2}{5} = 5\frac{3}{5}\%$$

Exam Hall Approach :

Let each side = 10

$$\text{Volume} = 10 \times 10 \times 10 = 1000$$

New length, breadth and height = 11, 12, 8

$$\text{New volume} = 11 \times 12 \times 8 = 1056$$

$$\text{Required percentage} = \frac{56}{1000} \times 100 = 5\frac{3}{5}\%$$

11. If $x\%$ of a total sum (A) is taken by 1st person, $y\%$ of remaining sum is taken by 2nd person, and $Z\%$ of remaining sum is taken by 3rd person, then if R be the remaining amount, then initial amount:

$$R \times [\frac{100}{100-x}] [\frac{100}{100-y}] [\frac{100}{100-z}]$$

12. On increasing/decreasing the cost of a certain article by $a\%$ and, a man can buy ' x ' kg article less/more in y rupees, then

$$\text{Increased cost} = \frac{ay}{100 \times x}$$

$$\text{Also, the initial cost} = \frac{ay}{(100 \pm a)x}$$

'+' for increase, '-' for decrease .

Example :- The reduction of 20% in the price of rice enables a person to obtain 50 kg more for Rs.450. Find the original price of rice per kg.

SSC CGL 02/12/2022 (2nd Shift)

(a) Rs.1 (b) Rs.2 (c) Rs.1.25 (d) Rs.2.25

Solution :-

Expenditure = price $\times$ consumption

Here , expenditure is constant in both cases . so, the ratio of price is inversely proportional to the ratio of consumption.

Price of rice $\Rightarrow 5 : 4$

Consumption $\Rightarrow 4 : 5$

Consumption increase = 1 unit $\rightarrow$ 50 kg

Initial consumption = 4 unit $\rightarrow$ 200 kg

Original price per k.g.(Rate)

$$= \frac{\text{Total expenditure}}{\text{Total consumption}} = \frac{450}{200} = 2.25 \text{ Rs./kg}$$

Exam hall approach :

original price per kg =

$$\frac{20 \times 450}{(100 - 20)50} = \frac{20 \times 450}{80 \times 50} = \frac{9}{4} = 2.25 \text{ Rs./kg}$$

Types of Questions :-

Simple Calculation Based

Example :- If 20% of $(A + B) = 30\%$ of $(A - B)$, then what percentage of B is equal to A ?

Solution :- 20% of $(A + B) = 30\%$ of $(A - B)$

$$\Rightarrow 2(A + B) = 3(A - B) \Rightarrow \frac{A}{B} = \frac{5}{1} = \frac{5}{1} \times 100 = 500\%$$

Example :- Z is increased successively by $11\frac{1}{9}\%$ and 10% to

get R. If R = 165, then what is the value of Z ?

SSC MTS 25/07/2022 (Morning)

(a) 145 (b) 115 (c) 125 (d) 135

Solution :- $11\frac{1}{9}\% = \frac{1}{9}$, $10\% = \frac{1}{10}$

$$Z \times \frac{10}{9} \times \frac{11}{10} = 165 \Rightarrow Z = 135$$

Example :- The length and breadth of a cuboid are increased by 10% and 20%, respectively, and its height is decreased by 20%. The percentage increase in the volume of the cuboid is:

SSC CHSL 14/10/2020 (Morning)

(a) $5\frac{4}{5}\%$ (b) $5\frac{1}{5}\%$ (c) $5\frac{2}{5}\%$ (d) $5\frac{3}{5}\%$

Solution :- Let each side = 10

Volume = $10 \times 10 \times 10 = 1000$
 New length, breadth and height = 11, 12, 8
 New volume = $11 \times 12 \times 8 = 1056$
 Required percentage = $\frac{1056 - 1000}{1000} \times 100 = \frac{56}{10} = 5\frac{3}{5}\%$

Example :- Two numbers are less than a third number by 37% and 30%, respectively. By What percentage is the second number more than the first?

SSC CHSL 6/8/2021 (Afternoon)

- (a) $\frac{100}{9}$ (b) $\frac{10}{9}$ (c) 10 (d) $\frac{100}{3}$

Solution :-

Let the third number = 100

Second number = 63

Third number = 70

$$\text{Required percentage} = \frac{70 - 63}{63} \times 100 = \frac{100}{9}$$

Fraction Based

Example :- If the numerator of a fraction is increased by 12% and its denominator is reduced by 8%, it gives a new fraction $\frac{16}{17}$. What is the original fraction?

Selection Post 27/06/2023 (3rd Shift)

- (a) $\frac{92}{119}$ (b) $\frac{23}{28}$ (c) $\frac{28}{23}$ (d) $\frac{119}{92}$

Solution :- Let the fraction be $\frac{a}{b}$

According to question,

$$\frac{a \times \frac{112}{100}}{b \times \frac{92}{100}} = \frac{16}{17}$$

$$\frac{a}{b} = \frac{16 \times 92}{17 \times 112} = \frac{1472}{1904} = \frac{92}{119}$$

Percentage Error Based

Example :- An accountant multiplied a number by $\frac{9}{2}$ instead of $\frac{2}{9}$. What is the approximate percentage error in his calculation?

SSC GD Constable 22/11/2021 (Afternoon)

- (a) 0.95% (b) 95% (c) 9.5% (d) 0.095%

Solution :- Let the no. be 18 i.e. LCM of (2, 9)

$$\text{Correct no} = 18 \times \frac{9}{2} = 81$$

$$\text{Incorrect no} = 18 \times \frac{2}{9} = 4$$

$$\text{Required percentage} = \frac{81 - 4}{81} \times 100 = \frac{77}{81} \times 100 = 95.06 \approx 95\%$$

Successive Increase/Decrease Based

Example :- Two successive increments of 30% each is by what percentage more than two successive decrements of 30% each? (Correct to two decimal places)

SSC CGL 20/07/2023 (2nd shift)

- (a) 32.54% (b) 28.15% (c) 25.25% (d) 35.29%

Solution :- According to question,

$$\text{Net increment \%} = 30 + 30 + \frac{30 \times 30}{100} = 69\%$$

$$\text{Net decrement \%} = -30 - 30 + \frac{30 \times 30}{100} = -51\% \text{ (decrement)}$$

$$\text{Required \%} = \frac{69 - 51}{51} \times 100 = 35.29\%$$

Example :- When the price of an item was reduced by 20% its sales increased by x%. If there is an increase of 25% in the receipt of the revenue, then the value of x is:

SSC CHSL 10/08/2021 (Evening)

- (a) 53.84% (b) 55.75% (c) 56.25% (d) 61.50%

Solution :-

Ratio → Initial : final

Price → 5 : 4

Sale → x : y

Revenue → 4 : 5

$$\text{Now, } x : y = 4 \times 4 : 5 \times 5 = 16 : 25$$

$$\% \text{ increase in sales} = \frac{25 - 16}{16} \times 100 = 56.25\%$$

Population Based

Example :- In the first year the population of a town decreased by 5% due to the Corona virus first wave. In the next year it decreased again by 5% due to the second wave and in the third year it increased by 5%. At the end of the third year the population was 9,47,625. What was the population at the beginning of the first year?

SSC CGL 08/12/2022 (3rd Shift)

- (a) 7,00,000 (b) 10,00,000 (c) 8,97,993 (d) 9,92,519

Solution :- Let the population at the beginning of the first year be x

According to question,

$$x \times \frac{19}{20} \times \frac{19}{20} \times \frac{21}{20} = 9,47,625$$

$$x \times \frac{7581}{8000} = 9,47,625 \Rightarrow x = \frac{9,47,625 \times 8000}{7581} = 10,00,000$$

Income and expenditure based

Income = Expenditure + Saving

Example :- Raman spends 75% of his income. If his income increases by 20% and expenditure also increases by 10%, then By what percent will Raman's savings increase?

SSC CHSL 08/08/2023 (4th Shift)

- (a) 35% (b) 40% (c) 50% (d) 45%

Solution :- Let income of Raman = 100

Income = Expenditure + Savings

Initial :- 100 = 75 + 25

Final :- 120 = 82.5 + 37.5

$$\% \text{ increase in Saving} = \frac{37.5 - 25}{25} \times 100 = 50\%$$

Example :- The salaries of P and Q together amount to ₹ 6,40,000. P and Q save 60% and 40%, respectively, of their individual salaries. If their expenditures are the same, then what is the ratio of P's salary to Q's salary ?

SSC MTS 14/09/2023 (3rd Shift)

(a) 4 : 3 (b) 2 : 3 (c) 3 : 2 (d) 3 : 4

Solution :- Let the salary of P = $100R_1$

and the salary of Q = $100R_2$

Expenditure of P = 40% of $100R_1 = 40R_1$

Expenditure of Q = 60% of $100R_2 = 60R_2$

According to the question,

$$40R_1 = 60R_2 \Rightarrow \frac{R_1}{R_2} = \frac{3}{2}$$

Required ratio = $100 \times 3 : 100 \times 2 = 3 : 2$

Examination Based

Example :- In an examination, Anita scored 31% marks and failed by 16 marks. Sunita scored 40% marks and obtained 56 marks more than those required to pass. Find the minimum marks required to pass.

SSC CGL 7/03/2020 (Morning)

(a) 3116 (b) 3944 (c) 7100 (d) 264

Solution :- Let M be the maximum marks

According to question:

31% of M + 16 = 40% of M - 56

9% of M = 72 $\Rightarrow$ M = 800

Passing marks = 40% of M - 56 = 320 - 56 = 264

Example :- In a test consisting of 140 questions, a candidate correctly answered 70% of the first 80 questions. What percentage of the remaining questions does the candidate need to correctly answer to score 60% in the test?

SSC CPO 11/12/19 (Morning)

(a) 40% (b) $45\frac{1}{3}\%$ (c) $46\frac{2}{3}\%$ (d) 35%

Solution :-

As per the question out of 140 the candidate answered 70% out of 80 questions correct

$$= \frac{(70 \times 80)}{100} = 56$$

The student has to answer questions to score 60% in the test

$$= 140 \times \frac{60}{100}$$

= 84 questions correct

$$\text{Required percentage} = \frac{(84 - 56)}{60} \times 100 = 46\frac{2}{3}\%$$

Example :- Find the approximate percentage of aggregate marks scored by Ramesh, who obtained 50%, 60% and 70% marks in three examinations with maximum marks 100, 130 and 150 respectively.

RRB NTPC CBT - I 12/03/2021 (Evening)

(a) 61.32 (b) 71.23 (c) 55.5 (d) 70.1

Solution :- Total obtain marks

$$= \frac{100 \times 50}{100} + \frac{130 \times 60}{100} + \frac{150 \times 70}{100} = 50 + 78 + 105 = 233$$

Total maximum marks = 100 + 130 + 150 = 380

$$\text{Then, Total aggregate \%} = \frac{233}{380} \times 100 = 61.32\%$$

Example :- In the school, there were 1400 boys and 600 girls. 40% of girls and 60% of boys passed the examination. What is the percentage of failed candidates ?

RRB NTPC CBT - I 16/02/2021 (Morning)

(a) 46% (b) 65% (c) 72% (d) 63%

Solution :- Boys = 1400, Girls = 600

$$40\% \text{ of girls} = 600 \times \frac{40}{100} = 240$$

$$60\% \text{ of boys} = 1400 \times \frac{60}{100} = 840$$

Total number of fail candidates = 60% of Girl + 40% of Boys
= 360 + 560 = 920

$$\text{Percentage of failed candidates} = \frac{920}{2000} \times 100 = 46\%$$

Election Based

Example :- In the election, 8% of the voters did not cast their votes. Two candidates contested the election in which the winning candidate got 48% of the total votes and won the election by 1100 votes. Find the total number of voters in the election, given that all the votes cast were valid?

SSC CGL 02/12/2022 (1st Shift)

(a) 25500 (b) 26500 (c) 27500 (d) 28500

Solution :- Let, the total no. of votes = 100%

Then, total votes cast = 100% - 8% = 92%

Now winner got 48% of total votes = 48%

Then, defeated candidates got total votes = 92% - 48% = 44%

Difference between winner and defeated candidates =

4% = 1100 votes

Total no. of voters = 100% = 27500

Example :- In a village election between three candidates, 80% of the voters cast their votes, out of which 3% votes were declared invalid. The winning candidate got 42,680 votes which were 55% of the total valid votes. Find the number of valid votes received by the candidate who stood third, given that he received only 10% of the total valid votes.

(a) 7760 (b) 7700 (c) 7670 (d) 7680

$$\text{Solution :- Total valid votes} = x \times \frac{80}{100} \times \frac{97}{100}$$

$$\text{Winner got votes} = x \times \frac{80}{100} \times \frac{97}{100} \times \frac{55}{100} = 42,680 \text{ votes}$$

$$\Rightarrow x = 1,00,000 \text{ votes}$$

Third one receive votes

$$= 1,00,000 \times \frac{80}{100} \times \frac{97}{100} \times \frac{10}{100} = 7760 \text{ votes}$$

Commission Based

Example :- A marketing agent earns a commission of 2% on first Rs. 2,00,000, 1.5% on next Rs 2,00,000 and 1% on the remaining amount of sales made in a month. If the sales achieved by the agent for the month of April 2018 are Rs. 5,68,000. The commission earned is

SSC CPO 15/03/2019 (Morning)

(a) Rs.8680 (b) Rs.7730 (c) Rs.8240 (d) Rs.7105

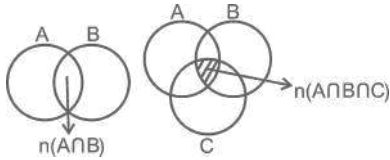
Solution :-

Total commission $\rightarrow$
 $200000 \times 2\% + 200000 \times 1.5\% + 168000 \times 1\% = 8680 \text{ Rs.}$

Venn Diagram Based

Venn diagram concept :

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$



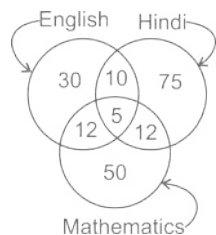
Example :- 500 candidates appeared in an examination comprising test in English, Hindi and Mathematics. 30 candidates failed in English only; 75 failed in Hindi only; 50 failed in Mathematics only; 15 failed in both English and Hindi; 17 failed in both Hindi and Mathematics; 17 failed in both Mathematics and English; 5 failed in all three tests.

What is the percentage of candidates who failed in at least two subjects ?

UPSC CDS - I (10/04/2022)

(a) 5.4% (b) 6.4% (c) 6.8% (d) 7.8%

Solution :-



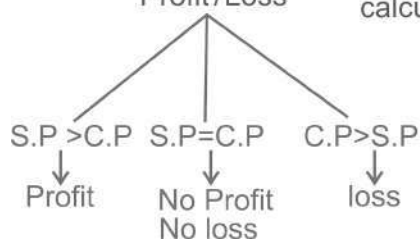
Number of candidates who failed in at least two subjects

$$= (12 + 10 + 12 + 5) = 39$$

$$\text{Required percentage} = \frac{39}{500} \times 100 = 7.8\%$$

Profit and Loss

Note:- Profit and loss always calculated on Cost Price



Profit %	Loss %
$\frac{\text{Profit}}{CP} \times 100$	$\frac{\text{loss}}{CP} \times 100$

Type - 1

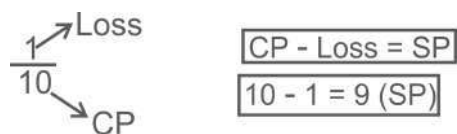
(a) When an item is sold at Profit :-

$$\text{Profit \%} = 10\% = \frac{1}{10}$$



(b) When an item is sold at Loss:-

$$\text{Loss \%} = 10\% = \frac{1}{10}$$



Example:- The cost price of an article is 5,000. What should the selling price of the article be so that a profit of 25% is earned?

RRC Group D 09/09/2022 (Afternoon)

- (a) ₹6,250 (b) ₹7,250
(c) ₹5,250 (d) ₹8,250

Solution:- SP on 25% profit = $5000 \times \frac{125}{100} = ₹6,250$

Exam hall approach :-

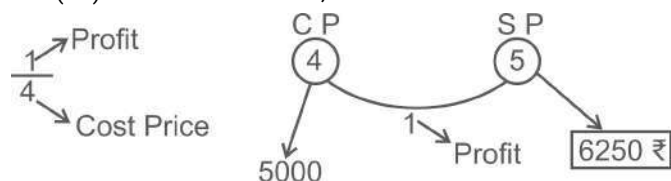
$$\text{Profit} = 25\% = \frac{25}{100} = \frac{1}{4} = \frac{\text{Profit}}{CP}$$

CP = 4 unit, profit = 1 unit

SP = 4 + 1 = 5 unit

4 unit = 5000 ⇒ 1 unit = 125

(SP) 5 unit = 5 × 125 = ₹ 6,250



Example:- A watch is purchased for ₹3,200 and sold for ₹2,700. Calculate the percentage loss up to two places of decimal.

SSC CGL 26/07/2023 (4th shift)

- (a) 15.18% (b) 15.62% (c) 16.12% (d) 16.50%

Solution:- CP = 3200, SP = 2700,

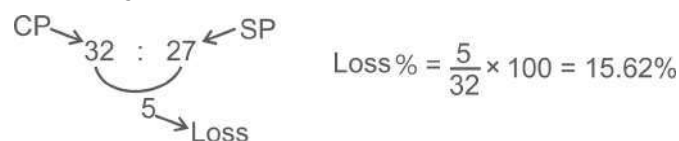
$$\text{Loss} = SP - CP \Rightarrow 3200 - 2700 = 500$$

$$\text{Loss\%} = \frac{3200-2700}{3200} \times 100 = \frac{500}{32} = 15.62\%$$

Exam hall approach: CP : SP = 3200 : 2700 ⇒ 32 : 27

$$\text{Loss} = 32 - 27 = 5$$

$$\text{Loss\%} = \frac{5}{32} \times 100 = 15.62\%$$



(c) When an item is sold at Profit on SP:-

$$\text{Profit\%} = 10\% = \frac{1}{10}$$



(d) When an item is sold at Loss on SP:-

$$\text{Loss\%} = 10\% = \frac{1}{10}$$



Example:- Parikh sold his pen at a profit of Rs 11. He calculated the profit percentage on selling price and found it to be 25%. the cost price (in Rs) of the pen is:

SSC MTS 13/08/2019 (Afternoon)

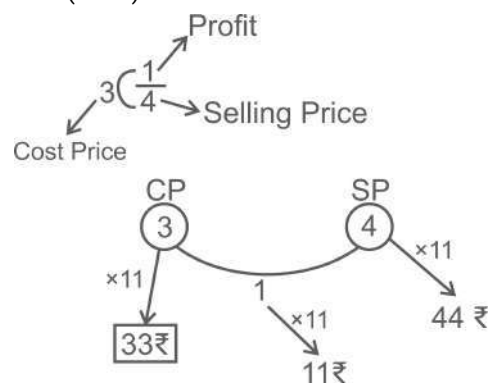
- (a) 33 (b) 24 (c) 36 (d) 44

Solution:- Fractional value of 25% = $\frac{1}{4}$

Profit (1 unit) = 11 Rs.

the SP = 4 units and

$$CP = (4 - 1)\text{units} = 3 \text{ units} = 3 \times 11 = 33 ₹$$



Type - 2

$$SP = CP \times \frac{100 \pm \text{profit / loss \%}}{100}$$

(CP = Cost price, SP = Selling price)

Example:- A person buys a table fan for ₹2,800 and sells it at a loss of 12%. What is the selling price of the table fan?

SSC MTS 04/09/2023 (2nd Shift)

(a) ₹2,466 (b) ₹2,468 (c) ₹2,462 (d) ₹2,464

Solution:-

$$CP = 2800, \text{ Loss\%} = 12$$

$$SP = 2800 \times \frac{100 - 12}{100} = 28 \times 88 = ₹2,464$$

Exam hall approach :-

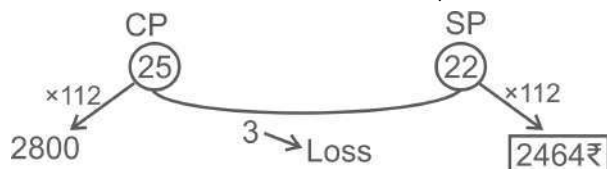
$$\text{Loss} = 12\% = \frac{12}{100} = \frac{3}{25} = \frac{\text{Loss}}{CP}$$

Let : CP = 25 , Loss = 3

$$SP = 25 - 3 = 22$$

$$25 \text{ unit} = 2800 \Rightarrow 1 \text{ unit} = 112$$

$$22 \text{ unit} = 22 \times 112 = 2464 \Rightarrow SP = ₹2,464$$



Example:- The cost price of an article is 5,000. What should the selling price of the article be so that a profit of 25% is earned?

RRC Group D 09/09/2022 (Afternoon)

(a) ₹6,250 (b) ₹7,250 (c) ₹5,250 (d) ₹8,250

Solution :-

$$\text{Required } SP = 5000 \times \frac{100 + 25}{100} = ₹6,250$$

Exam hall approach :-

$$\text{Profit} = 25\% = \frac{25}{100} = \frac{1}{4} = \frac{\text{Profit}}{CP}$$

Let: CP = 4 unit , Profit = 1 unit

$$SP = 4 + 1 = 5 \text{ unit}$$

$$4 \text{ unit} = 5000 \Rightarrow 1 \text{ unit} = 125$$

$$(SP) 5 \text{ unit} = 5 \times 125 = ₹6,250$$

Type - 3

"To find out the purchase price of an item by selling it at profit or loss %"

$$CP = SP \times \frac{100}{100 \pm \text{profit/loss \%}}$$

Note :- In case of profit you must add and in case of loss subtract in above formula.

Example :- A person bought a car and sold it for Rs.3,00,000 If he incurred a loss of 20%, then how much did he spend to buy the car?

UPSC CSAT (04/10/2020)

(a) Rs. 3, 60,000 (b) Rs. 3, 65,000

(c) Rs. 3, 70,000 (d) Rs. 3, 75,000

Solution :-

$$CP \text{ of car} = 3,00,000 \times \frac{100}{100 - 20} = ₹3,75,000$$

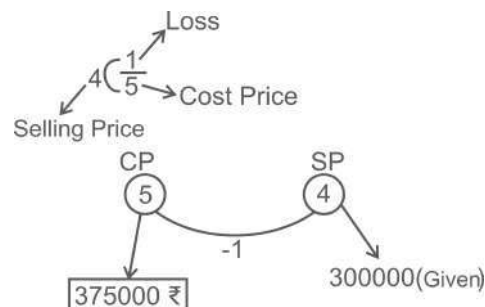
Exam hall approach :-

$$\text{Loss} = 20\% = \frac{1}{5}$$

Let: CP = 5 unit, Loss = 1 unit, SP = 4 unit

$$4 \text{ unit} = 300000 \Rightarrow 1 \text{ unit} = 75000$$

$$(CP) 5 \text{ unit} = 5 \times 75000 = ₹3,75,000$$



Example :- A woman sold her earphone for Rs. 2,000 and got a percentage profit equal to the numerical value of cost price. The cost price of the earphone is:

SSC CHSL 03/06/2022 (Morning)

(a) Rs. 500 (b) Rs. 200 (c) Rs. 600 (d) Rs. 400

Solution :-

Let the CP of earphone = ₹x

Then profit % = x

$$\text{ATQ, } x = 2000 \times \frac{100}{100 + x}$$

$$100x + x^2 = 2000 \times 100$$

$$(2000 - x) \times 100 = x^2$$

Now, putting the different value of x in the given options to satisfy the result we get x = 400

$$(2000 - 400) \times 100 = 400^2$$

$$160000 = 160000 \text{ (LHS = RHS)}$$

So, CP of earphone = ₹400

Exam hall approach :-

In this type of question , go through option :-

Option (d) → Let C.P. = 400 Rs. and

Profit % = 400%

$$\text{Then, S.P.} = 400 + 400 \times 400\% = 2000 \text{ Rs.}$$

Type - 4

When instead of amount, number of goods sold are given :-

On selling x article the profit or loss is equal to selling price of y articles, then

$$\text{Profit \%} = \frac{100 \times y}{x - y}, \text{ Loss \%} = \frac{100 \times y}{x + y}$$

Example :- By selling 8 items, a shopkeeper gains the selling price of 2 items. The gain percentage of the shopkeeper is:

SSC MTS 05/09/2023 (3rd Shift)

(a) 25% (b) $16\frac{2}{3}\%$ (c) $33\frac{1}{3}\%$ (d) 30%

$$\text{Solution :- Profit \%} = \frac{100 \times y}{x - y}$$

$$\text{Profit \%} = \frac{100 \times 2}{8 - 2} = \frac{200}{6} = 33\frac{1}{3}\%$$

Exam hall approach :-

According to the question

$$8 \text{ SP} - 8 \text{ CP} = 2 \text{ SP}$$

$$\frac{CP}{SP} = \frac{3}{4}$$

$$\text{So, profit \%} = \frac{1}{3} \times 100 = 33\frac{1}{3}\%$$

Type - 9

Dishonest Shopkeeper/Trader

If a trader profess to sell goods at cost price but uses false weights, then

$$\text{Profit percentage} = \frac{\text{Error}}{\text{True value} - \text{Error}} \times 100,$$

where Error = True value - False value



While Selling

Example:- A dishonest dealer professes to sell his goods at cost price, but uses false weights, and thus gains 20%. For one kilogram, how many grams of watts does he use?

RRC Group D 26/08/2022 (Evening)

(a) 750.5 gm (b) 708.06 gm (c) 833.33 gm (d) 785.5 gm

Solution:-

Let error = x gram

$$\text{Profit percentage} = \frac{\text{Error}}{\text{True value} - \text{Error}} \times 100$$

$$20 = \frac{x}{1000 - x} \times 100$$

$$20000 - 20 \times x = 100x$$

$$x = \frac{20000}{120} = 166.66 \text{ gram}$$

Therefore he watts he uses = $1000 - 166.66 = 833.33$ gm

Exam hall approach :-

C.P : S.P

100 : 120

X gm : 1000 gm

In ₹ 120 → 1000 gm

In ₹ 100 → $\frac{1000 \times 100}{120} = 833.33$ gm

While Buying

Example :- In a defective balance the weight of left edge is 100 gm more than its right edge. A shopkeeper while purchasing keeps weights on left edge and while selling keeps weights on the right edge. He only uses weight of 1 kg. If he sells the things on marked price, then his profit is ?

CG TET 2016 PAPER 2 (Math & science)

(a) $\frac{100}{9}\%$ (b) $\frac{200}{11}\%$ (c) $\frac{100}{11}\%$ (d) $\frac{200}{9}\%$

Solution :-

$$\frac{S.P._{\text{actual}}}{C.P._{\text{actual}}} = \frac{S.P._{\text{given}}}{C.P._{\text{given}}} \times \frac{\text{True Value}}{\text{False value}}$$

$$\frac{S.P._{\text{actual}}}{C.P._{\text{actual}}} = \frac{1100}{1000} \times \frac{1000}{900} = \frac{11}{9}$$

$$\text{profit}\% = \frac{2}{9} \times 100 = \frac{200}{9}\%$$

Exam hall approach :-

C.P. : S.P.

$$1000 : 1100$$

$$900 : 1000$$

$$\text{Overall} \rightarrow 9 : 11 \Rightarrow \text{Profit}\% = \frac{2}{9} \times 100 = \frac{200}{9}\%$$

Type - 10

If a shopkeeper used to sell his articles at $x\%$ loss on cost price but uses y grams instead of z grams, then:

$$\text{profit or loss \% is} = [(100 - x) \frac{z}{y} - 100]\%$$

* +ve or -ve sign as per Profit or loss.

Example :- A dishonest merchant sells goods at a 12.5% loss on the cost price, but uses 28 g weight instead of 36 g. What is his percentage profit or loss ?

SSC CGL 2022 Tire - I (Held On : 01 Dec 2022 shift 1)

(a) 6.25% loss (b) 12.5% gain (c) 18.75% gain (d) 10.5% loss

Solution :-

Method - 1

$$\text{profit or loss \%} = [(100 - x) \frac{z}{y} - 100]\%$$

$$= [(100 - 12.5) \frac{36}{28} - 100]\% = [87.5 \times \frac{36}{28} - 100]\%$$

$$= [112.5 - 100]\% = 12.5\%$$

Method - 2

$$\frac{S.P._{\text{actual}}}{C.P._{\text{actual}}} = \frac{S.P._{\text{given}}}{C.P._{\text{given}}} \times \frac{\text{True Value}}{\text{False value}}$$

$$\frac{S.P._{\text{actual}}}{C.P._{\text{actual}}} = \frac{7}{8} \times \frac{36}{28} = \frac{9}{8} \Rightarrow \text{profit}\% = \frac{1}{8} \times 100 = 12.5\%$$

Exam hall approach :-

$$\text{Fractional value of } 12.5\% = \frac{1}{8}$$

C.P : S.P.

12.5% → 8 : 7

Faulty weight → 28 : 36

Final ratio → 8 : 9

$$\text{Profit \%} = \frac{S.P. - C.P.}{\text{cost price}} \times 100 = \frac{9 - 8}{8} \times 100 = 12.5\%$$

Type - 11

If a man purchases 'a' items for Rs. x and sells 'b' items for Rs. y , then his profit % or loss % is given by:

$$\text{profit \% or loss \%} = \frac{ay - bx}{bx} \times 100\%$$

Example :- A shopkeeper bought toffees at a rate of 10 for ₹15 and sold them at a rate of 16 for ₹40. Find his profit percentage. (correct to two decimal place)

SSC CGL 21/04/2022 (Morning)

(a) 65.05% (b) 33.33% (c) 50.55% (d) 66.67%

Solution :-

$$\text{profit}\% = \frac{10 \times 40 - 16 \times 15}{16 \times 15} \times 100$$

$$= \frac{400 - 240}{240} \times 100 = \frac{160}{240} \times 100 = 66.67\%$$

Exam hall approach :-

CP of 10 toffee = ₹15
 CP of 16 toffee = $\frac{15}{10} \times 16 = 24$
 SP of 16 toffee = ₹40
 Ratio → C.P. : S.P.
 24 : 40 or 3 : 5
 Profit % = $\frac{2}{3} \times 100 = 66.67\%$

Type - 12

Successive Loss/Profit %

(a) Successive profit or When a article is first sold at $x\%$ profit then again sold at $y\%$ profit, overall change in original value

$$= +x\% + y\% + \frac{xy}{100} \%$$

Example :- Anil sells a car to Bina at a profit of 20%. Bina sells it to Karan at a profit of 25%. If Karan pays ₹2,25,000 for it, then, find the CP of the car for Anil .

SSC MTS 12/09/2023 (1st Shift)

(a) ₹ 1,75,000 (b) ₹ 2,00,000 (c) ₹ 1,50,000 (d) ₹ 1,25,000

Solution :-

$$\text{overall change} = +20\% + 25\% + \frac{20 \times 25}{100} \% = 50\%$$

$$150\% = ₹2,25,000$$

$$1\% = \frac{₹2,25,000}{150} = 1500$$

$$100\% = 150000$$

$$\text{C.P of car for Anil} = \text{Rs } 1,50,000$$

(b) When a number is first sold at $x\%$ profit then again sold at $y\%$

$$\% \text{ loss, overall change in original value} = +x\% - y\% - \frac{xy}{100} \%$$

Example :- Raju sold an article to Ravi at 10% gain. Ravi sold the article to Ramu at a loss of 10%. If Ramu paid ₹1,485 to Ravi, find the selling price of Raju to Ravi.

SSC MTS 01/09/2023 (1st Shift)

(a) ₹1,200 (b) ₹1,500 (c) ₹1,650 (d) ₹1,600

Solution :-

$$\text{loss percentage} = 10 - 10 - \frac{10 \times 10}{100} = -1\% \text{ (loss)}$$

Let the CP of the article for Raju = 100 unit

So, SP of Raju = CP of Ravi = 99 unit

$$99 \text{ unit} = 1485$$

$$100 \text{ unit (CP of Raju)} = \frac{1485}{99} \times 100 = 1500$$

$$\text{Required amount} = 1500 \times \frac{11}{10} = 1650$$

Exam hall approach :-

According to the question,

C.P of raju = x

$$x \times \frac{11}{10} \times \frac{9}{10} = ₹ 1485 \Rightarrow x = ₹ 1500$$

$$\text{Required amount} = 1500 \times \frac{11}{10} = ₹ 1650$$

(c) Successive loss or When a number is first sold at $x\%$ loss then again sold at $y\%$ loss, overall change in original value

$$= -x\% - y\% + \frac{xy}{100} \%$$

Example:- The marked price of an item was ₹5,000. While selling the item, the seller offered successive discounts of 15% and 5%. What was the final selling price of the item?
 SSC MTS 12/07/2022 (Evening)

(a) ₹ 4,730 (b) ₹ 4,370.50 (c) ₹ 4,250 (d) ₹ 4,037.50

Solution:-

$$\begin{aligned} \text{overall change in original value} &= -15\% - 5\% + \frac{15 \times 5}{100} \% \\ &= -19.25 \% \end{aligned}$$

$$100 \text{ units} = 5,000$$

$$80.75\% \text{ units} = \frac{5000}{100} \times 80.75 = 4,037.50 \text{ Rs}$$

(d) When a number is first sold at $x\%$ loss then again sold at $y\%$ profit, overall change in original value = $-x\% + y\% - \frac{xy}{100} \%$

Type - 13

Value added tax (VAT) or overhead/repairing expenses

In case of value added tax (VAT) of Rs. x , or in case of any repairment cost on the article of Rs. x

Then total CP = actual CP + VAT or repairment cost

Example:- Vinay bought a flat for ₹4,50,000. He spent ₹10,000 on its painting and repairs. Then he sold it for ₹ 4,25,500. Find his gain or loss percentage.

DMRC CRA 17/02/2020 (2nd shift)

(a) Loss 6.5% (b) Loss 7.5% (c) Gain 7% (d) Gain 6%

Solution :-

According to the question,

$$\text{Actual CP of the flat} = 450000 + 10000 = ₹ 460000$$

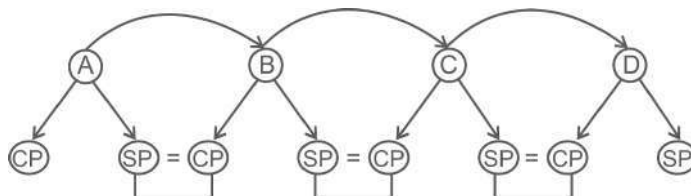
$$\text{Loss} = 4,60,000 - 4,25,500 = 34500$$

$$\text{loss\%} = \frac{34500}{460000} \times 100 = 7.5\%$$

Type - 14

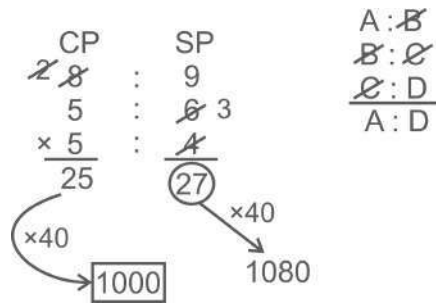
If A sells an article to B at a% profit, B sells that article to C at b% profit, and C pays Rs. x ,

$$\text{then amount paid by A} = x \left[\frac{100}{100 + a} \right] \left[\frac{100}{100 + b} \right]$$



Example :- A sells an article to B at 12.5% profit while B sells it to C at 20% profit and C sells it to D at 20% loss. If the cost price of D is 1080 Rs then find the cost price of A. ?

Solution :-



$$(27 - 12)\% = ₹ 900 \Rightarrow 1\% = \frac{900}{15} = ₹ 60$$

Cost price (100%) = ₹6,000

Example :- A sells an item at 20% profit to B, B sells the same at 10% profit to C and receives 1,32,000. Had C purchased the same item from A, he would have spent 5% less than what he spent with B. What profit would A have made then?

SSC CHSL 13/10/2020 (Morning)

(a) Rs 25,400 (b) Rs 24,450 (c) Rs 25,540 (d) Rs 24,540

Solution :-

$$\text{Amount paid by A} = 132000 \left[\frac{100}{100 + 20} \right] \left[\frac{100}{100 + 10} \right]$$

$$= 132000 \times \frac{5}{6} \times \frac{10}{11} = 100000$$

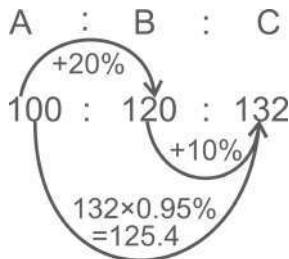
spent 5% less than what he spent with B

$$132000 \times \frac{95}{100} = 125400$$

$$125400 - 100000 = 25400$$

Exam hall Approach :-

Let CP of A = 100 unit



According to the question,

CP of C (132 unit) = 132000 ₹

Profit earned by A if he had sold the item to C (25.4 unit)
= 25400 ₹

Type - 15

A man sells items at a profit/loss of $x\%$. If he had sold it for Rs. M more, he would have gained/loss $y\%$ then,

$$\text{C.P.} = \left[\frac{M}{y \pm x} \right] \times 100$$

+ sign is used when one is profit and the other is loss.

- sign is used when both are either profit or loss.

Example :- A seller sold a bag at a profit of 12%. If he had sold it for ₹900 more, he would have gained 27%. What is the CP of the bag?

SSC CHSL 17/08/2023 (2nd Shift)

(a) ₹6,000 (b) ₹6500 (c) ₹5560 (d) ₹7000

Solution :-

$$\text{C.P.} = \left[\frac{900}{27 - 12} \right] \times 100 = \frac{900}{15} \times 100 = 6,000$$

Exam hall Approach :-

According to the question,

Discount

Key Points :-



1. Marked Price :- The price on the label of a product, also called printed price or advertising price.

2. Discount :- Amount of rebate on a fixed price is called discount.

3. Selling Price :- Marked Price – Discount

Let us say Rs. 100 is the marked price (MP). and Discount is 10% i.e. 10 Rs

Therefore, selling price = 100 – 10 = Rs. 90

Type - 1

Discount :- Discount is always calculated on Marked price.

$$\text{Discount \%} = \frac{M.P. - S.P.}{M.P.} \times 100$$

$$\text{Or, } \frac{\text{Discount}}{\text{Marked Price}} \times 100$$

Example :- An article is sold for Rs. 2160 after earning a profit of 20%. If the Marked price of an article is Rs. 600 more than its cost price, then find the discount percentage.

SBI PO Mains (02/01/2022)

(a) 10 (b) 20 (c) 30 (d) 40 (e) 50

Solution :- Let, CP = 100 unit

Then, SP = 120 unit

but, SP = Rs. 2160 (given)

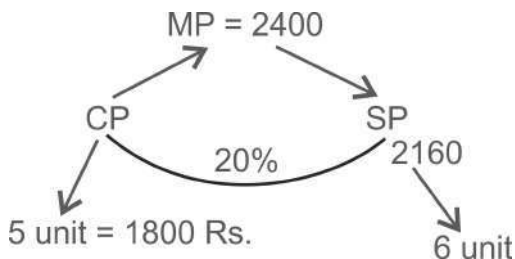
therefore, 120 unit = 2160 $\Rightarrow$ 1 unit = 18

Then, CP = Rs. 1800

Now, MP = 1800 + 600 = Rs. 2400

Discount = MP - SP = 2400 - 2160 = 240

And discount% = $\frac{240}{2400} \times 100 = 10\%$



Type - 2

$$\text{Selling Price} = \frac{\text{Marked Price} \times (100 - \text{Discount}\%)}{100}$$

Example :- The marked price of an item was Rs.480 but the shopkeeper offered a discount of 5%. At what price does he finally sell the item ?

SSC CHSL 21/03/2023 (4th Shift)

(a) Rs.435 (b) Rs.450 (c) Rs.466 (d) Rs.456

Solution :-

$$\text{Selling Price} = \frac{480 \times (100 - 5\%)}{100} = \frac{480 \times 95}{100} = \text{Rs. 456}$$

Exam hall approach :-

$$\text{Selling price of the article} = 480 \times \frac{95}{100} = 456 \text{ Rs.}$$

Type - 3

(a) Buy x get y free :- Discount% = $\frac{y}{x+y} \times 100\%$

Example :- If 3 coconuts are offered free on purchase of 12 coconuts, priced ₹25 each, what is the effective discount on each coconut ?

SSC CGL 2022 06/12/ 2022

(a) 20% (b) 24% (c) 20.83% (d) 15%

Solution :- Due to offer he get 3 coconuts free ,

Price for 3 coconuts = 25 $\times$ 3 = Rs.75

Total coconuts = 12 + 3 = 15 coconuts

Actual price for 15 coconuts = 25 $\times$ 15 = Rs. 375

$$\text{Effective discount} = \frac{75}{375} \times 100 = 20\%$$

Exam hall approach :- Net discount = $\frac{\text{free coconut}}{\text{total coconut}}$

$$= \frac{3}{12+3} \times 100 = \frac{3}{15} \times 100 = 20\%$$

(b) Buy x get y :- Discount% = $\frac{y-x}{y} \times 100\%$

Example :- During festivals, a banner on a shop displays, 'Pay for 3 and get 5. The discount percentage offered is:

SSC MTS 02/11/2021 (Morning)

(a) $166\frac{2}{3}\%$ (b) 60% (c) 40% (d) $66\frac{2}{3}\%$

$$\text{Solution :-} \text{Discount \%} = \frac{5-3}{5} \times 100 = 40\%$$

Type - 4

$$\frac{CP}{MP} = \frac{100 - \text{Discount \%}}{100 \pm \text{profit/loss \%}}$$

Example :- The face value of a book is Rs. 1000. A book seller offers a discount of 10% on this. If he still earns 20% profit then what will be the cost of the book ? (In rupees)

RPF Constable 18/01/2019 (Evening)

(a) 740 (b) 750 (c) 760 (d) 770

$$\text{Solution :-} \frac{CP}{1000} = \frac{100 - 10\%}{100 + 20\%} = \frac{3}{4} \Rightarrow CP = 750$$

Exam hall approach :-

C.P. S.P. M.P.

5 : 6

9 : 10

45 : 54 : 60

$$60 = 1000,$$

$$45 \text{ (C.P.)} = \frac{1000}{60} \times 45 = 750$$

Type - 5

Net Profit/Loss Percentage

Note :- In this type of question, we do not need to calculate cost price or selling price. We can directly use successive formulas.

$$\text{Net Profit / Loss} = M\% - D\% - \frac{M\% \times D\%}{100}$$

M% = markup price, D% = discount

Example :- A tradesman marks his goods at 26% above the cost price. He allows his customer a discount of 12% on the marked price. His profit per cent is:

SSC CPO 2022 09/11/2022

(a) 10.88% (b) 11.08% (c) 10.50% (d) 10%

Solution :- Net profit% = $26 - 12 - \frac{26 \times 12}{100}$

$$= 14 - \frac{312}{100} = 14 - 3.12 = 10.88\%$$

Example :- A shopkeeper marks up his goods by 30% and then gives a discount of 10% on the marked price. Apart from this, he uses a faulty balance which reads 1 kg for 870 g. What is his net profit percentage?

SSC CHSL 2021 24/05/2022

(a) 38.42% (b) 34.84% (c) 34.48% (d) 38.24%

Solution :- profit % = $30 - 10 - \frac{30 \times 10}{100} = 20 - 3 = 17\%$

Ratio → C.P. : S.P.
100 : 117

Faulty weight → 870 : 1000

29 : 39

$$\text{net profit \%} = \frac{10}{29} \times 100 = 34.48\%$$

Exam hall approach :-

Ratio → C.P. : S.P.

30% markup → 10 : 13

10% discount → 10 : 9

Faulty weight → 870 : 1000

Final → 87 : 117 = 29 : 39

$$\text{Net profit \%} = \frac{39 - 29}{29} \times 100 = \frac{10}{29} \times 100 = 34.48\%$$

Type - 6

Successive Discount

Successive discount and overall discount :-



When two or more than two discounts are given, it is called a Successive Discount.

Formula :- $(x + y - \frac{xy}{100})\%$

Example:- What is the net discount, if two successive discounts of 10% and 12% are given?

RRC Group D 22/09/2022 (Morning)

(a) 21.4% (b) 20.8% (c) 23.2% (d) 22.0%

Solution :-

$$\text{Net discount} = 10 + 12 - \frac{10 \times 12}{100} \Rightarrow 22 - 1.2 = 20.8\%$$

Exam hall approach :-

SP	CP
10	9
25	22
250	198

$$\text{Required percentage} = \frac{250 - 198}{250} \times 100 = 20.8\%$$

Example:- What will be an equivalent discount for three successive discounts of 20%, 20%, and 10%?

SSC CHSL 17/03/2018

Solution :- Single equivalent discount

$$\Rightarrow (x + y + z - \frac{xy + yz + zx}{100} + \frac{xyz}{10000})\%$$

$$\Rightarrow (20 + 20 + 10 - \frac{20 \times 20 + 20 \times 10 + 10 \times 20}{100} + \frac{20 \times 20 \times 10}{10000})\%$$

$$\Rightarrow (50 - \frac{400 + 200 + 200}{100} + \frac{4000}{10000})\% = (50 - \frac{800}{100} + \frac{4000}{10000})\%$$

$$\Rightarrow (50 - 8 + 0.4)\% = 42.4\%$$

Exam hall Approach :- $20\% = \frac{1}{5}$, $10\% = \frac{1}{10}$

$$\text{equivalent discount} = \frac{4}{5} \times \frac{4}{5} \times \frac{9}{10} = \frac{72}{125}$$

$$= \frac{72}{125} \times 100 = 57.6\%$$

$$\text{So, equivalent discount} = 100 - 57.6 = 42.4\%$$

Example :- The marked price of an article is ₹550. It is sold for ₹317.90, after giving three successive discounts of 15%, x% and 20% on its marked price. If it is sold by giving a single discount of 2x%, then its selling price will be:

SSC GD Constable 17/11/2021 (Morning)

(a) 440 (b) 385 (c) 495 (d) 484

Solution :- ATQ,

$$550 \times \frac{17}{20} \times \frac{4}{5} \times x = 317.90$$

$$374x = 317.90 \Rightarrow x = \frac{317.90}{374} = \frac{17}{20}$$

$$x = \frac{20 - 17}{20} \times 100 = \frac{3}{20} \times 100 = 15\%$$

$$\Rightarrow 2x = 30\%$$

$$\text{So, the required SP of an article} = 550 \times 70\% = ₹385$$

Type - 7

Trade Discount and Cash Discount

In case of Trade Discount and Cash Discount, first deduct the Trade Discount and then calculate the Cash Discount on the remaining amount.

Example :- A retailer permits a trade discount of 25% followed by a cash discount of 8% on the already discounted price of his goods, and still makes a net gain of 17.3% of the cost price. At what percentage above the cost price of the goods is the marked price fixed ?

RRC Group D 02/09/2022 (Morning)

(a) 50% (b) 60% (c) 75% (d) 70%

Solution :-

Let: MP = 100 ,

$$\text{value after trade discount} = 100 \times \frac{75}{100} = 75$$

$$\text{value after cash discount} = 75 \times \frac{92}{100} = 69$$

$$\text{Net gain of 17.3\% of the cost price} = \frac{17.3}{100}$$

$$\text{So, SP} = 117.3 \Rightarrow 117.3 = 69$$

$$\text{CP} = \frac{69}{117.3} \times 100 = \frac{1000}{17}$$

$$\text{MP} - \text{CP} \Rightarrow 100 - \frac{1000}{17} = \frac{700}{17}$$

Percentage above the cost price of the goods

$$= \frac{\frac{700}{17}}{\frac{1000}{17}} \times 100 = 70\%$$

Exam hall approach :- Two successive discounts,

$$\text{overall discount} = 25 + 8 - \frac{25 \times 8}{100} = 31\%$$

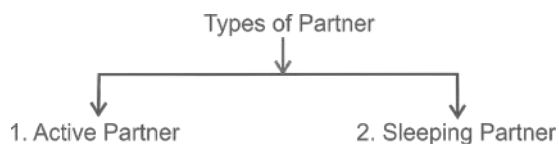
$$\frac{C.P.}{M.P} = \frac{100 - D}{100 + P} = \frac{100 - 31}{100 + 17.3} = \frac{69}{117.3}$$

$$\begin{array}{cc} \text{(M.P)} & \text{(C.P)} \\ 117.3 & 69 \\ 1173 & 690 \\ 17 & 10 \\ \hline & 7 \end{array}$$

$$\text{Required \%} = \frac{7}{10} \times 100 = 70\%$$

Partnership

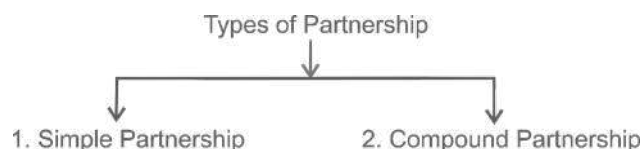
When two or more persons join hands for a business to attain profits is called **Partnership** and the persons are called **partners**.



1. Active Partners :- The partner who manages the business is known as **Active Partner** or **working partner**.

2. Sleeping Partner :- A partner, who simply invests his money in the business but does not manage it voluntarily is referred as **Sleeping partner**.

Every partner invests some amount of money for a certain time to help the partnership firm to get profits.



1. Simple partnership :- If all partners invest their different amounts of capital (money) for the same period or the same capital for a different period, then such a partnership is called a simple partnership.

2. Compound partnership :- If all the partners invest their different capitals (money) for different periods, then such a partnership is called a compound partnership. In Compound partnership partners' profit not only depends on their investments but also on the time period.

* Basics Terms :-

P = Profit

C = Capital

I = Investment

T = Time

Relation Between Profit, Capital And Time

$$\text{Profit} = \text{Capital} \times \text{Time}$$

$$P = C \times T, \quad C = \frac{P}{T}, \quad T = \frac{P}{C}$$

Example:- X, Y and Z invested Rs.4,000, Rs.6,000 and Rs.8,000 respectively in a business. They invested the money for 3, 2 and 4 years respectively. If the total profit is Rs.56,000, then what is the share of Y in the profit?

SSC CHSL 15/03/2023 (2nd Shift)

(a) Rs.18,000 (b) Rs.12,000 (c) Rs.15,000 (d) Rs.9,000

Solution:-

Ratio $\rightarrow$ X : Y : Z

Amount $\rightarrow$ 2 : 3 : 4

Time $\rightarrow$ 3 : 2 : 4

profit $\rightarrow$ 6 : 6 : 16

the share of Y = $56,000 \times \frac{6}{28} = \text{Rs } 12,000$

Type - 1

Active Partner and Sleeping Partner

Active partner gets some extra percentage from the profit for managing the business

Example:- A is a working partner and B is a sleeping partner in a business. A invested ₹30,000 and B invested ₹50,000. A receives 20% of the profit for managing the business and the rest is divided in proportion to their capital. What does B get out of the profit of ₹10,000?

SSC CHSL 02/08/2023 (2nd Shift)

(a) ₹5,500 (b) ₹5,000 (c) ₹4,500 (d) ₹4,000

Solution:-

A receive 20% of the profit for managing the business .

Remaining profit excluding A's managing profit = $10,000 \times 80\% = 8,000$

B's profit = $8,000 \times \frac{5}{8} = 5,000 \text{ Rs.}$

Type - 2

When Capital Investment is the same for the Different Time Periods

$$\text{Profit Ratio} = P_1 : P_2 : P_3 = T_1 : T_2 : T_3$$

{Where P_1, P_2, P_3 is profit and T_1, T_2, T_3 is time}

Example:- A and B enter into a partnership with ₹ 10 lakhs each. After 3 months, A withdraws half of his capital, and then after another six months puts it back. The profit at the end of the year is ₹ 14 lakhs, how much should A get?

DDA ASO 18/09/2017 (Afternoon)

(a) ₹ 4 lakhs (b) ₹ 10 lakhs (c) ₹ 6 lakhs (d) ₹ 3 lakhs

Solution:-

ratio $\rightarrow$ A : B

Investment $\rightarrow$ 10 : 10

Profit $\rightarrow \{(10 \times 3) + (5 \times 6) + (10 \times 3)\} : (10 \times 12)$

$\Rightarrow 90 : 120 = 3 : 4$

Required money = $14 \times \frac{3}{3+4} = 6 \text{ lakh}$

Type - 3

When Time is same for the Different Capital Investment

$$\text{Profit Ratio} = P_1 : P_2 : P_3 = C_1 : C_2 : C_3$$

{Where P_1, P_2, P_3 is profit and C_1, C_2, C_3 is Capital}

Example :- Aman and Ravi invested Rs. 24000 and Rs. 8000 for a period of 2 year. After 2 year, they earned Rs. 24000. What will be the shares of Aman and Ravi out of this earning?

RRB Clerk Pre 07/08/2022 (2nd Shift)

- (a) Rs. 36,000, Rs. 12,000
 (b) Rs. 18,000, Rs. 6,000
 (c) Rs. 25,000, Rs. 40,000
 (d) Rs. 20,000, Rs. 12,000
 (e) None of these

Solution :- Ratio of profit = Ratio of investment $\times$ Time

According to the question

$$\text{Ratio of Profit of Aman and Ravi} = 24000 \times : 8000 = 3 : 1$$

$$\text{Share of Aman} = 24000 \times \frac{3}{4} = \text{Rs. } 18000$$

$$\text{Share of Ravi} = 24000 \times \frac{1}{4} = \text{Rs. } 6000$$

Type - 4

(a) When capital investment X_1, X_2 and X_3 is different for the different time period T_1, T_2 and T_3 .

$$\text{Profit Ratio} = P_1 : P_2 : P_3 = X_1 T_1 : X_2 T_2 : X_3 T_3$$

Example :- A, B and C did certain investments and the ratio of their time periods is 3 : 2 : 7 respectively. Ratio of the profits of A, B and C is 4 : 3 : 14 respectively. What is the ratio of the investments of A, B and C ?

SSC CGL Tier II (02/03/2023)

- (a) 1 : 3 : 4 (b) 7 : 9 : 11 (c) 8 : 9 : 12 (d) 2 : 3 : 11

Solution:-

Let, the investments of A, B and C be x, y, z respectively.

Profit = Total Investment $\times$ Total time

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{Profit} \rightarrow 3x : 2y : 7z = 4 : 3 : 14$$

$$\Rightarrow x : y : z = \frac{4}{3} : \frac{3}{2} : \frac{14}{7} = 8 : 9 : 12$$

(b) When capitals increased or decreased in different time periods :-

Let's say in a partnership Between P and Q, P invests Rs. C_1 for a time period of T_1 but Q invests Rs. C_2 for a time period of T_2 and Rs. C_3 for a time period of T_3 .

$$\text{Profit Ratio} = \{C_1 \times T_1 : (C_2 \times T_2 + C_3 \times T_3)\}$$

Example :- A, B and C started a business with capital in the ratio of 8 : 6 : 5. After 8 months A withdrew 25% of his capital while C increased 20% of his investment if at the end of the year total profit of the business is Rs. 28,000 then what is the profit share of C?

IBPS RRB PO Pre 06/08/2023 (4th Shift)

- (a) 12000 (b) 7000 (c) 4000 (d) 8000 (e) 6000

Solution :- Let the capital of A, B and C is 80R, 60R and 50R respectively.

Ratio of profit = Ratio of investment $\times$ Time

According to the question

Ratio of profit of A, B and C

$$= (80R \times 8 + 60R \times 4) : (60R \times 12) : (50R \times 8 + 60R \times 4)$$

$$\Rightarrow 11 : 9 : 8$$

$$\text{Profit share of C} = 28,000 \times \frac{8}{28} = 8,000$$

Type - 5

When someone Join or Left a Business

Example:- Anil and Sunil started a business investing equal amounts. Anil left after 9 months. They earned an annual profit of ₹28,000. What is Sunil's share of annual profit ?

RRC Group D 22/09/2022 (Morning)

- (a) ₹16,000 (b) ₹15,000 (c) ₹13,000 (d) ₹12,000

Solution:-

Anil : Sunil

Investment $\rightarrow$ 1 : 1

Time $\rightarrow$ 9 : 12

Profit $\rightarrow$ 3 : 4

Total profit (4 + 3) = 7 unit $\rightarrow$ 28000 (given)

$$\text{So share of Sunil} = \frac{28000}{7} \times 4 = ₹16000$$

Example :- Rahul and Raman started a business by investing Rs. 16000 and Rs 12000 respectively. After 4 months, Rahul left the business and Mohan joined the business by investing Rs.20000. At the end of the year, there was a profit of Rs.46000. What will be the share of Mohan in the profit ?

SSC MTS 20/06/2023 (Afternoon)

- (a) Rs.28700 (b) Rs.20000
 (c) Rs. 36800 (d) Rs.23400

Solution :-

Ratio $\rightarrow$ Rahul : Raman : Mohan

Investment $\rightarrow$ 16 : 12 : 20

Time $\rightarrow$ 4 : 12 : 8

Profit $\rightarrow$ 4 : 9 : 10

Profit (4 + 9 + 10) units = Rs. 46000

$$\text{Share of Mohan (10) units} = \frac{46000}{23} \times 10 = \text{Rs. } 20000$$

Type - 6

(Tax/donation based)

Example:- A and B invested money in a business in the ratio of 7 : 5. If 15% of the total profit goes for charity, and A's share in the profit is Rs.5,950, then what is the total profit?

SSC CGL 18/07/2023 (4th shift)

- (a) Rs.12,500 (b) Rs.12,000
 (c) Rs.10,500 (d) Rs.11,750

Solution:- Let the investment of A and B be 7x and 5x.

If investment time is equal, then the ratio of profit is equal to the ratio of investment.

According to the question,

$$7x = 5950 \text{ Rs.}$$

$$12x = 5950 \times \frac{12}{7} = 10200 \text{ Rs.}$$

$$\text{Total profit} = 10200 \times \frac{100}{(100 - 15)} = 12000 \text{ Rs.}$$

Simple Interest

Simple interest is a method to calculate the amount of interest charged on a sum at a given rate and for a given period of time.

Some important Terms

1) Principal : The principal is the amount that was initially borrowed (loan) from the bank or invested. It is denoted by P.

2) Rate: It is the rate of interest at which the principal amount is borrowed or given for a certain time. It is denoted by R.

3) Time: Time is the duration for which the principal amount is given to someone. Time is denoted by T.

4) Amount: Sum of Principal and Simple interest.

Basic Formulae :-

1. Simple Interest on a sum (principal) of money at R% per annum for T years is given by:

$$\text{Simple interest (S.I.)} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100} = \frac{PRT}{100}$$

$$2. (\text{Principal}) P = \frac{SI \times 100}{R \times T}, (\text{Time}) T = \frac{SI \times 100}{R \times P},$$

$$(\text{Rate}) R = \frac{SI \times 100}{P \times T}$$

3. Amount (A) = Principal + Simple interest(S.I.)

Type - 1

When we have to find Simple Interest

Example :- Find the simple interest on ₹8,000 at the rate of 10% per annum in 6 years.

SSC MTS 12/09/2023 (2nd Shift)

(a) ₹4,200 (b) ₹4,400 (c) ₹4,800 (d) ₹4,600

Solution:-

Total interest in 6 years = $6 \times 10\% = 60\%$

Simple interest = $8000 \times 60\% = ₹4,800$

Exam hall approach:

$\frac{1}{10} \rightarrow$ Interest
 $\frac{1}{10} \rightarrow$ Principal

10 unit $\rightarrow$ 8000 Rs.

6 unit $\rightarrow$ 4800 Rs.

When we have to find Amount

Example :- Ramesh invested ₹1,232 at 5% p.a. rate of simple interest in a bank. What amount will he get after 3 years?

RRC Group D 22/08/2022 (Evening)

(a) ₹1,285.80 (b) ₹2,145.80 (c) ₹1,848.80 (d) ₹1,416.80

Solution :- Principal = ₹1232,

Rate% = 5%, Time = 3 years

$$S.I = \frac{1232 \times 5 \times 3}{100} = ₹184.80$$

$$\text{Amount} = P + S.I = 1232 + 184.80$$

$$= ₹1,416.80$$

Exam hall approach :

$$\text{Rate for 3 years} = 5 \times 3 = 15\% = \frac{3}{20}$$

$$20 \text{ unit} = ₹1,232 \Rightarrow \text{Amount (23 unit)} = \frac{1232}{20} \times 23$$

$$\text{Amount} = ₹1,416.80$$

When we have to find Principal

Example :- A sum at simple interest of 5.5% per annum amounts to ₹7,503 in 4 years, Find the sum (in ₹).

SSC MTS 14/09/2023 (3rd Shift)

(a) 5,350 (b) 6,150 (c) 5,980 (d) 5,520

Solution :- Rate for 4 years at S.I. = $5.5\% \times 4 = 22\%$

$$122\% = ₹7503$$

$$(\text{Principal}) 100\% = \frac{7503}{122} \times 100 = ₹6150$$

When we have to find Time

Example :- In what time will ₹3,720 amount to ₹5,282.4 at 12% simple interest per annum?

SSC CGL 27/07/2023 (3rd shift)

(a) 3 years (b) $3\frac{1}{2}$ years (c) $5\frac{1}{2}$ years (d) 5 years

Solution :-

$$\text{Rate} = 12\% = \frac{12}{100}$$

$$\text{Ratio of principal and Amount} = 3,720 : 5,282.4 = 37200 : 52824 = 100 : 142$$

$$\text{Now, the required time} = \frac{142 - 100}{12}$$

$$= \frac{42}{12} = 3\frac{1}{2} \text{ yrs}$$

When we have to find Rate

Example :- The simple interest received on a sum for 20 years is $\frac{3}{5}$ of the sum. What is the annual rate of interest?

SSC MTS 15/06/2023 (Morning)

(a) 3 percent (b) 4 percent (c) 5 percent (d) 6 percent

Solution :- Let the sum be 100 units.

According to question,

$$\text{Simple Interest} = 100 \times \frac{3}{5} = 60 \text{ unit}$$

$$60 = \frac{100 \times 20 \times r}{100}$$

$$\text{Rate of interest (r)} = 3\%$$

Example :- A certain sum of money lent out at simple interest amounts to ₹12,600 in 2 years and ₹16,200 in 4 years. The rate percent per annum is:

SSC MTS 11/09/2023 (3rd Shift)

(a) 25% (b) $12\frac{6}{7}\%$ (c) $17\frac{1}{2}\%$ (d) 20%

Solution :- Interest of one year

$$= \frac{16200 - 12600}{2} = 1800$$

$$\text{Principal} = 12600 - 1800 \times 2$$

$$\Rightarrow 12600 - 3600 = 9000$$

$$\text{Rate} = \frac{1800}{9000} \times 100 = 20\%$$

Exam hall approach :-

$$\begin{array}{l} 2 \text{ years} \rightarrow 12,600 \\ 4 \text{ years} \rightarrow 16,200 \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} \rightarrow \text{difference of SI} = 3600$$

$$2 \text{ year simple interest} = 3600$$

$$1 \text{ year simple interest} = 1800$$

$$\text{Principal} = 12600 - 3600 = 9000$$

$$\text{Rate} = \frac{1800}{9000} \times 100 = 20\%$$

Important Note :-

1). Notice that in the formula, $SI = \frac{P \times R \times T}{100}$, we are just calculating $(T \times R)\%$ of P.

2). Simple Interest increases linearly with the number of years i.e. to get SI for 1 year, you can directly divide given interest by the number of years.

Example :- If SI is Rs. 400 for 4 years, then it will be Rs. 200 for 2 years.

Type - 2

*** If a certain sum becomes n times of itself in T years on simple interest, then -**

$$R\% = \frac{(n-1)}{T} \times 100$$

Example :- A sum of money becomes 7 times in 16 years at simple interest per annum. Find the per annum rate of interest.

SSC MTS 05/09/2023 (3rd Shift)

(a) 25.8% (b) 39.2% (c) 20.3% (d) 37.5%

Solution :-

$$R\% = \frac{(n-1)}{T} \times 100 = \frac{(7-1)}{16} \times 100 \Rightarrow R = \frac{600}{16} = 37.5\%$$

*** If a certain sum becomes n times of itself at R% per annum on simple interest, then -**

$$T = \left(\frac{n-1}{R} \times 100 \right) \text{ years}$$

Example :- In how many least number of complete years will a sum of money become more than 4 times of itself at the rate of 14 percent per annum on simple interest ?

SSC MTS 10/05/2023 (Evening)

(a) 20 years (b) 25 years (c) 29 years (d) 22 years

Solution :- Amount > 4 × Principal

So interest should be > 3P

$$T > \frac{(n-1)}{R} \times 100 \text{ years}$$

$$T > \frac{(4-1)}{14} \times 100 \text{ years} = \frac{150}{7} \text{ years} = 21.42 \text{ years}$$

Required minimum complete years = 22 years

Type - 3

*** When principal become n time at certain rate in certain time**

Example :- A sum of money doubles itself in 4 years at simple interest. In how much time will it become 7 times at the same rate of interest ?

SSC CHSL Tier II (26/06/2023)

(a) 25 years (b) 23 years (c) 24 years (d) 21 years

Solution :- Let principal be ₹ x

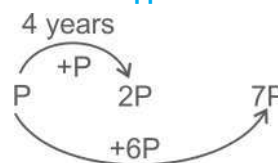
According to the question,

Simple interest in 4 years = ₹ x

Require time for 6x interest = 24 years

So, money became 7 times in 24 years

Exam hall approach :-



p → 4 years

6p → 6 × 4 years = 24 years

Type - 4

*** Simple Interest ∝ Principal, Rate, Time**

Example :- What will be the ratio of simple interest earned by a certain amount at the same rate of interest for 8 years and 20 years ?

SSC MTS 16/05/2023 (Evening)

(a) 5 : 7 (b) 2 : 5 (c) 3 : 5 (d) 4 : 3

Solution :- Let principle = p, Time = t, Rate = r%

Simple interest (S_1, S_2)

$$S_1 = \frac{p \times r \times 8}{100}, S_2 = \frac{p \times r \times 20}{100}$$

Required ratio $S_1 : S_2 = 8 : 20 = 2 : 5$

Type - 5

*** Principal Invested in Parts at different rates**

$$\frac{p_1 \times r_1 \times t_1}{100} + \frac{p_2 \times r_2 \times t_2}{100} = SI$$

Example :- A sum of Rs 15,600 is invested partly at 7% per annum and the remaining at 9% per annum simple interest. If the total interest at the end of 3 years is Rs3,738. How much money was invested at 7% per annum ?

SSC CPO 13/03/2019 (Morning)

(a) 7,800 (b) 7,900 (c) 7,600 (d) 7,700

Solution :- Let, the amount invested at the rate of 7% = x

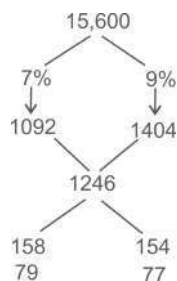
According to the question,

$$\frac{x \times 7 \times 3}{100} + \frac{(15600 - x) \times 9 \times 3}{100} = 3738$$

$$7x + 140400 - 9x = 124600 \Rightarrow x = 7900$$

Exam hall approach :-

$$\text{Interest for 1 year} = \frac{3,738}{3} = \text{Rs. 1246}$$



$(79 + 77) \text{ unit} = 15,600 \Rightarrow 156 \text{ unit} = 15,600$

Sum invested at 7% (79 unit) = $\frac{15600}{156} \times 79 = \text{Rs. } 7900$

Example :- A person invested a sum of Rs. 18,600 at $x\%$ p.a. and another sum that is twice the former at $(x + 2)\%$ p.a., both at a simple interest. If the total interest earned on both investments for $3\frac{1}{2}$ years is Rs 23,110.50, then the rate of interest p.a. on the second investment is. SSC CPO 11/12/2019(Morning)

(a) 11% (b) 10.5% (c) 13% (d) 12.5%

Solution :- We know that,

Rs 18,600 at $x\%$ and Rs. 37,200 at $(x + 2)\%$

As per question,

$$\Rightarrow 18600 \times \frac{7}{2} \times \frac{x}{100} + 37200 \times \frac{7}{2} \times \frac{(x + 2)}{100} = \frac{231105}{10}$$

$$\Rightarrow 186 \times \frac{7}{2} [x + 2x + 4] = \frac{231105}{10}$$

$$\Rightarrow 3x + 4 = \frac{2201}{62} \Rightarrow x = 10.5\%$$

So, the second rate is $x + 2 = 12.5\%$

Type - 6

Increase or decrease in Rate/time

Example :- A money lender finds that due to the fall in the annual rate of interest from 18% to 15%, his yearly income diminishes by Rs.750. His capital is:

Selection Post 30/06/2023 (1st Shift)

(a) Rs.25,500 (b) Rs.25,000 (c) Rs.5,000 (d) Rs.2,500

Solution :-

Principal $\times$ (difference in rates) = 750 Rs.

Principal $\times$ (18% - 15%) = 750 Rs.

$$\text{Principal} = \frac{750 \times 100}{3} = 25000 \text{ Rs.}$$

Example :- A sum of ₹800 amounts to ₹1,040 in 3 years at simple interest. If the interest rate is increased by 3%, find the amount.

SSC MTS 14/09/2023 (2nd Shift)

(a) ₹1,102 (b) ₹1,212 (c) ₹1,112 (d) ₹1,012

Solution :- According to the question,

$$\text{S.I} = 1040 - 800 = ₹ 240$$

$$\text{S.I} = \frac{p \times r \times t}{100} \Rightarrow 240 = \frac{800 \times r \times 3}{100}$$

(rate) $r = 10\%$

New rate = $10 + 3 = 13\%$

$$\text{Required amount} = 800 + \frac{800 \times 13 \times 3}{100} = ₹1,112$$

Exam hall approach :-

3 year interest = $1040 - 800 = 240$

1 year interest = 80

$$\text{Rate} = \frac{80}{800} \times 100 = 10\%$$

Increased rate = $10 + 3 = 13\%$

Increased rate for 3 years = 39%

$$\text{Required amount} = \frac{139}{100} \times 800 = ₹1,112$$

Example :- An amount of ₹P was put at simple interest at a certain rate for 4 years. If it had been put at a 6% higher rate for the same period, it would have fetched ₹600 more interest. What is the value of 2.5 P?

SSC CHSL 10/08/2023 (4th Shift)

(a) ₹ 3,750 (b) ₹6,250 (c) ₹ 4,850 (d) ₹ 2,500

$$\text{Solution :-} \frac{\text{Principal} \times 6 \times 4}{100} = 600 \text{ ₹}$$

$$\text{Principal} = \frac{600}{24} \times 100 = 2500 \text{ ₹}$$

$$\text{Now, } 2.5 \text{ P} = 2500 \times 2.5 = 6250 \text{ ₹}$$

Type - 7

Successive Interest Rate

Example :- A sum of money earns a simple interest at 7.25% per annum for the first eight years, at 8.5% for the next six years, and at 6.5% for the final four years. If the total interest earned during these eighteen years was ₹35,100, what was the original sum invested (in ₹)?

SSC CGL 20/07/2023 (2nd shift)

(a) 25,800 (b) 25,500 (c) 26,400 (d) 26,000

$$\text{Solution :-} \text{Simple Interest} = \frac{P \times R \times T}{100}$$

According to question,

$$35100 = \frac{P \times 7.25 \times 8}{100} + \frac{P \times 8.5 \times 6}{100} + \frac{P \times 6.5 \times 4}{100}$$

$$\Rightarrow 35100 = \frac{135P}{100} \Rightarrow P = 26000 \text{ Rs.}$$

Exam hall approach :-

Interest for first eight years = $7.25 \times 8 = 58\%$

Interest for next six years = $8.5 \times 6 = 51\%$

Interest for final four years = $6.5 \times 4 = 26\%$

Total interest = $58 + 51 + 26 = 135\%$

$121\% = 35100$

$$100\% (\text{Principal}) = \frac{35100}{135} \times 100 = 26000 \text{ Rs.}$$

Example :- What will be the total amount payable on ₹ 25,000 in 2 years if the rate of simple interest in successive years is 4% and 5% respectively?

SSC CHSL 04/08/2023 (1st Shift)

(a) ₹26,250 (b) ₹28,250 (c) ₹26,000 (d) ₹27,250

Solution :- Total interest rate = $5 + 4 = 9\%$

$$\text{Total amount} = 25,000 \times \frac{109}{100} = 27,250 \text{ ₹}$$

Example :- Tanya gives _____ to Sabir at simple interest with a condition that the rate of interest will increase by 2% per year. The Total interest after 2 years is Rs. 10,240 and the rate of interest for first year is 7%.

SBI PO Pre 19/12/2022 (1st Shift)

- (a) Rs.36,000 (b) Rs.38,000 (c) Rs. 40,000
(d) Rs. 42,000 (e) Rs. 64,000

Solution :-

Total rate of interest for 2 years = 7% + 9% = 16%

$\Rightarrow 16\% = 10,240$

100%(Principal) = 64,000

Type - 8

When Simple interest are equal

Example :- The simple interest earned on ₹7,000 in 2 years at the rate of R% per annum equals to the simple interest earned on ₹5,000 at the rate of 5% per annum in 14 years. The value of R (in percentage) is:

Matriculation Level 27/06/2023 (Shift - 1)

- (a) 28% (b) 15% (c) 25% (d) 20%

Solution :- Equating simple interest :-

$$\frac{7000 \times R \times 2}{100} = \frac{5000 \times 5 \times 14}{100}$$

$$R = \frac{5 \times 5 \times 14}{7 \times 2} = 25\%$$

Type - 9

When Amounts are equal

Example :- If the amount obtained by A by investing ₹9,100 for three years at a rate of 10% p.a. on simple interest is equal to the amount obtained by B by investing a certain sum of money for five years at a rate of 8% p.a. on simple interest, then 90% of the sum invested by B (in ₹) is

SSC CGL Tier II (29/01/2022)

- (a) 8,450 (b) 7,605 (c) 8,540 (d) 7,800

Solution :- Amount obtained by A

$$= 9100 + \frac{9100 \times 10 \times 3}{100} = 11830$$

$$\text{Amount obtained by B} = P + \frac{P \times 8 \times 5}{100} = \frac{7}{5} P$$

According to the question,

$$11830 = \frac{7}{5} P \Rightarrow P = \frac{11830 \times 5}{7} = 8450$$

90% of 8450 = Rs. 7605

Exam hall approach :-

Let, the Principal invested by B is P.

According to question,

$$9100 \times 1.3 = P \times 1.4$$

$$P = 6500 \times 1.3 = 8450 \text{ Rs.}$$

$$90\% \text{ of } P = 8450 \times 0.9 = 7605 \text{ Rs.}$$

NTPC CBT II (13/06/2022) 2nd Shift

- (a) 22 (b) 20 (c) 11 (d) 10

Solution :- As per question,

$$80,000 + \frac{80,000 \times 24 \times T}{100} = 91,000 + \frac{91,000 \times 20 \times T}{100}$$

$$\Rightarrow \frac{80,000 \times 24 \times T}{100} - \frac{91,000 \times 20 \times T}{100} = 11,000$$

$$\Rightarrow 19,200T - 18,200T = 11,000 \Rightarrow 1000T = 11,000 \Rightarrow T = 11$$

Type - 10

True Discount

Example :- What is the present worth of ₹1,100 due in 2 years at 5% simple interest per annum?

SSC CGL 26/07/2023 (3rd shift)

- (a) ₹3,000 (b) ₹2,000 (c) ₹1,000 (d) ₹1,500

Solution :- Simple interest = $\frac{P \times R \times T}{100}$

According to question,

$$1100 - P = \frac{P \times 5 \times 2}{100}$$

$$1100 - P = \frac{P}{10} \Rightarrow P = \frac{11000}{11} = 1000 \text{ Rs.}$$

Example :- Suresh borrows ₹80,000 at 24% per annum simple interest and Ramesh borrows ₹91,000 at 20% per annum simple interest. In how many years will their amounts of debts be equal?

Compound Interest

Compound interest is the resulting interest that is generated on the combination of the principal amount and the accumulated interest from previous periods.

Parameter	Simple Interest	Compound Interest
Formula	$S.I. = \frac{(P \times R \times T)}{100}$	$C.I. = P(1 + \frac{R}{100})^t - P$
Return Amount	Lesser	Higher
Principal Amount	Constant	Varying during the entire borrowing period
Interest Charged	Interest charged on principal amount	Interest charged on principal and accumulated interest.
Growth	The Growth remains quite uniform	The growth increases quite rapidly

Key - Points :-

1. For one year or one term simple interest and compound interest are the same.
2. In simple interest you divide the simple interest by the number of years to calculate one year interest but in compound interest this is not applicable.
3. Compound Interest is the interest on amount (principal + Interest); Symbol we will use 'CI'.

Example :- Let us say one person borrows Rs.100 for 3 years at the rate of 10% annually. What amount should he have to return after 3 years ?

Solution :- Here is the approximation technique to solve this problem:

$$100 \xrightarrow{10\%}{10} 110 \xrightarrow{10\%}{11} 121 \xrightarrow{10\%}{12.1} 133.10$$

Compound interest will be = $133.10 - 100 = 33.1$ Rs.

Or

The above concept can also be written as :

$$\text{Amount} = \frac{100 \times 110 \times 110 \times 110}{100 \times 100 \times 100} = 133.10$$

C.I. = Amount - Principal

$$C.I. = 133.10 - 100 = 33.10$$

$$\text{Or, Amount} = \text{Principal} (1 + \frac{r}{100})^t = 100 (1 + \frac{10}{100})^3$$

$$= 100 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10} = 133.10$$

$$C.I. = 33.10$$

Note :- Notice that in CI, interest is calculated on the amount at the end of every year while in SI interest for each year is calculated on the Principal amount.

4. If rate of interest is different for different year, let us say 10% for 1st year, 20% for 2nd year, and 30% for 3rd year then amount will be calculated as follows:

$$100 \xrightarrow{10\%}{10} 110 \xrightarrow{20\%}{22} 132 \xrightarrow{30\%}{39.6} 171.60$$

$$\text{Amount} = \frac{100 \times 110 \times 120 \times 130}{100 \times 100 \times 100} = ₹171.60$$

Different Methods To Calculate C.I.

There are different types of methods to solve CI problems. We use different methods depending on what is asked in questions and which type of data questions have. We will further understand this through examples.

- Formula method
- Tree method
- Ratio method
- Successive method

Now we will take one question from each method and try to solve it.

Formula method :- We use this formula when we have given CI and principal to find answers quickly.

CI = Amount - Principal

$$\text{Amount} = \text{Principal} (1 + \frac{r}{100})^t$$

Example :- The interest, compounded annually, on a sum of ₹3,240 after two years is ₹370. What is the rate of interest, correct up to two decimal places?

SSC MTS 02/11/2021 (Evening)

- (a) 5.56% (b) 8.56% (c) 6.56% (d) 7.56%

Solution :- we know that, $A = P (1 + \frac{r}{100})^t$

Here P = 3240 and A = 3240 + 370 = 3610.

$$3610 = 3240 (1 + \frac{r}{100})^2$$

$$\Rightarrow \frac{3610}{3240} = (1 + \frac{r}{100})^2$$

$$\Rightarrow (1 + \frac{r}{100}) = \sqrt{\frac{3610}{3240}} = \frac{19}{18}$$

$$\Rightarrow (1 + \frac{r}{100}) = (1 + \frac{1}{18}) \Rightarrow r = \frac{100}{18}$$

Clearly, $r = 5.56\%$

Exam hall approach :- Amount = (3240 + 370) = ₹3610

Ratio - principal : amount

$$3240 : 3610$$

$$\sqrt{324} : \sqrt{361} = 18 : 19$$

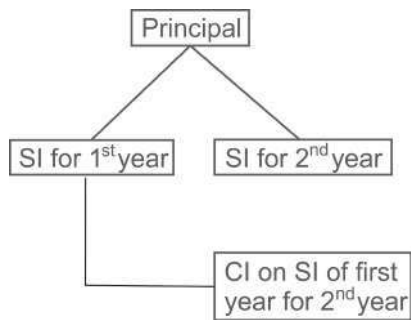
$$\text{Rate\%} = \frac{1}{18} \times 100 = 5.56\%$$

Tree Method

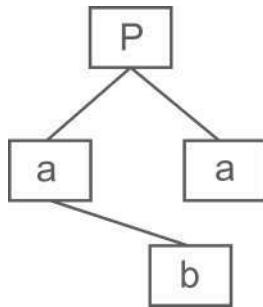
Nowadays the use of this method has reduced. But this method proves to be more effective when we are finding the difference of CI between second and third year or third and fourth year etc.

For 2 years cycle tree Solution Structure

For 2 years cycle tree Solution Structure

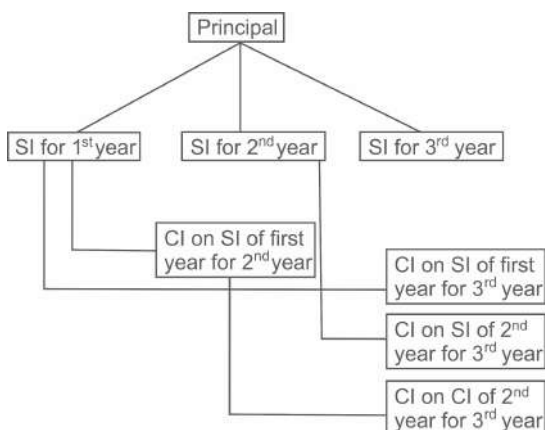


For 2 years

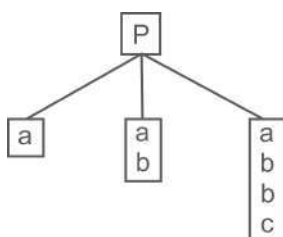


2 year CI	$a + a + b$
Amount after 2 year	$p + (2a + b)$
2 year difference of CI and SI	b
2nd year CI	$a + b$

For 3 years cycle tree Solution Structure



For 3 year cycle



First year interest	a
2nd year interest	$a + b$
3rd year interest	$a + 2b + c$

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3 year SI	$3a$
3 year CI	$3a + 3b + c$
Difference of 3 year CI and SI	$3b + c$

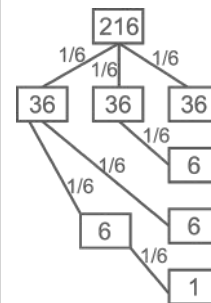
Example :- If the difference between CI for 3rd year and 2nd year is Rs. 420 on a certain sum of money at the rate of $16\frac{2}{3}\%$ for 3 years, find the principal.

(a) 12960 (b) 22060 (c) 32060 (d) 12560

Solution :- $16\frac{2}{3}\% = \frac{1}{6}$

$$\text{Rate} = \left(\frac{1}{6}\right)^3 = \frac{1}{216}$$

Let principal = 216 unit



Difference of CI of 2nd year and 3rd year = $(6 + 1) = 7$ unit

According to the question,

7 unit = 420 $\Rightarrow$ 1 unit = Rs. 60

so, principal (216) unit = $216 \times 60 = \text{Rs. } 12960$

Ratio Method

We prefer ratio method when rates are given in the fraction

Such rate like - $12\frac{1}{2}\%$, $16\frac{2}{3}\%$, $6\frac{1}{4}\%$

Example :- The principal that amounts to ₹ 4,913 in 3 years at $6\frac{1}{4}\%$ per annum compound interest compounded annually is:

Haryana CET 05/11/2022 (2nd Shift)

(a) ₹4,085 (b) ₹4,096 (c) ₹3,096 (d) ₹4,076

Solution :- Fraction $(6\frac{1}{4}\%) = \frac{1}{16}$

Ratio $\rightarrow$ Principle : Amount

1st year $\rightarrow$ 16 : 17

2nd year $\rightarrow$ 16 : 17

3rd year $\rightarrow$ 16 : 17

Final ratio $\rightarrow$ 4096 : 4913

Amount = 4913 units

4913 units = ₹4913 $\Rightarrow$ 1 unit = ₹1

So, principle (4096) units = ₹4096

Successive Method

We prefer this method when conversion of rate into fraction is tough. Also as per the requirement of the question, we can use this method where it is a little difficult to apply the ratio method. Such rates are - 3%, 6%, 7%, 17%, 30% etc.

Example :- Ram and Dipti each invested a sum of ₹8000 for a period of two years at 30% compound interest per annum.

However, while for Ram the interest was compounded annually, for Dipti it was compounded every eight months. How much more will Dipti receive as interest compared to Ram at the end of the two year period ?

NTPC CBT II (10/05/2022) 1st Shift

- (a) ₹312 (b) ₹320 (c) ₹304 (d) ₹296

Solution :- For Ram :

P = 8000, Rate = 30%, Time = 2 years

$$\text{Amount} = 8000 \times \frac{13}{10} \times \frac{13}{10} = \text{Rs. } 13520$$

For Dipti :

$$P = 8000, \text{Rate} = 30 \times \frac{8}{12} \% = 20\% = \frac{1}{5}$$

$$\text{Cycle} = 2 \times \frac{12}{8} = 3 \text{ (3 cycle of 8 - 8 months)}$$

$$\text{Amount} = 8000 \times \frac{6}{5} \times \frac{6}{5} \times \frac{6}{5} = \text{Rs. } 13824$$

$$\text{Difference} = 13824 - 13520 = \text{Rs. } 304$$

Exam hall approach :- Rate For ram = 30%

Effective rate of 30% = 69% (for 2 years)

$$\text{Rate For Dipti} = 30 \times \frac{8}{12} \% = 20\%$$

Effective rate of 20% = 72.8% (for 3 cycle)

$$\text{Difference} = 72.8\% - 69\% = 3.8\%$$

$$= 8000 \times \frac{3.8}{100} = \text{Rs. } 304$$

Type - 1

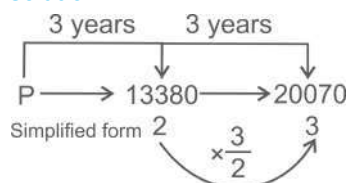
Calculate principal

Example :- A certain sum of money invested at compound interest becomes Rs 13,380 after 3 years and Rs 20,070 after 6 years, when the interest is compounded annually. Find the Sum ?

RRC Group D 23/08/2022 (Evening)

- (a) ₹ 8,505 (b) ₹7,020 (c) ₹8,920 (d) ₹7,007

Solution :-



A principal becomes 13380 after 3 years and 20070 after 6 years.

$$13380 : 20070$$

$$2 : 3$$

$$P \times \frac{3}{2} = 13380 \Rightarrow P = 2 \times 4460$$

$$\text{Principal} = ₹ 8,920$$

Type - 2

Calculate Rate

Example :- A sum of ₹14,375, when invested at r% interest per year compounded annually, amounts to ₹16,767 after two years. What is the value of r?

SSC CHSL 10/06/2022 (Evening)

- (a) 9 (b) 6 (c) 7 (d) 8

Solution :-

Principle = 14,375

Amount = 16,767

$$16,767 = 14,375 \left(1 + \frac{r}{100}\right)^2$$

$$\Rightarrow \frac{16767}{14375} = \left(1 + \frac{r}{100}\right)^2$$

$$\Rightarrow \frac{729}{625} = \left(1 + \frac{r}{100}\right)^2 \Rightarrow 1 + \frac{r}{100} = \frac{27}{25} \Rightarrow r = 8\%$$

Exam Hall Approach :

P	A
14375	16767
625	729
25^2	27^2

$$\text{Rate of interest} = \frac{2}{25} \times 100 = 8\%$$

Type - 3

Calculate Time

Example :- At the rate of 10% compound interest per annum, in how many years, ₹1,000 will become ₹1,331 ?

UKPSC CSAT (03/04/2022)

- (a) 2 (b) 2.5 (c) 3 (d) 3.5

Solution :-

According to the question,

$$1331 = 1000 \left(1 + \frac{10}{100}\right)^n$$

$$\frac{1331}{1000} = \left(\frac{11}{10}\right)^n \Rightarrow \left(\frac{11}{10}\right)^3 = \left(\frac{11}{10}\right)^n$$

So, $n = 3$ years

Exam hall approach :

Principal	Amount
1000	1331
Simplified form	
10^3	11^3
Here T = 3 Yr	

Type - 4

If the rate of interest of 1st year, 2nd, 3rd year are r_1 , r_2 and r_3 and so on then,

$$\text{Amount} = P \left[1 + \frac{r_1}{100}\right] \left[1 + \frac{r_2}{100}\right] \left[1 + \frac{r_3}{100}\right] \text{----- and so on}$$

Example :- Rs. 56000 is invested at the rate of 5% for 1st year, at 10% for 2nd year and at 15% for the 3rd year. Find the total compound interest earned in these 3 years.

AFCAT 12/02/2022 (1st Shift)

- (a) Rs.18283 (b) Rs.18382 (c) Rs.18832 (d) Rs.18238

Solution :- Principal = Rs. 56000

$$\text{Amount} = 56000 \times \left(1 + \frac{5}{100}\right) \times \left(1 + \frac{10}{100}\right) \times \left(1 + \frac{15}{100}\right)$$

$$\text{Amount} = 56000 \times \frac{105}{100} \times \frac{110}{100} \times \frac{115}{100}$$

$$= \text{Rs } 74382$$

$$\text{Now, CI} = A - P = 18382$$

Type - 5

Difference Between CI and SI

$$1. \text{ Difference of CI and SI for 2 years} = (CI - SI)_{2\text{yrs}} = P\left(\frac{R}{100}\right)^2$$

$$2. \text{ Difference of CI and SI for 3 years} = (CI - SI)_{3\text{yrs}} = P\left(\frac{R}{100}\right)^2 \left(\frac{300 + R}{100}\right) \\ = (CI - SI)_{3\text{yrs}} \times 100^3 = PR^2(300 + R)$$

Where P = Principal, R = rate of interest

Example :- On a certain sum, the difference between compound interest and simple interest at 10% per annum for 2 years is Rs.250. The sum is _____.

SSC CPO 09/11/2022 (Morning)

(a) Rs.25,450 (b) Rs.26,550 (c) Rs.25,000 (d) Rs.26,000

Solution :- We Know that

$$\text{Difference of CI and SI for 2 years} = (CI - SI)_{2\text{yrs}} = P\left(\frac{R}{100}\right)^2$$

According to question

$$250 = P\left(\frac{10}{100}\right)^2 \Rightarrow 250 = P\left(\frac{10 \times 10}{100 \times 100}\right) \Rightarrow P = \text{Rs.}25000$$

For 2 years Cycles/years

Rate	Compound interest (CI)	CI - SI
5%	10.25%	0.25%
10%	21%	1%
15%	32.25%	2.25%
20%	44%	4%
25%	56.25%	6.25%
30%	69%	9%

For 3 years Cycles/years

Rate	Compound interest (CI)	CI - SI
5%	15.76%	0.76%
10%	33.10%	3.10%
15%	52.09%	7.09%
20%	72.80%	12.80%
25%	95.31%	20.31%
30%	119.70%	29.70%

Type - 6

Pascal Rule of CI

1	1	1	1	1	→ 1 Year CI
1	2	3	4	5	→ 2 Year CI
1	3	6	10	15	→ 3 Year CI
1	4	10	20	30	→ 4 Year CI
1	5	15	35	70	→ 5 Year CI

Example :- The simple interest on a certain sum for 3 years at 12% per annum is Rs. 6750. What is the compound interest (in Rs.) on the same sum for 2 years at 20% per annum, if interest is compounded half-yearly? (round off to the nearest Rs.)

SSC CHSL 2020 (Held on : 12 april 2021 shift 3)

(a) 7729 (b) 8702 (c) 8000 (d) 6750

Solution :- Rate = 12%

Simple interest for 3 years (36% of principal) = 6750 Rs.

$$\text{Principal}(100\%) = \frac{6750}{36} \times 100 = 18750.$$

$$\text{Now, new rate} = \frac{20}{2} = 10\%$$

Time = 2 × 2 = 4 cycle

Four year ⇒ **Code**

10% of 18750 1875 → **4**

10% of 1875 187.5 → **6**

10% of 187.5 18.75 → **4**

10% of 18.75 1.875 → **1**

$$\text{Total compound interest} = (4 \times 1875) + (6 \times 187.5) + (4 \times 18.75) + 1.875 \\ = 8701.875 \approx 8702$$

Type - 7

Multiplying Factor

(a) If an amount become 'n' times in 't' years at the rate of compound interest then, time taken to becomes 'n^x' times for the same amount is equal to 't × x' years.

Example :- If a certain sum becomes two times itself in 6 years at compound interest in case of annual compounding, then the number of years in which it will become eight times of itself at the same rate of interest under annual compounding is:

RRC Group D 09/09/2022 (Morning)

(a) 18 years (b) 36 years (c) 24 years (d) 12 years

Solution :- Amount becomes 2 times in 6 years. It will become

$$8 \text{ times} = (2^3) \rightarrow 6 \times 3 = 18 \text{ years.}$$



(b) If a sum 'P' becomes 'x' times of itself in 't' years on CI, then

$$\text{Rate \%} = [x^{\frac{1}{t}} - 1] \times 100$$

Example :- At what rate percent compound interest, does a sum of money become 2.25 times of itself in 2 years?

Solution :-

$$\text{We know that Rate \%} = [x^{\frac{1}{t}} - 1] \times 100$$

According to question

$$\text{Rate \%} = [2.25^{\frac{1}{2}} - 1] \times 100$$

$$\text{Rate \%} = [1.5 - 1] \times 100$$

$$\text{Rate \%} = [0.5] \times 100 = 50 \%$$

(c) On compound interest If sum becomes A in 'a' years and B in 'b' years then,

$$1. \text{ If } b - a = 1, \text{ then } R\% = \left(\frac{B}{A} - 1\right) \times 100\%$$

2. If $b - a = 2$, then $R\% = (\sqrt{\frac{B}{A}} - 1) \times 100\%$

3. If $b - a = 3$, then $R\% = (\sqrt[3]{\frac{B}{A}} - 1) \times 100\%$

4. If $b - a = n$, then $R\% = [(\frac{B}{A})^{\frac{1}{n}} - 1] \times 100\%$

Example :- A sum of money becomes ₹6,400 in 3 years and ₹8,100 in 5 years on compound interest compounded annually. Find the rate of interest per annum.

SSC MTS 01/09/2023 (2nd Shift)

(a) $12\frac{3}{8}\%$ (b) $12\frac{1}{2}\%$ (c) $12\frac{6}{8}\%$ (d) $12\frac{5}{8}\%$

Solution :- $6400 = P(1 + \frac{r}{100})^3$ (eq 1)

$8100 = P(1 + \frac{r}{100})^5$ (equation 2)

Divide equation (2) by equation (1) we get

$$(1 + \frac{r}{100})^2 = \frac{81}{64}$$

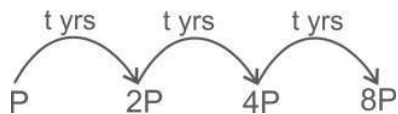
$$(1 + \frac{r}{100}) = \frac{9}{8} \Rightarrow r = 12\frac{1}{2}\%$$

Exam hall approach :- $a = 3, b = 5 \Rightarrow b - a = 2$

So, $R\% = (\sqrt{\frac{8100}{6400}} - 1) \times 100\% = \frac{1}{8} \times 100\% = 12\frac{1}{2}\%$

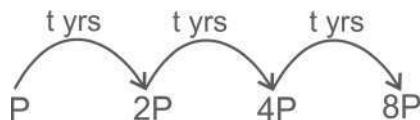
(d) Double Money Rule or 69 Rule or 72 Rule

69 Rule :- When rate is $R\%$



then, time (t) = $(\frac{69}{R} + 0.37)$ years

72 Rule :- When rate is $R\%$



then, time (t) = $\frac{72}{R}$ years

Note :- 69 rule gives more accurate result in comparison of 72 rule.

Example :- A certain sum is invested at 26% p.a, interest compounded annually. In how many years (approx) will it double?

SSC CHSL 01/06/2022 (Morning)

(a) 4 (b) 3 (c) 2 (d) 5

Solution :- using 69 rule :-

time (t) = $(\frac{69}{R} + 0.37)$ years

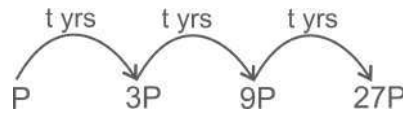
So, time = $\frac{69}{26} + 0.37 = 2.65 + 0.37 = 3$ years (approx)

Using 72 rule :

time (t) = $\frac{72}{R}$ years = $\frac{72}{26}$ years = 2.77 years = 3 years (approx)

(e) Triple Money Rule or 114 Rule

114 Rule :- When rate is $R\%$



then, time (t) = $\frac{114}{R}$ years

Type - 8

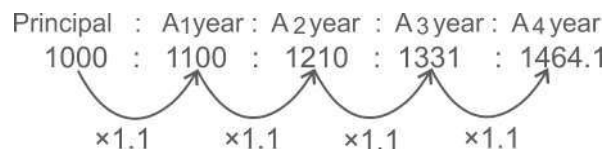
Let Principal = 1000, Rate = 10% per annum

Time = 4 years

Rate = 10% per annum or $\frac{1}{10}$

So, we can say that Principal = 10 and Amount = 11

Multiplying factor = $\frac{11}{10} = 1.1$



Example :- A Sum of Rs. 4000 is lent at 10% p.a., interest compounded annually. What is the difference between the compound interest for the 2nd year and 3rd year?

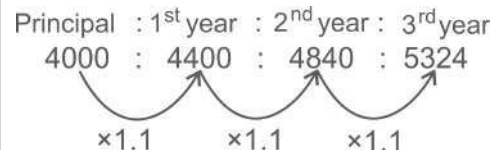
SSC MTS 08/10/2021 (Morning)

(a) Rs.45 (b) Rs.46.50 (c) Rs.45.40 (d) Rs. 44

Solution :- We know that

Rate = 10% per annum or $\frac{1}{10}$

Multiplying factor = $\frac{11}{10} = 1.1$



Here, 2nd year compound = $4840 - 4400 = \text{Rs.}440$

3rd year compound = $5324 - 4840 = \text{Rs.} 484$

So, 3rd year CI - 2nd year CI = $484 - 440 = \text{Rs.} 44$

Exam hall Approach :

3rd year CI - 2nd year CI = $4000 \times \frac{1}{10} \times \frac{11}{10} \times \frac{1}{10} = 44$

Hint : $4000 \times \frac{1}{10} \times \frac{11}{10} \times \frac{11}{10} - 4000 \times \frac{1}{10} \times \frac{11}{10}$

$4000 \times \frac{1}{10} \times \frac{11}{10} [\frac{11}{10} - 1]$

$4000 \times \frac{1}{10} \times \frac{11}{10} \times \frac{1}{10} = \text{Rs.} 44$

Type - 9

When Rate Compounded Monthly

Important Formulas :

Amount = A, Principal = P, Compound Interest = CI

Rate of interest = R, Time = number of cycles = n

1. CI = Amount - Principal

2. When interest is compounded annually :

$A (\text{in CI}) = P(1 + \frac{R}{100})^n$

3. When interest is compounded half-yearly :

$$A \text{ (in CI)} = P \left(1 + \frac{\frac{R}{2}}{100} \right)^{2n}$$

4. When interest is compounded quarterly :

$$A \text{ (in CI)} = P \left(1 + \frac{\frac{R}{4}}{100} \right)^{4n}$$

5. When interest is compounded monthly :

$$A \text{ (in CI)} = P \left(1 + \frac{\frac{R}{12}}{100} \right)^{12n}$$

6. When the interest is compounded annually and time period is given in fraction, say for $3\frac{2}{5}$ years:-

$$A \text{ (in CI)} = P \left(1 + \frac{R}{100} \right)^3 \times \left(1 + \frac{\frac{2}{5}R}{100} \right)$$

7. When different interest rates for different years are given, say $R_1\%$, $R_2\%$, $R_3\%$, and $R_4\%$ for 1st, 2nd, 3rd and 4th year

respectively :

$A \text{ (in CI)} =$

$$P \left(1 + \frac{R_1}{100} \right) \times \left(1 + \frac{R_2}{100} \right) \times \left(1 + \frac{R_3}{100} \right) \times \left(1 + \frac{R_4}{100} \right)$$

8. Equated monthly installment (EMI) concept :

$$\text{Amount borrowed} = \frac{x}{\left(1 + \frac{R}{100} \right)^n}$$

where x = Installment (EMI)

9. When the interest is compounded 8 monthly, time period is 2 years and the interest rate is 12% p.a., then

For 12 months, interest rate = 12%,

so for 8 months, interest rate = $\frac{8 \times 12}{12} = 8\%$

Here, time period = 2 years = 24 months

So, $\frac{24}{8} = 3$, so we need to conduct 3 cycles each of interest rate 8% in order to calculate required CI.

When rate compounded	Rate(r)	Time (t)
Annually	$R\%$	t years
Half yearly	$\frac{R}{2}\%$	$2t$ years
Eight monthly	$\frac{2R}{3}\%$	$\frac{3}{2}t$ years
Quarterly	$\frac{R}{4}\%$	$4t$ years
monthly	$\frac{R}{12}\%$	$12t$ years

10. Effective C.I. rate for two interest rates = $(a + b + \frac{ab}{100})\%$, where a and b are interest rates.

If $a = b$, For 2 year cycle = $2a + \frac{a^2}{100}$

i.e. 2% and 5% = $(2 + 5 + \frac{2 \times 5}{100})\% = 7 + 0.1 = 7.1\%$

i.e. 3% and 3% = $2 \times 3 + \frac{3^2}{100} = 6 + 0.09 = 6.09\%$

11. Effective C.I. rate for three interest rates

$$= (a + b + c + \frac{ab + bc + ca}{100} + \frac{abc}{10000})\%$$

where a , b and c are interest rates.

If $a = b = c$ then interest rate = $(3a + \frac{3a^2}{100} + \frac{a^3}{10000})\%$

$$= 3a + 3a^2 + a^3$$

i.e. 5%, 5% and 5% = $3 \times 5 + \frac{3 \times 5^2}{100} + \frac{5^3}{10000} = 15.7625\%$

Or $3 \times 5 + 3 \times 5^2 + 5^3 = 15 + 75 + 125 = 15.7625\%$

i.e. 10%, 10% and 10% = $3 \times 10 + \frac{3 \times 10^2}{100} + \frac{10^3}{10000} = 33.10\%$

Or $3 \times 10 + 3 \times 10^2 + 10^3 = 30 + 300 + 1000 = 33.1000 = 33.10$

12. Effective rates :-

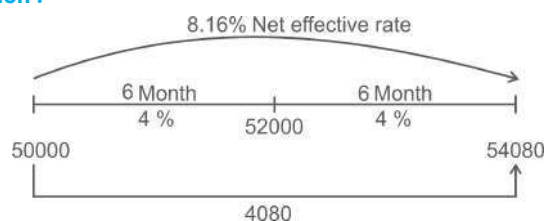
Years → Rate ↓	2nd	3rd	4th
5%	10.25%	15.7625%	21.550625%
10%	21%	33.1%	46.41%
15%	32.25%	52.0875%	-
20%	44%	72.8%	107.36%
25%	56.25%	95.3125%	-
30%	69%	119.70%	-

Example :- Find the compound interest on ₹50,000 for one year at 8% per annum when compounded half yearly.

SSC CHSL 02/08/2023 (2nd Shift)

(a) ₹3,900 (b) ₹3,880 (c) ₹4,080 (d) ₹3,950

Solution :-



$$\text{Effective interest rate} = 4 + 4 + \frac{4 \times 4}{100} = 8.16\%$$

$$\text{Compound interest} = 50,000 \times \frac{8.16}{100} = \text{Rs. } 4080$$

Example :- What will be the compound interest (nearest to Rs 1) on a sum of Rs 25,000 for 2 years at 12% p.a., if the interest is compounded 8-monthly?

SSC CGL 04/06/2019 (Evening)

(a) Rs 6,394 (b) Rs 6,439 (c) Rs 6,493 (d) Rs 6,349

Solution :- Interest is compounded 8 monthly

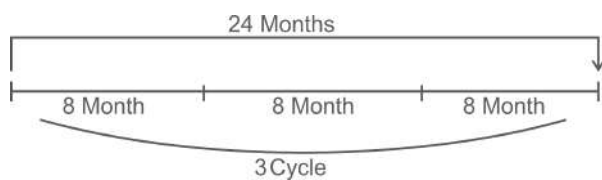
$$\text{So, Effective Rate} = 12 \times \frac{8}{12} = 8\%$$

or 12 month = 12%,

So 8 month = 8%

$$\text{Effective time Period} = 2 \times \frac{12}{8} = 3$$

Or 2 Year = 24 month



Fractional value of 8% = $\frac{2}{25}$

For three compounding cycle :-

Principal : Amount

$$(25)^3 : (27)^3$$

$$15625 : 19683$$

Now, 15625 units = Rs. 25000 $\Rightarrow$ 1 unit = 1.6 Rs.

C.I. = (19683 - 15625) = 4058 unit

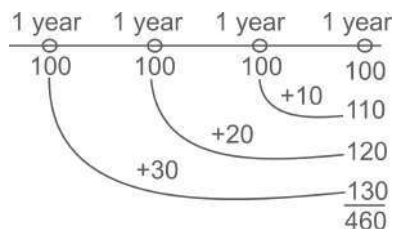
C.I. = $4058 \times 1.6 \approx$ Rs. 6493

Installment

Simple Interest Installments

If a sum is paid in 4 years of equal installments of 100 rs at 10% SI

Then



1st installment will clear = 130

2nd installment will clear = 120

3rd installment will clear = 110

4th installment will clear = 100

460

Hence total debt clear from four equal installments of 100 rs in four years = 460

Important Formula

$$\text{Annual Installment in case of SI} = \frac{100A}{100t + \frac{rt(t-1)}{2}}$$

Where, A = Amount borrowed,

t = Time period (number of installment), r = Interest rate

Example 1 : A computer is available for ₹39,000 on cash payment or ₹19,000 as cash payment followed by five monthly installments of ₹4,200 each. What is the rate of interest per annum under the installment plan?

SSC CHSL 02/08/2023 (2nd Shift)

- (a) $20\frac{19}{29}\%$ (b) $20\frac{17}{29}\%$ (c) $20\frac{20}{29}\%$ (d) $20\frac{18}{29}\%$

Solution :-

Remaining amount = 39,000 – 19,000 = 20,000 ₹

$$\text{Amount} = 20,000 + \frac{20,000 \times R \times 5}{12 \times 100} = 20,000 + \frac{250R}{3}$$

Sum of Amounts of these installments =

(4200 + S.I. on 4200 for 4 months) + (4200 + S.I. on 4200 for 3 months) + (4200 + S.I. on 4200 for 2 months) + (4200 + S.I. on 4200 for 1 month) + 4200

$$= (4200 \times 5) + (\text{S.I. on 4200 for 10 months})$$

$$= 21000 + \frac{4200 \times R \times 10}{12 \times 100} = 21000 + 35R$$

Now, equating amount :-

$$20,000 + \frac{250R}{3} = 21000 + 35R$$

$$\Rightarrow \frac{145R}{3} = 1,000 \Rightarrow R = \frac{3000}{145} = \frac{600}{29} = 20\frac{20}{29}\%$$

Example 2 : What annual installment will discharge a debt of ₹5,664 in 4 years at 12% simple interest ?

SSC CGL 02/12/2022 (3rd Shift)

- (a) ₹1,230 (b) ₹1,210 (c) ₹1,200 (d) ₹1,220

Solution: Using formula,

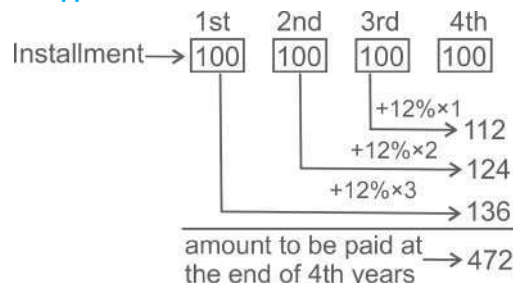
$$x = \frac{100A}{100t + \frac{rt(t-1)}{2}}$$

....(where A = Amount borrowed, r = rate and

t = no. of installments)

$$x = \frac{566400}{100 \times 4 + \frac{12 \times 4(4-1)}{2}} = \frac{566400}{100 \times 4 + 72} = \frac{566400}{472} = \text{Rs. 1200}$$

Exam hall Approach:-



According to the question,

472 units = Rs. 5664

Then, 100 units = Rs. 1200

Compound Interest Installment

1) Equated monthly installment (EMI) :-

Amount borrowed = $\frac{x}{(1 + \frac{R}{100})^n}$, where x = installment (EMI)

Example:- Find the amount which is needed to be paid each year to clear the debt of ₹2,10,000 in 2 years at 10% compound interest rate.

$$\text{Solution:- } 210000 = \frac{x}{(1 + \frac{10}{100})^2} + \frac{x}{(1 + \frac{10}{100})}$$

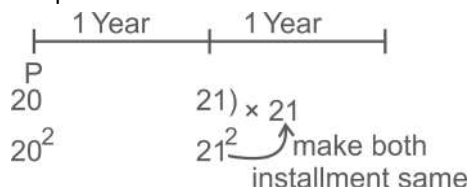
$$\Rightarrow 210000 = \frac{x}{(\frac{11}{10})^2} + \frac{x}{(\frac{11}{10})}$$

$$\Rightarrow 210000 = \frac{100x}{121} + \frac{10x}{11} \Rightarrow 210000 = \frac{100x + 110x}{121}$$

$$\Rightarrow 210000 = \frac{210x}{121} \Rightarrow x = ₹1,21,000$$

2) Installments of compound interest :-

If a sum is due for two years of two equal installments at 5% compound interest rate then



Example:- Find the amount which is needed to be paid each year to clear the debt of ₹2,10,000 in 2 years at 10% compound interest rate.

$$\text{Solution:- Given } R = 10\% = \frac{10}{100}$$

Principal : installment

1st year - 10×11 : 11×11

$$\text{2nd year - } 10^2 : 11^2$$

$$\text{-----}$$
$$210 : 242$$

Each installment (per year) = 121 unit

Principal(210 unit) = ₹2,10,000

Now, you can calculate the amount needed to be paid each year i.e. 121 unit

$$= \frac{210000}{210} \times 121 = ₹1,21,000$$

Mixture and Alligation

Mixture :- When two or more than two substances mixed together in any ratio to produce a product is called mixture .

Mean price :- the average of all the given prices of the known items is called Mean Price .

Formula

1) If from x litre of liquid A, p litre of mixture is withdrawn and the same quantity of liquid B is added. Again from mixture, q litre mixture is withdrawn and the same quantity of liquid B is added. Again from mixture, r litre is withdrawn and same quantity of liquid B is added, then

$$\text{In final mixture, liquid A} = x \left(\frac{x-p}{x} \right) \left(\frac{x-q}{x} \right) \left(\frac{x-r}{x} \right) \dots$$

$$\Rightarrow \text{Liquid B in final mixture} = x - (\text{liquid A in final mixture})$$

2.) There is $x\%$ water in 'a' unit of the mixture of sugar and water. The quantity of water vapourised such that decreased in the percentage of water is from $x\%$ to $y\%$ is given by

$$\text{Required quantity of vapourized water} = \frac{a(x-y)}{y} \text{ unit}$$

3.) The amount of milk is $x\%$ in 'M' litre mixture. How much water should be mixed in it so that percentage amount of milk would be $y\%$?

$$\text{Amount of water} = \frac{M(x-y)}{y}$$

Mixture (Mixing Concept)

A mixture can be derived by mixing any combination of two or more ingredients to get a desired quantity.

Type - 1

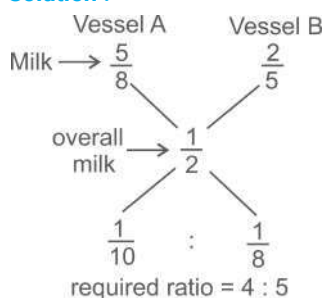
Mixing two different mixture to obtain a single mixture

Example:- Milk and water in two vessels A and B are in the ratio 5 : 3 and 2 : 3, respectively. In what ratio should the liquids in both the vessels be mixed to obtain a new mixture in vessel C containing half milk and half water ?

SSC CHSL 04/08/2023 (1st Shift)

(a) 3 : 4 (b) 4 : 5 (c) 2 : 3 (d) 3 : 5

Solution :-



Type - 2

Mixing some parts of two different mixture to obtain a single mixture

Example:- In two alloys, the ratio of Aluminium to Zinc are 5 : 6 and 3 : 5. If 242 kg of the first alloy and 144 kg of the second alloy are mixed, then the ratio of Aluminium and Zinc in the new alloy will be:

SSC CHSL 07/08/2023 (4th Shift)

(a) 76 : 117 (b) 82 : 111
(c) 93 : 100 (d) 68 : 125

Solution :- In 1st alloy :-

$$\text{Aluminium} = 242 \times \frac{5}{11} = 110 \text{ kg}$$

$$\text{Zinc} = 242 - 110 = 132 \text{ kg}$$

In 2nd alloy :-

$$\text{Aluminium} = 144 \times \frac{3}{8} = 54 \text{ kg}$$

$$\text{Zinc} = 144 - 54 = 90 \text{ kg}$$

$$\text{Required ratio} = 110 + 54 : 132 + 90 = 82 : 111$$

Exam hall approach: $\frac{242}{144} = \frac{121}{72}$

$$\begin{array}{lcl} \text{Aluminium} & : & \text{Zinc} \\ \text{A} \rightarrow & 5 & : 6 = 11 \times 8 \times 121 \rightarrow 11 \\ \text{B} \rightarrow & 3 & : 5 = 8 \times 11 \times 72 \rightarrow 9 \end{array}$$

$$\begin{array}{lcl} \text{C} \rightarrow & 55 + 27 & : 66 + 45 \\ \Rightarrow & 82 & : 111 \end{array}$$

Type - 3

Mixing three different mixture to obtain a single mixture

Example:- Mixture of milk and water in three bottles of equal capacity is in the ratio 4 : 1, 4 : 3 and 1 : 2 respectively. These three bottles are emptied into a large bottle. What will be the ratio of milk and water respectively in the large bottle?

SSC MTS 15/06/2023 (Morning)

(a) 179 : 136 (b) 152 : 165
(c) 198 : 175 (d) 133 : 145

Solution:- According to question,

Total quantity of milk in large bottle

$$\begin{aligned} &= \frac{4a}{5} + \frac{4a}{7} + \frac{a}{3} \\ &= \frac{84a + 60a + 35a}{105} = \frac{179a}{105} \end{aligned}$$

Total quantity of water in large bottle

$$\begin{aligned} &= \frac{a}{5} + \frac{3a}{7} + \frac{2a}{3} \\ &= \frac{21a + 45a + 70a}{105} = \frac{136a}{105} \end{aligned}$$

$$\text{Required ratio} = \frac{179a}{105} : \frac{136a}{105} = 179 : 136$$

Exam hall approach:

$$\begin{array}{lcl} \text{Milk} & : & \text{water} \\ \text{bottle}_1 & \rightarrow & 4 : 1 = 5 \times 7 \times 3 \rightarrow 21 \\ \text{bottle}_2 & \rightarrow & 4 : 3 = 7 \times 5 \times 3 \rightarrow 15 \\ \text{bottle}_3 & \rightarrow & 1 : 2 = 3 \times 5 \times 7 \rightarrow 35 \end{array}$$

Large bottle $\Rightarrow 84 + 60 + 35 : 21 + 45 + 70$
 $\Rightarrow 179 : 136$

Type - 4

Mixing some parts of three different mixture to obtain a single mixture

Example:- There are three cups of mixture of milk and water in ratio 4 : 5, 5 : 6, 9 : 7. 36 liters of first and 44 liters of second cup are taken. How much quantity from third cup is to be taken so that final mixture from the three cups will have milk and water in ratio 1 : 1 ? (in litres)

SSC MTS 10/05/2023 (Afternoon)

(a) 32 (b) 64 (c) 54 (d) 60

Solution:-

1st cup	2nd cup	3rd cup
Milk:Water	Milk:Water	Milk:Water
(4:5)=36 liter	(5:6)=44 liter	(9:7)=16x liter
16 liter, 20 liter	20 liter, 24 liter	9x liter, 7x liter

Quantity taken

According to the question,

$$\frac{16 + 20 + 9x}{20 + 24 + 7x} = \frac{1}{1} \Rightarrow 2x = 8 \Rightarrow x = 4$$

quantity from third cup is to be taken

$$= (9x + 7x) = (16x) = 64 \text{ liter}$$

Type - 5

Concentration :- Concentration is the percentage of a particular quantity in the full mixture.

Example:- In a mixture ratio of water and milk = 1 : 7

Then, concentration of milk in mixture = $\frac{7}{1+7} \times 100 = 87.5\%$

Example:- Concentration of water = $\frac{1}{1+7} \times 100 = 12.5\%$

Or concentration of water = (100 - Concentration of milk)
 $= (100 - 87.5) = 12.5\%$

Example:- If 36 litres of 20% milk solution is mixed with 48 litres of 10% milk solution, then what will be the percentage concentration of milk in the final solution?

Higher Secondary 27/06/2023 (Shift - 3)

(a) 14.28% (b) 15.48% (c) 12.38% (d) 13.36%

Solution:- According to question,

Total quantity of milk in final solution

$$= \frac{36 \times 20}{100} + \frac{48 \times 10}{100} = 12 \text{ liter}$$

Total quantity of water in final solution

$$= 84 - 12 = 72 \text{ liter}$$

$$\text{Required \%} = \frac{12}{(72 + 12)} \times 100 = 14.28\%$$

Type - 6

Addition or Removal of some quantity

Example:- A 40 - litre mixture contains 25% alcohol and 75% water. If 10 litres of water are added to the mixture, the percentage of alcohol in the new mixture is:

SSC CGL 25/07/2023 (4th shift)

(a) 27% (b) 18% (c) 20% (d) 25%

Solution:- Alcohol = 10L and Water = 30 L

Now,

Alcohol = 10L and Water = 30 + 10 = 40 L

$$\text{Required \%} = \frac{10}{10 + 40} \times 100 = 20\%$$

Example:- A mixture contains liquid A and B in the ratio 5 : 3, respectively. If 3 litres of liquid B is added to it, the ratio of liquid A to liquid B becomes 7 : 6. What is the quantity of liquid A in the mixture ?

SSC CHSL 17/08/2023 (3rd Shift)

(a) $7 \frac{2}{3}$ litres (b) $9 \frac{1}{3}$ litres

(c) $11 \frac{2}{3}$ litres (d) $13 \frac{1}{3}$ litres

Solution:- According to the question,

$$\frac{5x}{3x + 3} = \frac{7}{6}$$

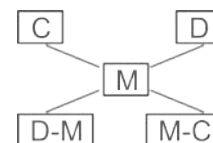
$$30x = 21x + 21 \Rightarrow x = \frac{21}{9} \text{ liter}$$

$$\text{Required quantity} = 5 \times \frac{21}{9} = 11 \frac{2}{3} \text{ litres}$$

Type - 7

Alligation

It is used to find the ratio in which two or more substances at their respective prices should be mixed to produce a mixture at a given price.



where

C = Cost price of first type

D = Cost price of second type

M = Mean price

(D - M) = quantity first type

(M - C) = quantity second type

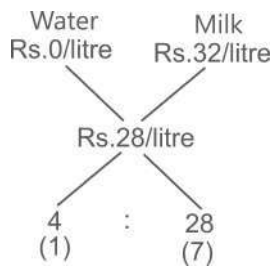
It is a method of average and involves the application of the weighted average concept.

Example:- In what ratio must water be mixed with milk, costing ₹32 per litre, in order to get a mixture costing ₹28 per litre?

SSC CGL 14/07/2023 (2nd shift)

(a) 8 : 1 (b) 1 : 7 (c) 1 : 8 (d) 7 : 1

Solution:-



Required ratio = 4 : 28 = 1 : 7

Type - 8

Increase/decrease of mixture

There is $x\%$ milk in 'a' unit mixture of milk and water. The amount of milk that should be added to increase the percentage of milk from $x\%$ to $y\%$ is given

Required quantity of milk = $\frac{a(x-y)}{100-y}$ unit

Example:- If 49 litres of a milk solution has 63% milk in it, then how much milk should be added to make the concentration of milk 70% in the solution?

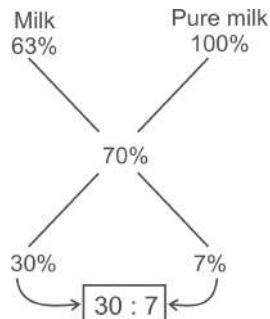
SSC CHSL 25/05/2022 Afternoon

(a) $11\frac{13}{30}$ (b) $14\frac{7}{30}$ (c) $21\frac{11}{30}$ (d) $13\frac{1}{30}$

Solution :-

$$\begin{aligned}\text{Required quantity of milk} &= \frac{a(x-y)}{100-y} = \frac{49(70-63)}{100-70} \\ &= \frac{49 \times 7}{30} = \frac{343}{30} = 11\frac{13}{30} \text{ litres}\end{aligned}$$

Exam hall approach:



ATQ, $\Rightarrow 30 \text{ unit} = 49 \text{ litres (Given)}$

$$\text{Then, } 7 \text{ unit} = \frac{49}{30} \times 7 = \frac{343}{30} = 11\frac{13}{30} \text{ litres}$$

Type - 9

Replacement

If only one process is repeated n times, then

Remaining Quantity of pure liquid = $x(1 - \frac{y}{x})^n$

where,

x = Quantity of original liquid,

y = Quantity to be carried out and replaced,

n = Number of times the operation is conducted

Example:- A 10-litre solution of milk and water contains 8 litres of milk. 2 litres of the solution is replaced by pure milk and mixed. The process is repeated two more times. How much milk (in litres) is present in the mixture so obtained ?

SSC CHSL 14/08/2023 (3rd Shift)

(a) 8.976 (b) 8.597 (c) 8.796 (d) 8.679

Solution:- Left quantity of water = initial quantity of water $\times$
 $(1 - \frac{\text{milk taken out}}{\text{total mixture}})^n = 2 \times (1 - \frac{2}{2+8})^3 = 2 \times (\frac{4}{5})^3 = 1.024$
 litre

Left quantity of milk = $10 - 1.024 = 8.976$ litre

Replacement of equal quantities

Example:- A can of water and milk mixture contains 60% milk. A part of this mixture is replaced by another mixture containing 50% milk and the percentage of milk was found to be 52%. The quantity of mixture replaced is:

SSC CGL 13/12/2022 (2nd Shift)

(a) $\frac{1}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) $\frac{2}{5}$

Solution:- Using Alligation method, we have :



The quantity of mixture replaced

$$= \frac{4}{1+4} = \frac{4}{5}$$

Replacement of unequal quantities

Example:- In a 64 litres mixture, the alcohol and water are in the ratio of 1 : 3 respectively. 32 litres of mixture is taken off and replaced with 16 litres of water. What is the new ratio of alcohol and water respectively ?

SSC MTS 19/05/2023 (Evening)

(a) 1 : 5 (b) 5 : 12 (c) 2 : 7 (d) 3 : 10

Solution:- $(1 + 3) \text{ unit} = 64 \text{ liter}$

1 unit = 16 liter

Initial quantity of alcohol and water

= 16 liter and 48 liter

After 32 liter of mixture taken out

$(1 + 3) \text{ unit} = 32 \text{ liter} \Rightarrow 1 \text{ unit} = 8 \text{ liter}$

Quantity of alcohol and water = 8 liter and 24 liter

After adding 16 liter water

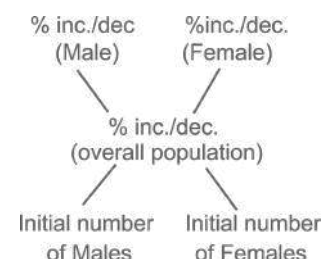
New ratio of alcohol and water

= $8 : (24 + 16) = 1 : 5$

Use of Mixture and Alligation in other chapter

Type - 1

When there is an overall increase or decrease in the population, the result comes in the ratio of the number of males and females.



Example:- In a certain year, the population of a city was 18000. If in the next year, the population of males increased by 5% and that of females increased by 7%, and the total population increased to 19200, then what was the ratio of the populations of males and females in that given year?

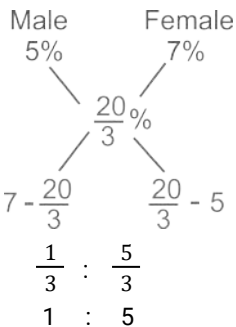
SSC CGL 20/07/2023 (1st shift)

(a) 2 : 5 (b) 1 : 5 (c) 4 : 3 (d) 3 : 5

Solution:-

$$\text{overall \%age increase in population} = \frac{19200 - 18000}{18000} \times 100$$

$$= \frac{1200}{18000} \times 100 = \frac{20}{3}\%$$



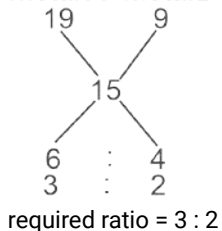
Example:- In an experiment, it was observed that metal A is 19 times heavier than water and metal B is 9 times heavier than water. In what ratio metal A and B mixed so that the mixture thus obtained be 15 times heavier than water?

SSC CHSL 11/08/2023 (3rd Shift)

(a) 2 : 3 (b) 1 : 3 (c) 3 : 1 (d) 3 : 2

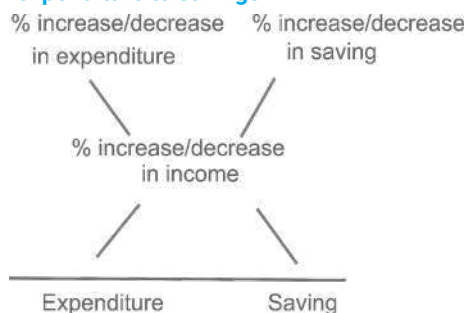
Solution:-

metal A metal B



Type - 2

when an increase or decrease in income is given, the result is the ratio of expenditure to savings.



Example:- Vikas spends 90% of his income. When his income increased by 25%, his expenditure increased by 20%.

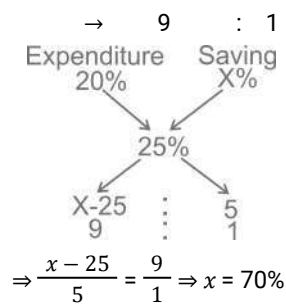
The percentage of increase in his savings is:

Higher Secondary 05/08/2022 (Shift - 2)

(a) 58% (b) 70% (c) 45% (d) 62%

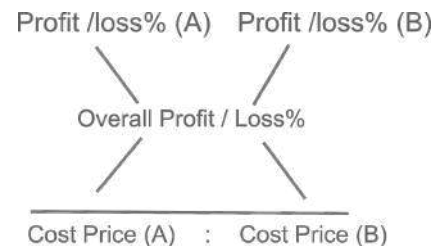
Solution :- $\Rightarrow 90\% = \frac{9(\text{spend})}{10(\text{income})}$

Ratio $\rightarrow$ Expenditure : Saving



Type - 3

In profit and loss when total profit /loss% is given. Then the result comes in proportion to the cost price of the first article and the second article.



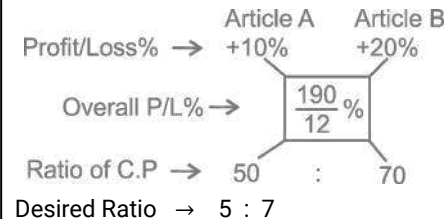
Example :- The sum of cost prices of A and B is Rs. 1200. The sum of the selling prices of A and B is Rs. 1390. If the profit of 10% and 20% is obtained on A and B respectively,

then find the ratio between the cost prices of A and B.

SSC MTS 19/08/2019 (Afternoon)

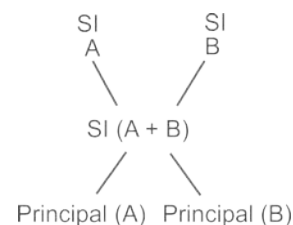
(a) 5 : 7 (b) 5 : 9 (c) 7 : 13 (d) 6 : 11

Solution :- Overall profit% = $\frac{1390 - 1200}{1200} \times 100 = \frac{190}{12}\%$



Type - 4

When total interest is given, then the result comes as the original principal of both the items.



Example :- Rajesh invested ₹10,000 by dividing it into two different investment schemes, A and B, at simple interest rates of 8% and 10%, respectively. If the total interest earned in 2 years is ₹1,680, the amount invested in scheme A is:

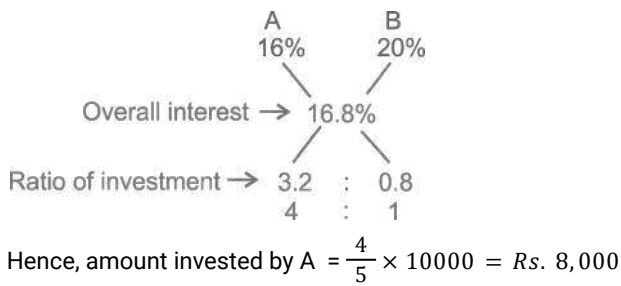
SSC CPO 03/10/2023 (1st Shift)

(a) ₹4,000 (b) ₹2,000 (c) ₹6,000 (d) ₹8,000

Solution :- S.I for 2 years of A = 16%

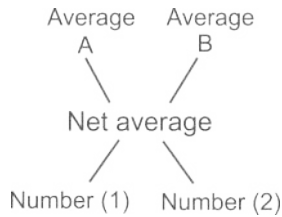
S.I for 2 years of B = 20%

Total rate % = $\frac{1680}{10000} \times 100 = 16.8\%$



Type - 5

When the overall average of the number is given, the result comes in the form of the ratio of first and second number.

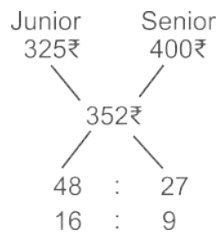


Example :- During a school excursion each student of junior school was charged ₹325 and each student of senior school was charged ₹400. If there were 80 students from junior school and the combined average amount charged per student was ₹352, then how many students from senior school went for the excursion?

SSC CGL 24/07/2023 (1st shift)

(a) 55 (b) 45 (c) 50 (d) 20

Solution :-

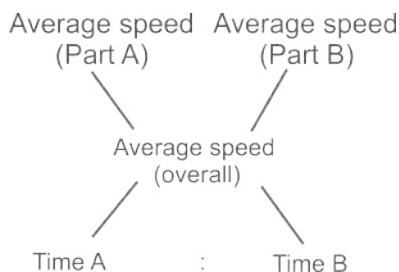


16 units = 80 junior students

9 units = $\frac{80}{16} \times 9 = 45$ senior students

Type - 6

When the total average speed is given, the result is proportional to time.

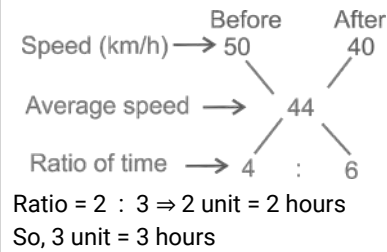


Example :- A man drives his car at a speed of 50 km/hr for 2 hours and then decides to slow down to 40 km/hr till the end of the journey. If his average speed for the journey was 44 km/hr, then for how much time he drives his car at speed 40 km/hr?

SSC MTS 16/05/2023 (Afternoon)

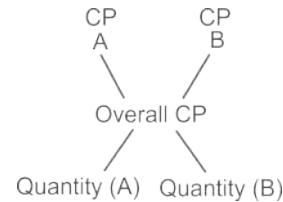
(a) 2 hours (b) 1 hour
(c) 3 hours (d) 4 hours

Solution :-



Type - 7

When, cost price of two articles and mixture are given, then, resultant is the ratio of first number and second number.

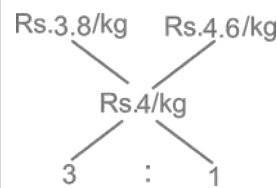


Example:- How many kilograms of sugar at ₹3.80 per kg must be mixed with 60 kg of sugar at ₹4.60 per kg, so that by selling the mixture at ₹5.20 per kg there may be a gain of 30%?

SSC MTS 01/09/2023 (3rd Shift)

(a) 170 kg (b) 120 kg (c) 160 kg (d) 180 kg

Solution:- Cost price of mixture = $5.2 \times \frac{10}{13} = ₹4/\text{kg}$



1 unit = 60kg

Required quantity (3 unit) = 180 kg

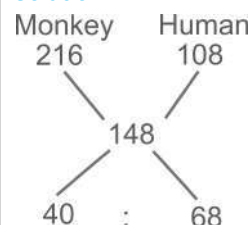
Miscellaneous

Example :- At a certain time in a park, the number of heads and the number of legs of monkeys and human visitors were counted, and it was found that there were 54 heads and 148 legs. Find the number of monkeys in the park.

SSC CGL Tier II (26/10/2023)

(a) 20 (b) 16 (c) 18 (d) 14

Solution :-



So, the ratio of the no of monkey and human = 40 : 68 = 10 : 17

10 + 17 = 27 unit = 54

10 unit = $\frac{54}{27} \times 10 = 20$

Average

An average or arithmetic mean of given data is the sum of the given observations divided by the number of observations.

$$\text{Therefore, Average (A)} = \frac{\text{Sum of the given observations (S)}}{\text{Number of Observations (N)}}$$

Example :- Average of 10, 15, 25 and 30

Solution:- Average = $\frac{10 + 15 + 25 + 30}{4} = \frac{80}{4} = 20$

Example :- Find the average of $3^{20}, 3^{30}, 3^{40}$.

Graduate Level 03/08/2022 (Shift - 3)

- (a) $(3^{20} + 3^{30} + 3^{40})$ (b) 3^{30}
 (c) $(3^{20} + 3^{30} + 3^{40})$ (d) $(3^{19} + 3^{29} + 3^{39})$

Solution :-

$$\begin{aligned} \text{Average of } 3^{20}, 3^{30} \text{ and } 3^{40} \\ = \frac{3^{20} + 3^{30} + 3^{40}}{3} = \frac{3(3^{19} + 3^{29} + 3^{39})}{3} = (3^{19} + 3^{29} + 3^{39}) \end{aligned}$$

Some important Formulae

- 1) Average of first 'n' natural numbers = $\left(\frac{n+1}{2}\right)$
- 2) Average of first 'n' even numbers = $(n+1)$
- 3) Average of first n odd numbers = n
- 4) Average of consecutive numbers = $\frac{\text{First number} + \text{Last number}}{2}$
- 5) Average of 1 to n odd numbers = $\frac{\text{Last odd number} + 1}{2}$
- 6) Average of 1 to n even numbers = $\frac{\text{Last even number} + 2}{2}$
- 7) Average of squares of first n natural numbers = $\frac{(n+1)(2n+1)}{6}$
- 8) Average of cubes of first n natural numbers = $\frac{n(n+1)^2}{4}$
- 9) Average of n multiples of any number = $\frac{\text{Number} \times (n+1)}{2}$
- 10) Combined mean or combined average :

If the mean of n_1 and n_2 observations are $\bar{x}_1$ and $\bar{x}_2$ respectively

$$\text{then, Combined mean} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

- 11) If the observations are in A.P. then average is the middle term.

(a) If number of terms are odd

13, 14, 15, 16, 17, 18, 19

↓
Avg.

$$\frac{13+19}{2} = \frac{14+18}{2} = \frac{15+17}{2} = 16$$

(b) If number of terms are even

19, 20, 21, 22, 23, 24, 25, 26, 27, 28

↓
Avg. = $\frac{23+24}{2} = 23.5$

Important Note:-

- 1) If all the numbers increase by 'a' then the average of the numbers will also increase by 'a'.

Example :- The average of 21 numbers is 60. If each number is increased by 20, then find the number by which the average is increased.

SSC MTS 16/05/2023 (Afternoon)

- (a) 12 (b) 18 (c) 15 (d) 20

Solution :- Average of 21 numbers = 60

If Each number is increased by 20 then average will be also increase by 20

- 2) If all the numbers decrease by 'a' then the average of the numbers will also decrease by 'a'.

Example :- The average of 24 numbers is 43. If we subtract a number 'M' from each number, then the average becomes 38. What was the value of M?

SSC MTS 04/05/2023 (Morning)

- (a) 12 (b) 5 (c) 7 (d) 10

Solution :- Average of 24 number = 43

After subtracting 'M' from each number,

then new average will be $\Rightarrow 43 - M = 38$

So, value of 'M' = $43 - 38 = 5$

- 3) If all the numbers are multiplied by 'a' then their average must also be multiplied by 'a'.

Example :- The average of 70 values is 40. If each value is multiplied by 20, what will be the changed average?

RRC Group D 26/08/2022 (Morning)

- (a) 800 (b) 60 (c) 90 (d) 1400

Solution :- The change in each number of a group leads to the same change in average of the group.

Then, the changed average = $40 \times 20 = 800$

Deviation Method

If we have to find average traditionally then simply

$$\text{Average} = \frac{\text{Sum of the given observations (S)}}{\text{Number of Observations (N)}}$$

but this method can be lengthy sometimes so to avoid this we use the Deviation method.

Step 1 - Consider any number in the range of these numbers as average.

Step 2 - Find the difference of average from each number i.e. deviation.

Step 3 - Add the deviation and divide it by the total number of observations.

Step 4 - Add or subtract (according to sign of deviation) the deviation from the average that we considered to get an accurate average.

Example :- Find the average of 45, 50, 46, 55, 40, 41, 52

Solution :- Let average = 46

45, 50, 46, 55, 40, 41, 52
 Deviation: -1, +4, 0, +9, -6, -5, +6

$$\text{Net deviation in average} = \frac{-1 + 4 + 0 + 9 - 6 - 5 + 6}{7} = \frac{+7}{7} = +1$$

Actual Average = $46 + 1 = 47$

Example :- Find the average of 45, 50, 46, 55, 40, 41, 52

Solution :- Let average = 50

45, 50, 46, 55, 40, 41, 52
Deviation: -5, +0, -4, +5, -10, -9, +2

$$\text{Net deviation in average} = \frac{-5 + 0 - 4 + 5 - 10 - 9 + 2}{7} \\ = \frac{-21}{7} = -3$$

Actual Average = 50 - 3 = 47

Run Rate and Bowling Average

Basically the term Run Rate means how many runs have been scored in one over of a cricket match. Higher run rate is good for batsmen.

$$\text{Run rate} = \frac{\text{Total runs scored}}{\text{Number of overs played}}$$

Example :- If the batsman has scored 225 runs in 9 overs, find his run rate.

$$\text{Solution :- Run rate} = \frac{225}{9} = 25 \text{ runs/over}$$

$$\text{Bowling Average} = \frac{\text{Total runs allowed}}{\text{Total number of wickets taken}}$$

Lower Bowling average is good for bowlers.

Example :- If the bowler has taken 3 wickets allowing 120 runs, find his bowling average.

$$\text{Solution :- Bowling Average} = \frac{120}{3} = 40 \text{ runs/wicket}$$

Example :- The average score of a cricketer for 10 matches is 40 runs. If the average for the first six matches is 44, then the average runs of the last four matches is:

RRC Group D 18/09/2022 (Evening)

(a) 20 (b) 38 (c) 42 (d) 34

Solution :- The average score of a cricketer for 10 matches = 40 runs

Then the sum of the scores for 10 matches = 400 runs

Average of first 6 matches = 44

Sum of the scores of 6 matches = 44 × 6 = 264

$$\text{Average of last four matches} = \frac{400 - 264}{4} = \frac{136}{4} = 34$$

Exam Hall approach :-

$$\text{Deviation} = \frac{(44 - 40) \times 6}{\text{last four match}} = \frac{24}{4} = 6$$

Average of last four matches = 40 - 6 = 34

(Note :- When the average is positive the net average decreases and when the average is negative the net average increases.)

Example :- A batsman scored 84 runs in his 12th innings, thereby improving his average score per innings by 4 runs. What is the average score per innings of the batsman after the 12th innings?

RRB NTPC CBT II (15/06/2022) 1st Shift

(a) 44 (b) 40 (c) 42 (d) 36

Solution :-

Let the average score of the batsman till 11th inning be A.

Total sum of runs in 11 innings = 11A

$$\text{ATQ, } \frac{11A + 84}{12} = A + 4 \Rightarrow A = 36$$

So, The average score after 12th innings = 36 + 4 = 40

Addition of New Observation

If the average is x and a number having value greater than x is added then the new average will be greater than x. Similarly, if a number having value less than x is added then the new average will be less than x.

Example :- The average of the first five natural numbers is 3. Find a new average after adding 6.

Solution :- The first five natural numbers are : 1, 2, 3, 4, 5.

$$\text{Sum of first five natural numbers} = \frac{6 \times 5}{2} = 15$$

As number 6 is greater than the old average, the new average is going to be greater than the old average.

$$\text{So, new average} = \frac{15 + 6}{5 + 1} = \frac{21}{6} = 3.5 \text{ which is greater than 3.}$$

Example :- The average weight of 10 parcels is 28.6 kg. The addition of a new parcel reduces the average weight by 0.6 kg. What is the weight (in kg) of the new parcel?

Selection Post 28/06/2023 (3rd Shift)

(a) 23 (b) 21 (c) 22 (d) 20

Solution :- Weight of new parcel

$$= 28.6 - (11 \times 0.6) = 22 \text{ kg.}$$

Removal of Existing Observation

If the average is x and a number having value less than x is removed then the new average will be greater than x. Similarly, if a number having value greater than x is removed then the new average will be less than x.

Example :- The average of the first five natural numbers is 3. Find a new average after removing 2.

Solution :- The first five natural numbers are : 1, 2, 3, 4, 5.

$$\text{Sum of first five natural numbers} = 3 \times 5 = 15$$

As number 2 is less than the old average, the new average is going to be greater than the old average.

$$\text{So, new average} = \frac{15 - 2}{5 - 1} = \frac{13}{4} = 3.25 \text{ which is greater than 3.}$$

Example :- 18 years is the average age of 36 boys and their professor. The average becomes 17 years if we leave the professor out. What is the age (in years) of the professor?

RRB NTPC CBT - I 01/02/2021 (Morning)

(a) 34 (b) 44 (c) 64 (d) 54

Solution :- Average age of 36 boys and professor = 18 years

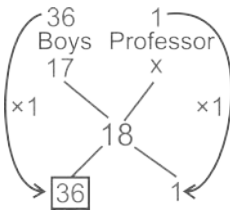
Sum of ages of 36 boys and professor

$$= 18 \times 37 = 666 \text{ years}$$

Average age of boys only = 17 years

Sum of ages of boys = $17 \times 36 = 612$ years
 Age of Professor = $666 - 612 = 54$ years

Exam hall approach :-



$$\text{Age (in years) of the professor (x)} = \frac{18 + 36}{1} = 54 \text{ years}$$

Example :- There are some children in a camp and their average weight is 40 kg. If 5 children with average weight 36 kg join the camp or if 5 children with average weight 43.2 kg leave the camp, the average weight of children in both cases is equal. How many children are there in the camp, initially?

SSC CGL 23/8/2021 (Evening)

- (a) 35 (b) 45 (c) 40 (d) 50

Solution :- Let number of children = x

$$\frac{40x + 5 \times 36}{x+5} = \frac{40x - 5 \times 43.2}{x-5}$$

$$(x-5)(40x+180) = (x+5)(40x-216)$$

$$40x^2 + 180x - 200x - 900 = 40x^2 - 216x + 200x - 1080$$

$$4x = 180, x = 45$$

Exam hall approach :-

Number	Average
n	40
+5	36
-5	43.2

$$(4 \times 5) = -20$$

$$(3.2 \times 5) = -16$$

According to question,

$$\Rightarrow 40 - \frac{20}{n+5} = 40 - \frac{16}{n-5}$$

$$\Rightarrow -\frac{20}{n+5} = -\frac{16}{n-5} \Rightarrow n = 45$$

Overlapping of Observation

Overlapping of one observation :-

Example :- The mean of 25 observations is 36. If the mean of its first 13 observations is 32 and the last 13 observations is 40, then what will be its 13th observation?

RRB NTPC CBT - I 19/01/2021 (Evening)

- (a) 38 (b) 36 (c) 23 (d) 40

Solutions :- Sum of 25 observations = $36 \times 25 = 900$

Sum of first 13 observations = $32 \times 13 = 416$

Sum of Last 13 observations = $40 \times 13 = 520$

Sum of first 13 observations + Sum of last 13 observations - 13th observation = Total sum of 25 observations

$$416 + 520 - 13\text{th observation} = 900$$

$$13\text{th observation} = 36$$

Exam hall Approach :-

Average of 25 observation = 36

Average of 1st 13 observation = 32

Average of last 13 observation = 40

Deviation = -4×13 and $+4 \times 13 = -52$ and $+52 = 0$

13th observation = $36 + 0 = 36$

Overlapping of Two observation :-

Example :- The average of 10 numbers is 26.5. If the average of the first 6 numbers is 25 and the average of the last 6 numbers is 29, then what is the average of the 5th and the 6th number?

AFCAT 15/02/2022 (1st Shift)

- (a) 29.2 (b) 29 (c) 29.5 (d) 28

Solution :- According to the question,

Sum of 1st 10 numbers = $10 \times 26.5 = 265$

Sum of 1st 6 numbers = $6 \times 25 = 150$

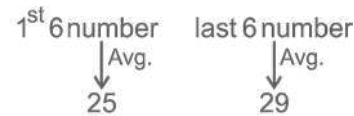
Sum of last 6 numbers = $6 \times 29 = 174$

Sum of 5th and 6th numbers = $(150 + 174) - 265 = 59$

$$\text{Required average} = \frac{59}{2} = 29.5$$

Exam hall Approach :-

Average = 26.5



Deviation: -1.5×6

Net deviation = $-9 + 15 = +6$

$$\text{average of the 5th and the 6th number} = 26.5 + \frac{6}{2} = 29.5$$

Data Misread

Example :- The average of 10 observations is 46. It was realized later that an observation was misread as 42 in the place of 142. The correct average is:

SSC GD Constable 16/11/2021 (Morning)

- (a) 52 (b) 46 (c) 58 (d) 56

Solution :-

$$\text{The correct average} = 46 + \frac{142 - 42}{10}$$

$$= 46 + \frac{100}{10} = 46 + 10 = 56$$

Example :- Renu's marks were wrongly entered as 99 in place of 9. Due to this error, the average marks of the class were 2.25 more than the actual average. What is the number of students in the class?

RRB NTPC CBT II (09/05/2022) 2nd Shift

- (a) 40 (b) 36 (c) 42 (d) 38

Solution :- Let there be " n " students

Increased marks due to the wrong entry = $99 - 9 = 90$

$$\text{ATQ, } \frac{90}{2.25} = n \Rightarrow n = 40$$

Replacement

Example :- The average height of a certain number of students in a group is 155.6 cm. If 12 students having an average height of 150.5 cm join the group and 7 students having an average height of 159 cm leave the group, the average height of the students in the group will decrease by 34 mm. What is the number of students, initially, in the group?

SSC CHSL 5/8/2021 (Morning)

- (a) 30 (b) 25 (c) 40 (d) 20

Solutions :- $34 \text{ mm} = 3.4 \text{ cm}$

Let the number of students = n

According to question,

$$n \times 155.6 + 12 \times 150.5 - 7 \times 159$$

$$= (n + 5) \times 152.2$$

$$3.4n + 693 = 761 \Rightarrow n = 20$$

Shifting of Observation

Example :- The average weight of students of section A and B having 40 students each is 45.5 kg and 44.2 kg respectively. Two students of section A having average weight 48.75 kg were shifted to B and 2 students of section B were shifted to section A, making the average weight of both the sections equal. What is the average weight (in kg) of the students who were shifted from section B to section A?

SSC CGL 23/08/2021 (Afternoon)

- (a) 34.5 (b) 35 (c) 35.75 (d) 34.25

Solution :- Difference in the average weight of the students
 $= 45.5 - 44.2 = 1.3$

$$\text{Total difference of the weight} = 1.3 \times 40 = 52$$

So 26 kg is needed to be transfer from section A to B

If two students are transferred then the difference in the weight = 13

So, the average weight of the students transferred from B to A
 $A = 48.75 - 13 = 35.75$

Example :- There are three positive numbers. If the average of any two of them is added to the third number, the resulting sums are 154, 148 and 132. The sum of the original three numbers is:

SSC MTS 07/10/2021 (Evening)

Solution :-

Let the three numbers be a , b and c .

According to question,

$$\frac{a+b}{2} + c = 154$$

$$\Rightarrow a + b + 2c = 308 \dots\dots(1)$$

$$\frac{b+c}{2} + a = 148$$

$$\Rightarrow b + c + 2a = 296 \dots\dots(2)$$

$$\frac{a+c}{2} + b = 132$$

$$\Rightarrow a + c + 2b = 264 \dots\dots(3)$$

Adding eqn (1), (2) and (3), we get

$$4(a + b + c) = 308 + 296 + 264$$

$$\Rightarrow 4(a + b + c) = 868 \Rightarrow a + b + c = 217$$

Exam hall approach :-

$$\text{Sum of numbers} = \frac{154 + 148 + 132}{2} = \frac{434}{2} = 217$$

Age Based

Example :- The average age of 6 members of a family is 25 years. If the youngest member is 4 years old, then what was the average age of the family just before the birth of the youngest member?

SSC CHSL 02/08/2023 (1st Shift)

- (a) 27.2 years (b) 24.8 years (c) 32.8 years (d) 25.2 years

Solution :- Total age of 6 member

$$= 25 \times 6 = 150 \text{ years}$$

$$\text{Total age before 4 year} = 150 - (6 \times 4) = 126 \text{ years}$$

$$\text{Required average age} = \frac{126}{5 \text{ members}} = 25.2 \text{ years}$$

Miscellaneous

Example :- Mean temperature of a city for a week is 28°C . If the mean temperature for Monday, Tuesday, Wednesday and Thursday is 27.5°C and the mean temperature for Thursday, Friday, Saturday and Sunday is 29°C , the temperature recorded for Thursday is.

RRB ALP Tier - I (29/08/2018) Evening

- (a) 30°Celsius (b) 29°Celsius (c) 31°Celsius (d) 28°Celsius

Solution :- Mean temperature of a city = 28°C

$$\text{Deviation in the average temperature for Monday to Thursday} = -0.5 \times 4 = -2^\circ\text{C}$$

$$\text{Deviation in the average temperature for Thursday to Sunday} = +1 \times 4 = 4^\circ\text{C}$$

Net deviation in temperature

$$= -2 + 4 = 2^\circ\text{C}$$

$$\text{So, the temperature recorded for Thursday} = 28 + 2 = 30^\circ\text{C}$$

Example :- An art exhibition was held in the month of november. The numbers of visitors on weekdays and on weekends were 450 and 750 respectively. If the first day of the month was Monday, then what was the average number of visitors in the month ?

SSC MTS 16/08/2019 (Evening)

- (a) 520 (b) 530 (c) 510 (d) 540

Solution:- Month is started with monday so total weekends
 $= 2 \times 4 = 8$

$$\text{Total number of visitors on weekends} = 750 \times 8 = 6000$$

$$\text{Total number of visitors on weekdays} = 450 \times 22 = 9900$$

$$\text{Desired average} = \frac{6000 + 9900}{30} = 530$$

Example :- There are 45 students in a hostel. If the number of students increases by 5, the total expenses of the mess increase by ₹30 per day while the average daily expenditure per head diminishes by ₹2. Find the original total daily expenditure of the mess.

SSC MTS 13/07/2022 (Evening)

- (a) ₹1,170 (b) ₹1,710 (c) ₹1,200 (d) ₹1,250

Solution :-

Let the average expenditure be x

Now according to the question

$$45x + 30 = 50(x - 2)$$

$$45x + 30 = 50x - 100$$

$$5x = 130 \Rightarrow x = 26$$

$$\text{original total daily expenditure} = 45x = 45 \times 26 = 1,170 \text{ Rs}$$

Work and Time

Time :- Time is the duration during which any activity or work happens or continues.

Work :- Work is a task or set of activities to achieve a certain result.

Efficiency :- Efficiency is inversely proportional to the Time taken when the amount of work done is constant.

$$\text{Efficiency} \propto \frac{1}{\text{time taken}}$$

The efficiency of work means, "How much work one person can do in one day (expressed in percentage)".

If A,B,C person can do the work in x days, y days, z days respectively then ratio of their efficiency -

$$A : B : C$$

Time $\rightarrow x : y : z$

Efficiency $\rightarrow \frac{1}{x} : \frac{1}{y} : \frac{1}{z}$

Example :- A, B and C are assigned to complete a work. If the ratio of time taken by A, B and C is 4 : 3 : 6. Find the ratio of their efficiency.

SSC MTS 10/05/2023 (Evening)

(a) 3 : 4 : 2 (b) 4 : 3 : 6 (c) 3 : 4 : 6 (d) 4 : 6 : 8

Solution :- According to the question,

$$A : B : C$$

Time $\rightarrow 4 : 3 : 6$

Efficiency $\rightarrow \frac{1}{4} : \frac{1}{3} : \frac{1}{6} \times 12 \text{ (LCM of 4,3,6)}$

Ratio of efficiency of A : B : C = 3 : 4 : 2

Basic Concept

If A completes a work in X days,	If B completes a work in Y days,
Work completed by him in 1 day = $\frac{1}{X}$	Work completed by him in 1 day will be = $\frac{1}{Y}$
one day work of A and B together $(\frac{1}{X} + \frac{1}{Y})$.	

Type - 1

- If A completes a work in X days, B can do the same work in Y days, then-

Total time taken by A and B to complete the work = $(\frac{XY}{X+Y})$ days.

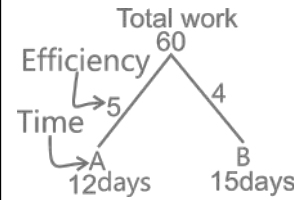
Example :- A can do a piece of work in 12 days and B can do the same work in 15 days. If both do the work together, then in how many days will the work be completed?

SSC CHSL 02/08/2023 (4th Shift)

(a) $7\frac{2}{3}$ (b) $6\frac{2}{3}$ (c) $5\frac{2}{3}$ (d) $8\frac{2}{3}$

Solution :- Required time = $\frac{A \times B}{A+B} = \frac{12 \times 15}{12+15}$
 $= \frac{12 \times 15}{27} = 6\frac{2}{3}$ days

Exam Hall Approach :-



Required time = $\frac{60}{9} = 6\frac{2}{3}$ days

Type - 2

- If A completes a work in X days, B can do the same work in Y days and C can complete the work in Z days then-

Total time taken by A, B and C to complete the work

= $(\frac{XYZ}{XY + YZ + ZX})$ days.

Example :- A, B and C can complete a piece of work in 20, 24 and 30 days, respectively. The number of days they take to finish it if they work together will be:

RRC Group D 09/09/2022 (Evening)

(a) 8 days (b) 5 days (c) 6 days (d) 7 days

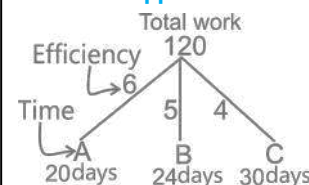
Solution :-

Total time taken by A, B and C to complete the work

= $(\frac{20 \times 24 \times 30}{20 \times 24 + 24 \times 30 + 30 \times 20})$ days

= $(\frac{20 \times 24 \times 30}{480 + 720 + 600})$ days = $(\frac{20 \times 24 \times 30}{1800}) = 8$ days

Exam Hall Approach:-



Required days = $\frac{120}{6+5+4} = 8$ days

Type - 3

Efficiency related Questions

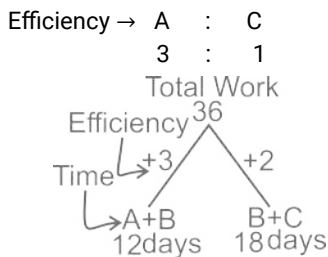
$$\text{Efficiency} \propto \frac{1}{\text{time taken}}$$

Example :- A and B together finish a task in 12 days. B and C together can finish it in 18 days. If A is three times more efficient than C, then in what time B alone would finish the same task?

RRB NTPC CBT - I 27/02/2021 (Evening)

(a) 22 days (b) 25 days (c) 24 days (d) 20 days

Solution :- According to question,



According to question

Ratio of the efficiency (A : B : C) = 1.5 : 1.5 : 0.5

Time taken by B alone to do the same work = $\frac{36}{1.5} = 24$ days

Type - 4

If A can do a work in x days and B is $y\%$ more efficient than A then -

Time taken by B alone = $x \times \frac{100}{100 + y}$ days

Example :- A can complete a certain job in 32 days. B is 60% more efficient than A. In how many days can B alone complete the same job?

RRC Group D 24/08/2022 (Morning)

(a) 10 days (b) 32 days (c) 15 days (d) 20 days

Solution :- Time taken by B alone

$$= 32 \times \frac{100}{100 + 60} \text{ days} = 20 \text{ days}$$

Exam hall approach :-

B is 60% more efficient than A.

Efficiency of A : B = 5 : 8

Total work = $5 \times 32 = 160$ units

B will do this work in = $\frac{160}{8} = 20$ days

Type - 5

- If A and B completes a work in X days, B and C can do the same work in Y days and C and A can complete the work in Z days then -

Total time taken by A, B and C to complete the work

$$= \left(\frac{2XYZ}{XY + YZ + ZX} \right) \text{ days.}$$

Example :- A and B can complete a piece of work in 20 days, B and C can complete the same piece of work in 12 days, while C and A can do it in 15 days. In how many days can A, B, and C together complete the same work?

RRC Group D 22/09/2022 (Evening)

(a) 15 (b) 5 (c) 10 (d) 12

Solution :-

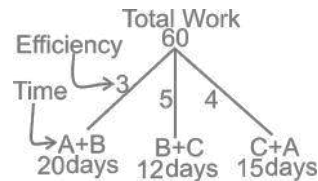
Total time taken by A, B and C to complete the work

$$= \left(\frac{2 \times 20 \times 12 \times 15}{20 \times 12 + 12 \times 15 + 15 \times 20} \right) \text{ days.}$$

$$= \left(\frac{2 \times 20 \times 12 \times 15}{240 + 180 + 300} \right) \text{ days} = \left(\frac{2 \times 20 \times 12 \times 15}{720} \right) \text{ days.}$$

$$= 10 \text{ days}$$

Exam Hall Approach :-



$$2(A + B + C) = 12$$

Efficiency of A + B + C = 6

Time taken by them to complete the work = $\frac{60}{6} = 10$ days

Type - 6

If A alone can do a work in x days and A and B together can do the same work in y days then B alone can do the same work in -

$$\left(\frac{xy}{x - y} \right) \text{ days.}$$

Example:- X and Y together can do a piece of work in 45 days. If Y alone can complete it in 180 days, then in how many days X alone can complete the work?

SSC MTS 11/05/2023 (Evening)

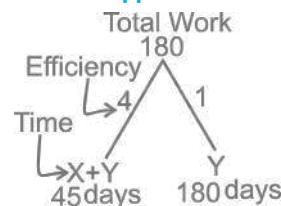
(a) 90 days (b) 40 days (c) 60 days (d) 80 days

Solution:-

Time taken by X alone to complete the work

$$= \left(\frac{180 \times 45}{180 - 45} \right) \text{ days} = \left(\frac{180 \times 45}{135} \right) \text{ days} = 60 \text{ days}$$

Exam Hall Approach:-



Efficiency of X = $4 - 1 = 3$ units

Required time = $\frac{180}{3} = 60$ days

Type - 7

- If A can Finish $\frac{p}{q}$ work in x days, then total time taken to finish the work by A = $\frac{q}{p} \times x$

Example :- A can complete $\frac{1}{3}$ of a work in 7 days and B can complete $\frac{2}{7}$ of the same work in 10 days. In how many days can both A and B together complete the work?

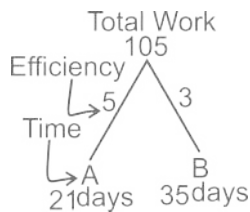
SSC CGL 25/07/2023 (2nd shift)

(a) $12\frac{3}{8}$ (b) $11\frac{7}{8}$ (c) $13\frac{1}{8}$ (d) $15\frac{1}{7}$

Solution :-

Time taken by A to complete the work = $\frac{3}{1} \times 7 = 21$ days

Time taken by B to complete the work = $\frac{7}{2} \times 10 = 35$ days



So, Time taken by A and B together to complete the work
 $= \frac{105}{8} = 13\frac{1}{8}$ days

Type - 8

- If A takes 'a' days more to complete a work than the time taken by (A + B) to do same work and B takes 'b' days more than the time taken by (A + B) to do the same work.

Then (A + B) do the work in $\sqrt{ab}$ days

Example :- When A alone does a piece of work, he takes 25 days more than the time taken by A and B to do the work together. On the other hand, B alone takes 16 days more than the time taken by A and B to do the work together. How many days will A and B, working together, take to do the work ?

SSC CPO 04/10/2023 (1st Shift)

- (a) 22 (b) 20 (c) 24 (d) 23

Solution :- Time taken to complete the work by A and B together $= \sqrt{25 \times 16} = 20$ days

Type - 9

Leaving the work after some days or before some days

Leaving after some days :-

Example :- X and Y can do a piece of work in 45 days and 40 days respectively. They begin to work together, but X leaves after n days and then Y completes the remaining work in 23 days. What is n equal to?

UPSC CDS - II (04/09/2022)

- (a) 8 (b) 9 (c) 10 (d) 12

Solution :-

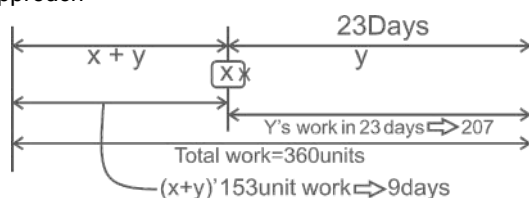


Efficiency of x = 8 unit

Efficiency of y = 9 unit

Efficiency of x + y = 17 unit

Note :- For Solving this type of question we use reverse approach



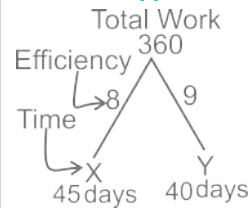
According to the question we know that y done the remaining work in 23 days

23 days = 207 unit (23 × 9)

Work done by x + y in n days = 360 - 207 = 153 unit

So that, value of 'n' = $\frac{153}{17} = 9$ days

Exam Hall approach :-



X and Y together do the work till n days

Then, work done by them in n days = (8 + 9)n = 17n

Remaining work = 360 - 17n

According to the question,

$$\frac{360 - 17n}{9} = 23 \Rightarrow 360 - 17n = 207 \Rightarrow n = \frac{153}{17} = 9$$

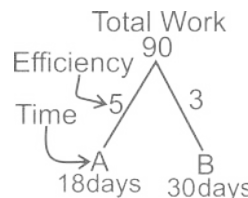
Leaving before some days :-

Example :- A and B can individually complete a piece of work in 18 days and 30 days, respectively. A and B started working together, but A left $16\frac{2}{3}$ days before the work is completed and B alone completed the rest of the work. For how many days did A work?

SSC CGL 20/07/2023 (1st shift)

- (a) 4 (b) 5 (c) 6 (d) 3

Solution :-

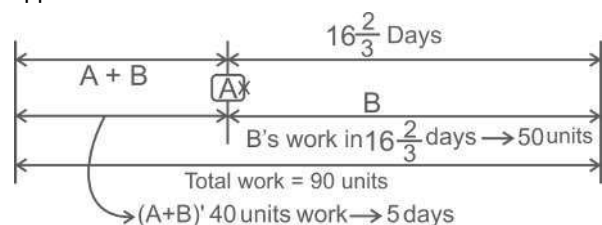


Efficiency of A = 5 unit

Efficiency of B = 3 unit

Total Efficiency of A + B = 8 unit

Note :- For Solving this type of questions, we use reverse approach



According to the question we know that B done the remaining work in $16\frac{2}{3}$ days

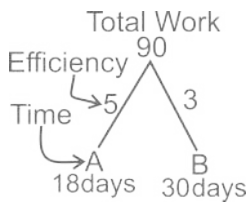
$$16\frac{2}{3} \text{ days} = 50 \text{ unit} \left(\frac{50}{3} \times 3 = 50 \right)$$

work done by (A + B) together = 90 - 50 = 40 unit

So that,

$$\text{Time taken to complete 40 units of work by (A + B)} = \frac{40}{8} = 5 \text{ days}$$

Exam hall approach :



Work done by B in $16\frac{2}{3}$ days = $\frac{50}{3} \times 3 = 50$ units

Remaining work = $90 - 50 = 40$ units

A and B work together for = $\frac{40}{5+3} = 5$ days

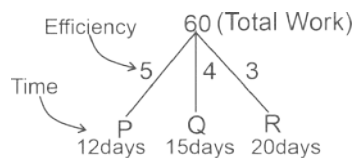
Mixed

Example :- P, Q and R can complete a work alone in 12, 15 and 20 days respectively. They started the work together. P left the work 8 days before the work was completed and Q left the work 5 days after P had left. R completed the remaining work alone. How many days will be required to complete the whole work?

SSC CHSL 15/03/2023 (4th Shift)

(a) $\frac{32}{3}$ days (b) 10 days (c) $\frac{28}{3}$ days (d) $\frac{25}{3}$ days

Solution :-



If P do not leave the work then he did the work in 8 days
= $8 \times 5 = 40$

If Q do not leave the work then he did the work in 3 days
= $3 \times 4 = 12$

Total work done by P and Q = $40 + 12 = 52$

So total work done by P, Q, R = $\frac{60 + 52}{5 + 4 + 3} = \frac{112}{12} = \frac{28}{3}$ days

So, will do total works in $\frac{28}{3}$ days

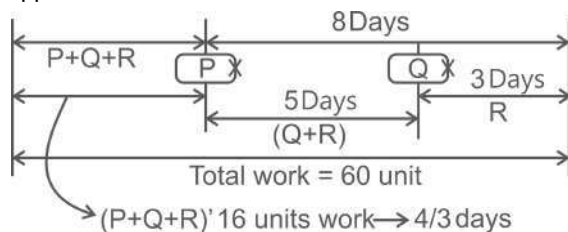
Exam Hall Approach :

Efficiency of P = 5 unit

Efficiency of Q = 4 unit

Efficiency of R = 3 unit

Note:- For Solving this type of questions, we use reverse approach



Work done by R in 3 days = $3 \times 3 = 9$ units

Work done by (Q + R) in 5 days = $5 \times 7 = 35$ units

So that, Work done by (P + Q + R) = $60 - 44 = 16$ units

Therefore, time taken to complete 16 units of work by

(P + Q + R) = $\frac{16}{12} = \frac{4}{3}$ days

Therefore, time taken to complete the total work = $3 + 5 + \frac{4}{3}$

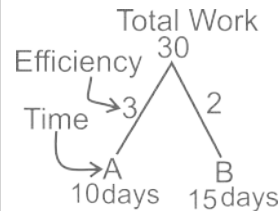
= $8 + \frac{4}{3} = \frac{28}{3}$ days

Type - 10

Working on Alternate days

Example:- A can complete a work in 10 days while B can complete the same work in 15 days. In how many days will the work be completed if they work alternatively? (Given : A starts the work)

Solution :-



Work done in 2 days = $3 + 2 = 5$ units,

5 units work = in 2 days

5×6 units work = $2 \times 6 = 12$ days.

Example :- To do a certain work, A and B work on alternate days, with B beginning the work on the first day. A can finish the work alone in 27 days. If the work gets completed in $9\frac{1}{2}$ days, then B alone can finish 5 times the same work with half efficiency in :

Solution :- As B starts the work, he will work for 5 days and A will work for $4\frac{1}{2}$ days,

Part of work done by A in $4\frac{1}{2}$ days = $\frac{9}{2 \times 27} = \frac{1}{6}$

So, Part of work done by B in 5 days = $1 - \frac{1}{6} = \frac{5}{6}$

Time taken by B to complete the work = $5 \times \frac{6}{5} = 6$ days.

Time taken by B to finish 5 times the work with half efficiency = $6 \times 5 \times 2 = 60$ days.

(As efficiency becomes $\frac{1}{2}$, number of days will be multiplied by 2 because efficiency and time are inversely proportional).

Type - 11

MDH Formula

MDH formula is used to compare the works of the same nature done at two different times using different manpower. The MDH formula can be represented as

$$\frac{M_1 D_1 H_1 E_1}{W_1} = \frac{M_2 D_2 H_2 E_2}{W_2}$$

Where,

M → Number of men/women,

D → Number of days,

H → Number of hours in a day,

E → Efficiency of 1 man,

W → Part of work done

Example :- 16 men can dig a ditch 24 m long in 18 days, working 6 hours a day. How many more men should be engaged to dig a similar ditch 42 m long in 9 days, each man

now working 12 hours per day?

Selection Post 30/06/2023 (3rd Shift)

(a) 14 (b) 10 (c) 12 (d) 9

Solution :- Let the required number of men be x , then

$$\Rightarrow \frac{16 \times 18 \times 6}{24} = \frac{x \times 9 \times 12}{42} \Rightarrow x = \frac{2 \times 2 \times 42}{3 \times 1 \times 2} = 28 \text{ men}$$

Extra required men = $28 - 16 = 12$ men

Example :- There are 300 men in a group. They have food for 30 days. After 10 days, 100 more men joined the group. Remaining food will last for how many days?

SSC MTS 04/05/2023 (Afternoon)

(a) 20 days (b) 10 days (c) 15 days (d) 12 days

Solution :- According to the question,

Let the numbers of days = x

$$300 \times 20 = (300 + 100) \times x \Rightarrow x = 15 \text{ days}$$

Example :- 20 women and 15 men together can complete a work in 6 days. It takes 150 days for a single woman to complete the same work. In how many days can a single man complete the work?

SSC CGL 25/07/2023 (4th shift)

(a) 450 (b) 225 (c) 350 (d) 210

Solution :- Equating the total work :-

$$(20W + 15M) \times 6 = (1W) \times 150$$

$$90M = 30W \Rightarrow \frac{M}{W} = \frac{1}{3}$$

Total work = $(1W) \times 150 = 3 \times 150 = 450$ units

Required number of days when a man can complete the work

$$\text{alone} = \frac{450}{1} = 450 \text{ days}$$

Example :- 4 men and 7 women can do a work in 8 days. 7 men and 4 women can do the same work in 5 days. Find the number of days in which 8 women can finish the work.

SSC CHSL 16/03/2023 (1st Shift)

(a) 60 Days (b) 55 Days (c) 40 Days (d) 45 Days

Solution :- Let efficiency of men be M

and women be W

According to the question,

$$8(4M + 7W) = 5(7M + 4W)$$

$$32M + 56W = 35M + 20W$$

$$36W = 3M \Rightarrow \text{efficiency of } \left(\frac{M}{W}\right) = \frac{12}{1}$$

Now, total work $\rightarrow 8(4M + 7W)$

$$= 8(4 \times 12 + 7 \times 1) \Rightarrow 440 \text{ unit}$$

$$\text{Then, Time taken by 8 women} = \frac{440}{8 \times 1} = 55 \text{ days}$$

Type - 12

If **A men** or **B boys** or **C women** can do a certain work in **m days** then **x men**, **y boys**, **z women** can do the same work in -

$$= \frac{m}{\frac{x}{A} + \frac{y}{B} + \frac{z}{C}} \text{ days}$$

Example :- 6 men or 2 women or 4 boys can finish a work in 77 days. How many days will 1 man, 1 woman and 1 boy together take to finish the same work?

SSC CHSL 02/06/2022(Evening)

(a) 86 (b) 80 (c) 84 (d) 82

Solution :- Time taken to finish the work

$$= \frac{77}{\frac{1}{6} + \frac{1}{2} + \frac{1}{4}} = \frac{77}{\frac{11}{12}} = \frac{77 \times 12}{11} = 84 \text{ days}$$

Exam hall approach:

$$6 \text{ men} = 2 \text{ women}, \frac{\text{men}}{\text{women}} = \frac{1}{3}$$

$$2 \text{ women} = 4 \text{ boys}, \frac{\text{women}}{\text{boys}} = \frac{2}{1}$$

Ratio - men : women : boy

Efficiency - 1 : 3 : 3

2 : 2 : 1

Final - 2 : 6 : 3

Total work = $2 \times 6 \times 77$ units

$$\text{Required day} = \frac{2 \times 6 \times 77}{(2 + 6 + 3)} = 84 \text{ days}$$

Type - 13

Wages Related Questions

$$\text{Efficiency} \propto \text{Wages} \propto \frac{1}{\text{Time taken}}$$

If **A** can do the work in x days and **B** can do the same work in y days and **C** can do the work in z days. If they work together and total wages is **W**, then -

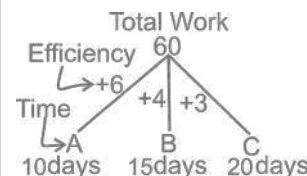
$$\text{Ratio of their wages} = \text{ratio of their efficiency} = \frac{1}{x} : \frac{1}{y} : \frac{1}{z}$$

Example :- A, B and C can do a work in 10 days, 15 days, and 20 days, respectively. They finished that work together and got ₹2,600 as wages. Find C's wage.

SSC CGL 20/07/2023 (2nd shift)

(a) ₹550 (b) ₹600 (c) ₹575 (d) ₹625

Solution :-



According to question,

Ratio of efficiency of A, B and C = $6 : 4 : 3$

$$\text{Share of C} = 2600 \times \frac{3}{6 + 4 + 3} = \text{Rs. } 600$$

Example:- X, Y and Z completed a work costing ₹3,400. X worked for 5 days, Y for 7 days and Z for 10 days. If their daily wages are in the ratio of 4: 5: 3, how much amount will be received by X?

SSC CGL 20/07/2023 (4th shift)

(a) ₹700 (b) ₹900 (c) ₹800 (d) ₹600

Solution :- Ratio of their wages :-

$$X : Y : Z = 4 \times 5 : 7 \times 5 : 10 \times 3$$

$$= 20 : 35 : 30$$

$$\text{Wages received by X} = \frac{20}{85} \times 3400 = ₹ 800$$

Pipe and Cistern

Pipes and Cisterns are just an application of Time and Work. Concept wise, it is one and the same. In the previous proportionality, Efficiency is replaced by Rate of filling. The equation in this case becomes

$$\text{Rate of filling} \propto \frac{1}{\text{time taken}}$$

Volume of water released or filled = Rate $\times$ Time

The concept of the percentage of work done can be used to solve most of the related questions.

A few basic points one needs to know to use the percentage concept are: When it is said that someone has done a work, it means he has done 100 % of the work. Hence, if A finishes work in 4 days, it means- in 4 days he will do 100% of the work. Hence, in one day he finishes 25% ($\frac{100}{4}$) of the work. Similarly, in 3 days he finishes 75 % of the work.

The complete work can also be considered as 1 unit. Then if A takes 4 days to finish a work, it means he can finish $\frac{1}{4}$ th of the work in 1 day.

Type - 1

When two taps A and B can fill a tank in x hours and y hours respectively, then time taken to fill the tank when opened together -

Or

Two pipe 'A' and 'B' can empty a tank in ' x ' hour and ' y ' hour respectively. If both pipe are open together, then time taken to empty the tank - $\frac{x \times y}{x + y}$

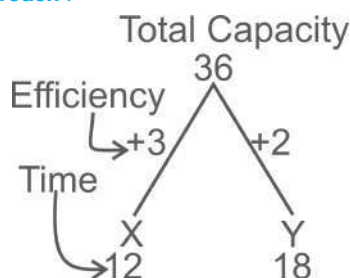
Example :- Two pipes X and Y can fill a tank in 12 minutes and 18 minutes, respectively. If both the pipes are opened simultaneously, how long will it take to fill the tank (in mins) ?
Selection Post 03/08/2022 (4th Shift)

- (a) 15 (b) $\frac{5}{36}$ (c) 14 (d) $7\frac{1}{5}$

Solution :-

$$\text{Required time} = \frac{xy}{x + y} = \frac{12 \times 18}{12 + 18} = \frac{36}{5} = 7\frac{1}{5} \text{ min}$$

Exam Hall Approach :



Time taken by (X + Y) to fill the whole tank

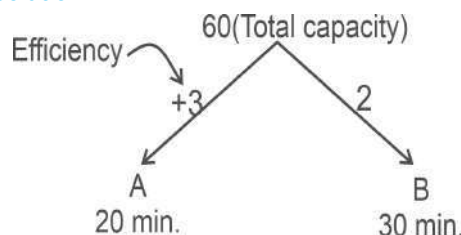
$$= \frac{36}{3 + 2} = \frac{36}{5} = 7\frac{1}{5} \text{ min}$$

Example :- Two pipes A and B fill tank in 20 min and 30 min. respectively. If A is kept open for only $\frac{1}{5}$ th of the total time and A and B are kept open for remaining time; then how much time is taken by both pipes to fill the tank.

SSC CPO 15/03/2019 (Evening)

- (a) $16\frac{5}{23}$ (b) $13\frac{1}{23}$ (c) $13\frac{5}{23}$ (d) $16\frac{1}{23}$

Solution :-



Let total time taken be $5x$.

According to question,

$$3x + (5 \times 4x) = 60 \Rightarrow 23x = 60 \Rightarrow x = \frac{60}{23}$$

Therefore,

$$\text{Total time} = 5x = 5 \times \frac{60}{23} = \frac{300}{23} = 13\frac{1}{23} \text{ minute}$$

Example :- One - fourth of a tank can be filled in 3 hours by pipe A and one-third of the same tank can be filled in 2 hours by pipe B. How long will it take for the tank to be filled if both pipes are kept open ?

SSC CPO 14/03/2019 (Morning)

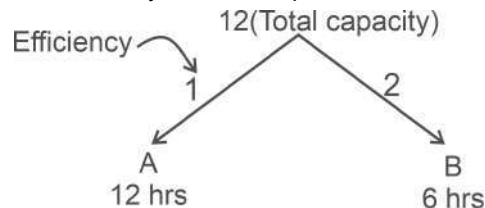
- (a) 5 h (b) 2 h (c) 4 h (d) $2\frac{1}{2}$ h

Solution :- Time taken by A to fill $\frac{1}{4}$ th of tank = 3 hr

Time taken by A to fill complete tank = $3 \times 4 = 12$ hr

Time taken by B to fill $\frac{1}{3}$ rd of tank = 2 hr

Time taken by B to fill complete tank = $2 \times 3 = 6$ hr



Let total capacity of tank = 12 unit

$$\text{Desired time} = \frac{12}{1 + 2} = 4 \text{ hours}$$

Type - 2

A pipe 'A' can fill a tank in ' x ' hours and 'B' can empty the tank in ' y ' hours .When both the pipes are opened together then.

Time taken to fill the tank = $\frac{xy}{y - x}$ hours ... [$y > x$]

Time taken to empty the tank = $\frac{xy}{x - y}$ hours ... [$x > y$]

Example :- An inlet pipe can fill an empty tank in 51 hours while an outlet pipe drains a completely-filled tank in 76.5 hours. If both the pipes are opened simultaneously when the tank is empty, in how many hours will the tank get completely

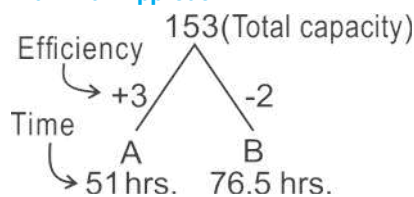
filled ?

SSC CHSL 08/08/2023 (1st Shift)

(a) 178.5 (b) 127.5 (c) 102 (d) 153

Solution :- Time taken to fill the tank

$$= \frac{51 \times 76.5}{76.5 - 51} = \frac{51 \times 76.5}{25.5} = 153 \text{ hr}$$

Exam Hall Approach :

Required time to fill the tank by A and B

$$= \frac{153}{3 - 2} = 153 \text{ hours}$$

Type - 3

Two pipes A and B can fill a tank in x hours and y hours respectively. If both the pipes are opened together, then the time after which pipe B should be closed so that the tank is full

in t hours = $[y (1 - \frac{t}{x})] \text{ hr}$

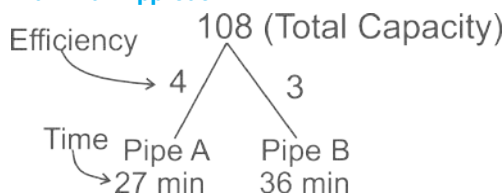
Example :- Two pipes A and B can fill a tank in 27 minutes and 36 minutes, respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 21 minutes?

DDA JE (Civil) 29/03/2023 (Evening)

(a) 9 minutes (b) 7 minutes (c) 6 minutes (d) 8 minutes

Solution :- Required time = $[y (1 - \frac{t}{x})]$

$$= [36 (1 - \frac{21}{27})] = 36 \times \frac{2}{9} = 8 \text{ min}$$

Exam Hall Approach :Work done by Pipe A in 21 min. = $21 \times 4 = 84$ unitsRemaining work = $108 - 84 = 24$ units

$$\text{Required time} = \frac{24}{3} = 8 \text{ min.}$$

Type - 4

If pipe A can fill a cistern in X hours, pipe B can fill the same cistern in Y hours, and pipe C can empty the full cistern in Z hours, then the time taken to completely fill the cistern, if Pipes

A, B and C are opened together. = $\frac{XYZ}{YZ + ZX - XY}$

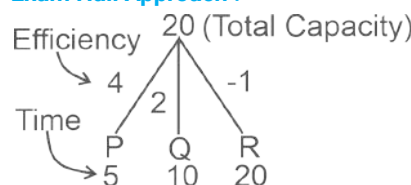
Example :- P can fill a tank in 5 hours. Q can fill the same tank in 10 hours. R can empty the same tank in 20 hours. How much time (in hrs) will all the three take together to fill the same tank ?

SSC CHSL 13/03/2023 (2nd Shift)

(a) 6 (b) 4 (c) 10 (d) 5

$$\text{Solution :- } t = \frac{5 \times 10 \times 20}{20 \times 10 + 20 \times 5 - 5 \times 10}$$

$$= \frac{1000}{200 + 100 - 50} = \frac{1000}{250} = 4 \text{ hr}$$

Exam Hall Approach :Total efficiency = $4 + 2 + (-1) = 5$ unit

$$\text{Total time taken by P, Q, and R to fill the tank} = \frac{20}{5} = 4 \text{ hours}$$

Type - 5

If a, b, call the taps are opened together, then the time required to fill / empty the tank will be :

$$\frac{1}{a} \pm \frac{1}{b} \pm \frac{1}{c} \pm \dots = \frac{1}{T}$$

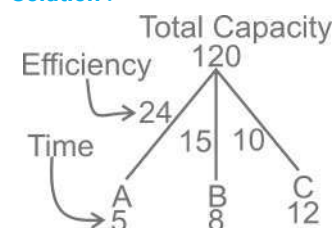
Where T is the required time.

Note :- Positive result = to fill the tank

Negative result = to empty the tank

Example:- Three pipes A, B and C can fill a tank in 5 hours, 8 hours and 12 hours, respectively. If all the pipes are opened at the same time, then the time taken to fill the tank is:

RRC Group D 05/09/2022 (Morning)

(a) $4\frac{22}{49}$ hour (b) $2\frac{22}{49}$ hour (c) $5\frac{22}{49}$ hour (d) $3\frac{22}{49}$ hour**Solution :-**

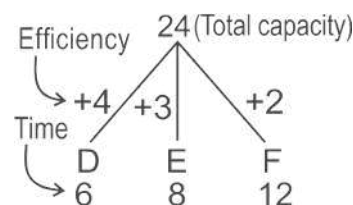
$$\text{Time taken by pipe (A + B + C) to fill the tank} = \frac{120}{49}$$

$$= 2\frac{22}{49} \text{ hours}$$

Example:- Three Pipes, D, E and F, can fill a tank in 6 min, 8 min and 12 min, respectively. All the pipes are opened simultaneously. And then pipes D and E are closed 3 minutes before the tank is full. In how much time (in min) will the tank be full ?

SSC MTS 07/07/2022 (Evening)

(a) 5 (b) 3 (c) 6 (d) 4

Solution :-

Since, the pipes D and E are closed 3 min before then, at that

time only pipe F is opened.

So, work done by pipe F to fill the tank

$$= 2 \times 3 = 6 \text{ unit}$$

$$\text{Remaining work} = 24 - 6 = 18 \text{ unit}$$

Now, the time taken by D, E, F together to

$$\text{complete the remaining work} = \frac{18}{9} = 2 \text{ min}$$

Therefore, the time taken to fill the whole tank

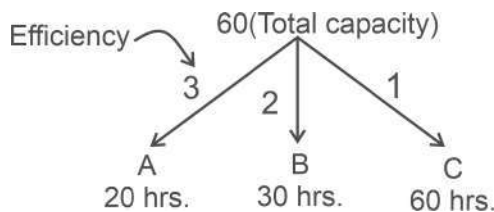
$$= 2 \text{ min} + 3 \text{ min} = 5 \text{ min}$$

Example :- Pipes A, B and C can fill a tank in 20, 30 and 60 hours, respectively. Pipes A, B and C are opened at 7 a.m., 8 a.m., and 9 a.m., respectively, on the same day. When will the tank be full?

SSC CGL Tier II (03/02/2022)

(a) 4:40 p.m. (b) 5:40 p.m. (c) 6:20 p.m. (d) 7:20 p.m.

Solution :-



Tank filled by A alone till 9 am = $2 \times 3 = 6$ units

Tank filled by B alone till 9 am = $1 \times 2 = 2$ units

Total tank filled 9 am = $6 + 2 = 8$ units

Remaining tank to be filled = $60 - 8 = 52$ units

Time taken by A, B and C together

$$= \frac{52}{3 + 2 + 1} = 8\frac{2}{3} \text{ hrs} = 8 \text{ hrs } 40 \text{ minutes}$$

So, the tank will be full at = 9 am + 8 hrs 40 min. = 5:40 p.m.

Type - 6

If a pipe fill a tank in x hours but it takes ' t ' more hours to fill it due to leakage in tank. If the tank is completely filled, then time required to empty (due to leakage) the tank?

$$= \frac{x(x+t)}{t}$$

Example:- A cistern can be filled in 9 hours. But due to a leak, it takes one hour more. If the cistern is full, in what time will it become empty due to the leak?

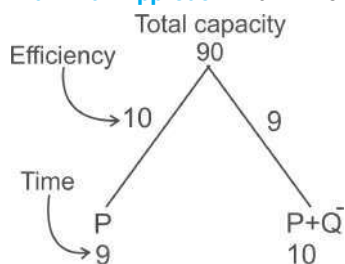
RRB JE 01/06/2019 (Evening)

(a) 60 hours (b) 30 hours (c) 90 hours (d) 75 hours

Solution :- Time required to take empty the full cistern

$$= \frac{x(x+t)}{t} = \frac{9(9+1)}{1} = 90 \text{ hr}$$

Exam Hall Approach : Tank filler pipe = P, leakage pipe = Q



So, Leak (Q) can empty 1 unit in 1 hour.

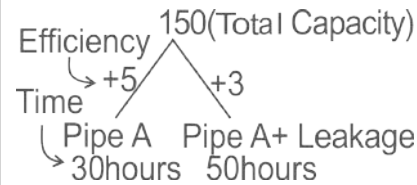
$$\text{Time taken by Leak (Q) to empty the tank} = \frac{90}{1} = 90 \text{ hours}$$

Example:- A pipe can fill a tank in 30 hours. Due to a leakage at the bottom, it is filled in 50 hours. How much time (in hrs) will the leakage take to empty the completely filled tank?

SSC CGL 24/07/2023 (2nd shift)

(a) 60 (b) 85 (c) 70 (d) 75

Solution :-



Efficiency of leakage = $5 - 3 = 2$ units

$$\text{Time taken by the leakage to empty the filled tank} = \frac{150}{2}$$

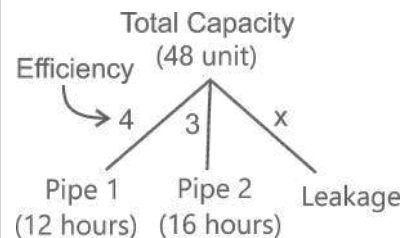
= 75 hours

Example:- Two pipes can fill a cistern in 12 hours and 16 hours, respectively. The pipes are opened simultaneously and it is found that due to leakage at the bottom, it takes 90 minutes more to fill the cistern. How much time will the leakage take to empty the completely filled tank?

SSC CPO 04/10/2023 (3rd Shift)

(a) $39\frac{13}{49}$ h (b) $36\frac{29}{49}$ h (c) $37\frac{15}{49}$ h (d) $38\frac{10}{49}$ h

Solution :-



According to the question,

$$\frac{48}{(7-x)} - \frac{48}{7} = \frac{3}{2} \Rightarrow 32x = 7(7-x)$$

$$\text{Efficiency of Leakage (x)} = \frac{49}{39} \text{ unit}$$

$$\text{Hence, required time} = \frac{48}{\frac{49}{39}} = \frac{48 \times 39}{49} = 38\frac{10}{49} \text{ days}$$

Type - 7

If two pipes A and B which can fill a tank, such that efficiency of A is x times of B and takes t min less/more than B to fill the tank

, then time taken to fill the tank by both pipe together = $\frac{xt}{(x^2 - 1)}$

Example :- Pipe A is 3 times faster than pipe B. Both pipes together fill a tank in 36 min. How much time will pipe A alone take to fill the tank?

RRC Group D 25/08/2022 (Morning)

(a) 80 min (b) 100 min (c) 156 min (d) 48 min

Solution :-

$$36 = \frac{3 \times t}{(9-1)} \Rightarrow t = 96 \text{ minutes (time difference)}$$

$$A : B = 3 : 1 \text{ (efficiency)} \Rightarrow A : B = 1 : 3 \text{ (time)}$$

$$2 = 96 \Rightarrow (A)1 = 48 \text{ minutes}$$

Exam Hall approach:-

Efficiency of A : B = 3 : 1

Total work = $(3 + 1) \times 36 = 144$

Time taken by pipe A alone to fill the tank = $144 \div 3 = 48$ min

Type - 8

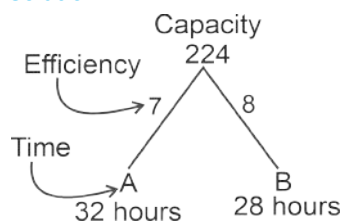
If the pipes move alternately :-

Example :- There are two inlets A and B connected to a tank. A and B can fill the tank in 32 h and 28 h, respectively. If both the pipes are opened alternately for 1 h, starting with A, then in how much time (in hours, to nearest integer) will the tank be filled ?

SSC CGL Tier II (03/03/2023)

(a) 22 (b) 30 (c) 36 (d) 24

Solution :-



A and B pipes are opened alternately for 1 h

Work done in 2 hour = $7 + 8 = 15$ units

Similarly, work done in 29 hour = $(15 \times 14) + 7 = 217$ units

In the 30th hour, work will be completed.

Type - 9

MDH Formula

MDH formula is used to compare the works of the same nature done at two different times using different manpower. The MDH formula can be represented as

$$\frac{M_1 D_1 H_1 E_1}{W_1} = \frac{M_2 D_2 H_2 E_2}{W_2}$$

Where,

M = denotes the number of men/women,

D = denotes the number of days,

H = denotes the number of hours in a day,

E = denotes the efficiency of 1 man,

W = denotes part of work done

Example:- 6 pipes, working 10 hours a day, can empty a cistern in 3 days. How many hours a day must 9 pipes work to empty the cistern in one day?

Delhi Police H.C.M. 17/10/2022 (Morning)

(a) 18 (b) 20 (c) 12 (d) 22

Solution :- Let the required time be x

ATQ,

$$6 \times 10 \times 3 = 9 \times 1 \times x$$

$$180 = 9x \Rightarrow x = \frac{180}{9} = 20$$

Type - 10

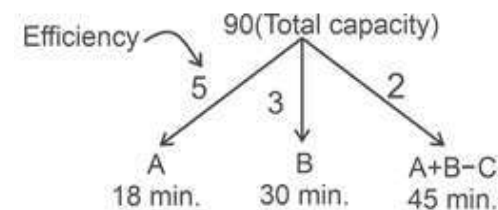
Example 1 :- Pipes A and B can fill a tank in 18 minutes and

30 minutes, respectively. Pipe C, attached to the tank, can drain off 125 litres of water per minute. If all the pipes are opened together, the tank is filled in 45 minutes. The capacity of the tank in litres, is:

SSC MTS 14/10/2021 (Evening)

(a) 1500 (b) 1375 (c) 1875 (d) 2000

Solution :-



Efficiency of C = $5 + 3 - 2 = 6$ unit

So, C can empty entire tank in = $\frac{90}{6} = 15$ minutes

Water drain by C in 1 minute = 125 litres

Water drain by C in 15 minute = $125 \times 15 = 1875$ litres

Capacity of tank = 1875 litres

Example:- Two pipes, A and B, can fill a tank of 1000 litres in 5 hours and 6 hours, respectively. If they are opened together, how many hours will they take to fill an empty tank of 1800 litres ?

RRC Group D 24/08/2022 (Evening)

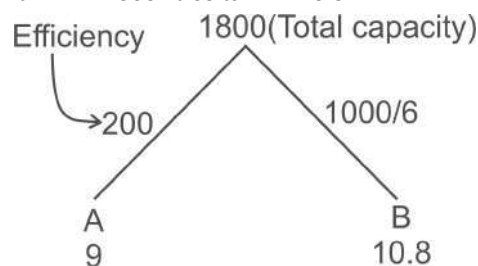
(a) $\frac{25}{11}$ (b) $\frac{54}{11}$ (c) $\frac{45}{11}$ (d) $\frac{52}{11}$

Solution :- Pipe A can fill 1000 litres in 5 hours

It will fill 1800 litres tank in 9 hr

Pipe B can fill 1000 litres in 6 hours

It will fill 1800 litres tank in 10.8 hr



$$\text{Efficiency of A and B} = 200 + \frac{1000}{6} = \frac{1100}{3}$$

$$\text{Pipe A and Pipe B will fill the tank in } \frac{1800 \times 3}{1100} = \frac{54}{11} \text{ hr}$$

Example:- There are three taps of diameter 2 cm, 3 cm and 4 cm, respectively. The ratio of the water flowing through them is equal to the ratio of the square of their diameters. The biggest tap can fill an empty tank alone in 81 min. If all the taps are opened simultaneously, then how long will the tank take (in min) to be filled?

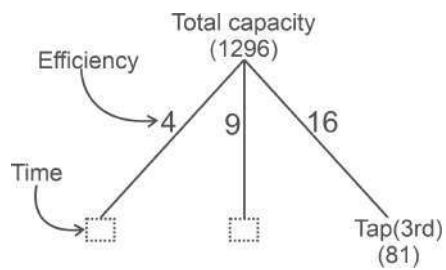
SSC CGL Tier II (02/03/2023)

(a) $34\frac{20}{29}$ (b) $64\frac{20}{29}$ (c) $54\frac{20}{29}$ (d) $44\frac{20}{29}$

Solution :- According to question,

Efficiency of three taps $\propto$ square of the diameter of the pipe

$$\begin{aligned} \text{Ratio of efficiency} &= (2)^2 : (3)^2 : (4)^2 \\ &= 4 : 9 : 16 \end{aligned}$$



$$\begin{aligned} \text{Total time taken by all three tap together} &= \frac{\text{total capacity}}{\text{total efficiency}} \\ &= \frac{1296}{(4 + 9 + 16)} = \frac{1296}{29} = 44\frac{20}{29} \text{ minute} \end{aligned}$$

Type - 11

Example:- Pipe K can fill a water tank in 10 hours. Pipe K is opened alone when the water tank is empty. When half of the water tank is filled by pipe K, then 4 more identical pipes are opened. What will be the total time taken to fill the water tank?
RRC Group D 07/10/2022 (Evening)

- (a) 6 hours 15 minutes (b) 7 hours 15 minutes
(c) 6 hours 45 minutes (d) 6 hours

Solution :- Let the total capacity of tank = 20 units

$$\text{Efficiency of Pipe K} = \frac{20}{10} = 2$$

$$\text{Time taken to fill the half tank} = \frac{10}{2} = 5 \text{ hours}$$

$$\text{Remaining capacity} = 20 - 10 = 10 \text{ units}$$

$$\text{Total efficiency with 4 more identical pipes}$$

$$= 2 + 2 + 2 + 2 + 2 = 10$$

$$\text{Time taken by all 5 pipes to fill remaining tank} = \frac{10}{10} = 1 \text{ hour}$$

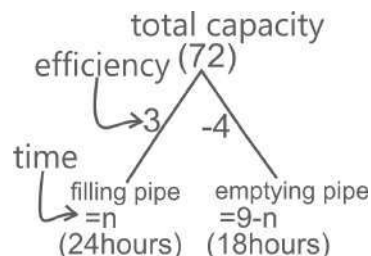
$$\text{Total time} = 5 + 1 = 6 \text{ hours}$$

Example:- Nine pipes are attached to a tank, of which some are emptying pipes and some are filling pipes. Each filling pipe can fill the empty tank in 24 hours, while each emptying pipe can empty the fully filled up tank in 18 hours. If the tank is full and all the pipes are opened simultaneously, the tank is emptied after 9 hours. The number of emptying pipes is:

UP Constable 26/10/2018 (1st Shift)

- (a) 6 (b) 5 (c) 4 (d) 3

Solution :-



ATQ ,

As tank is whole filled that is 72 units

Now both the pipes are opened

And it is emptied in 9 hours

$$72 + [(3 \times n) - 4(9 - n)] \times 9 = 0,$$

$$[(3 \times n) - 4(9 - n)] \times 9 = -72$$

$$[3n - 36 + 4n] = -8 \Rightarrow 7n = -8 + 36 \Rightarrow 7n = 28 \Rightarrow n = 4$$

$$\text{The number of emptying pipes} = 9 - n = 9 - 4 = 5$$

Speed, Time and Distance

Speed is defined as the distance travelled in unit time.

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Unit Conversion

km/h to m/s	$\text{Km/h} \xrightarrow{\times \frac{5}{18}} \text{m/sec}$ i.e. $90 \text{ km/h} = 90 \times \frac{5}{18} \text{ m/s} = 25 \text{ m/s}$
m/s to km/h	$\text{m/sec} \xrightarrow{\times \frac{18}{5}} \text{Km/h}$ i.e. $25 \text{ m/s} = 25 \times \frac{18}{5} \text{ km/h} = 90 \text{ km/h}$

Note:-

$$1 \text{ Hr} = 60 \text{ min} = 3600 \text{ sec}$$

$$1 \text{ yard} = 3 \text{ ft}$$

$$1 \text{ mile} = 1.609 \text{ km}$$

Example :- A bullet travels a distance of 90 m in 0.2 sec. Find its speed in km/hr.

RRB ALP Tier - II (23/01/2019) Morning

(a) 125 (b) 1,620 (c) 162 (d) 1,250

Solution :- Speed of bullet = $\frac{90}{0.2}$

$$= 450 \text{ m/s} = 450 \times \frac{18}{5} = 1620 \text{ km/h}$$

- When using the unit of speed in km/h, keep in mind that the unit of time should also be in hours. When the unit of speed is used in m/s, the unit of time should also be in seconds. That is, the units of speed and time are the same.

Example :- The speed of a car is 90 km/h. The distance covered by the car in 7 Seconds will be:

SSC MTS 05/09/2023 (1st Shift)

(a) 172 metres (b) 175 metres (c) 170 metres (d) 180 metres

Solution :- Required distance

$$= 90 \times \frac{5}{18} \times 7 = 175 \text{ meters}$$

Example :- A man travels from Delhi to Chandigarh at a speed of 50 km/h covering a distance of 300 km. Find the time taken?

Solution :- Here, we are given, Distance(D) = 300 km and Speed(S) = 50 km/h.

$$\therefore \text{Time}(T) = \frac{D}{S} = \frac{300}{50} = 6 \text{ hours}$$

Some Important Cases

Case 1 :- If Distance $\rightarrow$ constant then Speed $\propto \frac{1}{\text{Time}}$

Case 2 :- If speed $\rightarrow$ constant then Distance $\propto$ time.

Case 3 :- If time $\rightarrow$ constant then Distance $\propto$ Speed .

Example :- For the same distance, speed in outward journey and return journey is 50 km/h and 60 km/h. Find the ratio of the time taken.

Solution :- Here, the distance is constant

So, Speed $\propto \frac{1}{\text{Time}}$

So, the required ratio

$$= \frac{1}{50} : \frac{1}{60} = \frac{6}{300} : \frac{5}{300} = 6 : 5$$

Example :- If the ratio among the time taken by three cars to travel a certain distance is 2 : 3 : 5, then the ratio of speeds of these cars will be:

SSC MTS 11/09/2023 (1st Shift)

(a) 10 : 6 : 15 (b) 15 : 9 : 8 (c) 10 : 15 : 8 (d) 15 : 10 : 6

Solution :-

Ratio of time taken = 2 : 3 : 5

As we know:- time $\propto \frac{1}{\text{speed}}$

$$\text{Hence, ratio of speeds} = \frac{1}{2} : \frac{1}{3} : \frac{1}{5} = 15 : 10 : 6$$

Example :- In the same time, find the ratio of the distance travelled by A and B at a speed of 6 km/h and 8 km/h respectively.

Solution :- Here, time $\rightarrow$ constant then Distance $\propto$ Speed.

Speed $\rightarrow$ 6 : 8

Distance $\rightarrow$ 6 : 8

Hence, the required ratio = 6 : 8 = 3 : 4

Average speed

a.) When different distance is travelled each time then,

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$\text{i.e. } \begin{array}{c} \overline{\text{d}_1 \quad \text{d}_2 \quad \text{d}_3} \\ \text{t}_1 = \text{d}_1 / \text{v}_1 \quad \text{t}_2 = \text{d}_2 / \text{v}_2 \quad \text{t}_3 = \text{d}_3 / \text{v}_3 \end{array}$$

$$\text{Ave. Speed} = \frac{\text{d}_1 + \text{d}_2 + \text{d}_3}{\frac{\text{d}_1}{\text{v}_1} + \frac{\text{d}_2}{\text{v}_2} + \frac{\text{d}_3}{\text{v}_3}}$$

Example :- A car covers the first 210 km at a speed of 70 km/h. It covered the next 170 km at a speed of 85 km/h. What is its average speed ?

SSC CHSL 08/08/2023 (4th Shift)

(a) 72 km/h (b) 76 km/h (c) 74 km/h (d) 68 km/h

Solution :- Average speed = $\frac{210 + 170}{\frac{210}{70} + \frac{170}{85}} = \frac{380}{5} = 76 \text{ km/h}$

Example :- A person covers 11 km at a speed of 3 km/h, 21 km at a speed of 5 km/h and 37 km at a speed of 10 km/h. Find the average speed for the entire journey.

Selection Post 28/06/2023 (4th Shift)

(a) $5\frac{335}{347}$ km/h (b) $5\frac{235}{247}$ km/h
(c) $5\frac{347}{335}$ km/h (d) $5\frac{135}{347}$ km/h

Solution :-

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time taken}} = \frac{11 + 21 + 37}{\frac{11}{3} + \frac{21}{5} + \frac{37}{10}}$$

$$= \frac{69 \times 30}{110 + 126 + 111} = \frac{2070}{347} = 5\frac{335}{347} \text{ km/hr.}$$

b.) When an object travels the same distance with different speeds a and b, then

$$\text{Average Speed} = \frac{2ab}{a+b}, \text{ where a and b are speed}$$

Example :- A person travels from point A to point B at a speed of 30 km/h and comes back at a speed of 20 km/h. What is his average speed?

Solution :- Here, distance travelled is same,

$$\text{So, average speed} = \frac{2 \times 30 \times 20}{30 + 20} = \frac{1200}{50} = 24 \text{ km/h}$$

When an object travels the same distance with different speeds a, b and c, then

$$\text{Average Speed} = \frac{3abc}{ab + bc + ac}, \text{ where a, b and c are speed}$$

Example :- Gopal travels from A to B at the speed of 5 km/h, from B to C at 10 km/h, and from C to D at 15 km/h. If AB = BC = CD, then find Gopal's average speed.

SSC CHSL 09/06/2022 (Afternoon)

(a) $8\frac{2}{11}$ (b) $70\frac{2}{11}$ (c) $60\frac{2}{11}$ (d) $9\frac{2}{11}$

Solution :-

$$\text{When distance is same Average speed} = \frac{3xyx}{x \times y + y \times z + z \times x}$$

Average speed of Gopal in the entire journey

$$= \frac{3 \times 5 \times 10 \times 15}{5 \times 10 + 10 \times 15 + 15 \times 5} = \frac{2250}{50 + 150 + 75} = \frac{2250}{275}$$

$$= \frac{90}{11} = 8\frac{2}{11} \text{ km/hr}$$

Type - 1

Example :- A takes 30 minutes more than B to cover a distance of 15 km at a certain speed. But if A doubles his

speed, he takes one hour less than B to cover the same distance. What is the speed (in km/h) of B?

SSC CGL 04/06/2019 (Afternoon)

(a) 6 (b) 5 (c) $6\frac{1}{2}$ (d) $5\frac{1}{2}$

Solution :-

Let the speed of A and B be a and b respectively,

$$\text{ATQ: } \frac{15}{a} - \frac{15}{b} = \frac{1}{2} \text{ --- (i)}$$

$$\text{and, } \frac{15}{b} - \frac{15}{2a} = 1 \text{ --- (ii)}$$

$$\text{Adding (i) and (ii) } \frac{15}{2a} = \frac{3}{2} \Rightarrow a = 5$$

Putting the value of a in eq (i), we get

$$3 - \frac{15}{b} = \frac{1}{2} \Rightarrow b = 6 \text{ km/h}$$

Exam Hall approach :-

Here, Distance is constant, so

Ratio of speed (A) = 1 : 2

Then ratio of time(A) = 2 : 1

Here time difference 1 unit = $\frac{3}{2}$ hours,

then taken time by A = 2 units = 3 hours

Now time taken by B = 2.5 hours

$$\text{Hence speed of B} = \frac{15}{2.5} = 6 \text{ km/h}$$

Type - 2

A and B started their journeys from X to Y and Y to X, respectively. After crossing each other, A and B completed the remaining part of the journey in 'a' hours and 'b' hours, respectively. If the Speed of A is x km/hr and speed of B is y km/hr then

$$\frac{x}{y} = \sqrt{\frac{b}{a}}$$

Example :- Two motorists travelling in opposite directions meet at the same point in between. After this they take 9 and 16 hours to reach their respective destinations. What is the ratio of their speeds?

RRB JE 28/05/2019 (Morning)

(a) 5 : 4 (b) 4 : 7 (c) 4 : 3 (d) 5 : 3

Solution :- We know that $\rightarrow \text{speed} \propto \frac{1}{T}$

$$\frac{S_1}{S_2} = \sqrt{\frac{T_2}{T_1}} \Rightarrow \frac{S_1}{S_2} = \sqrt{\frac{16}{9}}$$

$$= \frac{S_1}{S_2} = \frac{4}{3}$$

Type - 3

When Distance is constant and two speed is given with total time or difference of time :-

The difference between time can be solved by some trick

Early, early case

' - '

Late , late case	' - '
Early , late case	' + '
Late , early case	' + '

a). Total distance travelled when a man covers a certain distance with s_1 km/hr and comes back with s_2 km/hr. He takes 't' hr to complete the whole journey is :



$$\text{Distance} = \frac{S_1 \times S_2}{S_1 + S_2} \times \text{total time}$$

Example :- A man travel a certain distance by train at the rate of 25 km/hr and walk back at the rate of 3 km/hr. The whole journey took 5 hr 36 min. What distance did he travel by train ?

Solution :- $D = \left(\frac{25 \times 3}{25 + 3} \right) \times \frac{28}{5} = 15 \text{ km}$

b). The total distance covered when a person travels a certain distance with s_1 km/h and he returns with a speed of s_2 km/h. If it takes 't' more hours to return than it takes to go:



$$\text{Distance} = \frac{S_1 \times S_2}{S_1 - S_2} \times \text{difference of time}$$

Example :- Aman drove from home to a town at the speed of 60 km/hr and on his return journey, he drove at the speed of 30 km/hr and also took an hour longer to reach home. What distance did he cover each way ?

SSC CHSL 17/03/2023 (1st Shift)

(a) 75 km (b) 55 km (c) 60 km (d) 80 km

Solution :-

$$\begin{aligned} \text{Distance}(D) &= \left(\frac{xy}{x-y} \right) t \text{ km} \\ &= \left(\frac{60 \times 30}{60 - 30} \right) \times 1 \text{ hr} = 60 \text{ km} \end{aligned}$$

Exam Hall approach :-

Speed = 60 : 30

Time = 1 : 2 (when distance is same then time is inversely proportional to the speed) 1 unit = 1 hour

Distance = speed $\times$ time $\Rightarrow 60 \times 1 = 60 \text{ km}$

Example :- A person driving at the speed of 45 km/h reaches office 1 minute early while she reaches 3 minutes late if driving at 40 km/hr. What is the distance (in km) that she is covering?

RRB ALP Tier - I (21/08/2018) Afternoon

(a) 24 (b) 30 (c) 32 (d) 28

Solution :-

Required distance

$$= \frac{45 \times 40}{45 - 40} \times \frac{(3 + 1)}{60} = \frac{1800}{5} \times \frac{4}{60} = 24 \text{ km}$$

Exam Hall approach :-

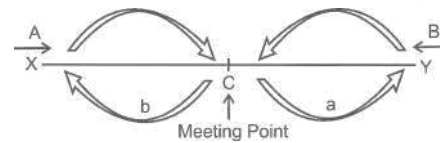
$$\text{Time difference} = 1 + 3 = 4 \text{ min} = \frac{4}{60} \text{ hr} = \frac{1}{15}$$

$$\frac{D}{40} - \frac{D}{45} = \frac{1}{15} \Rightarrow D \times \frac{45 - 40}{40 \times 45} = \frac{1}{15} \Rightarrow D = \frac{40 \times 3}{5} = 24 \text{ km}$$

Type - 4

A and B started their journeys from X to Y and Y to X , respectively . After crossing each other at point C , A and B completed the remaining part of the journey in 'a' hours and 'b' hours , respectively.

Time when they meet at point C (T) = $\sqrt{a \times b}$



Example :- A and B started their journeys at 9 a.m from X to Y and Y to X , respectively. After crossing each other at point C, A and B completed the remaining part of the journey in '16' min and '36' min , respectively. Find out the time when they meet at C ?

Solution :- $T = \sqrt{16 \times 36} = 24 \text{ Min}$

Meeting time at C = 9:24 Min

Example :- A and B started moving at the same time towards each other from points X and Y, respectively. After meeting on the way, A and B took 5.4 hours and p hours, respectively to reach Y and X, respectively. If the speed of B is 50% more than that of A, then what is the value of p?

RRC Group D 02/09/2022 (Afternoon)

(a) 2.7 (b) 1.8 (c) 2.4 (d) 3.6

Solution :-

A : B

Speed $\rightarrow$ 2 : 3

$$\text{After meeting, } \frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

$$\Rightarrow \sqrt{\frac{p}{5.4}} = \frac{2}{3} \Rightarrow p = \frac{4}{9} \times 5.4 \Rightarrow p = 2.4 \text{ hr.}$$

Type - 5

When actual speed is given :-

A person covers a certain distance, if he increases his speed by S_1 km/hr then he takes t_1 minute less time but if he reduces his speed by S_2 km/hr then how much delay will he reach if his actual speed is S Km/hr.

Example :- A person covers a certain distance at a certain speed. If he increases his speed by 5 km/h, he can walk 12 min early . If he reaches first and reduces his speed by 5

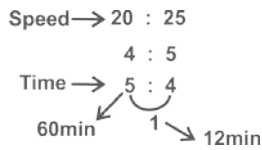
km/h, then how late will he reach? If his actual speed is 20 Km/h.

RRB NTPC CBT 2 Level -6 (9 May 2022)

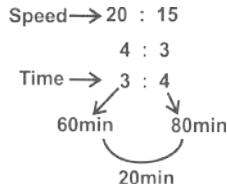
(a) 36 minutes (b) 50 minutes (c) 30 minutes (d) 20 minutes

Solution :- According to the question

The distance is constant so, $s \propto \frac{1}{t}$



Now,



That person will arrive 20 minutes late.

Type - 6

A person covers a certain distance if he increases his speed by S_1 km/hr then he takes t_1 min less time but if he reduces his speed by S_2 km/hr then he takes t_2 min more time then Find out the actual speed (S) ?

$$D = \frac{S \times (S + S_1) \times \frac{t_1}{60}}{S_1} = \frac{S \times (S - S_2) \times \frac{t_2}{60}}{S_2}$$

Actual time

Example :- A person covered a certain distance at some speed. If he moves 6 km/hr faster, he takes 30 minutes less. If he slows down by 4 km/hr, he takes 30 minutes more. What is his original speed (in km/hr)?

Delhi Police Constable 02/12/2020 (Shift 2)

(a) 21 (b) 24 (c) 28 (d) 36

Solution :-

$$D = \frac{S \times (S + S_1)}{S_1} \times \frac{t_1}{60} = \frac{S \times (S - S_2)}{S_2} \times \frac{t_2}{60}$$

$$\Rightarrow \frac{S \times (S + 6)}{6} \times \frac{30}{60} = \frac{S \times (S - 4)}{4} \times \frac{30}{60}$$

$$\Rightarrow \frac{S + 6}{6} = \frac{S - 4}{4} \Rightarrow 4S + 24 = 6S - 24 \Rightarrow 2S = 48 \Rightarrow S = 24$$

Type - 7

Distance travelled by different medium :-

Example :- A man travelled a distance of 47 km in 6 hours. He travelled partly on foot at the rate of $6\frac{1}{2}$ km/h and partly on bicycle at the rate of $8\frac{1}{2}$ km/h. The distance travelled on foot is:

SSC MTS 11/10/2021 (Afternoon)

(a) 13 km (b) 15 km (c) 12 km (d) 16 km

Solution :-

Let the distance travelled on foot = x,

so distance travelled on bicycle = $47 - x$

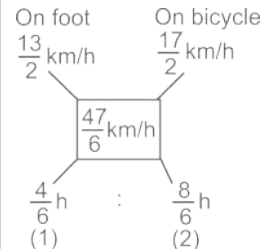
$$6 = \frac{x}{\frac{13}{2}} + \frac{47 - x}{\frac{17}{2}} \Rightarrow 6 = \frac{2x}{13} + \frac{2(47 - x)}{17}$$

$$\Rightarrow 6 = \frac{2x}{13} + \frac{94 - 2x}{17} \Rightarrow 6 = \frac{34x + 1222 - 26x}{221}$$

$$\Rightarrow 1326 = 8x + 1222 \Rightarrow 8x = 104 \Rightarrow x = 13 \text{ km}$$

Exam Hall approach:-

Using alligation method



Here 3 unit = 6 hours

So time taken on foot = 1 unit = 2 hours

Hence distance covered on foot

$$= 2 \times \frac{13}{2} = 13 \text{ km}$$

Type - 8

Speed with stoppage:-

Without stoppage, the average speed of a train is 'u' and with stoppage average speed of train is 'v'.

Then stoppage time per hour

$$= \frac{\text{Without stoppage speed} - \text{with stoppage speed}}{\text{Without stoppage speed}} = \frac{u - v}{u}$$

Example :- The speed of a train including stoppages is 75 kmph and excluding stoppages, the speed of the train is 90 kmph. For how many minutes does the train stop per hour?

SSC CHSL 13/03/2023 (3rd Shift)

(a) 10 minutes (b) 15 minutes (c) 20 minutes (d) 11 minutes

Solution :-

Formula :- Stoppage time per hour

$$= \frac{\text{Without stoppage} - \text{with stoppage}}{\text{Without stoppage}}$$

$$\Rightarrow \frac{90 - 75}{90} \text{ hr} = \frac{1}{6} \times 60 = 10 \text{ min.}$$

Example :- A car starts running at an initial speed of 60 km/h, with its speed increasing every hour by 10 km/h. How many hours will it take to cover a distance of 1210 km?

Selection Post 30/06/2023 (2nd Shift)

(a) 12 (b) 8 (c) 10 (d) 11

Solution :-

Distance traveled in 1st hour = 60 km.

According to question,

Series of Distance will be covered = 60, 70, 80, ...

$$\Rightarrow 60 + 70 + 80 + \dots = 1210$$

$$\text{Sum of A.P.}(S_n) = \frac{n}{2}[2a + (n - 1)d]$$

$$2420 = 10n[12 + n - 1]$$

$$242 = n[n + 11]$$

$$242 = 11 \times [11 + 11] \Rightarrow n = 11$$

So, it will take 11 hours to cover a distance of 1210 km

Example :- Two men do the same journey by travelling at the speed of 18 km/h and 24 km/h, respectively. If one man takes 20 minutes more than the other man, then the distance (in km) of the total journey is:

SSC CHSL 04/08/2023 (1st Shift)

(a) 24 (b) 20 (c) 18 (d) 21

Solution :-

Ratio = 1st person : 2nd person

Speed = 18 : 24

Time = 24 : 18

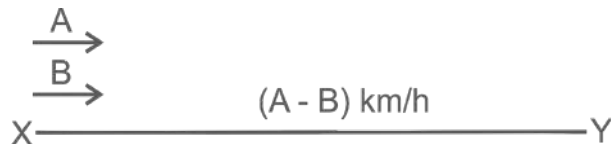
Difference in time (6 units) = 20 min.

Final time (18 units) = 1 hour

Total journey time = 24 × 1 hour = 24 km

Relative Speed

(i). Same Direction :-

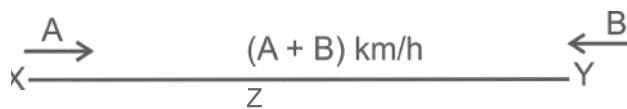


i.e. A and B are going to the same point Z at speeds of 80 km/h and 60 km/h respectively.

Relative speed of A with respect to B will be

= 80 - 60 = 20 km/h.

(ii) Opposite direction :-



i.e. A is moving towards point Z at 60 km/h and B is going away from point Z at 80 km/h.

Relative speed of A with respect to B

= 60 + 80 = 140 km/h.

Example :- A policeman sees a thief from a distance of 185 m. The thief starts running away and the policeman chases him. The thief and the policeman run at speeds of 8m/s and 9m/s respectively. What is the distance between them after 2 minutes?

SSC MTS 05/09/2023 (2nd Shift)

(a) 65 m (b) 92 m (c) 120 m (d) 130 m

Solution :- Relative speed = 9 - 8 = 1 m/s

1 second = 1 meter

2 minute (120 second) = 120 meter

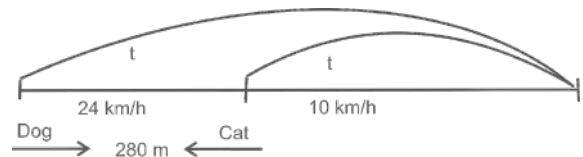
Required distance = 185 - 120 = 65 meter

Example :- A dog saw a cat at a distance of 280 m. The cat at once ran with the speed of 10 km/h and the dog also ran to catch it with the speed of 24 km/h. How much time will the dog take to catch the cat ?

RRB NTPC CBT - I 04/02/2021 (Evening)

(a) 1.3 min (b) 1.2 min (c) 1.4 min (d) 1.5 min

Solution :-



Time is equal in both the cases

$$\frac{D_D}{D_C} = \frac{S_D}{S_C} = \frac{24}{10} = \frac{12}{5}$$

Difference of distance = 12 - 5 = 7 unit

7 unit = 280 m

Distance of cat = 5 unit

$$= \frac{280}{7} \times 5 = 200 \text{ m} = 0.2 \text{ km}$$

$$\text{Time} = \frac{0.2}{10} = \frac{1}{50} \text{ hour} = \frac{60}{50} = 1.2 \text{ minutes}$$

Exam hall approach:-

Relative speed b/w dog and cat = 24 - 10 = 14 km/h

$$14 \times \frac{5}{18} = \frac{70}{18} \text{ m/sec}$$

$$\text{Time} = \frac{280}{\frac{70}{18}} = 4 \times 18 = 72 \text{ sec} = 1.2 \text{ minutes}$$

Example :- Two cyclists start at the same time from opposite ends of a 200 km long straight road. One cyclist travels at a speed of 20 km/h and the other at a speed of 30 km/h. How long will it take them to meet?

SSC MTS 11/09/2023 (1st Shift)

(a) 5 hours (b) 4 hours (c) 5.5 hours (d) 4.5 hours

Solution :-

Relative speed = 20 + 30 = 50 km/h

$$\text{So, required time} = \frac{200}{50} = 4 \text{ hour}$$

(1) When the distance is constant :-

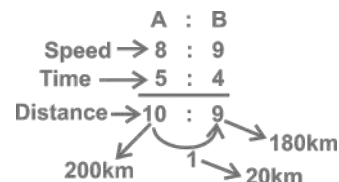
(2) When the speed is constant :-

(3) When time is stable :-

(4) When no value is constant :-

Example :- The speed of B is 12.5% more than the speed of A. They walk for 5 hours and 4 hours respectively. If A will walk 20km more than B then find the distance covered by B.

Solution :-



Example :- one person total trip $\frac{2}{15}$ part by rail, $\frac{9}{20}$ part by bus and the remaining 10 km. Covers distance on foot. then find the total distance

Solution :-

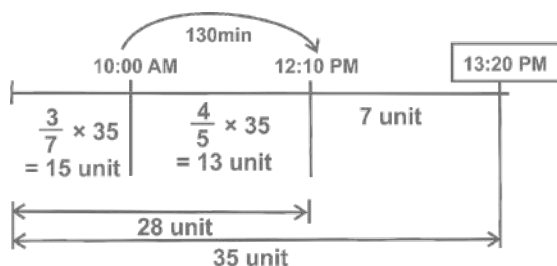
Train	Bus	Walk
$\frac{2}{15} \times 60$	$\frac{9}{20} \times 60$	$= 10 \text{ km}$
$= 8 \text{ unit}$	$= 27 \text{ unit}$	

Here,
 LCM (total distance) of 15 and 20 = 60 units
 Distance covered by foot = Total distance - (Distance covered by train bus) = $60 - 35 = 25$ units
 Now,
 $25 \text{ units} = 10 \text{ km}$
 Hence, $60 \text{ units} = 24 \text{ km}$

Example :- A person starts his journey in the morning. He started the journey at 10:00 am He covers

$\frac{3}{7}$ parts and on the same day at 12:10 he covers 80% of the journey then find the starting and ending time of the journey.

Solution :-



$13 \text{ units} = 130 \text{ minutes}$
 Therefore, $7 \text{ units} = 70 \text{ minutes} = 1 \text{ hour } 10 \text{ m.}$
 And $15 \text{ units} = 150 \text{ minutes} = 2 \text{ hours } 30 \text{ m.}$
 Hence, starting time of journey = 07:30 AM
 Last time of journey = 13:20 PM

When two objects start moving from a fixed place at different times t_1 and t_2 and with different speeds s_1 and s_2 respectively, then find-

1. At what time will the two objects meet each other?
2. At what distance will the two objects meet each other

For solution:-

Step 1:- We will first find the distance covered by the object in the given interval of time.

Step 2:- By the time the second object starts moving at its scheduled time, the first object will have covered some distance. Hence the second object will cover this distance only.

Example :- Ram starts from point A at 8 a.m. and reaches point B at 2 p.m. on the same day. On the same day, Raju starts from point B at 8 a.m. and reaches point A at 6 p.m. on the same day. Both points A and B are separated by only a straight line track. At what time they both meet?

SSC CGL 12/12/2022 (3rd Shift)

(a) 11:45 a.m. (b) 9:42 a.m (c) 10:42 a.m. (d) 12:42 p.m.

Solution :-

Time taken by Ram = 6 hrs

Time taken by Raju = 10 hrs

The ratio of the speed of Ram and Raju

$$= 10 : 6 = 5 : 3 \text{ (speed } \propto \frac{1}{\text{time}})$$

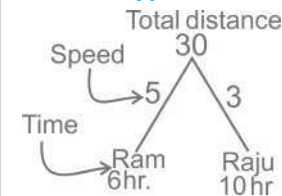
Relative speed of person when traveling in opposite direction
 $= 5 + 3 = 8 \text{ km/hr}$

Distance covered by both person = $5 \times 6 = 30 \text{ km}$

$$\text{Time of meeting} = \frac{30}{8} = 3 \text{ hr } 45 \text{ min}$$

So, both person will meet each other at 11:45 am i.e. (8 + 3:45)

Exam hall approach :-



Relative speed = $5 + 3 = 8$

$$\text{time to meet} = \frac{30}{8} = \frac{15}{4} = 3\frac{3}{4} \text{ hr (3.45hr)}$$

So, both person will meet each other at 11:45 am i.e. (8 + 3:45)

Example :- Anil and Tirath drove between two points A and B 192 km apart. Anil started the journey from point A at 8:20 a.m.; drove at a speed of 64 km/h; reached point B and immediately returned to A at the same speed. Tirath started the journey from point A at 9:50 a.m.; drove at a speed of 96 km/h; reached point B and immediately returned to A at the same speed. At what time did Anil and Tirath first meet each other?

SSC CGL 21/07/2023 (4th shift)

(a) 11:38 a.m. (b) 11:28 a.m. (c) 11:43 a.m. (d) 11:33 a.m.

Solution :-

$$\text{Time taken by Anil to reached point B} = \frac{192}{64} = 3 \text{ hours}$$

So, Anil reached B at $8:20 + 3 = 11:20 \text{ a.m.}$

Now, distance covered by Tirath till

$$11:20 \text{ a.m.} = 1.5 \times 96 = 144 \text{ km}$$

so, the distance between Anil and Tirath at 11:20 a.m.

$$= 192 - 144 = 48 \text{ km}$$

Relative speed of Anil and Tirath

$$= 64 + 96 = 160 \text{ km/hr}$$

Required time of meeting

$$= \frac{48}{160} \times 60 = 18 \text{ minute}$$

Time at which they meet = $11:20 + 18 \text{ min} = 11:38 \text{ a.m.}$

Example :- The distance between two places A and B are 140 km. Two cars x and y start simultaneously from A and B, respectively. If they move in the same direction, they meet after 7 hours. If they move towards each other, they meet after one hour. What is the speed (in km/h) of car y if its speed is more than that of car x?

SSC CGL Tier II (03/02/2022)

(a) 60 (b) 100 (c) 80 (d) 90

Solution :-

Let the speed of car x = p km/h

and speed of car y = q km/h

According to the question,

For same direction case,

$$q - p = \frac{140}{7} = 20 \text{(1)}$$

For opposite direction case,

$$q + p = \frac{140}{1} = 140 \dots\dots(2)$$

On solving equation (1) and (2), we get

$$q = 80 \text{ km/h}$$

So, speed of car y = 80 km/h

$$\text{Distance covered by B per jump} = \frac{35}{5} = 7 \text{ meters}$$

Now, A jump 8 times : B jump 6 times

$$\text{Speed of A} = 8 \times 5 = 40$$

$$\text{Speed of B} = 7 \times 6 = 42$$

Required Ratio :- 20 : 21

Example :- Ajit Singh left from place P at 9 : 30 am for place Q, and David Raj left place Q at 1 : 30 for place P. The distance between them is 416 km. If Ajit Singh's speed is 44 km/h and David Raj's speed is 52 km/h, then at what time will they meet each other?

SSC CHSL 11/08/2021 (Afternoon)

(a) 4.30 pm (b) 5 pm (c) 6 pm (d) 4 pm

Solution :-



Distance travelled by Ajit in first 4 hours (from 9:30 to 1:30)

$$= 44 \times 4 = 176 \text{ km}$$

$$\text{Remaining distance} = 416 - 176 = 240 \text{ km}$$

$$\text{Relative speed} = 44 + 52 = 96 \text{ km/h}$$

$$\text{Time} = \frac{240}{96} = 2.5 \text{ hours}$$

$$1 : 30 + 2.5 \text{ hours} = 4 \text{ pm}$$

Example :- An airplane took off from the starting point 45 minutes later than the scheduled time. The destination was 2100 km away from the starting point. To reach on time, the pilot had to increase the speed by 40% of its usual speed. What was the increased speed (in km/h)?

SSC MTS 22/10/2021 (Morning)

(a) 1870 (b) 2520 (c) 2940 (d) 1120

Solution :-

$$0\% = \frac{2}{5}, \text{ let initial speed}$$

= 5 units, then increased speed = 7 units

Initial : New

$$\text{Ratio of speed} = 5 : 7$$

$$\text{Ratio of time} = 7 : 5$$

Clearly, 2 units of time will cover the delay of 45 minutes.

$$\text{So, 5 units of time} = \frac{45}{2} \times 5 \text{ minutes}$$

New or increased speed

$$= \frac{\text{total distance}}{\text{new time}} \Rightarrow \frac{2100 \times 60 \times 2}{45 \times 5} = 1120$$

Miscellaneous

Example :- A bus starts from a bus station after every 30 minutes in the same direction. The speed of each bus is the same. Z is riding a bike in the opposite direction at a speed of 25 km/h. Z meets each bus in 20 minutes. What is the speed of each bus?

SSC MTS 19/07/2022 (Evening)

(a) 50 km/h (b) 75 km/h (c) 30 km/h (d) 60 km/h

Solution :- let the speed of the bus be x km

According to question,

Distance travelled by bus in (30 - 20) min. = 10 min. = distance travelled by Z in 20 min.

$$\text{Distance covered by bus in } \frac{10}{60} \text{ hour}$$

$$= \text{distance travelled by Z in } \frac{20}{60} \text{ hour} \Rightarrow \left(\frac{10}{60}\right) x = 25 \times \frac{20}{60}$$

$$\Rightarrow x = \frac{500}{10} = 50 \text{ km/h.}$$

Example :- A monkey climbs a 12 meters high slippery pillar. In his first minute, he climbs 2 meters and in the next minute, he slips one meter down. In this way, how much time will he take to reach the top of the pillar?

UPPSC CSAT 2020

(a) 10 min (b) 21 min (c) 12 min (d) 11 min

Solution :-

On first minute monkey climb = 2 m

On the second minute it slips = 1 m

For every two minute, it climbs 1 m

So, average speed = 1 m/2 min

For 10 meters = 20 min.

For the last 2 m jump add 1 min

So time taken 20 + 1 = 21 min

Monkey takes 21 minutes to reach the top of the pole.

Example :- A is chasing B in the same interval of time. A jumps 8 times, while B jumps 6 times. But the distance covered by A in 7 jumps is the same as that of B in 5 jumps. The ratio between the speeds of A and B is _____.

SSC CGL 08/12/2022 (1st Shift)

(a) 32 : 24 (b) 30 : 56 (c) 20 : 21 (d) 28 : 36

Solution :- Distance covered by A in 7 jumps is equal to distance covered by B in 5 jumps.

Let the distance be 35 meters.

According to Questions,

$$\text{Distance covered by A per jump} = \frac{35}{7} = 5 \text{ meters}$$

Boat and Stream

Important Terms

1). Still Water Speed : Speed of a thing in water which is not flowing.

2). Speed of stream : The speed at which water is flowing.

3). Downstream speed : Speed of a thing, moving in the same direction as that of the stream .

$$\text{Downstream speed} = \text{speed of boat in still water} + \text{Speed of stream}$$

4). Upstream Speed : Speed of a boat, moving in opposite direction as that of the current.

$$\text{Upstream speed} = \text{speed of boat in still water} - \text{Speed of stream}$$

Example:- The speed of a boat in still water is 15 km/h, and the speed of the current is 5 km/h, in how much time (in hours) will the boat travel a distance of 60 km upstream and the same distance downstream?

SSC CPO 25/11/2020 (Evening)

(a) 12 (b) 9 (c) 10 (d) 20

Solution:- speed of a boat in still water = 15 km/h

speed of the current = 5 km/h

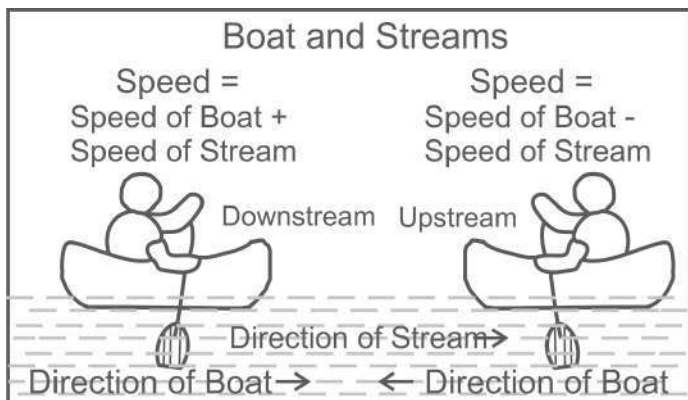
Downstream speed = 15 + 5 = 20 km/h

Upstream speed = 15 - 5 = 10 km/h

Time for downstream = $\frac{60}{20} = 3\text{hr}$

Time for upstream = $\frac{60}{10} = 6\text{hr}$

Total time = 3 + 6 = 9 hr



Type - 1

(a) Still water speed = $\frac{\text{Downstream speed} + \text{Upstream Speed}}{2}$

Example:- The speed of a boat when travelling downstream is 48 km/h, whereas when travelling upstream it is 32 km/h. What is the speed of the boat in still water?

Selection Post 28/06/2023 (2nd Shift)

(a) 80 km/h (b) 30 km/h (c) 40 km/h (d) 20 km/h

Solution:- Speed of boat in still water

$$= \frac{\text{Downstream speed} + \text{Upstream Speed}}{2} = \frac{48 + 32}{2} = 40 \text{ km/hr.}$$

b) Speed of the current = $\frac{\text{Downstream speed} - \text{Upstream Speed}}{2}$

Example:- A person can row downstream 49 km in 7 hours and upstream 36 km in 9 hours. What is the speed of the current?

Selection Post 02/08/2022 (2nd Shift)

(a) 1.5 km/h (b) 2 km/h (c) 2.5 km/h (d) 3 km/h

Solution:- Downstream speed = $\frac{49}{7} = 7 \text{ km/hr}$

Upstream speed = $\frac{36}{9} = 4 \text{ km/hr}$

$$\text{Speed of the current} = \frac{\text{Downstream speed} - \text{Upstream Speed}}{2} = \frac{7 - 4}{2} = 1.5 \text{ km/hr}$$

Type - 2

a) If a boat takes equal time to travel 'x' km downstream and 'y' km upstream then ,

$$\frac{\text{Speed of boat}}{\text{Speed of Stream}} = \frac{(x + y)}{(x - y)}$$

Example:- A man takes half time in rowing a certain distance downstream than upstream. What is the ratio of the speed in still water to the speed of current?

UPSC CSAT (04/10/2020)

(a) 1 : 2 (b) 2 : 1 (c) 1 : 3 (d) 3 : 1

Solution:- Let the time taken to row a certain distance upstream be 2x. So, time taken to row a certain distance downstream = x.

$$\frac{\text{speed in still water}}{\text{speed of current}} = \frac{(x + y)}{(x - y)} = \frac{2x + x}{2x - x} = \frac{3x}{x} = 3 : 1$$

b) If the time taken by the boat to row same distance in downstream is T_1 and in upstream is T_2 . then the ratio of speed of the boat to the speed of stream :-

$$\frac{\text{Speed of boat}}{\text{Speed of Stream}} = \frac{(T_2 + T_1)}{(T_2 - T_1)}$$

Example:- A boat covers a certain distance against the stream in 9 hours 36 min and it covers the same distance along the stream in 6 hours. What is the ratio of the speed of the boat in still water to that of the stream ?

SSC CGL 12/12/2022 (2nd Shift)

(a) 13 : 3 (b) 9 : 2 (c) 11 : 6 (d) 8 : 5

Solution :- Let the speed of the boat in still water and the speed of stream be x and y respectively.

Downstream speed (D) = (x + y) km/hr

Upstream speed (U) = (x - y) km/hr

As we know, Time $\propto \frac{1}{\text{speed}}$ (when distance is constant)

D : U

$$\text{Time} \rightarrow 6 : \frac{48}{5} = 5 : 8$$

$$\text{Speed} \rightarrow 8 : 5$$

$$\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{x+y}{x-y} = \frac{8}{5}$$

$$\frac{x}{y} = \frac{8+5}{8-5} = \frac{13}{3} \text{ (using componendo and dividendo)}$$

Required ratio = 13 : 3

Exam hall approach: $T_1 = 9\text{ hr } 36\text{ min} = \frac{48}{5}$, $T_2 = 6\text{ hr}$

$$\frac{x}{y} = \frac{\frac{48}{5} + 6}{\frac{48}{5} - 6} = \frac{78}{18} = \frac{13}{3}$$

Type - 3

Note :- (In the following formulas B = speed of boat and S = speed of stream)

a) If it takes T hours more to go to a point upstream than downstream for the same distance. Then

$$\text{Distance between the two places} = \frac{T(B^2 - S^2)}{2S}$$

Example:- The speed of a boat in still water is 20 km/h, while the river is flowing at a speed of 8 km/h and the time taken to cover a certain distance upstream is 6 h more than the time taken to cover the same distance downstream. Find the distance.

SSC CPO 03/10/2023 (3rd Shift)

(a) 126 km (b) 125 km (c) 120 km (d) 336 km

Solution:- Let the covered distance = x km

Speed of the boat = 20 km/hr

Speed of the water = 8 km/hr

According to question,

$$\frac{x}{12} - \frac{x}{28} = 6 \Rightarrow \frac{16x}{12 \times 28} = 6 \Rightarrow x = 126 \text{ km}$$

Exam hall approach :-

$$\text{Distance between the two places} = \frac{T(B^2 - S^2)}{2S}$$

$$= \frac{6(20^2 - 8^2)}{2 \times 8} = \frac{6(400 - 64)}{2 \times 8} = \frac{6 \times 336}{2 \times 8} = 126 \text{ km}$$

b) If the total time taken by the boat to row a distance of D and reach back to its initial points is T then

$$\text{Distance between the two places} = \frac{T(B^2 - S^2)}{2B}$$

Example:- A person is sailing in a river at 12 km/hr against the flow of a river which is flowing at 6 km/hr. If it takes one hour to reach back on its starting point then find one way distance?

AFCAT 12/02/2022 (1st Shift)

(a) 8 km (b) 9 km (c) 10 km (d) 11 km

Solution:- Speed of upstream ($x - y$) = 12 km/h

Speed of current (y) = 6 km/h

Then, speed of boat in still water (x) = 12 + 6 = 18 km/h

$$\text{Distance between the two places} = \frac{T(B^2 - S^2)}{2B}$$

$$= \frac{1(18^2 - 6^2)}{2 \times 18} = \frac{288}{2 \times 18} = 8 \text{ km}$$

Type - 4

Average Speed

Average speed when a boat cover a certain distance and return

$$\text{to the starting point} = \frac{(B^2 - S^2)}{B} = \frac{(B + S)(B - S)}{B} \text{ km/hr}$$

Or

$$\text{average speed} = \frac{2 \times \text{downstream speed} \times \text{upstream speed}}{\text{downstream} + \text{upstream}} \text{ km/hr}$$

Example:- A boat in a still flowing river travels a certain distance downstream at a speed of 12 km/h and then covers the same distance in the opposite direction at a speed of 24 km/h. Find the average speed for the total journey.

RRB ALP Tier - II (23/01/2019) Morning

(a) 20 km/h (b) 15 km/h (c) 18 km/h (d) 16 km/h

$$\text{Solution:- average speed} = \frac{2 \times S_1 \times S_2}{S_1 + S_2}$$

$$\frac{2 \times 12 \times 24}{12 + 24} = \frac{2 \times 12 \times 24}{36} = 16$$

Therefore, the average speed for the journey = 16 km/h

Type - 5

Time Related Question

Example:- The speed of a boat in still water is 30 km/h and the speed of the stream is 7.5 km/h. What is the time in hours taken by the boat in the stream to go from point A to B and then back to A covering a total distance of 90 km?

SSC MTS 05/09/2023 (1st Shift)

(a) 3.2 (b) 2.8 (c) 3.4 (d) 3.0

Solution :- Let total time taken is T hours

$$T = \frac{45}{30 + 7.5} + \frac{45}{30 - 7.5}$$

$$\Rightarrow \frac{45}{37.5} + \frac{45}{22.5} = 1.2 + 2 = 3.2 \text{ hours}$$

Example:- The ratio of the speed of the boat in still water to the speed of the stream is 8 : 3 and the boat can travel 22 km downstream in 24 minutes. Then, how much time will the boat take to cover 50 km upstream?

SSC CHSL 13/03/2023 (2nd Shift)

(a) 3 hrs (b) 4 hrs (c) 2.5 hrs (d) 2 hrs

Solution :- Let the speed of boat in still water and stream = 8x and 3x respectively

Speed of the boat to travel 22 km downstream in 24 min

$$\Rightarrow (8x + 3x) = \frac{22}{\frac{24}{60}} = 11x = 55 \text{ km/h} \Rightarrow x = 5 \text{ km/h}$$

Then, speed of boat in still water and stream = 40km/h and 15km/h respectively

$$\text{therefore, boat take to cover 50 km upstream} = \frac{50}{40 - 15} = 2 \text{ hours}$$

Example:- A boat can travel 20 km downstream in 50 min. If the speed of water is 20% of the speed of the boat downstream, how much time (in min) a boat will take to travel 120 km upstream?

SBI PO Mains (02/01/2022)

(a) 450 (b) 675 (c) 240 (d) 500 (e) None of these

Solution :- speed of boat in downstream

$$(x + y) = \frac{20}{50} \times 60 = 24 \text{ km/h}$$

$$\text{Speed of current (y)} = 24 \times \frac{20}{100} = 4.8 \text{ km/h}$$

$$\text{Speed of boat in still water (x)} = 24 - 4.8 = 19.2 \text{ km/h}$$

$$\text{Then, speed of upstream (x - y)} = 19.2 - 4.8 = 14.4 \text{ km/h}$$

Now, time taken by boat to travel 120 km

$$\text{upstream} = \frac{120}{14.4} \times 60 = 500 \text{ minute}$$

Type - 6

Return Based Question

Example:- A ship is 208 km away from the coast. There is a hole in it through which 6 liters of water is filled in 5 minutes. The ship also has a water tap which out 60 liters of water per hour. Then find the speed of the ship. If a rescue ship coming towards the ship at a speed of 7 km/h can reach it before it sinks? 156 liters of water is enough to sink ship.

Solution :- ship speed = x km/h

$$5 \text{ min} = 6 \text{ liter} \Rightarrow 60 \text{ min} = 72 \text{ liter}$$

$$\text{out water from ship in 1 hour} = 60 \text{ liter}$$

$$\text{Then, filled water in 1 hour} = 12 \text{ liter}$$

$$12 \text{ liter} = 1 \text{ hr} \Rightarrow 156 \text{ liter} = 13 \text{ hr}$$

$$\text{Distance b/w both ship} = 208 \text{ km,}$$

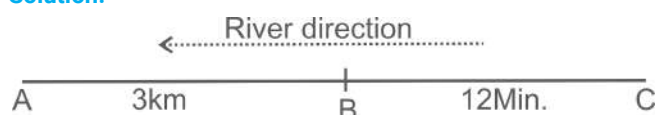
$$\text{Speed of rescue ship} = 7 \text{ km/h}$$

$$\text{Relative speed of both ship} = (x + 7) \text{ km/h}$$

$$(x + 7) = \frac{208}{13} \Rightarrow (x + 7) = 16 \Rightarrow x = 9 \text{ km/h}$$

Example:- suddenly her cap flew off and started flowing with the current. The boat continued to move against the current for 12 minutes. When Sakshi realized that her cap had blown off, she turned the boat and started sailing toward the cap. He caught the cap at the same place where he started walking. If his cap flew at a distance of 3 km from the starting point, then find the speed of the stream.

Solution:-



Let the speed of the boat be x km/hour and Speed of the current be y km/hour

Let the time for the boat to return from C to A be 't' minute

The hat started floating back from B to A. 12 minutes before the boat turned back,

so the hat's return took (12 minutes + t).

Distance BC = Time taken by boat upstream x Speed of boat upstream BC = 12(x - y)

Similarly,

$$\text{Distance AC} = t(x + y)$$

$$\text{Distance AB} = (t + 12)y$$

$$\text{Distance AB} + \text{BC} = \text{distance AC}$$

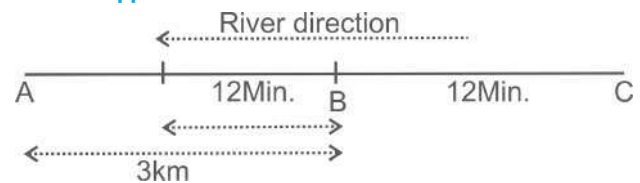
$$(t + 12)y + 12(x - y) = t(x + y) \text{ Time, } t = 12$$

As the hat's return took (12 minutes + t). Total time 24 min

$$= \frac{24}{60} = \frac{2}{5} \text{ hours}$$

$$\text{Speed} = \frac{3}{(\frac{2}{5})} = 7.5 \text{ km/hour}$$

Exam hall approach:



Distance from starting point to the point where cap fell = 3 km.

Time will be twice = $2 \times 12 = 24$ minutes or $\frac{2}{5}$ hour.

$$\text{Speed} = \frac{3}{(\frac{2}{5})} = 7.5 \text{ km/hour}$$

Train

$$\text{Distance} = \text{Speed} \times \text{Time}$$

We just need to remember this formula. If two values are given the third can be found easily.

Unit Conversion

km/h to m/s	$\text{Km/h} \xrightarrow{\times \frac{5}{18}} \text{m/sec}$ i.e. $90 \text{ km/h} = 90 \times \frac{5}{18} \text{ m/s} = 25 \text{ m/s}$
m/s to km/h	$\text{m/sec} \xrightarrow{\times \frac{18}{5}} \text{Km/h}$ i.e. $25 \text{ m/s} = 25 \times \frac{18}{5} \text{ km/h} = 90 \text{ km/h}$

Note:- 1 Hr = 60 min = 3600 sec
 1 yard = 3 ft
 1 mile = 1.609 km

Type - 1

Train Crossing One Man/Pole/Platform/Tunnel/Train

(a) Train crossing a person or pole :-

In such cases the length of the person or pole is taken to be negligible. To cross the person the train has to cover a distance equal to its length.

Example : What is the time taken by a 450 m long train running at 54 km/h to cross a man standing on a platform ?

NTPC CBT II (09/05/2022) 1st Shift

- (a) 25 seconds (b) 32 seconds
 (c) 28 seconds (d) 30 seconds

Solution :- Here, Speed = 54 km/h = $54 \times \frac{5}{18} \text{ m/s} = 15 \text{ m/s}$.

$$\text{Required time} = \frac{\text{Distance}}{\text{Speed}} = \frac{450}{15} = 30 \text{ seconds}$$

(b) Train crossing a platform or tunnel :-

In such a case the distance covered by train to cross the platform will be equal to sum of length of train and platform.

Example : Length of a train is 330 metres and it is moving at the speed of 72 km/hr. In how much time will it takes cross a platform of length 710 metres?

SSC CGL Tier II (08/08/2022)

- (a) 52 seconds (b) 56 seconds
 (c) 72 seconds (d) 64 seconds

Solution:- Here, Speed = 72 km/hr = $72 \times \frac{5}{18} = 20 \text{ m/s}$

Distance = Length of train + Length of platform

$$= 330 + 710 = 1040 \text{ m}$$

$$\text{Required time} = \frac{1040}{20} = 52 \text{ seconds}$$

Type - 2

Relative Speed in same direction

(a) Train crossing another train moving in the same direction :-
 In such cases the distance covered by one train to cross the other train will be equal to the sum of lengths of the two trains.

$$\text{Relative Speed} = \frac{\text{Relative Distance}}{\text{Relative Time}}$$

Example : A train traveling at 80 km/h crosses another train traveling in the same direction at 26 km/h in 30 seconds. What is the combined length of both the trains ?

SSC CHSL 08/08/2023 (4th Shift)

- (a) 400 m (b) 450 m (c) 550 m (d) 350 m

Solution :- Relative speed = $80 - 26 = 54 \text{ km/h}$

According to question,

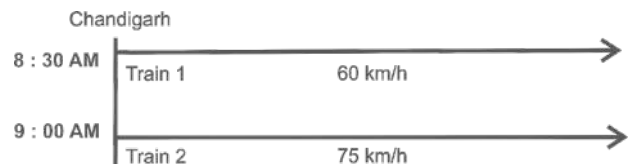
$$54 \times \frac{5}{18} = \frac{L1 + L2}{30}$$

$$L1 + L2 = 15 \times 30 = 450 \text{ m}$$

(b) When two train start their journey at different times:-

Example : Two trains start their journey from Chandigarh at 8:30 AM and 9:00 AM. Their speed is 60 Km/h and 75 Km/h respectively. Then find out at what time both the trains will meet each other and at what distance from Chandigarh. .

Solution :-



Now, time = $\frac{1}{2}$ hours, speed = 60 Km/h

Hence, distance covered by the first train = $\frac{1}{2} \times 60 = 30 \text{ km}$

$$\text{Meeting time} = \frac{\text{distance}}{\text{speed}} = \frac{30}{75 - 60} = 2 \text{ hours}$$

Both trains will meet each other at 11:00 am (9:00 + 2hr)

Total time = 2.5 hr

$$\text{Distance} = 2.5 \times 60 = 150 \text{ km}$$

(c) When a train crosses a running person/man :-

Example :- A train 228 meters long is running at a speed of 125 km/hr. In how much time (in seconds) will this train take to cross a person running at a speed of 17 km/hr in the same direction in which it is moving?

SSC MTS 14/09/2023 (3rd Shift)

- (a) 9.2 (b) 7.6 (c) 6.2 (d) 8.5

Solution:- Relative speed = $(125 - 17) = 108 \times \frac{5}{18} = 30 \text{ m/s}$

$$\text{Time required} = \frac{228}{30} = 7.6 \text{ seconds}$$

(d) Train crossing a man sitting in another train moving in the same direction :-

In such a case, the distance covered by the train to cross the man will be equal to the length of the train.

The length of the second train will not be considered :-

Example : Two trains are moving in the same direction at the speed of 36 km/hr and 48 km/hr. The time taken by the faster train to cross a man sitting in the slower train is 33 seconds. What will be the length of the faster train?

SSC CGL 07/12/2022 (4th Shift)

- (a) 770 metres (b) 90 metres
(c) 110 metres (d) 180 metres

Solution :- Let the length of the train be x m

$$\frac{x}{48 - 36} \times \frac{18}{5} = 33 \Rightarrow \frac{3x}{10} = 33 \Rightarrow x = 110$$

So, the length of the train is 110 metres.

Relative Speed in Opposite direction

(a) Train crossing another train moving in the opposite direction:-

In such a case the distance covered by one train to cross the other train will be equal to the sum of lengths of the two trains.

Example : Two trains that are 125 metres and 99 metres in length, respectively, are running in opposite directions at the speeds of 40 km/h and 32 km/h, respectively. In how much time will they be completely clear of each other from the moment they meet?

SSC CHSL 17/08/2023 (1st Shift)

- (a) 10.2 sec (b) 11.2 sec (c) 9.2 sec (d) 12.2 sec

Solution :- Here, Relative speed of first train with respect to

$$\text{second train} = (40 + 32) \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$\begin{aligned} \text{Distance} &= \text{Length of first train} + \text{Length of second train} \\ &= 125 + 99 = 224 \text{ meters} \end{aligned}$$

$$\text{Required Time} = \frac{224}{20} = 11.2 \text{ seconds}$$

(b) When two train start their journey from two different places towards each other :-

Find Time when they meet :-

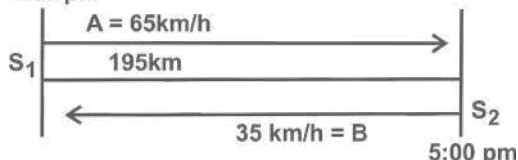
Example : S_1 and S_2 are two stations which are 195 km apart. A train starts from S_1 at 4 : 00 pm and moves towards S_2 at the speed of 65 km/h. Another train B starts from S_2 at 5 : 00 pm and moves towards S_1 at the speed of 35 km/h. At what time will the two trains meet?

SSC CGL 21/07/2023 (3rd shift)

- (a) 6 : 06 p.m (b) 6 : 30 p.m (c) 6 : 15 p.m. (d) 6 : 18 p.m

Solution :-

4:00 pm



Distance travelled by train A in 1 hour = $65 \times 1 = 65$ km
Remaining distance = $195 - 65 = 130$ km

Their relative speed = $65 + 35 = 100$ km/hr.

$$\text{Time taken to meet} = \frac{130}{100} = 1.3 \text{ hr or } 1 \text{ hr. } 18 \text{ min.}$$

$$\text{Time at which they meet} = 5:00 \text{ pm} + 1 \text{ hr. } 18 \text{ min.} = 6 : 18 \text{ pm}$$

Find distance :-

Example :- The distance between two stations A and B is 200km. A train runs from A to B at a speed of 75 km/hr, while another train runs from B to A at a speed of 85 km/hr. What will be the distance between the two train (in km) 3 minutes before they meet?

SSC CGL 21/04/2022(Morning)

- (a) 5 (b) 8 (c) 10 (d) 6

Solution :- Relative speed of two train is sum of their speed because they are running towards each other

$$\text{Relative speed} = 75 \text{ km/h} + 85 \text{ km/h} = 160 \text{ km/h}$$

$$\text{Time} = 3 \text{ minutes} = \frac{3}{60} \text{ hours}$$

$$\text{Distance} = \text{speed} \times \text{time} = 160 \times \frac{3}{60} = 8 \text{ km}$$

(c) Train crossing a man sitting in another train moving in opposite direction :-

In such a case the distance covered by the train to cross the man will be equal to the length of the train.

Length of the second train will not be considered :-

Example : Renu was sitting inside train (A), which was travelling at 50 km/h. Another train, B, whose length was three times the length of A crossed her in the opposite direction in 15 seconds. If the speed of train B was 58 km/h, then the length of train A (in m) is :

SSC CGL Tier II (12/09/2019)

- (a) 210 (b) 180 (c) 160 (d) 150

Solution :- Let the length of train A = L

$\Rightarrow$ the length of train B = $3L$

Relative speed of the trains = $(50 + 58) \text{ km/h}$

According to the question

$$\frac{3L}{(50 + 58) \times \frac{5}{18}} = 15 \Rightarrow 3L = 15 \times 30 \Rightarrow L = 150 \text{ m}$$

Miscellaneous

Example :- A locomotive engine without any wagon can go at a speed of 50 km per hour and its speed diminishes by a quantity which varies as the square root of the number of wagons attached. If with 25 wagons its speed is 35 km per hour, then what is the greatest number of wagons that can be attached, if the speed is NOT to fall below 11 km per hour?

RRB NTPC CBT - I 07/04/2021 (Morning)

- (a) 169 (b) 144 (c) 196 (d) 225

Solution :- Speed of the engine when no wagon is attached is 50 km/h

The rate of diminishes $(50 - S) \propto \sqrt{N}$

N is the number of wagon attached to the engine or $50 - S = K \sqrt{N}$

$$\text{When } N = 25, S = 35 \text{ km/h} \Rightarrow 50 - 35 = K\sqrt{25} \Rightarrow K = \frac{15}{5} = 3$$

If the speed $S = 11 \text{ km/h}$

$$\text{Then, } (50 - 11) = 3 \times \sqrt{N}$$

A/Q

$$\Rightarrow (50 - 11) = 3 \times \sqrt{N} \Rightarrow 39 = 3 \times \sqrt{N}$$

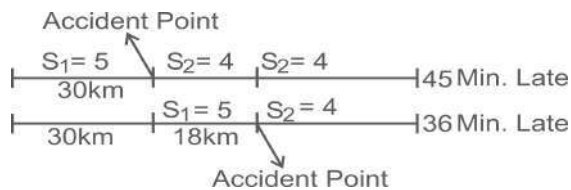
$$\Rightarrow 13 = \sqrt{N} \Rightarrow N = 13^2 = 169$$

Example :- A train service got disrupted due to an accident after a travel of 30 km. The speed was reduced to four-fifth to the original speed. This caused it a delay of 45 min. If the accident had happened 18 km farther, it would have been running late for 36 min. What was the original speed of the train?

RRB NTPC CBT - I 02/02/2021 (Evening)

(a) 30 km/h (b) 60km/h (c) 50km/h (d) 40km/h

Solution :-



Total time saved = $45 - 36 = 9$ minutes

For 18 km distance

Distance in both cases are equal = 18 km

$$\Rightarrow \frac{S_1}{S_2} = \frac{T_2}{T_1} = \frac{4}{5}$$

Difference of time = 1 unit = 9 minutes

$$T_2 = 4 \text{ unit} = 36 \text{ minutes} = \frac{36}{60} = \frac{3}{5} \text{ h}$$

$$\text{Speed} = \frac{D}{T} = \frac{18}{\frac{3}{5}} \times 5 = 30 \text{ km/h}$$

Example :- Two guns are fired from the railway station at an interval of 8 minutes. A police officer sitting in the train reaching the station noticed that there was an interval of 7 minutes between the sounds of two guns. If the velocity of sound is 350 m/sec. At what speed (km/h) was the train coming to the station?

(a) 162 (b) 90 (c) 120 (d) 180

Solution :- Time interval = 8 minutes – 7 minutes = 1 minutes (60 seconds)

As we know,

Distance covered by train in 7 minutes = Distance covered by

sound in 60 seconds = 350×60

7 minutes = (7×60) seconds

$$\text{Speed of train} = \frac{350 \times 60}{7 \times 60} \times \frac{18}{5} = 180 \text{ km/hr}$$

Linear/Circular Race

Important terms used in race :-

Linear race :- it is a straight path such that the starting and finishing point don't coincide with each other.

Circular race :- it is a circular path such that the starting and finishing point coincide with each other.

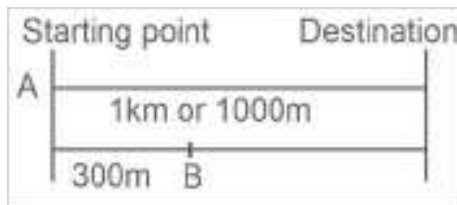
Dead Heat or Deadlock :- A dead heat situation is when all participants reach the finishing point at the same time.

Head-start :- when a racer gets a start ahead of the starting point. It is called head start.

Trail :- when a racer starts the race behind the starting point or start the race behind the other racer.

• Some cases of linear race :-

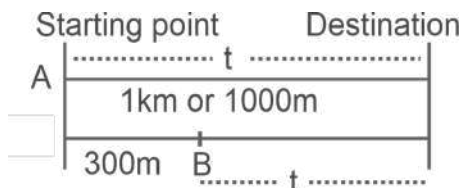
In a 1 km race, A gives a start of 300 meters to B.



A starts the race from the starting point, whereas, B starts 300 metres ahead of A.

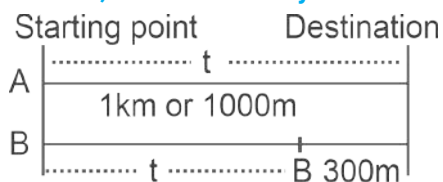
To cover a race of 1000 metres in this case, A will have to cover 1000 metres while B will have to cover only $(1000 - 300) = 700$ metres.

• When the same race ends in a dead heat. :-



A dead heat situation is when all participants reach the finishing point at the same instant of time.

• In a race of 1 km, A wins over B by 300 meter :-



At the same time, A reached the winning point, whereas, B is behind A by 300 m.

To cover a race of 1000 metres in this case, A has covered 1000 metres while B has covered only $(1000 - 300) = 700$ metres.

Example :- In a 1610 m race, Amarjeet reaches the final point in 91 seconds and Dinesh reaches the final point in 115 seconds. By how much distance does Amarjeet beat Dinesh?

Selection Post 30/06/2023 (3rd Shift)

(a) 326 m (b) 340 m (c) 336 m (d) 330 m

Solution :- Distance covered by Dinesh = 1610 m

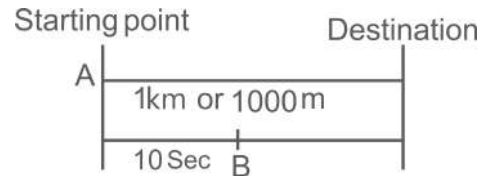
Time taken by Dinesh = 115 seconds

Time taken by Amarjeet to finish the race = 91 sec.

Distance traveled by dinesh in 91 seconds = $\frac{1610}{115} \times 91$
 $= 14 \times 91 = 1274$ m.

So, amarjeet beat dinesh by $(1610 - 1274) = 336$ m

• In a race of 1 km, A gives B a start of 10 sec :-



A will start 10 seconds after B starts from the starting point. Both A and B will reach the finishing point at the same time.

Example :- A gives B a head-start of 10 seconds in a 1500 m race and both finish the race at the same time. What is the time taken by A (in minutes) to finish the race if the speed of B is 6 m/s?

SSC CPO 11/11/2022 (Afternoon)

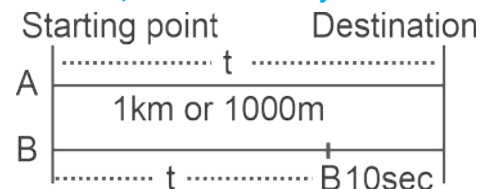
(a) 3 (b) 4 (c) 8 (d) 5

Solution :- Speed of B = 6 m/s

Time taken by B to complete the race = $\frac{1500}{6} = 250$ sec

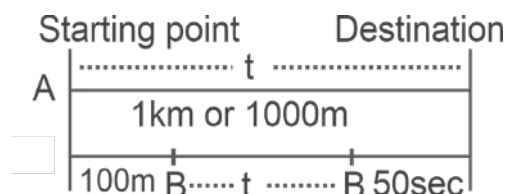
So, time taken by A to complete the race
 $= 250 - 10 = 240$ sec or 4 minutes

• In a race of 1 km, A wins over B by 10 sec :-



Here, time required by B to finish the race is $(t + 10)$ seconds.

• In a race of 1 km, A gives a start of 100 m to B and still wins over B by 50 sec. :-



Example :- In a linear race of 1000 m, A beats B by 50 m or 5 seconds. What is the difference between the speeds (in m/s) of A and B?

SSC CGL 06/12/2022 (3rd Shift)

(a) $1\frac{10}{19}$ (b) $\frac{10}{19}$ (c) $\frac{9}{19}$ (d) $\frac{9}{10}$

Solution :- Ratio of distance covered by A and B respectively
 $= 1000 : 950$

Speed of B = $\frac{50 \text{ m}}{5 \text{ seconds}} = 10$ m/s

Time taken by B to cover 950m is equal to time taken by A to cover 1000m.

Time (B) = $\frac{950}{10} = 95$ sec.

Now, speed of A = $\frac{1000 \text{ m}}{95 \text{ sec.}} = \frac{200}{19}$ m/s

Required difference = $\frac{200}{19} - 10 = \frac{10}{19}$ m/s

Example :- A and B take part in a 100 m race. A runs at 5 km/hr. A gives B a start of 8 m and still beats him by 8

seconds. Find the speed of B.

RRB JE 02/06/2019 (Afternoon)

(a) 5.15 km/hr (b) 4.14 km/hr (c) 4.20 km/hr (d) 4.25 km/hr

Solution :- According to the question,

$$\text{Time taken by A to complete the 100m race} = \frac{100}{5 \times \frac{5}{18}}$$

$$= 72 \text{ second}$$

$$\text{Distance covered by B} = 100 - 8 = 92 \text{ m.}$$

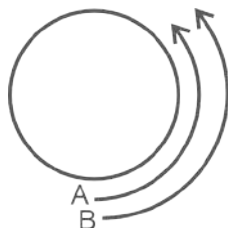
$$\text{Time taken by B} = 72 + 8 = 80 \text{ sec.}$$

$$\text{Therefore the speed of B} = \frac{92}{80} \times \frac{18}{5} = 4.14 \text{ km/h}$$

Circular race concepts

- Let A and B be two racers running in a circular path of length L with speed S_1 and S_2 respectively. ($S_1 > S_2$)

Case 1 :- when both racer running in the same direction



Same direction

(I). Time taken by both racer to meet for the 1st time at the starting point = LCM of $(\frac{L}{S_1}, \frac{L}{S_2})$

Example :- Zeenat and Nasreen participate in a circular race of 1.2 km. They start the race at a speed of 18 km/h and 27 km/h. respectively, from the same point at the same time. After how much time (in minutes) will they meet at the starting point for the first time, if they are running in the same direction ?

SSC CHSL 02/08/2023 (1st Shift)

(a) 9 (b) 7 (c) 10 (d) 8

Solution :-

Time taken by Zeenat to complete the circular race =

$$\frac{1.2}{18} \times 60 = 4 \text{ min.}$$

Time taken by Nasreen to complete the circular race =

$$\frac{1.2}{27} \times 60 = \frac{8}{3} \text{ min.}$$

$$\text{Hence, required time} = \text{LCM}(4, \frac{8}{3}) = 8 \text{ min.}$$

(II). Time taken to meet for the 1st time at anywhere on the track = $(\frac{L}{S_1 - S_2})$

Example :- Two persons A and B are running around a circular track of 1200 m long with respective speeds of 27 kmph and 36 kmph. They start from the same point at the same time and are running in the same direction. When will they meet for the first time (in seconds) on the track?

SSC CHSL 14/08/2023 (2nd Shift)

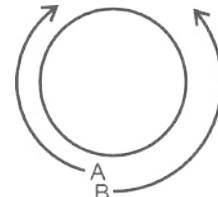
(a) 420 (b) 240 (c) 480 (d) 360

Solution :- Relative speed = $(36 - 27) \text{ km/h} = 9 \text{ km/h}$

The time they met for the first time on the track

$$= \frac{1200}{9 \times \frac{5}{18}} = 480 \text{ second}$$

Case 2 :- when both racer running in the opposite direction



Opposite direction

(I). Time taken to meet for the 1st time at the starting point = LCM of $(\frac{L}{S_1}, \frac{L}{S_2})$

Example :- In a circular race of 900m length, A and B start with speeds 27 km/hr and 36 km/hr starting at the same time from the same point. When will they meet for the first time at the starting point when running in the opposite direction?

Solution :- Time taken by A to reach start point

$$= \frac{900}{27 \times 5} \times 18 = 120 \text{ sec}$$

Time taken by B to reach start point

$$= \frac{900}{36 \times 5} \times 18 = 90 \text{ sec.}$$

$$\text{LCM of 90 and 120} = 30 \times 3 \times 4 = 360 \text{ sec.}$$

$$\text{Required time} = \frac{360}{60} = 6 \text{ min}$$

(II). Time taken to meet for the 1st time at anywhere on the track = $(\frac{L}{S_1 + S_2})$

Example :- In a circular race of 2500 m, a man and a woman start from a point towards opposite directions with speeds of 37 km/h and 35 km/h, respectively. After how much time from the start of the race will they meet for the first time?

SSC CHSL 02/06/2022(Morning)

(a) 2 min 40 sec (b) 2 min 30 sec

(c) 2 min 5 sec (d) 2 min 20 sec

Solution :- Relative speed when both are in opposite directions = $37 + 35 = 72 \text{ km/hr}$

$$\text{Total distance} = \frac{2500}{1000} = 2.5 \text{ km}$$

$$\text{First time when they will meet} = \frac{2.5}{72} = \frac{5}{144} \text{ hr} = \frac{5}{144} \times 60 = 2 \text{ min } 5 \text{ sec}$$

Example :- A and B are running on a circular track of length 1400 metres. The speed of A is 33 m/sec and the speed of B is 47 m/sec. They start from the same point at the same time in the opposite direction. After how much time will they meet for the second time?

SSC CHSL 17/08/2023 (2nd Shift)

(a) 33 sec (b) 38 sec (c) 35 sec (d) 32 sec

Solution :- Total length of circular track = 1400 meter

Relative speed of A and B = $33 + 47 = 80 \text{ m/sec}$

Time taken to meet for 1st time

$$= \frac{1400}{80} = \frac{35}{2} \text{ seconds}$$

Hence, Time taken to meet for 2nd time

$$= \frac{35}{2} \times 2 = 35 \text{ seconds}$$

Number of meeting points

- All the distant meeting points are equidistant in a circular path and the same goes for time.
- It is independent of the length of the track
- To get the no. of different meeting points, find the simplest ratio of the two speeds i.e. $x : y$

Case 1:- No. of meeting points when two runners running in same direction = $x - y$

Case 2:- No. of meeting points when two runners running in opposite direction = $x + y$

Example :- Anjali and Babita are running on a circular track in opposite directions from same time at same point with speeds of 8 m/sec and 6 m/sec, respectively. If the length of the circular track is 960 m, how many times they will meet?
SSC CHSL 09/06/2022 (Evening)

- (a) 7 (b) 6 (c) 12 (d) 14

Solution :- Number of distinct points at which they (the bodies in circular motion) will meet can be determined by finding out the reduced ratios of their speed ($a : b$)

As they are moving in opposite direction Ratio of their speeds = $4 : 3$

So, they will meet at $4 + 3 = 7$ distinct points.

Example :- Two persons started running on a circular track simultaneously with speeds of 20 m/s and 30 m/s in opposite directions. If the circumference of the circular track is 100 m, then find at how many distinct points they will cross each other?

SSC CGL 19/07/2023 (2nd shift)

- (a) 3 (b) 2 (c) 10 (d) 5

Solution :- According to question,

Ratios of speed $\Rightarrow 2 : 3$

No of meeting points when two runner running in opposite direction = $(2 + 3) = 5$

Some Other concept

Example :- In a 1500 m race, X beats Y by 100 m and X beats Z by 240 m. By what distance does Y beat Z in the same race?
SSC CGL 18/07/2023 (3rd shift)

- (a) 160 m (b) 140 m (c) 150 m (d) 200 m

Solution :-

$$\begin{array}{rcl} Z & : & X & : & Y \\ & & 1500 & : & 1400 \\ 1260 & : & 1500 & & \end{array}$$

$$126 : 150 : 140$$

Then, Y beat Z by $= (140 - 126) = 14$ units

According to the question,

$$140 \text{ units} = 1500 = \frac{1500}{140} \times 14 = 150 \text{m}$$

So, Y beat Z by 150m .

in a race of 600 m, B can beat C by 60 m. By how many metres will A beat C in a race of 400 m ?

SSC CGL 08/12/2022 (3rd Shift)

- (a) 70 (b) 68 (c) 76 (d) 72

Solution :- The ratio of the distance covered by A and B in a race of 500m = $500 : 450 = 10 : 9$

The ratio of the distance covered by B and C in a race of 600m = $600 : 540 = 10 : 9$

on balancing the above ratio, we have :

$$\begin{array}{rcl} A & : & B & : & C \\ 10 & & 9 & &) \times 40 \\ & & 10 & & 9) \times 36 \end{array}$$

$$400 : 360 : 324$$

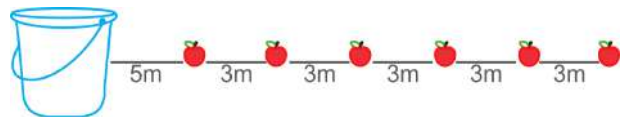
Clearly, we can see that A beat C in a race of 400 m by 76m i.e. $400 - 324$

Example :- In a race, a competitor has to collect 6 apples which are kept in a straight line on a track and a bucket is placed at the beginning of the track which is a starting point. The condition is that the competitor can pick only one apple at a time, run back with it and drop it in the bucket. If he has to drop all the apples in the bucket, how much total distance he has to run if the bucket is 5 meters from the first apple and all other apples are placed 3- meters apart ?

UPSC CSAT (07/08/2016)

- (a) 40 m (b) 50 m (c) 75 m (d) 150 m

Solution :-



Distance covered by competitor to pick 1st apple = $2 \times 5 = 10 \text{ m}$

Distance Covered by competitor to pick 2nd apple = $2 (5 + 3) = 16 \text{ m}$

Distance Covered by competitor to pick 3rd apple = $2 (5 + 3 \times 2) = 22 \text{ m}$

Distance Covered by competitor to pick 4th apple = $2 (5 + 3 \times 3) = 28 \text{ m}$

Distance Covered by competitor to pick 5th apple = $2 (5 + 3 \times 4) = 34 \text{ m}$

Distance Covered by competitor to pick 6th apple = $2 (5 + 3 \times 5) = 40 \text{ m}$

So, the total distance covered by competitor =

$$10 + 16 + 22 + 28 + 34 + 40 = 150 \text{ m}$$

Example :- In a linear race of 500 m, A can beat B by 50 m and

Vedic Maths

Vedic Maths is a collection of techniques/formula solving mathematical problem sets in a fast and easy way.

Addition Tricks

Method :- 1

Step 1 : Find out the number which is closest to the 10s multiple because it is easier to add those numbers.

i.e. 7, 8, 9 close to 10

21, 22, 23 close to 20

67, 68, 69, close to 70

97, 98, 99, close to 100 and so on.

Step 2 : Add the numbers which are the multiples of 10s

Step 3 : Add/Subtract the deficiency of numbers.

Example :- Add 38 and 97.

Solution :-

Step 1 : 38 and 97 are closer to 40 and 100 respectively.

Step 2 : Add 40 and 100 = 140

Step 3 : Both numbers have deficiency of 2 and 3 respectively.
So $140 - 5 = 135$

Example :- Add 82 and 167.

Solution :-

Step 1 : 82 and 167 are closer to 80 and 170 respectively.

Step 2 : Add 80 and 170 = 250

Step 3 : 82 has 2 extra and 167 has 3 deficiency.
So $250 + 2 - 3 = 249$

Example :- Add 146 and 467.

Solution :-

Step 1 : 146 and 467 are closer to 150 and 470 respectively.

Step 2 : Add 150 and 470 = 620

Step 3 : Both numbers have deficiency of 4 and 3 respectively.
So $620 - 7 = 613$

Method :- 2

Add ones with ones, tens with tens, hundreds with hundreds, thousand with thousand, and so on.

Example :- $357 + 576 + 457$

$(300 + 500 + 400) + (50 + 70 + 50) + (7 + 6 + 7) = 1200 + 170 + 20$

Hundredth place Tens place Ones place 1390

Method :- 3

Addition When numbers are close to each other :-

Step 1 : Double the smaller number.

Step 2 : Find the gap between the original number.

Step 3 : Now add both the results.

Example :- $63 + 71$

Step 1 : double the smaller number i.e. $63 \times 2 = 126$

Step 2 : gap between the original number i.e. $71 - 63 = 8$

Step 3 : Add both the results i.e. $126 + 8 = 134$

Example :- $615 + 632$

Step 1 : double the smaller number i.e. $615 \times 2 = 1230$

Step 2 : gap between the original number i.e. $632 - 615 = 17$

Step 3 : add both the results i.e. $1230 + 17 = 1247$

Subtraction Tricks

Method :- 1

Example :- $357 + 576 - 475$

$(300 + 500 - 400) + (50 + 70 - 70) + (7 + 6 - 5) = 400 + 50 + 8$

Hundredth place Tens place Ones place 458

Method :- 2

When double digit subtracted from its reversed number :-

Step 1 : Find the difference between the digits.

Step 2 : multiply the difference by 9

Example :- $63 - 36$

Step 1 : Find the difference between the digits i.e. $6 - 3 = 3$

Step 2 : Multiply the difference by 9 i.e. $3 \times 9 = 27$

Multiplication Tricks

(a) For 2 digit Multiplications :-

Method :- 1

Example :- 81×92

Step 1 : Multiplying vertically the units place

$$\begin{array}{r} 81 \\ \times 92 \\ \hline \end{array}$$

Step 2 : Multiplying in a cross pattern and adding the results.

For example, $8 \times 2 = 16$ and $9 \times 1 = 9$ and $16 + 9 = 25$.

$$\begin{array}{r} 81 \\ \times 92 \\ \hline 52 \end{array}$$
 2 carry

Step 3 : Multiplying the tens digit vertically.

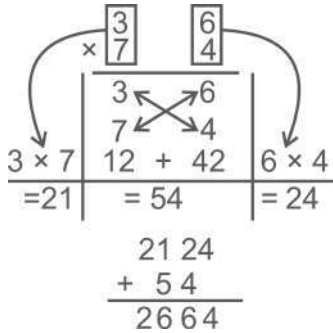
For example, $8 \times 9 = 72 + 2$ (carry) = 74

$$\begin{array}{r} 81 \\ \times 92 \\ \hline 7452 \end{array}$$

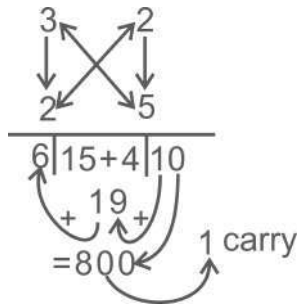
Example :- 29×81

$$\begin{array}{r} 29 \\ \times 81 \\ \hline 2349 \end{array}$$

Example :- 36×74



Example :- 32×25



Method :- 2

When even number is multiplied by number ending with 5 :-

Step 1 : Double the number which ends with 5.

Step 2 : Given even number is divided by 2

Step 3 : Now multiplying both the resultants

Example :- Find $36 \times 65 = ?$

Step 1 : Double the number which is ending with 5
i.e. $65 \times 2 = 130$

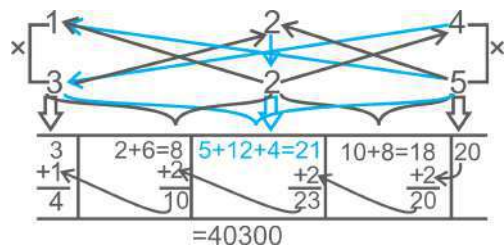
Step 2 : Given Even number is divided 2 .so, $\frac{36}{2} = 18$

Step 3 : Multiply both the resultant = $130 \times 18 = 2340$

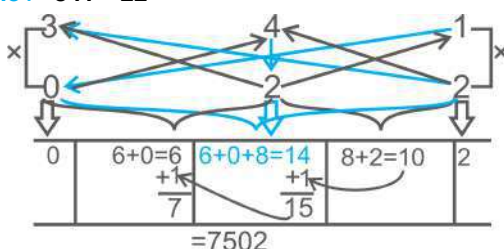
(b) For 3 digit Multiplications :-

Method :- 1

Example :-



Example :- 341×22



Method :- 2

Multiplication of Two Numbers in which one Number Contains only 9 :-

Step 1 : The number being multiplied by 9's is first reduced by 1 and that result will be the 1st part.

Step 2 : subtract the resultant from the number consisting of 9.

Step3 : combine both the numbers.

Example :- 423×999

Step 1 : The number being multiplied by 9's is first reduced by 1 and that result will be the 1st part i.e. $423 - 1 = 422$

Step 2 : subtract the resultant from number consist of 9 i.e. $999 - 422 = 577$

Step 3 : combine both the numbers i.e. 422,577

Method :- 3

Multiplying 2 three digit number having zero between them

$203 \times 809 :-$

$2 \times 8 = 16, 2(9) + 3(8) = 42, 3 \times 9 = 27$

Now, combining all three outcomes, we get : 164227

(c) Multiplying numbers which are close to powers of 10 :-

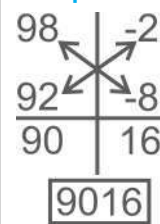
Step 1 : Find how much the number is more or less than the power of 10 and write it on the right side of the vertical line.

Step 2 : Either cross subtract or cross add. This will form the first part of the final answer.

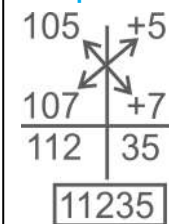
Step 3 : Multiply the right side of the vertical line and this shall form the right side of the result.

1). Here base is 100

Example :- 98×92



Example :- 105×107



2). Here base is 1000.



(d) Multiplication with 5, 25, 125, 625

$5 \rightarrow \frac{10}{2}, 25 \rightarrow \frac{100}{4}, 125 \rightarrow \frac{1000}{8}, 625 \rightarrow \frac{10000}{16}$

1). Multiplication of a Number by 5

$N \times 5$ means $N \times \frac{10}{2}$

$36 \times 5 = 36 \times \frac{10}{2} = 180$

$78 \times 5 = 78 \times \frac{10}{2} = 390$

$128 \times 5 = 128 \times \frac{10}{2} = 640$

$946 \times 5 = 946 \times \frac{10}{2} = 4730$

2). Multiplication of a Number by 25

As we know that $25 = \frac{100}{4}$

Step 1 : Divide the number by 4

Step 2 : Use (00) with the result .

Example :- Find 748×25

Step 1 : Divide the number by 4 $\Rightarrow \frac{748}{4} = 187$

Step 2 : Use (00) with the resultant $\Rightarrow 18700$

3). Multiplication of a Number by 125

$N \times 125$ means $N \times \frac{1000}{8}$

$$64 \times 125 = 64 \times \frac{1000}{8} = 8000$$

$$512 \times 125 = 512 \times \frac{1000}{8} = 64000$$

$$48 \times 125 = 48 \times \frac{1000}{8} = 6000$$

4). Multiplication of a Number by 625

$N \times 625$ Means by $N \times \frac{10000}{16}$

$$64 \times 625 = 64 \times \frac{10000}{16} = 40,000$$

$$80 \times 625 = 80 \times \frac{10000}{16} = 50,000$$

(e) Multiplying by 101 or 1001 :-

$$42 \times 101 = 4242$$

$$53 \times 101 = 5353$$

$$87 \times 1001 = 87087$$

$$92 \times 1001 = 92092$$

$$123 \times 1001 = 123123$$

$$456 \times 1001 = 456456$$

(f) Multiplying an even number by a decimal number which is a multiple of 5 :-

$$54 \times 1.5 = 54 \times \frac{3}{2} = 81$$

$$42 \times 2.5 = 42 \times \frac{5}{2} = 105$$

$$86 \times 3.5 = 86 \times \frac{7}{2} = 43 \times 7 = 301$$

$$44 \times 4.5 = 44 \times \frac{9}{2} = 198$$

(g) Multiplication of an even number by 15 :-

$N \times 15 \Rightarrow (\text{adding } N \text{ to half of itself}) \times 10$

$$56 \times 15 \Rightarrow (56 + 28) \times 10 = 840$$

$$78 \times 15 \Rightarrow (78 + 39) \times 10 = 1170$$

(h) Multiplying a number with twice of itself :-

$N \times 2N = 2N^2$

$$17 \times 34 = 2(17)^2 = 2(289) = 578$$

$$21 \times 42 = 2(21)^2 = 2(441) = 882$$

Division**(a) Division with same double digit number (11 , 22 , 33 , 44 etc)**

In this type of division , the number are like when we add first and last digit of the number , we get the middle number i.e 792

$$(7 + 2) = 9$$

Example :- Find $792 \div 55$

Pick the first and last digit of the number and divide by 5

$$\frac{72}{5} = 14.4$$

Example :- Find $682 \div 77$

Pick the first and last digit of the number and divide by 7

$$\frac{62}{7} = 8.85$$

(b) Division of number with 25

As we know that , $25 = \frac{100}{4}$

Example :- Find $35 \div 25$

Just 4 time the given number and place decimal after 2 place from the right hand side

$$35 \times 4 = 140 = 1.4$$

Example :- Find $354 \div 25$

Just 4 time the given number and place decimal after 2 place from the right hand side

$$354 \times 4 = 1416 = 14.16$$

(c) Division of number with 5

As we know that $5 = \frac{10}{2}$

Example :- Find $47 \div 5$?

Double the number and place the decimal after one place from the right hand side .

$$47 \times 2 = 94 = 9.4$$

Example :- Find $576 \div 5$?

Double the number and place the decimal after one place from the right hand side .

$$576 \times 2 = 1152 = 115.2$$

(d) Division of number with 0.125

As we know that $0.125 = \frac{1}{8}$

Example :- Find $235 \div 0.125$

Just 8 times the given number

$$235 \times 8 = 1880$$

Example :- Find $28 \div 0.125$?

Just 8 times the given number

$$28 \times 8 = 224$$

(e) Division of number with 9 , 99 , 999

If there is one 9 and there is one digit in the given number .

Example :- Find $(4 \div 9) = 0.44444.....$

If there are two 9 and there are two digits in the given number .

Example :- Find $53 \div 99 = 0.535353.....$

If there are three 9 and two digits in the given number.

Example :- $76 \div 999 = 0.076076076.....$

Tricks for Finding Square

(a) Square of Numbers with Unit Digit 5 :

Formula : 'By one more than previous one'

$(AB)^2 = A \times (A + 1)$ ending with 25 Here (B = 5 Always)

$25^2 = 2 \times (2 + 1) \dots 25 = 2 \times 3 \dots 25 = 625$

$35^2 = 3 \times 4 \dots 25 = 1225$

$65^2 = 6 \times 7 \dots 25 = 4225$

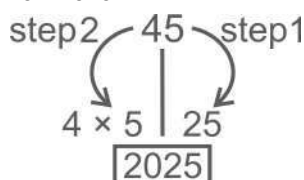
$135^2 = 13 \times 14 \dots 25 = 18225$

Example :- $(45)^2$

Step 1 - For last part simply put $(5)^2 = 25$

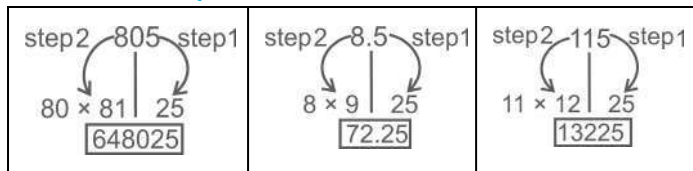
Step 2 - Multiply 4 with the next number i.e. 5

So, $45 \times 45 = 2025$

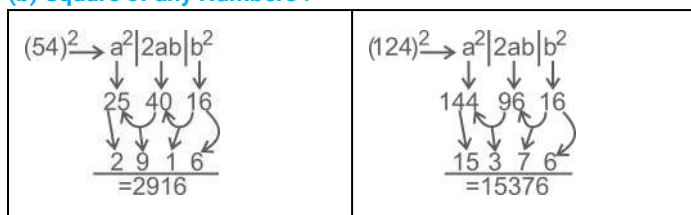


Note :- This method is applicable to numbers of any size.

Some more Examples :-



(b) Square of any Numbers :



Note :- While combining them start from the right end.

(c) Square of any Numbers near 50 and below 50 :

Step 1 - Check the difference of number from 50

Step 2 - Subtract the difference from 25 and write the first part from the left.

Step 3 - Write square of difference as first part from right

Example :-

$(43)^2$
 Step 1 : $50 - 43 = 7$
 Step 2 : $25 - 7 = 18$
 Step 3 : $18 | 49 \rightarrow 1849$

Note :- squares of numbers below 50 will always be below 2500 so our base for subtraction will be 25.

(d) Square of any Numbers near 50 and above 50 :-

Step 1 - Check the difference of 50 from the number.

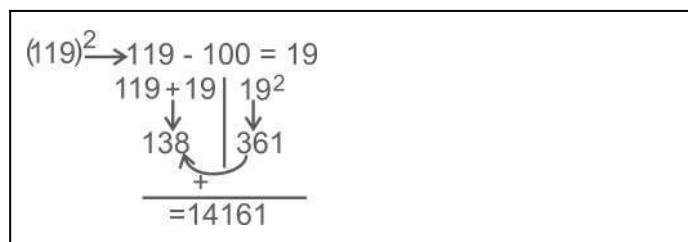
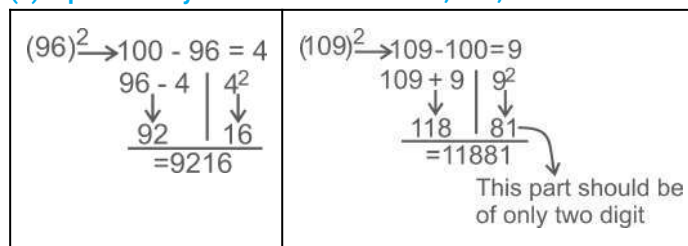
Step 2 - Add the difference in 25 and write as the first part from left.

Step 3 - Write square of difference as first part from right

Example :-

$(58)^2$
 Step 1 : $58 - 50 = 8$
 Step 2 : $25 + 8 = 33$
 Step 3 : $33 | 64 \rightarrow 3364$

(e) Square of any Numbers near base 10, 100, 1000 etc :-

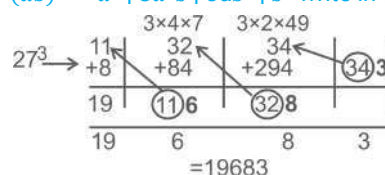


Trick to finding the Cube

Type - 1

We know that $(a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$

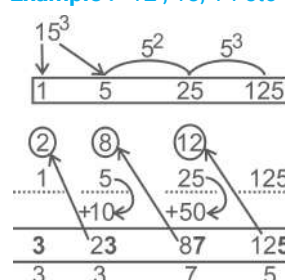
$(ab)^3 = a^3 | 3a^2b | 3ab^2 | b^3$ write in this form



Type - 2

When the first digit is 1.

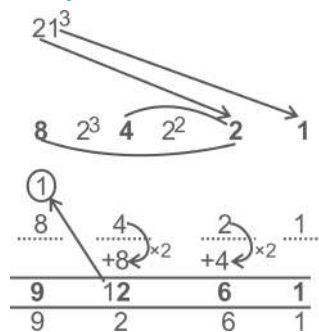
Example :- 12, 13, 14 etc



Type - 3

When the last digit is 1.

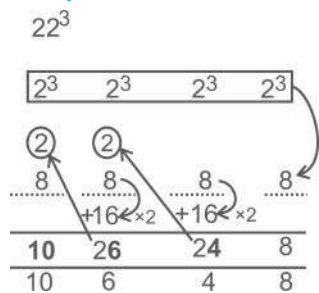
Example :- 21, 31, 41 etc



Type - 4

When both digits are the same.

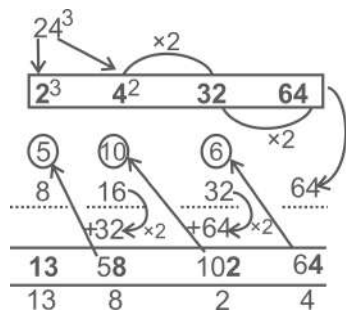
Example :- 22, 33, 44 etc



Type - 5

Mixed number

Example :- 24, 26, 46 etc



Square Root and Cube Root

Square Root

The square root of a number is a value that, when multiplied by itself, gives the original number.

i.e. 36 has two square roots 6 and -6;

$6^2 = 36$ and $-6^2 = 36$, hence we write $\sqrt{36} = \pm 6$

1). Square root by Factorization method :-

Example:- Find square root of 1089

Solution:- First find the factors of 1089.

$$\begin{array}{r} 3 \overline{)1089} \\ \underline{3 363} \\ 11 121 \\ \underline{11 121} \\ 0 \end{array}$$

$$1089 = 3 \times 3 \times 11 \times 11$$

$$\sqrt{1089} = 3 \times 11 = 33$$

2). Square root by Division method :-

Example :- Find square root of 2304.

Solution :- First find the factors of 2304.

$$\begin{array}{r} 48 \\ 4 \overline{)2304} \\ \underline{+4 16} \\ 88 704 \\ \underline{704} \\ 0 \end{array}$$

Note:- Make group of two digit from right $\sqrt{2304} = 48$

3). Find square root with some tricks :-

Exam Hall Approach :-

Example :- Find square root of 2304.

Solution :-

Step 1 : Take the unit digit of given number ,then write those number whose square have same unit digit as the given number

$$\begin{array}{c} 2304 \\ \swarrow \quad \searrow \\ 2^2 = 4 \quad 8^2 = 64 \end{array}$$

Step 2 : Now neglect the last two number from right side , consider first two number i.e 23

Step 3 : Then write the numbers (4 and 5) whose square contains 23. i.e. $4^2 < 23 < 5^2$

Note : Always choose the smaller number

$$\begin{array}{c} 23 \overline{)04} \text{ Eliminate} \\ \swarrow \quad \searrow \\ 4^2 = 16 \quad 5^2 = 25 \end{array}$$

Now we have two values 42 and 48

Step 4 : Multiply the smaller number with its successive number ($a \times (a + 1)$) i.e. $4 \times 5 = 20$

Now, $20 < 23$

Note: if the product is smaller then we will take larger number i.e.48 and if the product is greater then we will take smaller number i.e. 42

$$\begin{array}{c} 23 \overline{)04} \text{ eliminate} \\ \searrow \\ 4(4+1) = 20 < 23 \end{array}$$

So the answer will be 48.

Example :- $\sqrt{676} = ?$

Solution :-

$$\begin{array}{c} 6 \overline{)76} \text{ eliminate} \\ \swarrow \quad \searrow \\ 2(2+1) = 6 = 6 \end{array}$$

Where 6 lies between squares of 2 and 3 , but we know that we always take a smaller number (2).

So that , we have two number 24 and 26

Now ($a \times (a + 1)$) = $2 \times 3 = 6$

Note : if the product($2 \times 3 = 6$) is equal to the remaining digit (6) after elimination then we will take larger number i.e.26

Find the square root of non perfect square number :-

Example :- Find the square root of 171

Solution :-

Step 1 : Break the number with nearest perfect square number

$$\text{i.e } \sqrt{171} = \sqrt{169 + 2}$$

Step 2 : As we know $\sqrt{169} = 13$

$$\begin{array}{c} \sqrt{169 + 2} \\ | \quad + \frac{2}{26} \\ 13 \quad \times 2 \end{array}$$

$$\Rightarrow \sqrt{171} = 13 + 0.076 = 13.076$$

Example :- Find the square root of 395

Solution :-

Step 1 : Break the number with nearest perfect square number

$$\text{i.e } \sqrt{395} = \sqrt{400 - 5}$$

Step 2 : As we know that $\sqrt{400} = 20$

$$\begin{array}{r} \sqrt{400 - 5} \\ | \quad \quad \quad \frac{5}{40} \\ 20 \quad - \quad \frac{5}{40} \\ \times 2 \end{array}$$

$$\Rightarrow \sqrt{395} = 20 - 0.125 = 19.875$$

Important Table

If any number ends with	Square root will end with
1	1 or 9
4	2 or 8
9	3 or 7
16	4 or 6
25	5

1). If any number ends with 1, its square root will end with 1 or 9, given that the number is a perfect square.

i.e. $\sqrt{81} = 9$, $\sqrt{121} = 11$

2). If any number ends with 4, its square root will end with 2 or 8, given that the number is a perfect square.

i.e. $\sqrt{324} = 18$, $\sqrt{144} = 12$

3). If any number ends with 5, its square root will end with 5, given that the number is a perfect square.

i.e. $\sqrt{225} = 15$, $\sqrt{625} = 25$

4). If any number ends with 6, its square root will end with 6, given that the number is a perfect square.

i.e. $\sqrt{256} = 16$, $\sqrt{676} = 26$

5). If any number ends with 9, its square root will end with 3 or 7, given that the number is a perfect square.

i.e. $\sqrt{49} = 7$, $\sqrt{169} = 13$

6). The unit digit of the square of any number is not 2, 3, 7 or 8.

7). The repeated sum of the digits of a perfect square is always 1, 4, 7 or 9.

Cube Root

Cube root of a number is a value which when multiplied by itself thrice or three times produces the original value.

i.e. the cube root of 27, denoted as $\sqrt[3]{27}$, is 3, because when we multiply 3 by itself three times we get $3 \times 3 \times 3 = 27 = 3^3$.

Example :- Find the cube root of 64.

Solution :- To find the cube root of 64, we need to use the prime factorisation method.

$$64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64 = 4 \times 4 \times 4 = 64 = 4^3$$

Now taking the cube root on both the sides,

we get ; $\sqrt[3]{64} = 4$

Example :- Find $\sqrt[3]{343}$

Solution :- By prime factorisation 343

$$= 7 \times 7 \times 7 = 343 = 7^3 \Rightarrow \sqrt[3]{343} = 7$$

N^3		Unit digit
1^3	1	1
2^3	8	8
3^3	27	7
4^3	64	4
5^3	125	5
6^3	216	6
7^3	343	3
8^3	512	2
9^3	729	9

Example :- Find the cube root of 6859

Solution :-

Step 1 : Neglect last three digit of given number

$$\begin{array}{c} 6 \quad 8 \quad 5 \quad 9 \\ \times \end{array}$$

Step 2: Now find the perfect cube number smaller than 6

i.e. $1^3 = 1$

Step 3: $\sqrt[3]{6859} = 19$

(Units digit of resultant number = The number, whose unit digit of the perfect cube is equal to the unit digit of the given number)

$$\begin{array}{c} 6 \quad 8 \quad 5 \quad 9 \\ \times \quad \quad \quad \searrow \\ \text{Unit digit of } 9^3 \end{array}$$

i.e. $9^3 = 729$ so unit digit of resultant value = 9

Example :- Find the cube root of 19683

Solution :-

Step 1 : Neglect last three digit of given number

$$\begin{array}{c} 19 \quad 6 \quad 8 \quad 3 \\ \times \end{array}$$

Step 2: Now find the perfect cube number smaller than 19

i.e. $2^3 = 8$

Step 3: $\sqrt[3]{19683} = 27$

(Units digit of resultant number = The number, whose unit digit of the perfect cube is equal to the unit digit of the given number)

$$\begin{array}{c} 19 \quad 6 \quad 8 \quad 3 \\ \times \quad \quad \quad \searrow \\ \text{Unit digit of } 7^3 \end{array}$$

i.e. $7^3 = 343$ so unit digit of resultant value = 7

Example :- Find the cube root of 274625

Solution :-

Step 1 : Neglect last three digit of given number

$$\begin{array}{c} 2 \quad 7 \quad 4 \quad 6 \quad 2 \quad 5 \\ \times \end{array}$$

Step 2 : Now find the perfect cube number smaller than 274

i.e $6^3 = 216$

Step 3 : $\sqrt[3]{274625} = 65$

(Units digit of resultant number = The number, whose unit digit of the perfect cube is equal to the unit digit of the given number)

Find the cube root of non perfect cube number :-

Example : Find the cube root of 345

Solution :

Step 1 : Break the number with nearest perfect cube number

i.e $\sqrt[3]{345} = \sqrt[3]{343 + 2}$

Step 2 : As we know $\sqrt[3]{343} = 7$

$$\begin{array}{r} \sqrt[3]{343 + 2} \\ \downarrow \\ 7 + \frac{2}{\textcircled{3} \times 7 \times 7} \\ \text{Fixed No} \\ \text{2 times} \end{array}$$

$\Rightarrow \sqrt[3]{345} = 7 + 0.0136 = 7.0136$

Square Mirrors

- $14^2 + 87^2 = 78^2 + 41^2$
- $15^2 + 75^2 = 57^2 + 51^2$
- $26^2 + 97^2 = 79^2 + 62^2$
- $27^2 + 96^2 = 69^2 + 72^2$

Conditions to be a perfect square :-

1. A number should not have 2, 3, 7, or 8 at its unit digit.
2. If the unit digit of a number 5, then the tens digit of its square is always 2.
3. If a square number ends with zero, then the number of zeroes must be in the multiple of 2. i.e. $20^2 = 400$, $100^2 = 10000$

Squares, Cubes and Square roots

Squares

$1^2 = 1$	$11^2 = 121$	$21^2 = 441$
$2^2 = 4$	$12^2 = 144$	$22^2 = 484$
$3^2 = 9$	$13^2 = 169$	$23^2 = 529$
$4^2 = 16$	$14^2 = 196$	$24^2 = 576$
$5^2 = 25$	$15^2 = 225$	$25^2 = 625$
$6^2 = 36$	$16^2 = 256$	$26^2 = 676$
$7^2 = 49$	$17^2 = 289$	$27^2 = 729$
$8^2 = 64$	$18^2 = 324$	$28^2 = 784$
$9^2 = 81$	$19^2 = 361$	$29^2 = 841$
$10^2 = 100$	$20^2 = 400$	$30^2 = 900$

Cubes

$1^3 = 1$	$11^3 = 1331$	$21^3 = 9261$
$2^3 = 8$	$12^3 = 1728$	$22^3 = 10648$
$3^3 = 27$	$13^3 = 2197$	$23^3 = 12167$
$4^3 = 64$	$14^3 = 2744$	$24^3 = 13824$
$5^3 = 125$	$15^3 = 3375$	$25^3 = 15625$
$6^3 = 216$	$16^3 = 4096$	$26^3 = 17576$
$7^3 = 343$	$17^3 = 4913$	$27^3 = 19683$
$8^3 = 512$	$18^3 = 5832$	$28^3 = 21952$
$9^3 = 729$	$19^3 = 6859$	$29^3 = 24389$
$10^3 = 1000$	$20^3 = 8000$	$30^3 = 27000$

Square Roots

$\sqrt{1} = 1$	$\sqrt{5} = 2.236$
$\sqrt{2} = 1.414$	$\sqrt{6} = 2.449$
$\sqrt{3} = 1.732$	$\sqrt{7} = 2.645$
$\sqrt{4} = 2$	$\sqrt{8} = 2.828$

Expression	Result	Expression	Result	Expression	Result
2^1	2	3^1	3	4^1	4
2^2	4	3^2	9	4^2	16
2^3	8	3^3	27	4^3	64
2^4	16	3^4	81	4^4	256
2^5	32	3^5	243	4^5	1024
2^6	64	3^6	729	5^1	5
2^7	128	3^7	2187	5^2	25
2^8	256	3^8	6561	5^3	125
2^9	512	3^9	19683	5^4	625
2^{10}	1024	3^{10}	59049	5^5	3125

Number Series

Basics of Series - In this chapter a series of numbers are given, in which they are arranged under a particular rule. These series are formed in many ways and different rules work in them. Additionally, sometimes one of the terms in this series is given incorrectly and it is necessary to find out which term is incorrect in the given question.

Types of Series Pattern

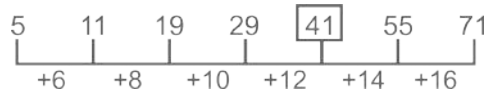
Type - 1 (BASED ON)

Addition / Subtraction

This pattern will be followed if the range of difference is less.

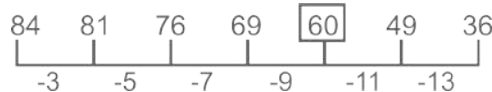
Example :- 5, 11, 19, 29, ?, 55, 71

Solution :- Logic → Series is based on addition



Example :- 84, 81, 76, 69, ?, 49, 36

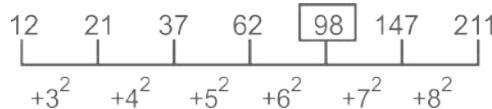
Solution :- Logic → Series is based on subtraction



Addition / Subtraction of Perfect Square

Example :- 12, 21, 37, 62, 98, 147, ?

Solution :- Logic → Series is based on a perfect square.

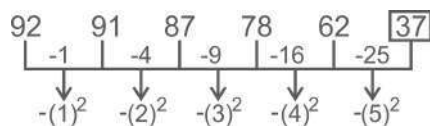


Example :- 92, 91, 87, 78, 62, ?

SSC MTS 12/09/2023 (3rd Shift)

(a) 37 (b) 38 (c) 35 (d) 36

Solution :-



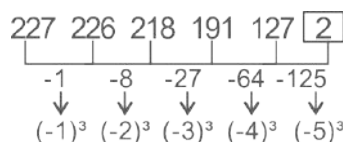
Addition / Subtraction of Perfect Cube

Example :- 227, 226, 218, 191, 127, ?

SSC MTS 12/09/2023 (2nd Shift)

(a) 3 (b) 4 (c) 2 (d) 1

Solution :-

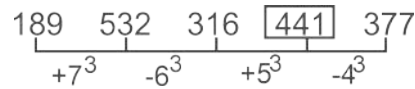


Example :- 189, 532, 316, ?, 377

SSC CGL Tier II (03/03/2023)

(a) 384 (b) 291 (c) 441 (d) 405

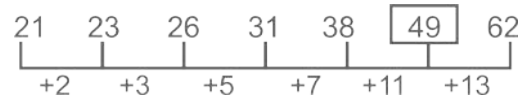
Solution :-



Addition / Subtraction of Prime Numbers

Example :- 21, 23, 26, 31, 38, ?, 62

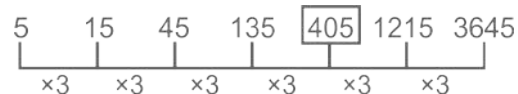
Solution :- Logic → Series is based on a prime number.



Multiplication / Division

Example :- 5, 15, 45, 135, ?, 1215, 3645

Solution :- Logic → Series is based on multiplication



Example :- 7500, 1500, 300, ?, 12

Solution :- Logic → Series is based on Division

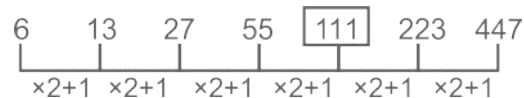


Multiplication And Addition

Example :- 6, 13, 27, 55, ?, 223, 447

Solution :-

Logic → Series is based on multiplication and addition.



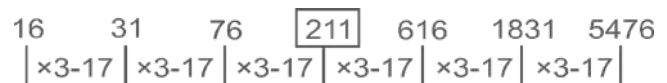
Multiplication And Subtraction

Example :- 16, 31, 76, ?, 616, 1831, 5476

SSC MTS 05/09/2023 (2nd Shift)

(a) 217 (b) 201 (c) 207 (d) 211

Solution :-



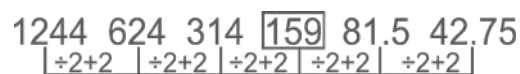
Division And Addition

Example :- 1244, 624, 314, ?, 81.5, 42.75

SSC CHSL 27/05/2022 (Evening)

(a) 156 (b) 159 (c) 155.5 (d) 157

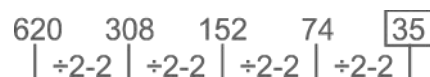
Solution :-



Subtraction And Division

Example :- 620, 308, 152, 74, ?

Solution :-



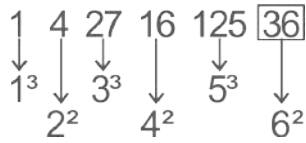
Type - 2

Perfect Cube Series

Example :- 1, 4, 27, 16, 125, ?

SSC CGL 24/07/2023 (2nd shift)

(a) 216 (b) 36 (c) 343 (d) 49

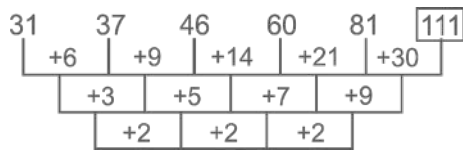
Solution :-

More Than One Tier Series

Example :- 31, 37, 46, 60, 81, ?

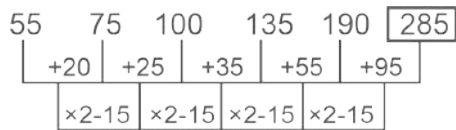
SSC CGL 24/07/2023 (4th shift)

(a) 117 (b) 106 (c) 122 (d) 111

Solution :-**Example :-** 55, 75, 100, 135, 190, ?

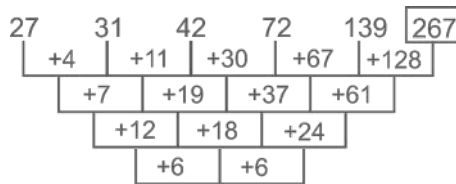
SSC CPO 10/11/2022 (Afternoon)

(a) 280 (b) 285 (c) 295 (d) 290

Solution :-**Example :-** 27, 31, 42, 72, 139, ?

SSC CHSL 06/06/2022 (Morning)

(a) 267 (b) 264 (c) 271 (d) 355

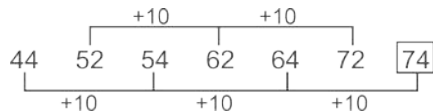
Solution :-

Alternate Mixed Series

Example :- 44, 52, 54, 62, 64, 72, ?

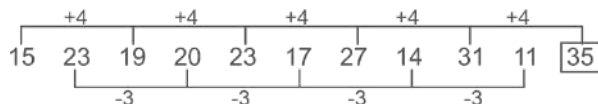
SSC MTS 04/05/2023 (Morning)

(a) 96 (b) 79 (c) 74 (d) 85

Solution :-**Example :-** 15, 23, 19, 20, 23, 17, 27, 14, 31, 11, ?

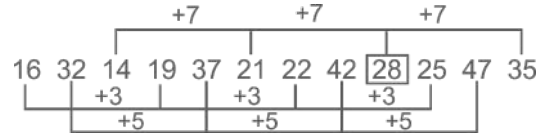
SSC MTS 11/05/2023 (Afternoon)

(a) 35 (b) 15 (c) 19 (d) 32

Solution :-**Example :-** 16, 32, 14, 19, 37, 21, 22, 42, ?, 25, 47, 35

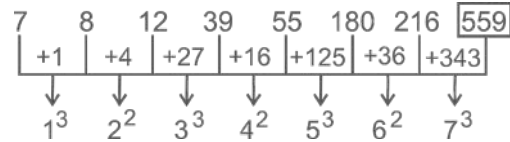
SSC MTS 12/07/2022 (Afternoon)

(a) 34 (b) 26 (c) 32 (d) 28

Solution :-**Example :-** 7, 8, 12, 39, 55, 180, 216, ?

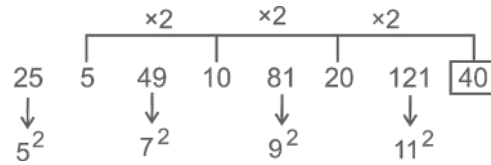
SSC CHSL 27/05/2022 (Evening)

(a) 555 (b) 559 (c) 595 (d) 599

Solution :-**Example :-** 25, 5, 49, 10, 81, 20, 121, ?

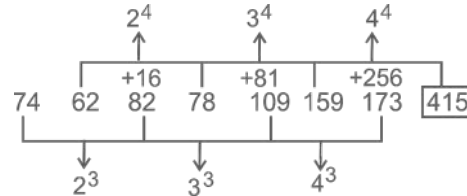
SSC CHSL 17/03/2020 (Evening)

(a) 30 (b) 40 (c) 25 (d) 35

Solution :-**Example :-** 74, 62, 82, 78, 109, 159, 173, ?

SSC CHSL 01/06/2022 (Morning)

(a) 223 (b) 375 (c) 415 (d) 295

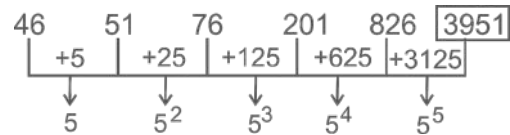
Solution :-

Increasing Power Series

Example :- 46, 51, 76, 201, 826, ?

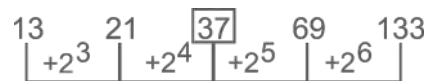
SSC CHSL 13/03/2023 (3rd Shift)

(a) 3757 (b) 3951 (c) 4853 (d) 3631

Solution :-**Example :-** 13, 21, ?, 69, 133

Graduate Level 02/08/2022 (Shift - 3)

(a) 52 (b) 37 (c) 33 (d) 41

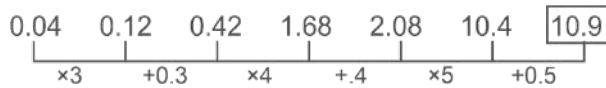
Solution :-

Decimal And Fractional

Example :- 0.04, 0.12, 0.42, 1.68, 2.08, 10.4, ?

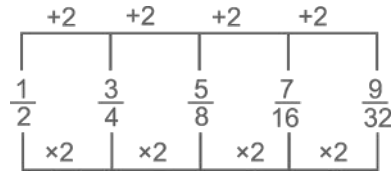
Graduate Level 27/06/2023 (Shift - 4)

(a) 12.3 (b) 12.9 (c) 11.8 (d) 10.9

Solution :-**Example :-** Which fraction comes next in the sequence

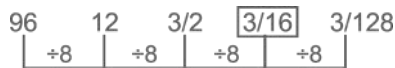
$$\frac{1}{2}, \frac{3}{4}, \frac{5}{8}, \frac{7}{16}, ?$$

(a) $\frac{9}{32}$ (b) $\frac{10}{17}$ (c) $\frac{11}{34}$ (d) $\frac{12}{35}$

Solution :-**Example :-** 96, 12, $\frac{3}{2}$, ?, $\frac{3}{128}$

SSC CPO 05/10/2023 (2nd Shift)

(a) $\frac{3}{12}$ (b) $\frac{3}{16}$ (c) $\frac{3}{64}$ (d) $\frac{1}{128}$

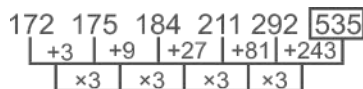
Solution :-

Mixed Series

Example :- 172, 175, 184, 211, 292, ?

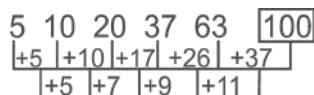
SSC CHSL 17/08/2023 (1st Shift)

(a) 535 (b) 537 (c) 534 (d) 532

Solution :-**Example :-** 5, 10, 20, 37, 63, ?

SSC CHSL 14/08/2023 (3rd Shift)

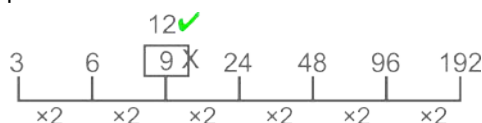
(a) 100 (b) 105 (c) 95 (d) 90

Solution :-

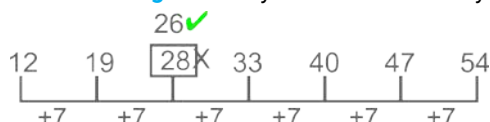
Type - 3

Wrong Number Series

In this type you will find two numbers wrong but on correcting one the other will automatically get corrected.

Example :- 3, 6, 9, 24, 48, 96, 192**Solution :- Logic** → every number is multiplied by 2 by the previous number.

Here, wrong term is 9, it should be replaced with 12.

Example :- 12, 19, 28, 33, 40, 47, 54**Solution :- Logic** → every number is added by 7.

Here, wrong term is 28, it should be replaced with 26.

Example :- Identify the number that does NOT belong to the following series.

1.5, 2, 3, 6, 18, 108, 1964

SSC MTS 21/07/2022 (Morning)

(a) 108 (b) 6 (c) 18 (d) 1964

Solution :- Here the next number is product of last 2 numbers

1.5, 2, 3, 6, 18, 108, 1964

$\times 1944$

Example :- In the given series one number is incorrect. Identify the INCORRECT number from among the options given.

3, 5, 7, 6, 10, 14, 12, 24, 28, 24, 40, 56

SSC CPO 13/12/2019 (Evening)

(a) 28 (b) 24 (c) 12 (d) 10

Solution :- Here 4 different series are given.

So 24 is wrong here.

Example :- In the following number series, two numbers have been put within brackets. Select the most appropriate option for these numbers in relation to their inclusion in the series.

1, 5, 17, (39), 65, 101, (145), 197

SSC CPO 09/12/2019 (Evening)

(a) The first bracketed number from the left is correct and the

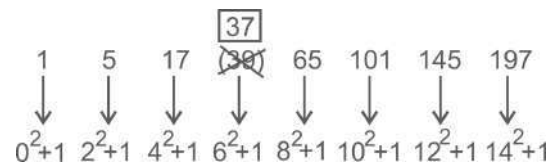
Second bracketed number from the left is incorrect.

(b) Both the bracketed numbers are incorrect.

(c) Both the bracketed numbers are correct.

(d) The first bracketed number from the left is incorrect and the

Second bracketed number from the left is correct.

Solution :-

So, 39 is wrong and it should be 37.

So, the first number in the bracket is wrong and the second is right.

Example :- Identify the number that is NOT related to the following series.

25, 28, 23, 30, 21, 33

SSC CPO 19/12/2019 (Morning)

(a) 21 (b) 30 (c) 23 (d) 33

Solution :-**Example :-** Identify the number that does NOT belong to the following series.

104, 108, 54, 58, 29, 31

SSC CPO 09/12/2019 (Morning)

(a) 29 (b) 54 (c) 58 (d) 31

Solution :-

Instead of 31 it should be 33.